1. ALL RESISTANCE VALUES ARE IN OHMS, 0.1 WATT +/- 5%.
2. ALL CAPACITANCE VALUES ARE IN MICROFARADS.
3. ALL CRYSTALS & OSCILLATOR VALUES ARE IN HERTZ.

J78 MLB CTO

LAST_MODIFIED=Mon Feb 24 09:56:42 2014

REV	ECN	DESCRIPTION OF REVISION	CK APPD	DATE
5	0002620513	ENGINEERING RELEASED		2014-02-24

DRAWING
TITLE=J17
ABBREV=DRAWING
LAST_MODIFIED=Mon Feb 24 09:56:42 2014

DRAWING TITLE: SCHEM,MLB CTO,J78

Apple Inc.

DRAWING NUMBER: 051-1635 SIZE: D
REVISION: 5.0.0
BRANCH: proto2
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SHEET: 1 OF 115

D

BOM Variants

BOM NUMBER	BOM NAME	BOM OPTIONS
985-1468	PCBA,MLB CTO,DEV,J78	DEVELOPMENT,J78_DEVEL
639-6799	PCBA,MLB CTO 3.5G,AMETHYST,VRAM_HYNIX,4GB,J78	J78,J78_COMMON,CPU:CTO,SSD:Y,GPU:AMETHYST,FB:4G_HYNIX,EEEE:FW6K
639-6160	PCBA,MLB CTO 3.5G,AMETHYST,VRAM_ELPIDA,4GB,J78	J78,J78_COMMON,CPU:CTO,SSD:Y,GPU:AMETHYST,FB:4G_ELPIDA,EEEE:FT04
639-6860	PCBA,MLB CTO 3.6G,AMETHYST,VRAM_HYNIX,4GB,J78	J78,J78_COMMON,CPU:CTO3.6,SSD:Y,GPU:AMETHYST,FB:4G_HYNIX,EEEE:FY53
639-6859	PCBA,MLB CTO 3.6G,AMETHYST,VRAM_ELPIDA,4GB,J78	J78,J78_COMMON,CPU:CTO3.6,SSD:Y,GPU:AMETHYST,FB:4G_ELPIDA,EEEE:FY54

Bar Code Labels / EEEE #'s

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
825-7896	1	LABEL,MLB,2D	EEEE_FRFN	CRITICAL	EEEE:FRFN
825-7896	1	LABEL,MLB,2D	EEEE_FRFL	CRITICAL	EEEE:FRFL
825-7896	1	LABEL,MLB,2D	EEEE_FW6K	CRITICAL	EEEE:FW6K
825-7896	1	LABEL,MLB,2D	EEEE_FT04	CRITICAL	EEEE:FT04
825-7896	1	LABEL,MLB,2D	EEEE_FY54	CRITICAL	EEEE:FY54
825-7896	1	LABEL,MLB,2D	EEEE_FY53	CRITICAL	EEEE:FY53

BOM Groups

BOM GROUP	BOM OPTIONS
J78_COMMON	COMMON,ALTERNATE,J78_COMMON1,J78_PROGPARTS,GPU_AMTH
J78_COMMON1	XDP,SPEAKERID,FBVDDQ_DFLT:1V5,RTCRST:Y
J78_PROGPARTS	SMC:PROG,BOOTROM:PROG,TBTROM:PROG,CIVROM:PROG,CAMROM:PROG,GPUROM:PROG
J78_DEVEL	XDP_CONN,LPCPLUS,DDRVREF_DAC,DEVEL_AUDIO,AP_ISNS:Y

J78 SCHEMATIC / PCB #'S

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
051-1635	1	SCH,MLB CTO,J78	SCH1	CRITICAL	J78
820-5029	1	PCBF,MLB CTO,J78	PCB1	CRITICAL	J78

J78 ALTERNATES

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
377S0147	377S0126		ALL	USB Diode Array
377S0155	377S0104		ALL	USB Diode
376S1218	376S1081		ALL	P/NCH DUAL FET
138S0803	138S0804		ALL	2.2UF CAPS SOFT
126-0161	126-0160		ALL	56UF CAPS,10X12 THD
197S0481	197S0480		Y1950	25MHX PCH XTAL
197S0479	197S0478		ALL	12MHZ CAM/NXP XTAL
372S0186	372S0185		ALL	Alternate Temp Diode
107S0251	107S0249		R5520	Alt 2mOhm sense
107S0254	107S0241		R5450,R5480	Alt 5mOhm sense
107S0255	107S0240		R5000,R5410	Alt 1mOhm sense
341S3912	341S3913		U3990	IC,ENETROM,ADESTO,V1,15,J78
138S0681	138S0638		ALL	10uF Caps
376S1106	376S0969		ALL	N-Ch FET
138S0860	138S0775		ALL	Single-source 1uF 402
138S0859	138S0788		ALL	Single-source 10uF
128S0398	128S0220		ALL	3.3V INPUT CAP
335S0812	335S0807		U5210	64 MBIT SPI SRL DUAL I/0 FLSH, SOIC8
378S0390	378S0140		ALL	DEGUG LEDS
378S0391	378S0140		ALL	DEBUG LEGS

C

CPUs

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
337S4531	1	HSW,SR14D,PRQ,C0,3.4,84W,4+2,1.2,6M,LGA	CPU	CRITICAL	CPU:ULT
337S00003	1	CPU,HSW,QF4G,3.5,QS,C0,4+2,84W,LGA	CPU	CRITICAL	CPU:CTO
337S00005	1	CPU,HSW,QF4D,3.6,QS,C0,4+2,84W,LGA	CPU	CRITICAL	CPU:CTO3.6

CPU SOCKET

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
511S0080	1	SOCKET.MOLEX,LGA1150,CPU-LF	U0500	CRITICAL	

B

ASICs

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
337S4541	1	IC,SR190,Z87,LPT-D,C2,PRQ,FCBGA708,23X22MM	U1100	CRITICAL	
337S4562	1	IC,QEWL,LPT-D,C2,SQS,FCBGA708,23X22MM	U1100	CRITICAL	PCH_TDP
338S1247	1	IC,TBT,FR-4C,A0,PRQ,CIO,SR1JC,FCBGA288	U2800	CRITICAL	
343S0616	1	IC,BCM57766A1,ENET&SD,8X8	U3900	CRITICAL	

GPU and VRAM

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
337S4654	1	IC,GPU,AMD,AMTHST,256-BIT,45X45,BGA2120	U8700	CRITICAL	GPU:AMETHYST
333S0685	4	IC,GDDR5,4GBIT,128MX32,5GBPS,GEMMA	U9200,U9250,U9300,U9350	CRITICAL	FB:4G_HYNIX
333S0685	4	IC,GDDR5,4GBIT,128MX32,5GBPS,GEMMA	U9400,U9450,U9500,U9550	CRITICAL	FB:4G_HYNIX
333S0766	4	IC,GDDR5,4GBIT,128MX32,5GBPS,GEMMA	U9200,U9250,U9300,U9350	CRITICAL	FB:4G_ELPIDA
333S0766	4	IC,GDDR5,4GBIT,128MX32,5GBPS,GEMMA	U9400,U9450,U9500,U9550	CRITICAL	FB:4G_ELPIDA

Programmable Parts

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
341S00032	1	IC,EFI,V0159,J78	U5210	CRITICAL	BOOTROM:PROG
335S0807	1	IC,64 MBIT SPI SERIAL FLASH	U5210	CRITICAL	BOOTROM:BLANK
341S3999	1	IC,SMC-B1,EXTERNAL,V2.21A13,P1,AM,J78	U5000	CRITICAL	SMC:PROG
338S1214	1	IC,SMC12-B1,40MHZ/50DMIPS,MCU,157BGA	U5000	CRITICAL	SMC:BLANK
341S00024	1	IC,T29,EPROM,FR,V23.1,J78A	U2890	CRITICAL	TBTROM:PROG
335S0915	1	IC,SRL SPI FLASH ROM,4MBIT,50MHZ,USON8	U2890	CRITICAL	TBTROM:BLANK
341S3912	1	IC,ENET ROM,NUMONYX,V1.15,J16/J16G/J17	U3990	CRITICAL	CIVROM:PROG
335S0854	1	1-MBIT,SPI FLASH ROM SOIC, HF	U3990	CRITICAL	CIVROM:BLANK
341S3778	1	IC,CAMERA FLASH,V7230,J16/J17	U4202	CRITICAL	CAMROM:PROG
335S0852	1	IC,FLASH,SPI,1MBIT,3V3	U4202	CRITICAL	CAMROM:BLANK
341S00009	1	IC,ROM,GFX,V1.0,J78	U9700	CRITICAL	GPUROM:PROG
335S0724	1	IC,CONF ROM,NEPT VBIOS,VTBD	U9700	CRITICAL	GPUROM:BLANK

A

SYNC_MASTER=J17_DINI SYNC_DATE=03/15/2013

PAGE TITLE

BOM Configuration

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DRAWING NUMBER 051-1635 SIZE D

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BRANCH proto2

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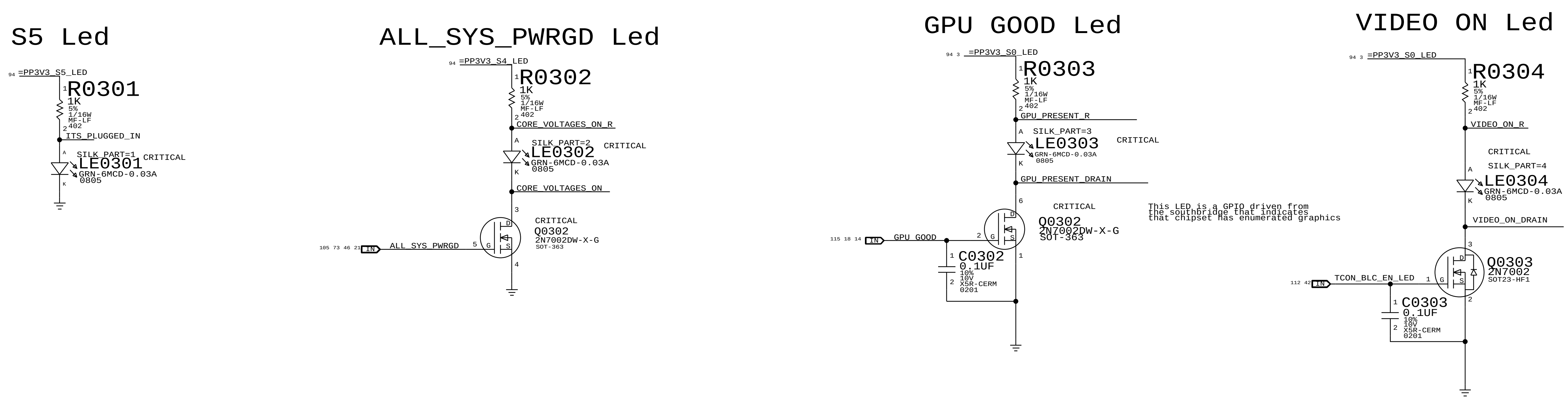

SYNC_MASTER=J78_MLB SYNC_DATE=02/21/2014

PAGE TITLE

DEBUG LEDS

Apple Inc.

DRAWING NUMBER 051-1635 SIZE D

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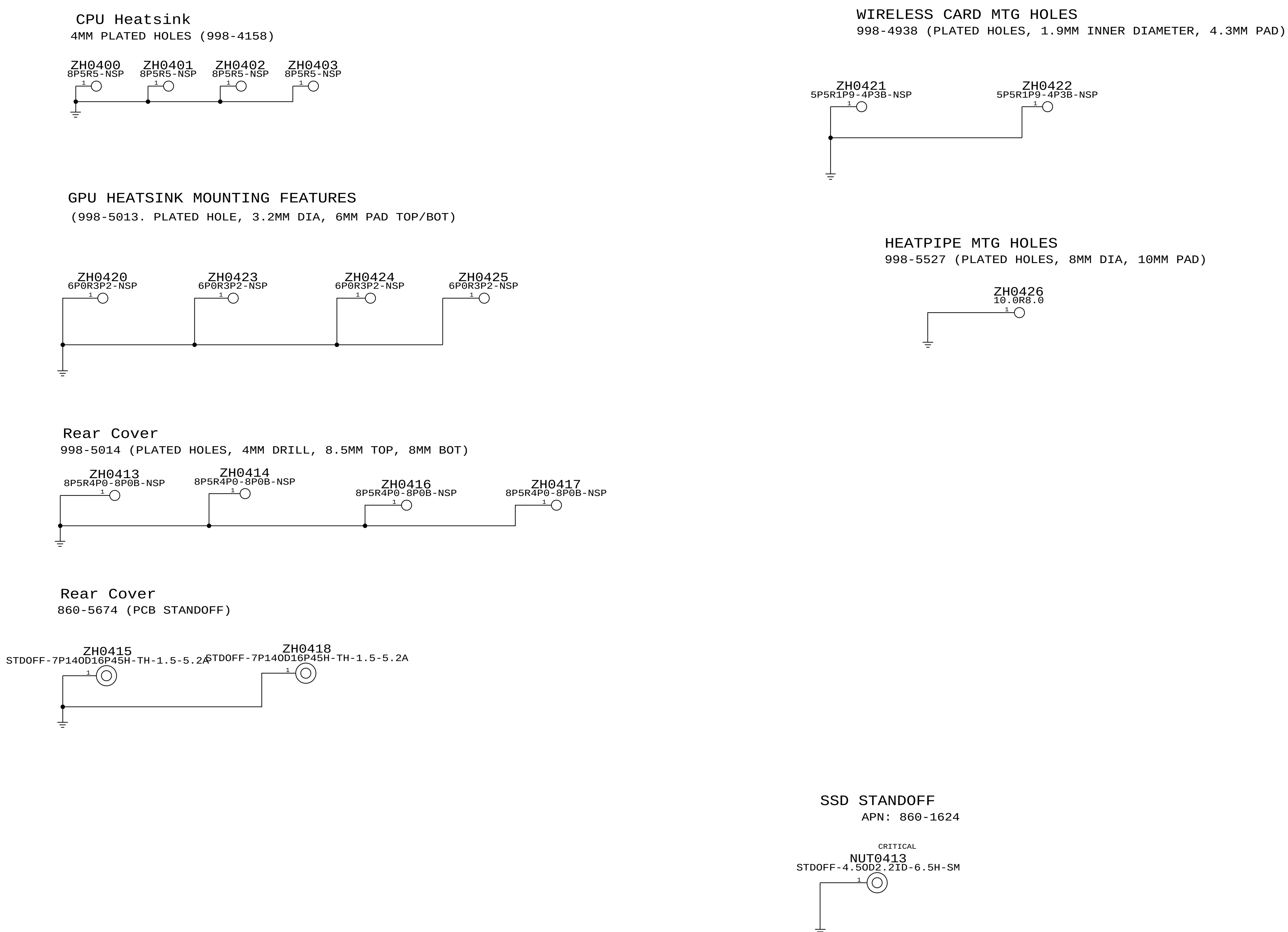

SYNC_MASTER=J78_MLB SYNC_DATE=02/21/2014

PAGE TITLE

Holes/PD parts

Apple Inc.

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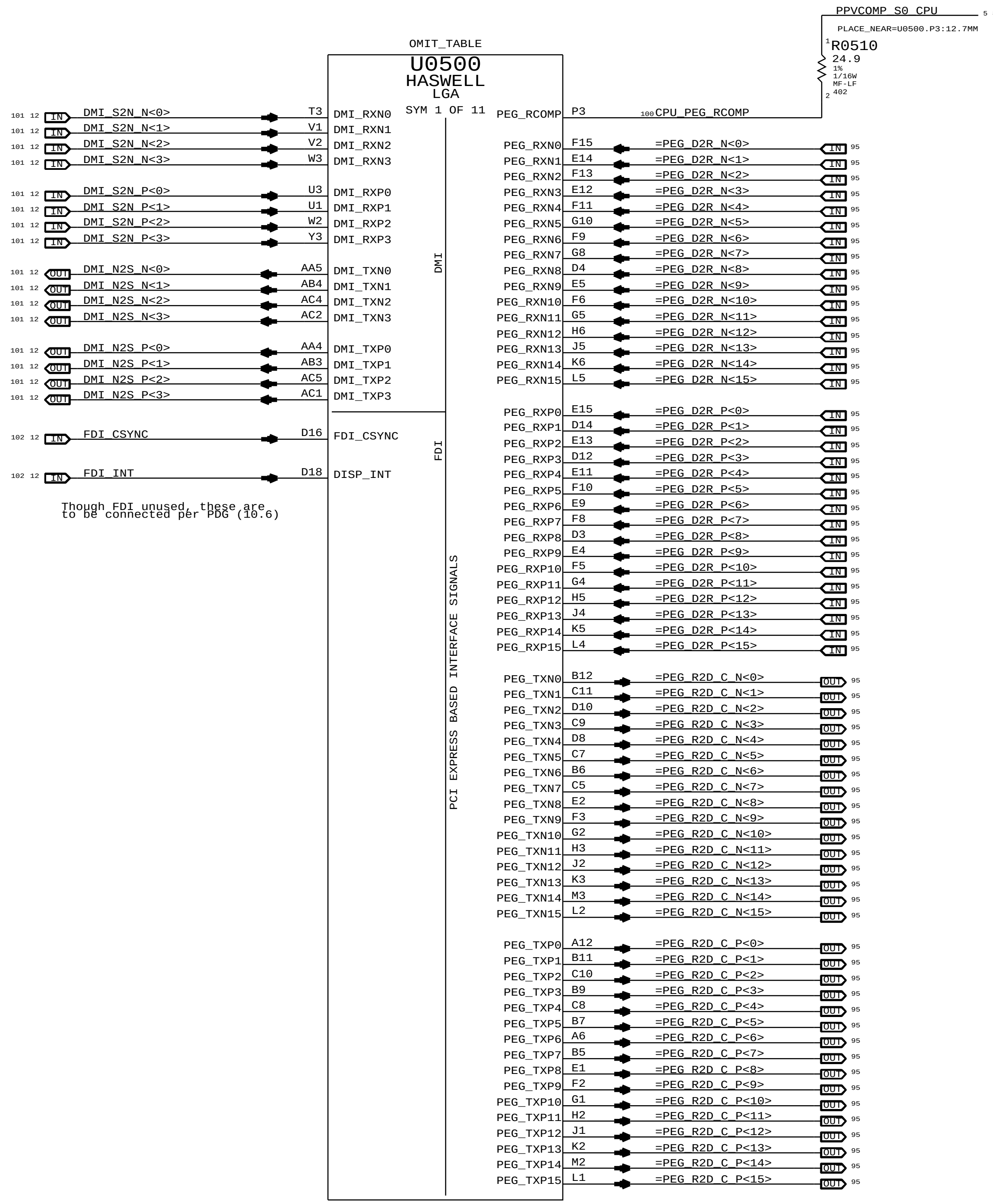

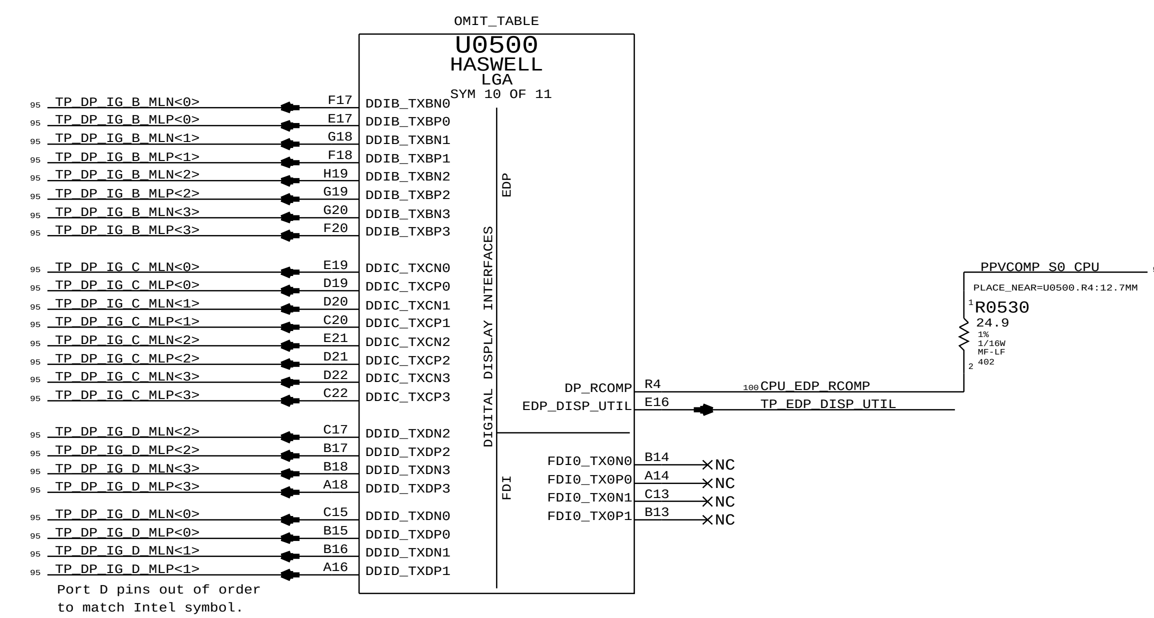

SYNC_MASTER=J78_MLB SYNC_DATE=02/21/2014

PAGE TITLE

CPU DMI/PEG/FDI/RSVD

DRAWING NUMBER 051-1635 SIZE D

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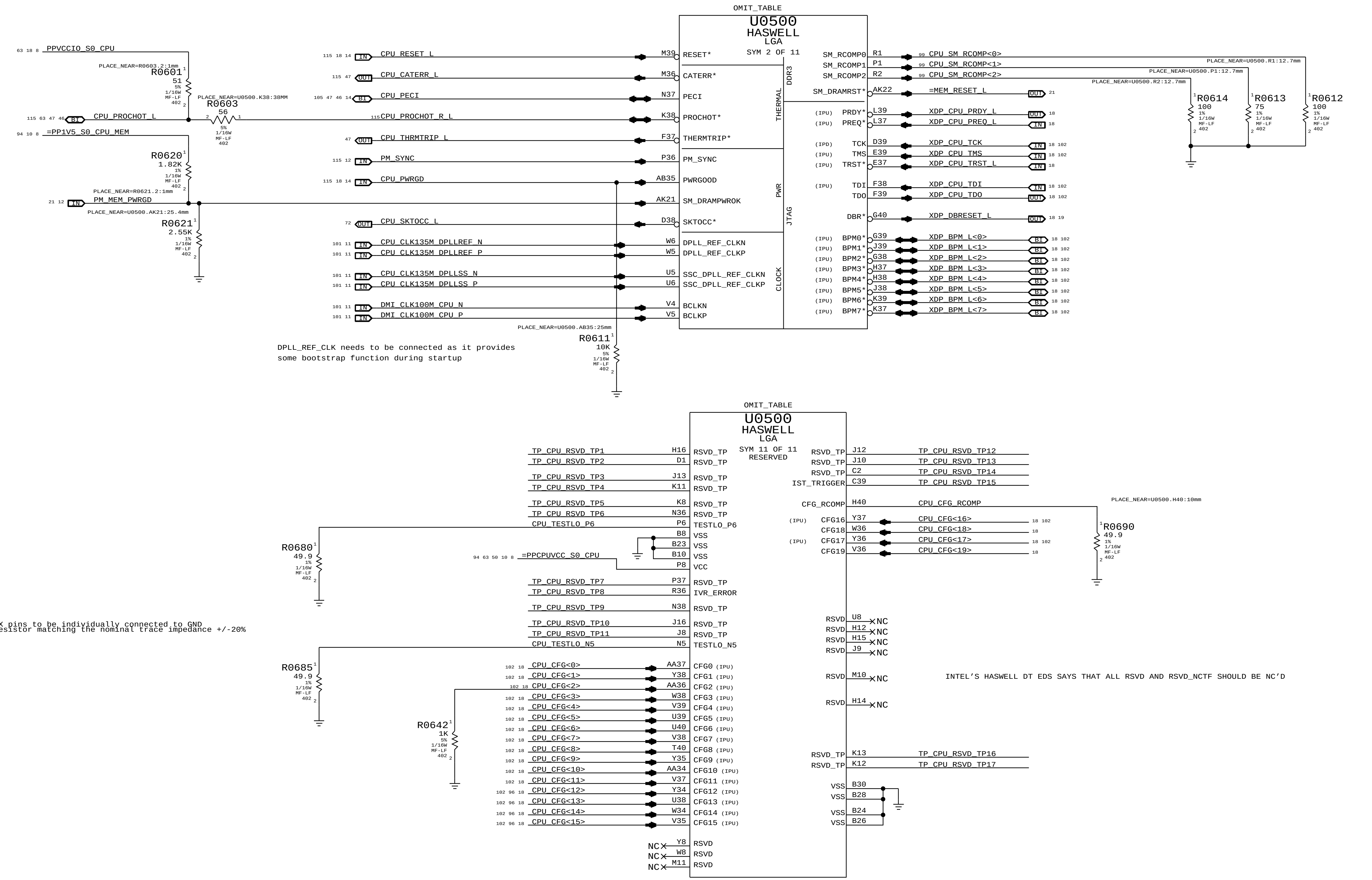

CFG [7] :PEG DEFER TRAINING	1 = (DEFAULT) IMMEDIATELY AFTER xxRESETB 0 = WAIT FOR BIOS	
CFG [6:5] :PCIE BIFURCATION	11 = 1 X16 (default) 10 = 2 X8 01 = RSVD 00 = X8, X4, X4	
CFG [4] :eDP ENABLE/DISABLE	1 = DISABLED(default) 0 = ENABLED	
CFG [3] :PCIE x4 LANE REVERSAL	1 = NORMAL OPERATION(default) 0 = LANES REVERSED	
CFG [2] :PCIE x16 LANE REVERSAL	1 = NORMAL OPERATION(default) 0 = LANES REVERSED	

SYNC_MASTER=J78_MLB SYNC_DATE=02/21/2014

PAGE TITLE

CPU Clock/Misc/JTAG/CFG

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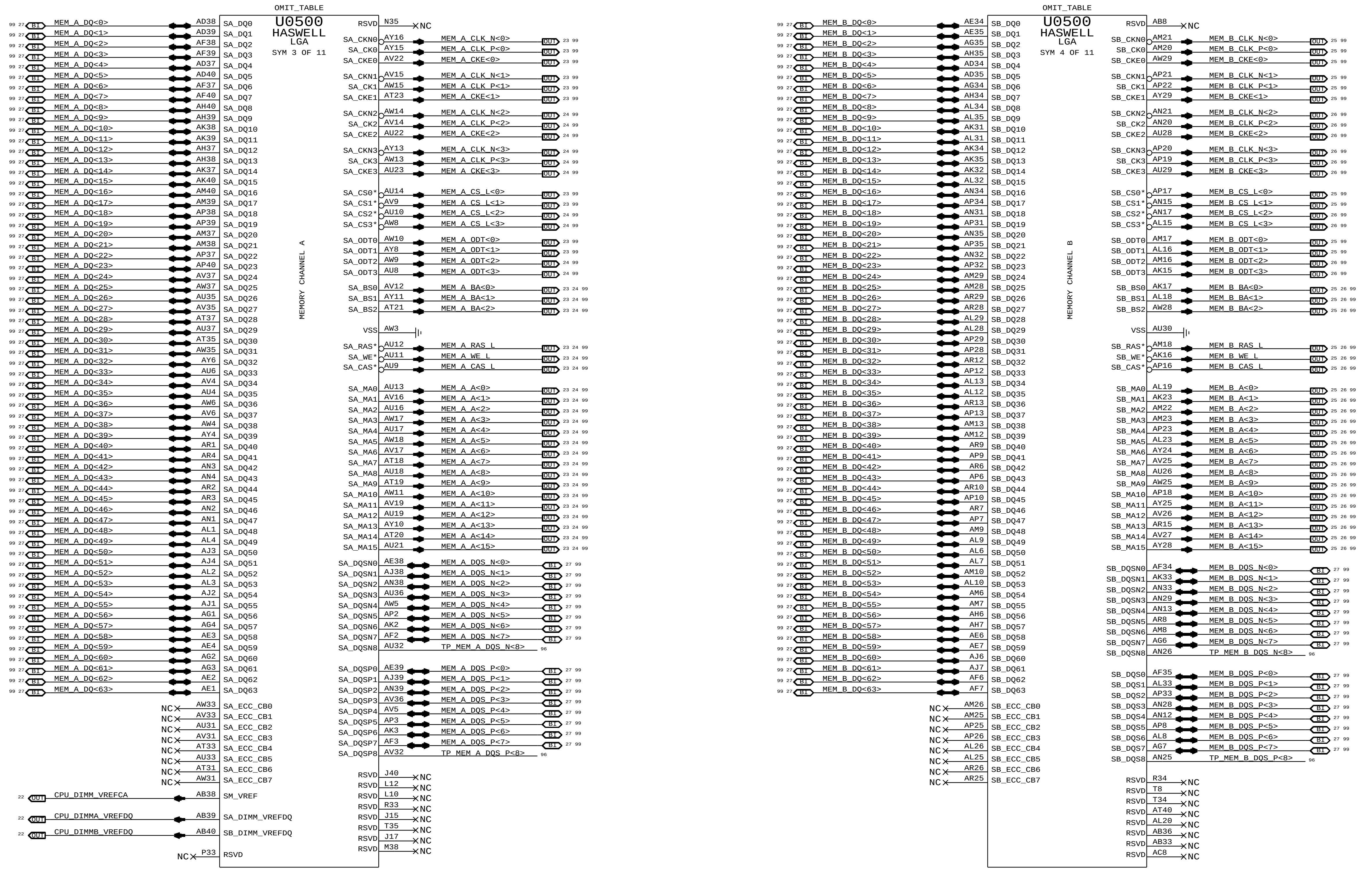

INTEL'S HASWELL DT EDS SAYS THAT ALL RSVD AND RSVD_NCTF SHOULD BE NC'D

SYNC_MASTER=J78_MLB SYNC_DATE=02/21/2014

PAGE TITLE

CPU DDR3 Interfaces

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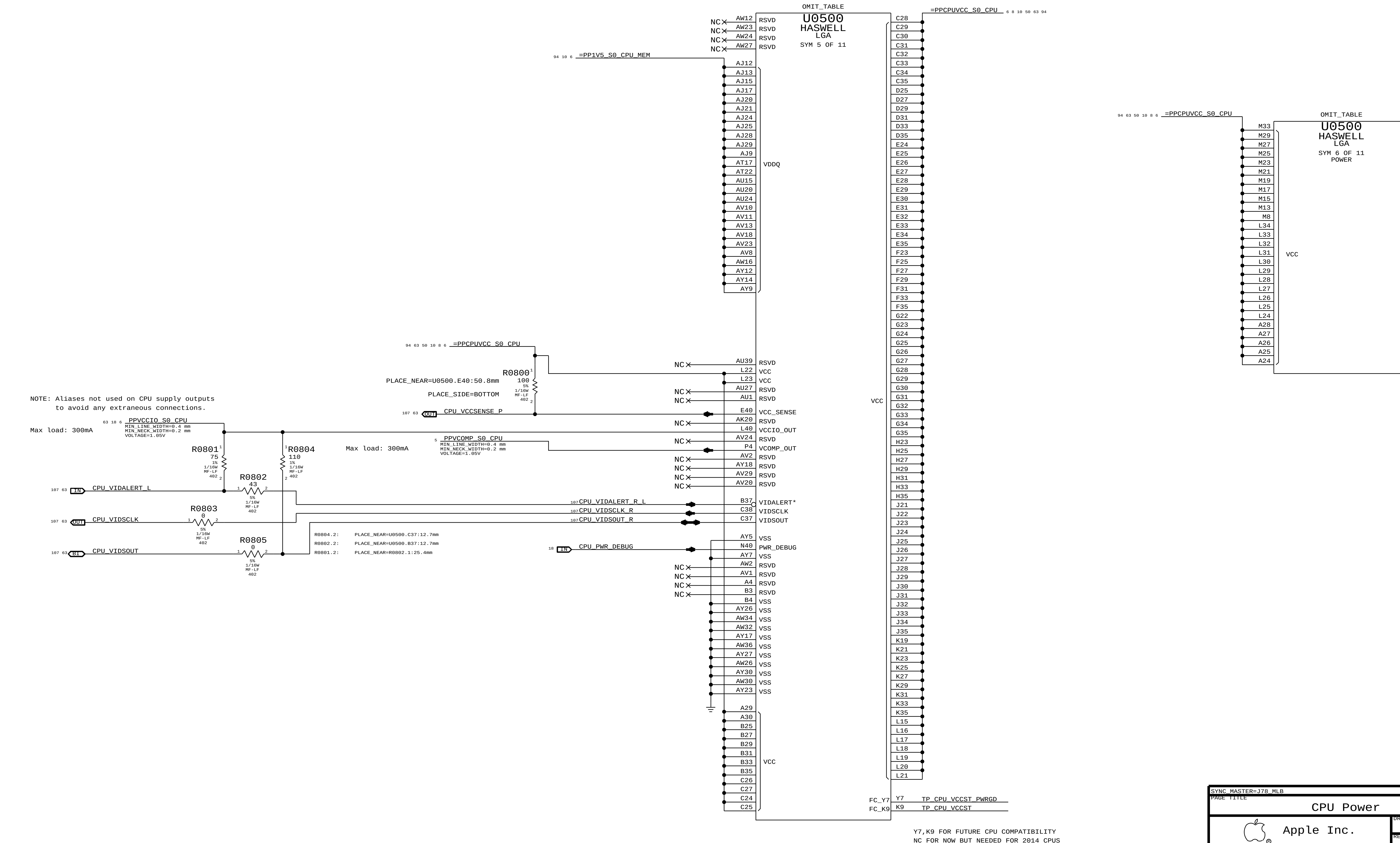

SYNC_MASTER=J78_MLB SYNC_DATE=02/21/2014

PAGE TITLE

CPU Power

Apple Inc.

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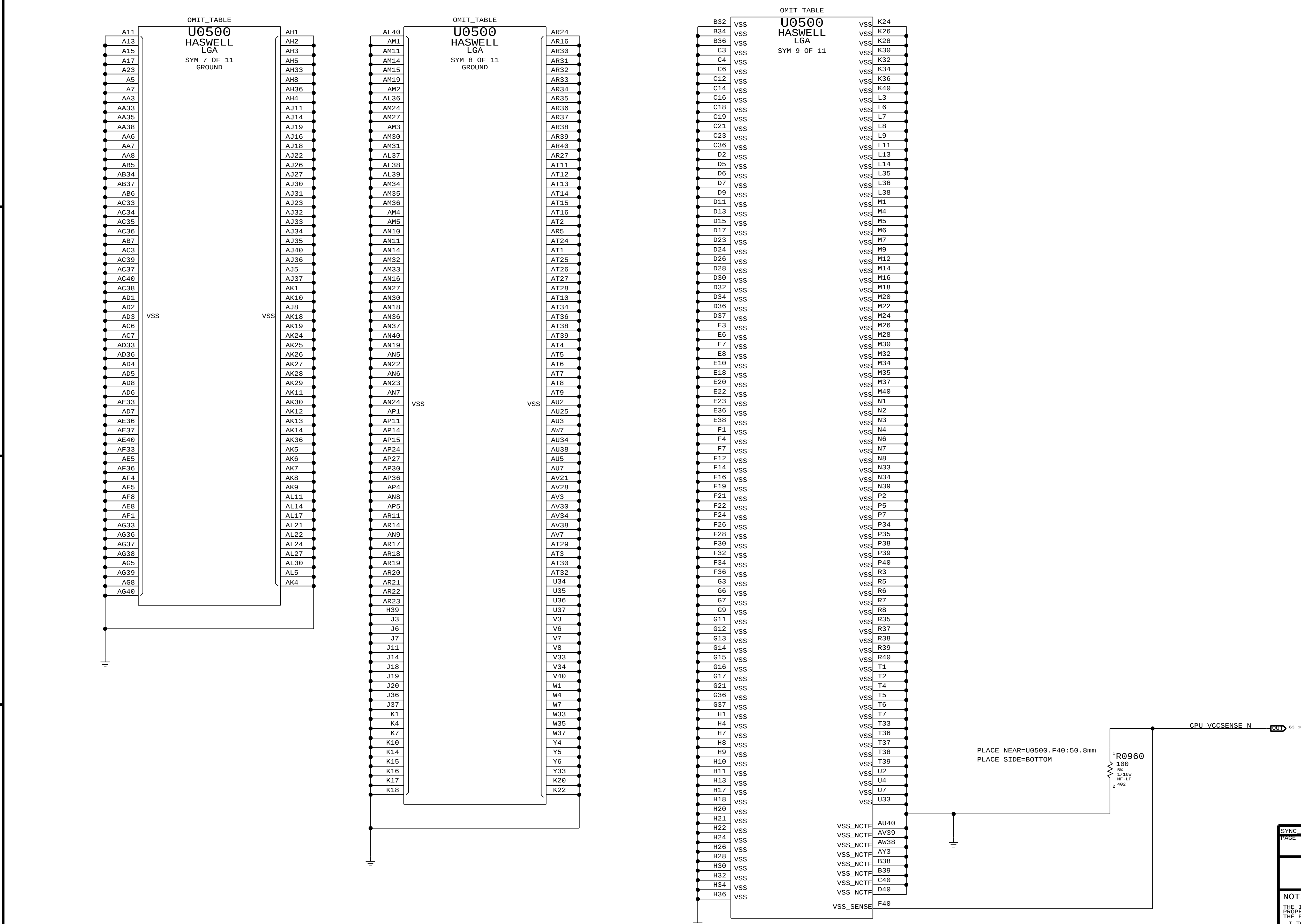

SYNC_MASTER=J78_MLB SYNC_DATE=02/21/2014

PAGE TITLE

CPU Ground

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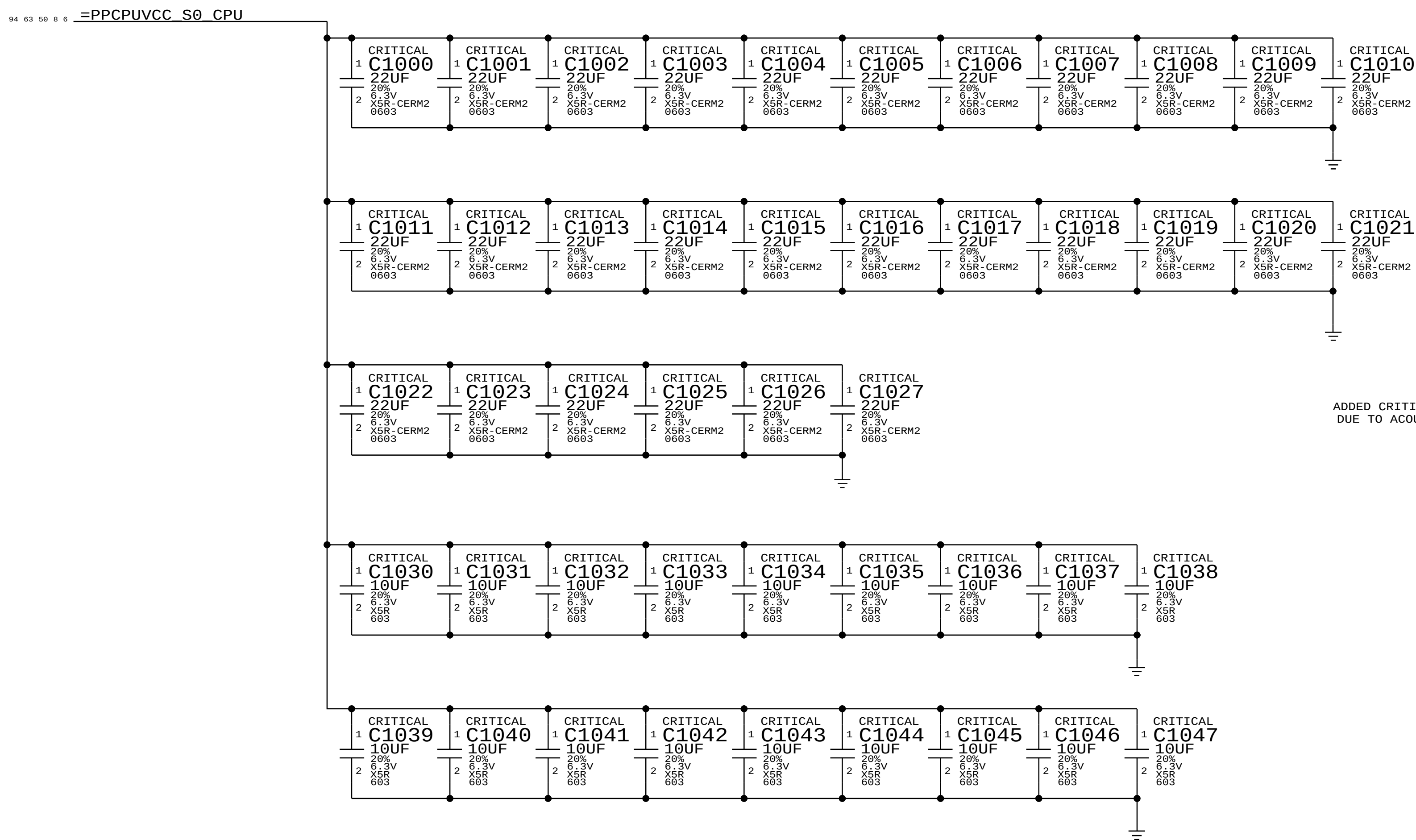
CPU VCORE DECOUPLING

Intel Recommendation:22x 22UF 0805,topside (18 inside cavity, 4 north of processor),8x 470uF bulk caps(5 stuffed,3 no-stuffed)
Apple Implementation:28x 22UF 0603 per Harold
18x 10UF 0603 placed inside socket cavity

Layout Note: These caps should be placed symmetrically on Top and Bottom sides.

BULK CAPS ON CPU VREG PAGE 71

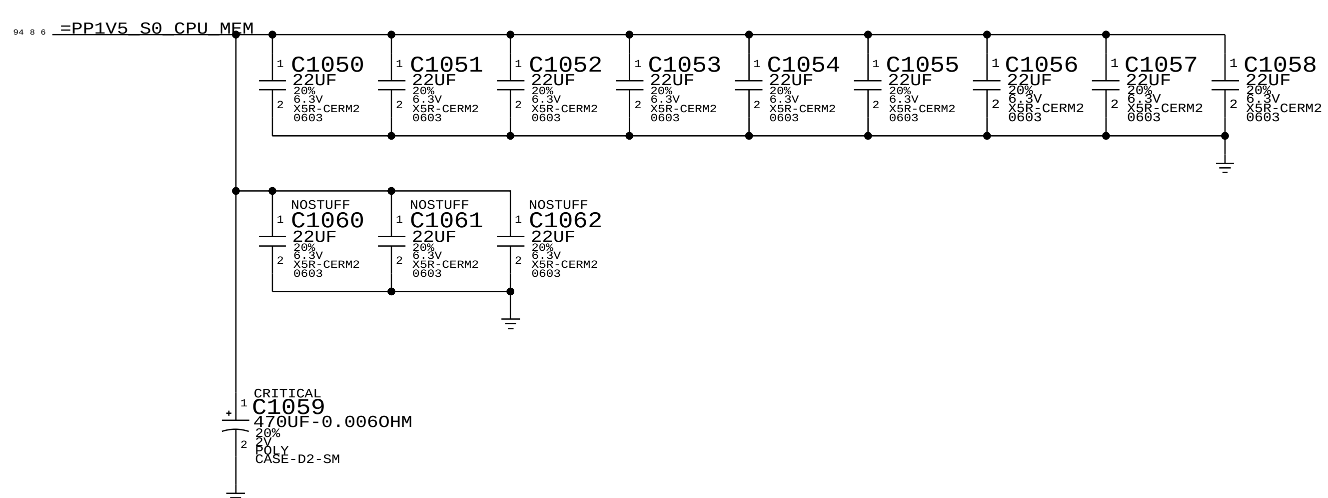


ADDED CRITICAL PROPERTY TO CPU CORE DECOUPLING
DUE TO ACOUSTICS CONCERNS

Memory (CPU VCCDDR) DECOUPLING

Intel Recommendation:9x 22UF 0805 near CPU power pins
Apple Implementation:9x 22UF 0603 per Harold

Layout Note: These caps should be placed symmetrically on Top and Bottom sides.



SYNC_MASTER=J78_MLB	SYNC_DATE=02/21/2014
PAGE TITLE	CPU DECOUPLING
Apple Inc.	DRAWING NUMBER 051-1635 SIZE D
	REVISION 5.0.0
NOTICE OF PROPRIETARY PROPERTY:	BRANCH proto2
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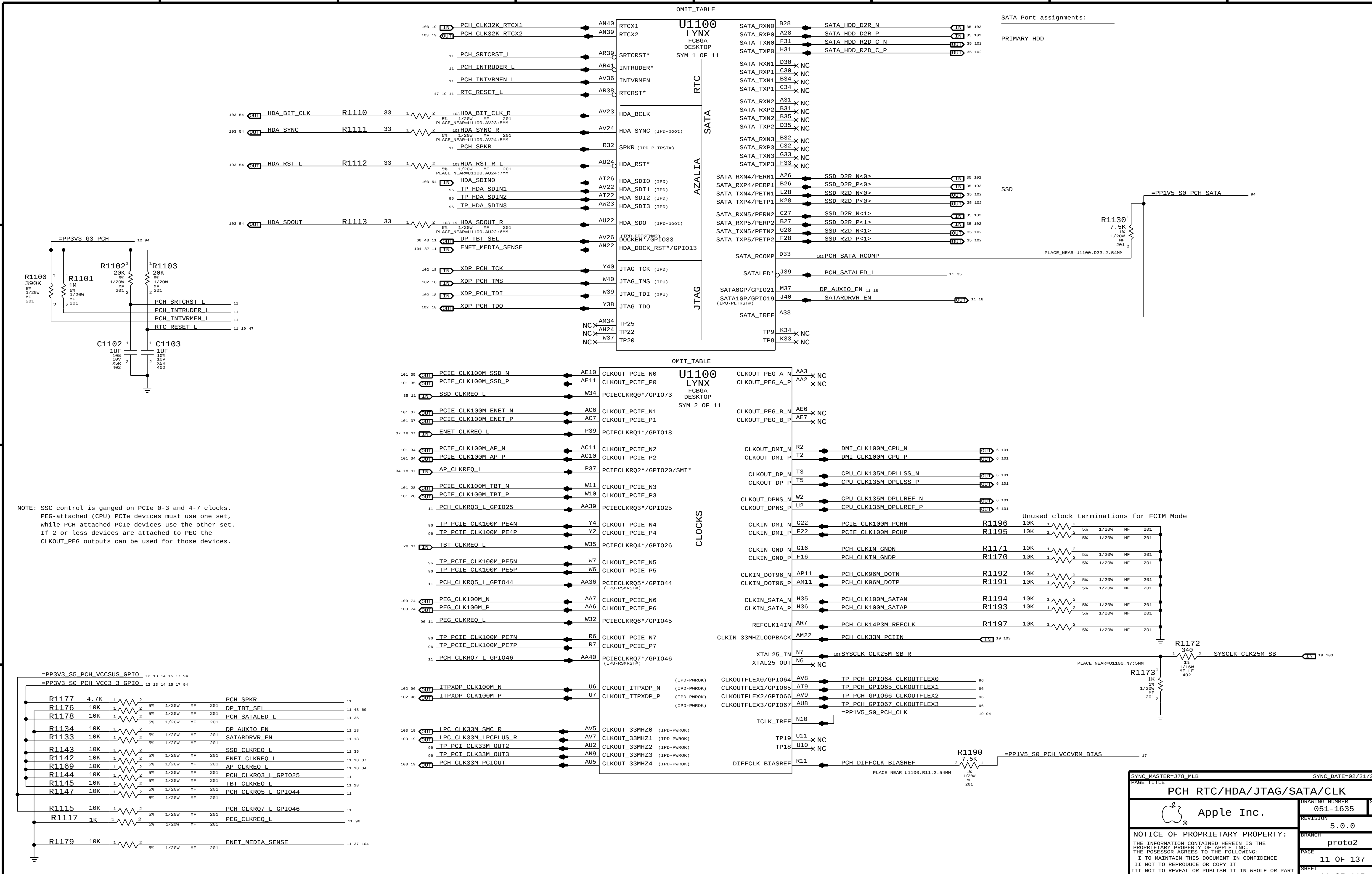
	SHEET 10 OF 115



NOTE: SSC control is ganged on PCIe 0-3 and 4-7 clocks.
PEG-attached (CPU) PCIe devices must use one set,
while PCH-attached PCIe devices use the other set.
If 2 or less devices are attached to PEG the
CLKOUT_PEG outputs can be used for those devices.

SYNC_MASTER=J78_MLB SYNC_DATE=02/21/2014

PAGE TITLE

PCH RTC/HDA/JTAG/SATA/CLK

Apple Inc.

DRAWING NUMBER	051-1635
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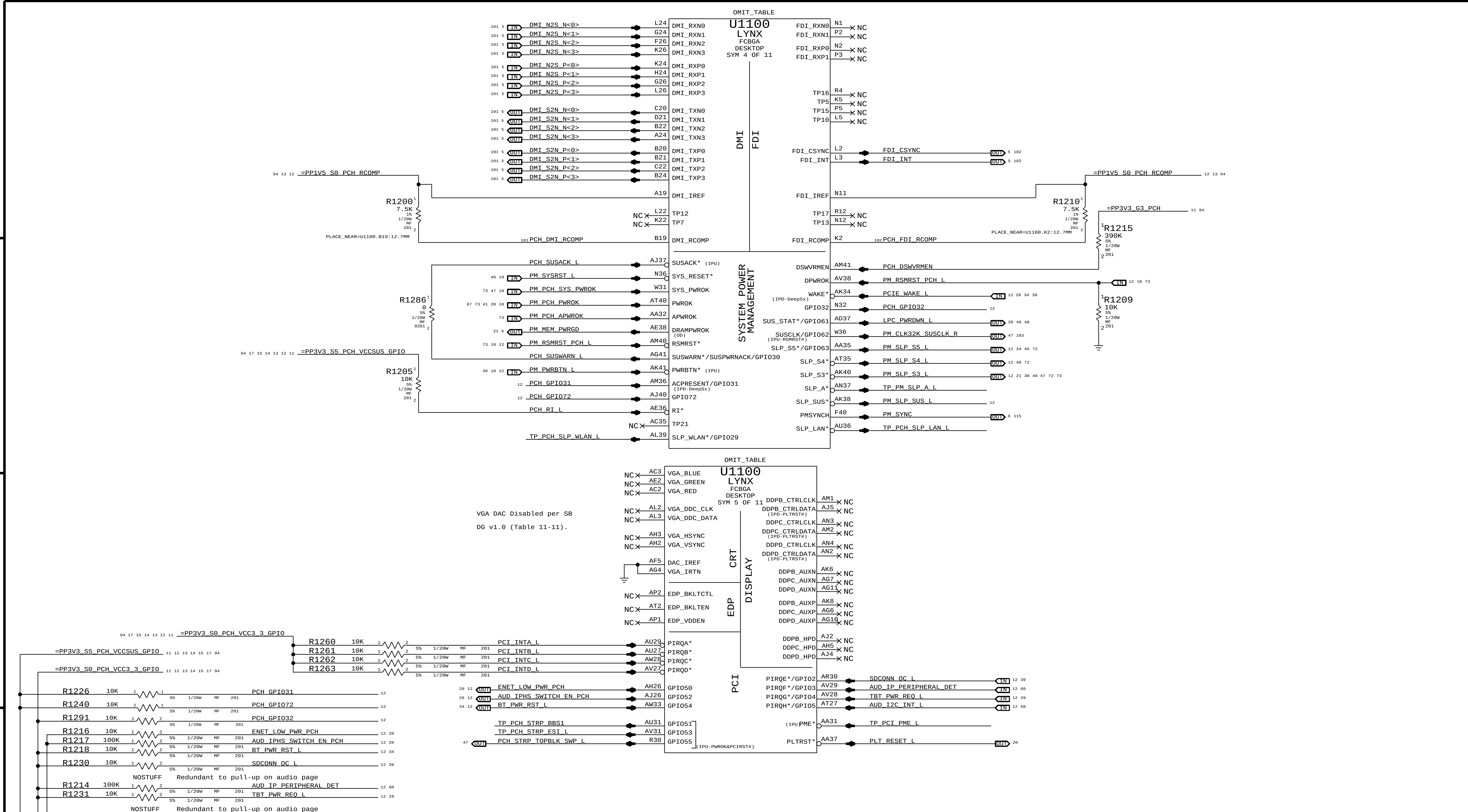

SYNC_MASTER=J78_MLB | SYNC_DATE=02/21/2014

PAGE TITLE

PCH DMI/FDI/PM/GFX/PCI

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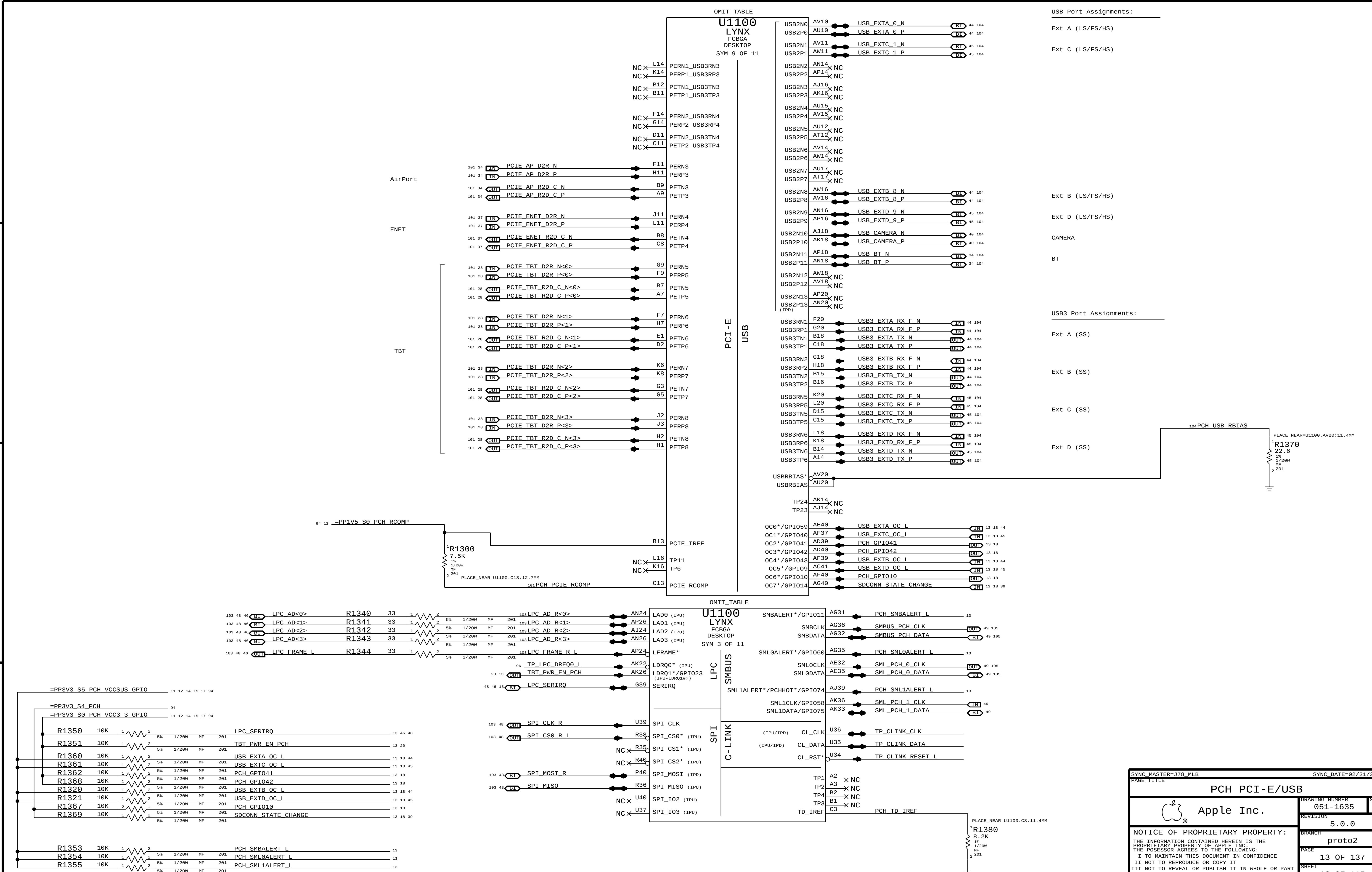

SYNC_MASTER=J78_MLB SYNC_DATE=02/21/2014

PAGE TITLE: PCH PCI-E/USB

Apple Inc.

DRAWING NUMBER: 051-1635 SIZE: D

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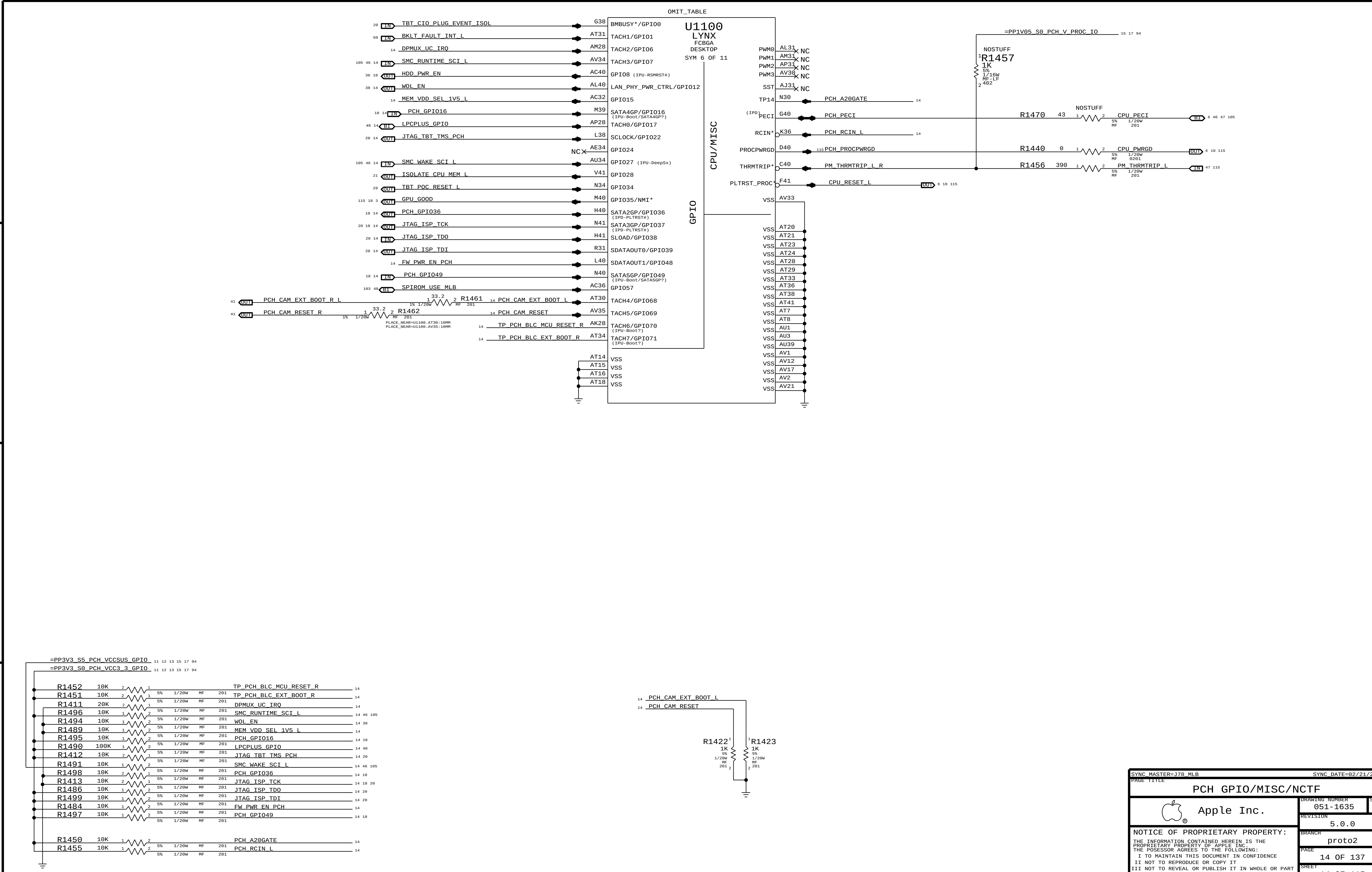

SYNC_MASTER=J78_MLB SYNC_DATE=02/21/2014

PAGE TITLE

PCH GPIO/MISC/NCTF

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DRAWING NUMBER 051-1635 SIZE D

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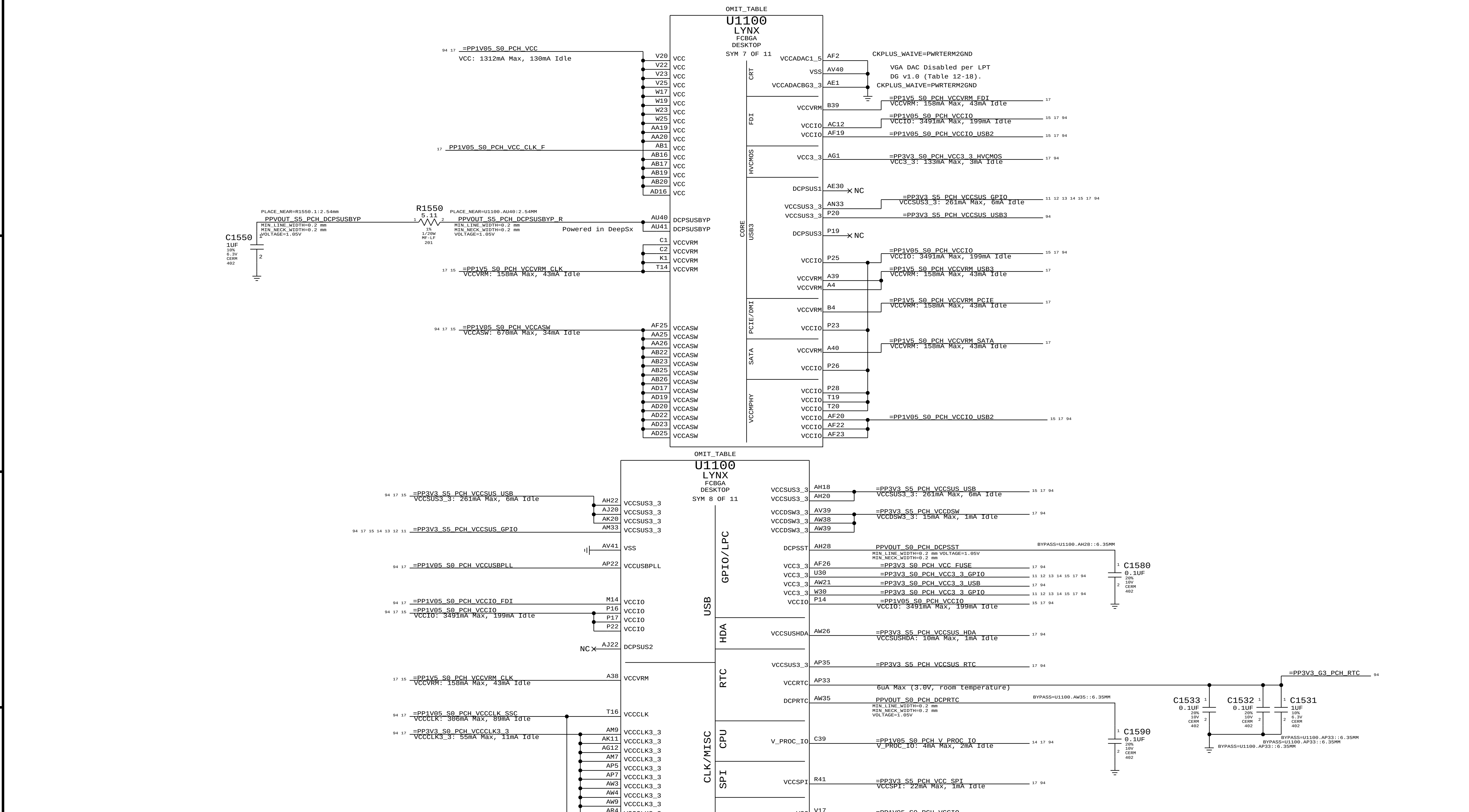

Current data from LPT EDS (doc #486708, Rev 1.0).

SYNC_MASTER=J78_MLB SYNC_DATE=02/21/2014

PAGE TITLE: PCH Power

Apple Inc.

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REVISION 5.0.0

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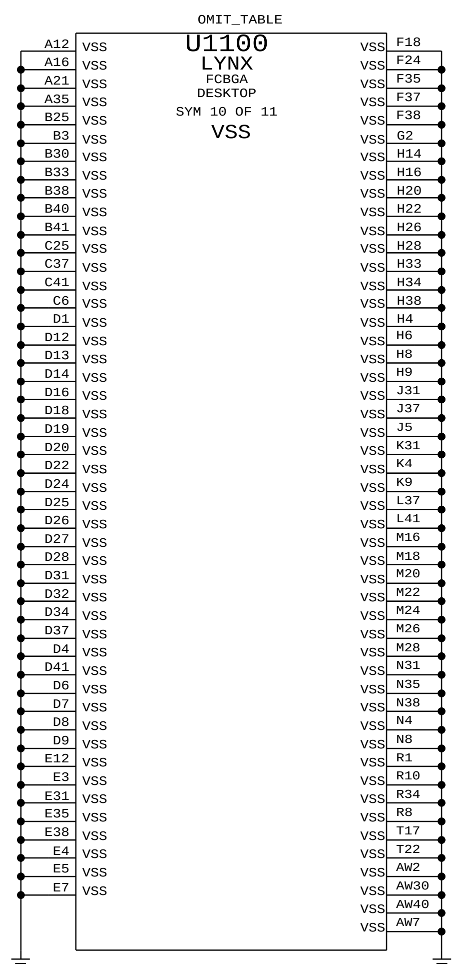

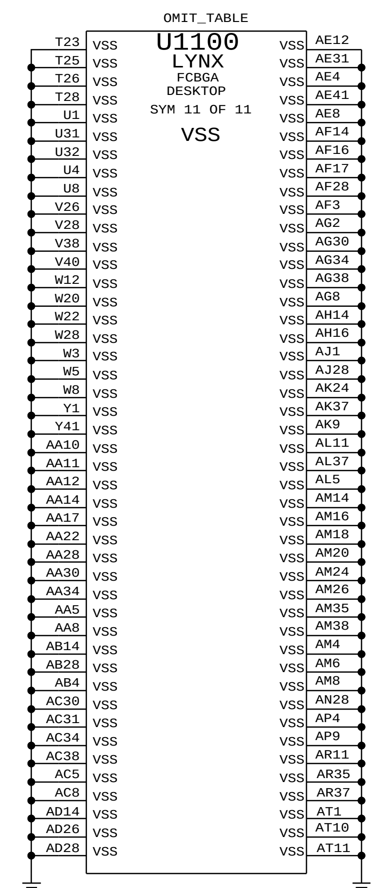

SYNC_MASTER=J78_MLB SYNC_DATE=02/21/2014

PAGE TITLE

PCH Grounds

Apple Inc.

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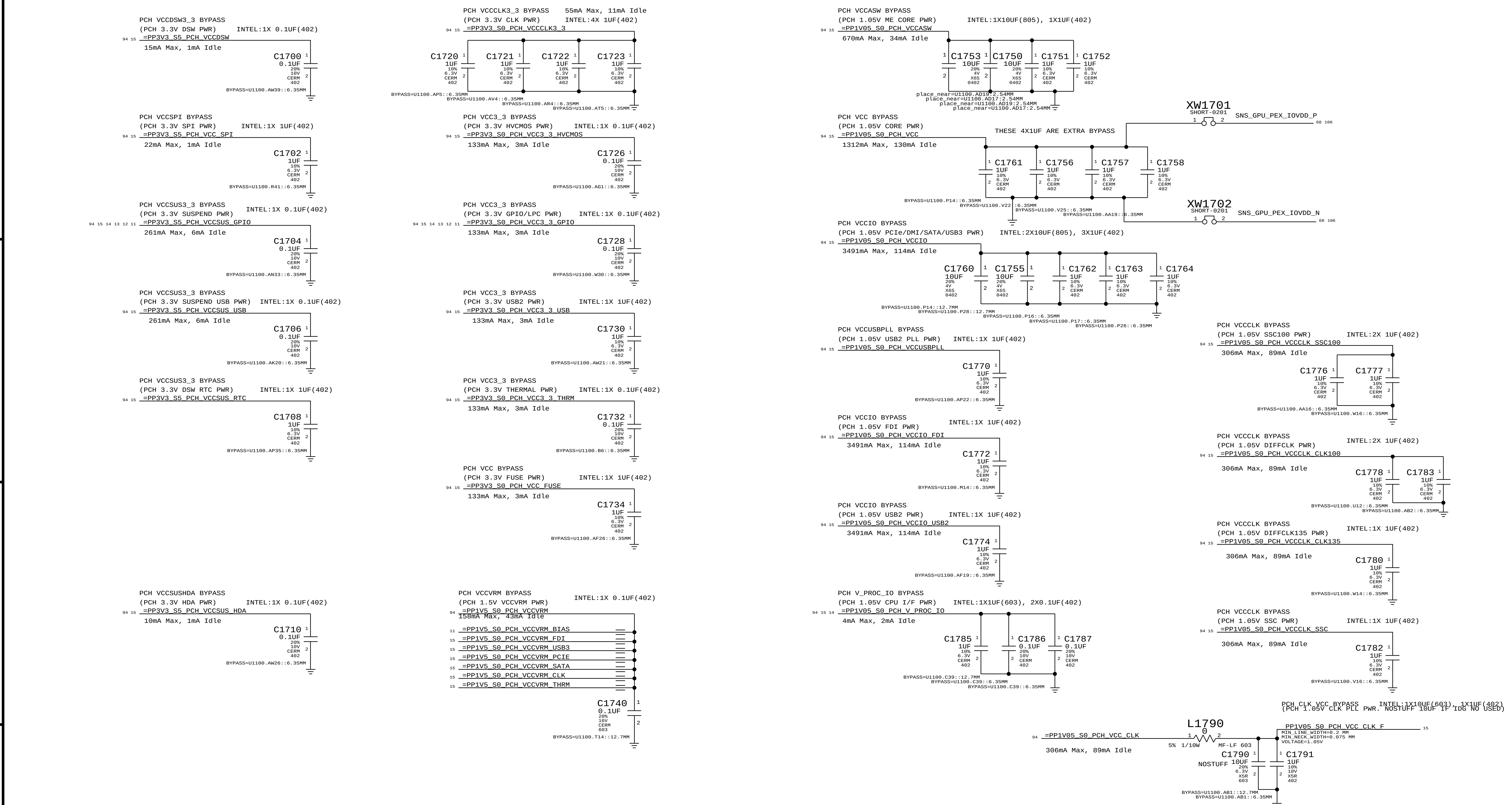
REVISION 5.0.0

BRANCH proto2



Current data from LPT EDS (doc #486708, Rev 1.0).

SYNC_MASTER=J78_MLB SYNC_DATE=02/21/2014

PAGE TITLE

PCH DECOUPLING

Apple Inc.

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BRANCH proto2

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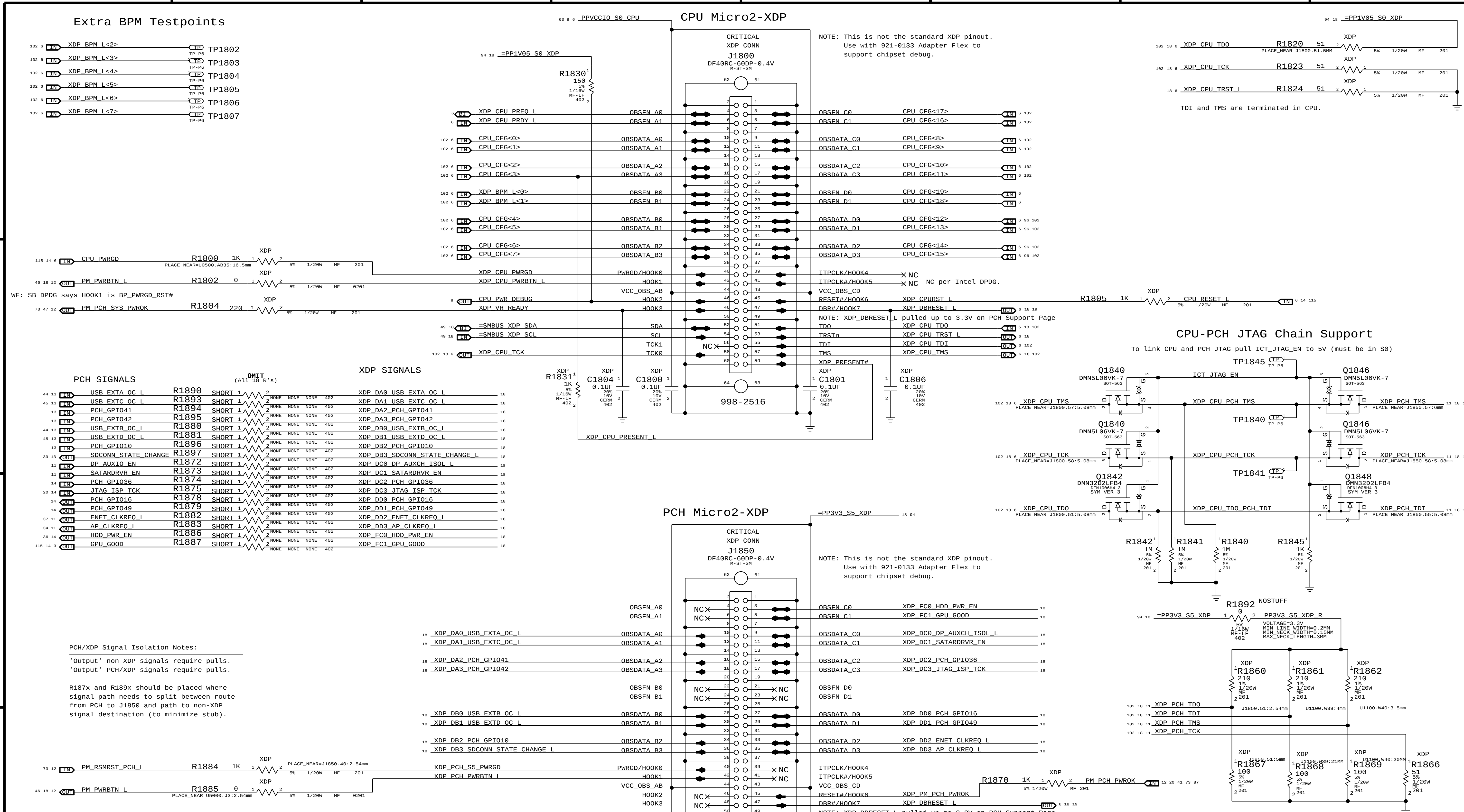

SYNC_MASTER=J78_MLB SYNC_DATE=02/21/2014

PAGE TITLE

CPU & PCH XDP

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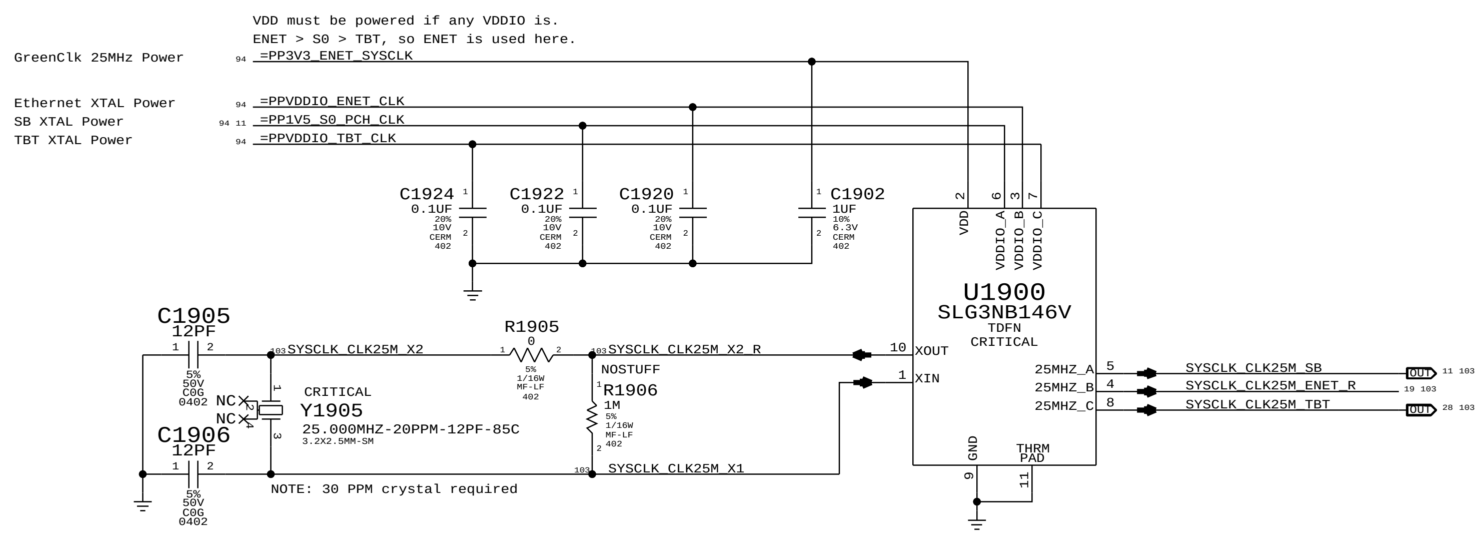
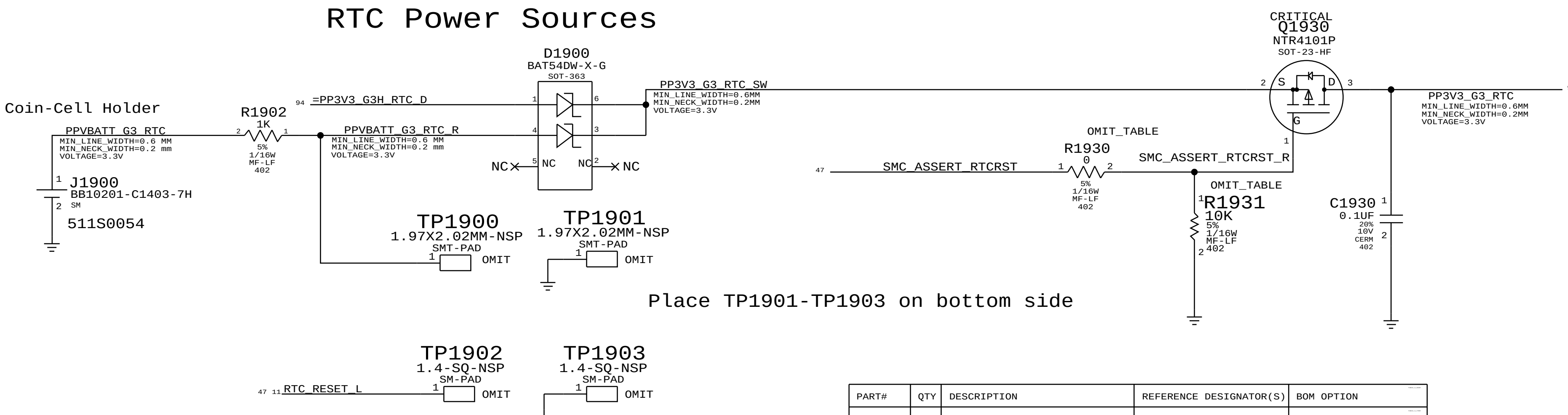
System 25MHz Clock Generator

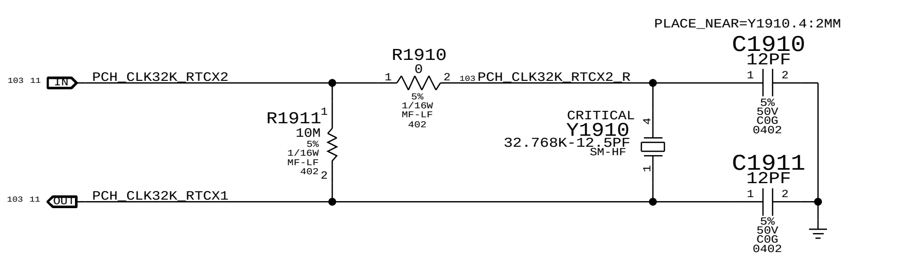


RTC Power Sources

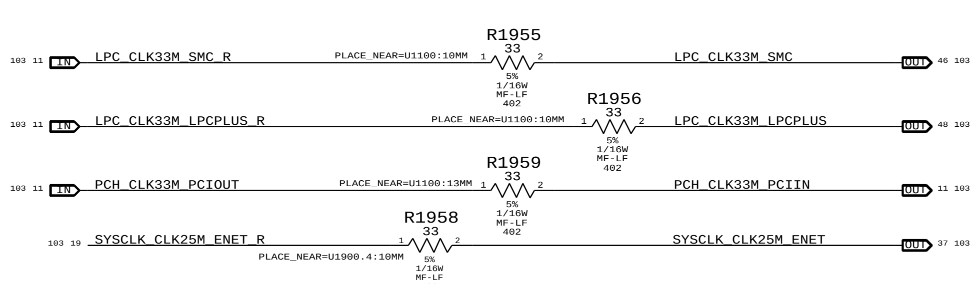


PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	BOM OPTION
116S0090	1	RES,10K OHM,402	R1930	RTCRST:Y
132S1059	1	CAP,0.1 UF,402	R1931	RTCRST:Y
116S0090	1	RES,10K OHM,402	R1931	RTCRST:N

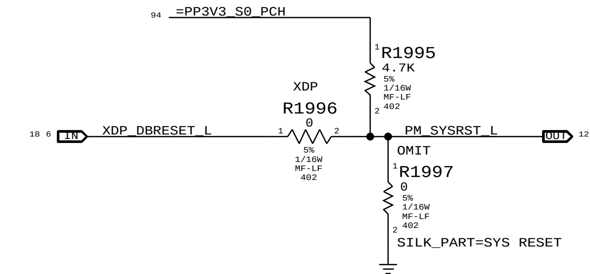
PCH RTC Crystal



Clock series termination

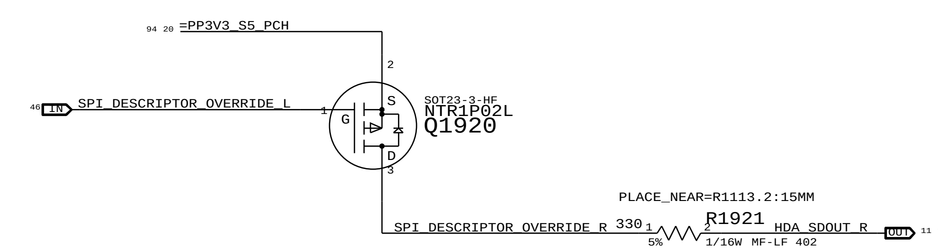


PCH Reset Button



PCH ME Disable Strap

PCH uses HDA_SDO as a power-up strap. If low, ME functions normally.
If high, ME is disabled. This allows for full re-flashing of SPI ROM.
SMC controls strap enable to allow in-field control of strap setting.



SYNC_MASTER=J78_MLB SYNC_DATE=02/21/2014

PAGE TITLE: Chipset Support

Apple Inc.

DRAWING NUMBER: 051-1635 SIZE: D

REVISION: 5.0.0

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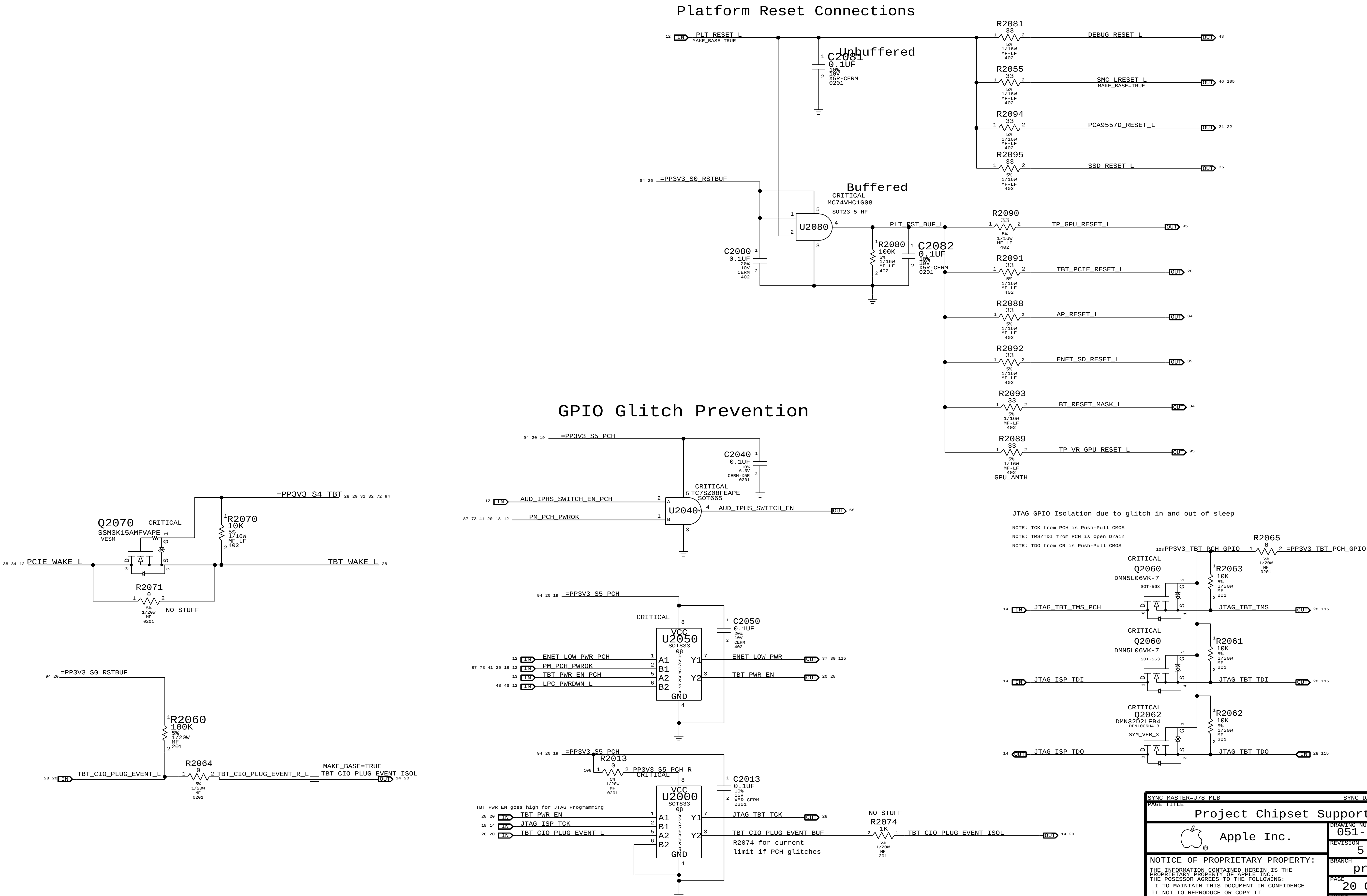

SYNC_MASTER=J78_MLB SYNC_DATE=02/21/2014

PAGE TITLE: Project Chipset Support

Apple Inc.

DRAWING NUMBER: 051-1635 SIZE: D

REVISION: 5.0.0

BRANCH: proto2

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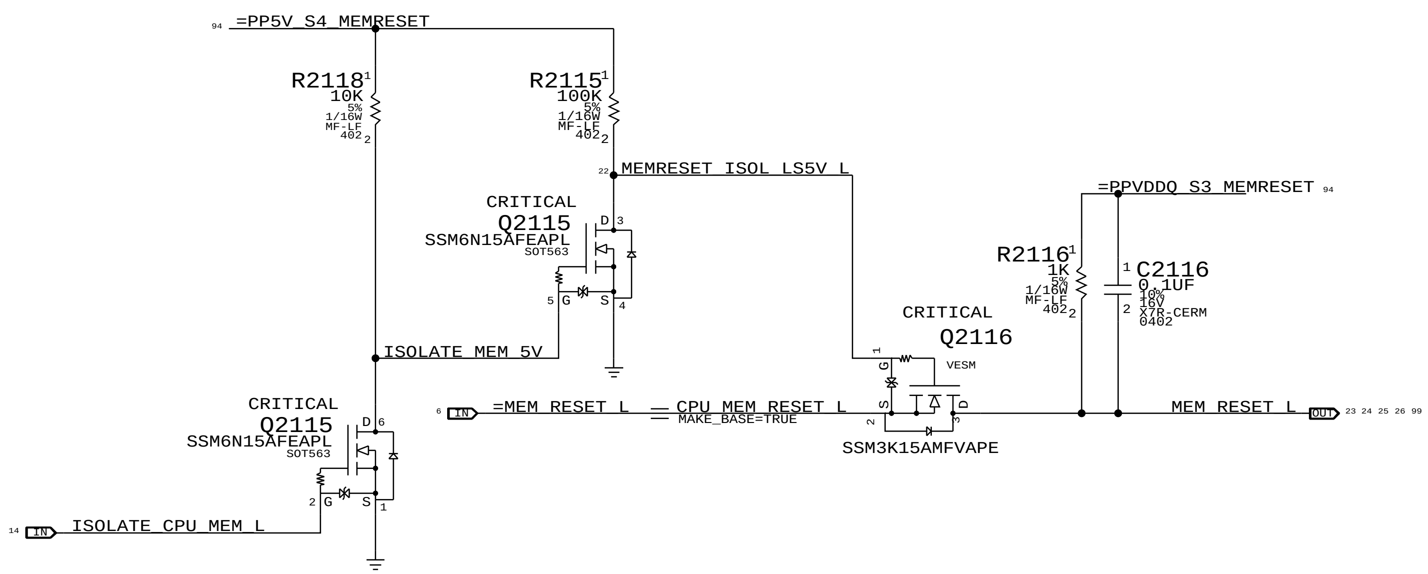
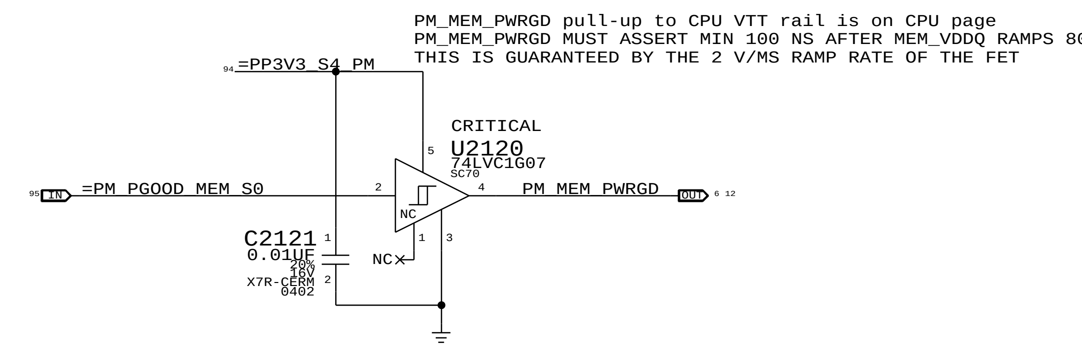
The circuit below handles CPU and VTT power during S0->S3->S0 transitions, as well as isolating the CPU's SM_DRAMRST# output from the SO-DIMMs when necessary.

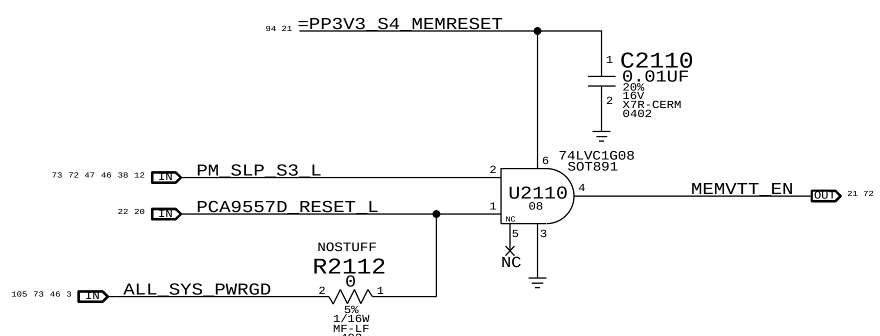
ISOLATE_CPU_MEM_L GPIO state during S3<->S0 transitions determines behavior of signals.
WHEN HIGH: CPU 1.5V remains powered in S3, VTT follows S0 rails, MEM_RESET_L not isolated.
WHEN LOW: CPU 1.5V follows S0 rails, VTT ensures clean CKE transition, MEM_RESET_L isolated.

MEMVTT_EN = PLT_RESET_L * PM_SLP_S3_L
MEM_RESET_L = !ISOLATE_CPU_MEM_L + CPU_MEM_RESET_L



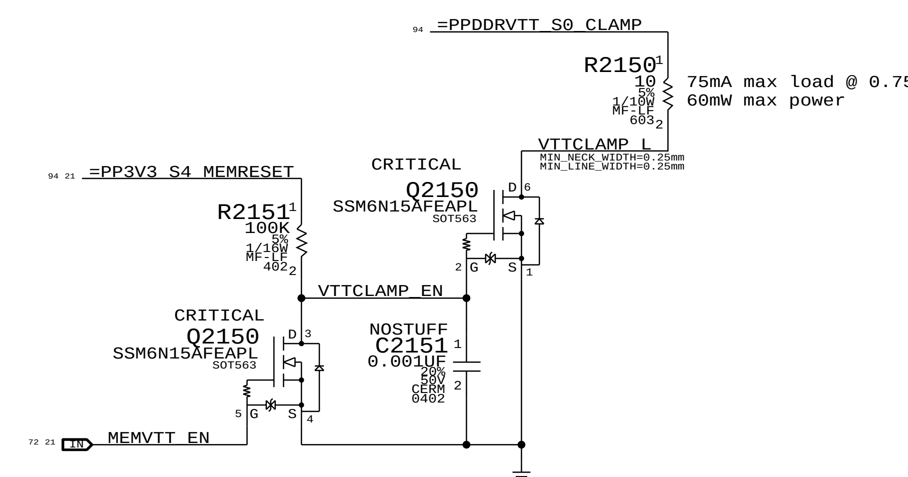
MEM S0 "PGOOD" FOR CPU





MEMVTT Clamp

Ensures CKE signals are held low in S3



	Step	ISOLATE_CPU_MEM_L	PLT_RESET_L	PM_SLP_S3_L	CPU_MEM_RESET_L	MEM_RESET_L	MEMVTT_EN
S0	0	1	1	1	1	CPU_MEM_RESET_L	1
to	1	0	1	1	1	1	1
S3	2	0	0	1	1	1	0
	3	0	0	0	X	1	0
to	4	0	0	1	X	1	0
	5	0	1	1	0 (*)	1	1
S0	6	0	1	1	1	1	1
	7	1	1	1	1	CPU_MEM_RESET_L	1

(*) CPU_MEM_RESET_L asserts due to loss of PM_MEM_PWRGD, must wait for software to clear before deasserting ISOLATE_CPU_MEM_L GPIO.

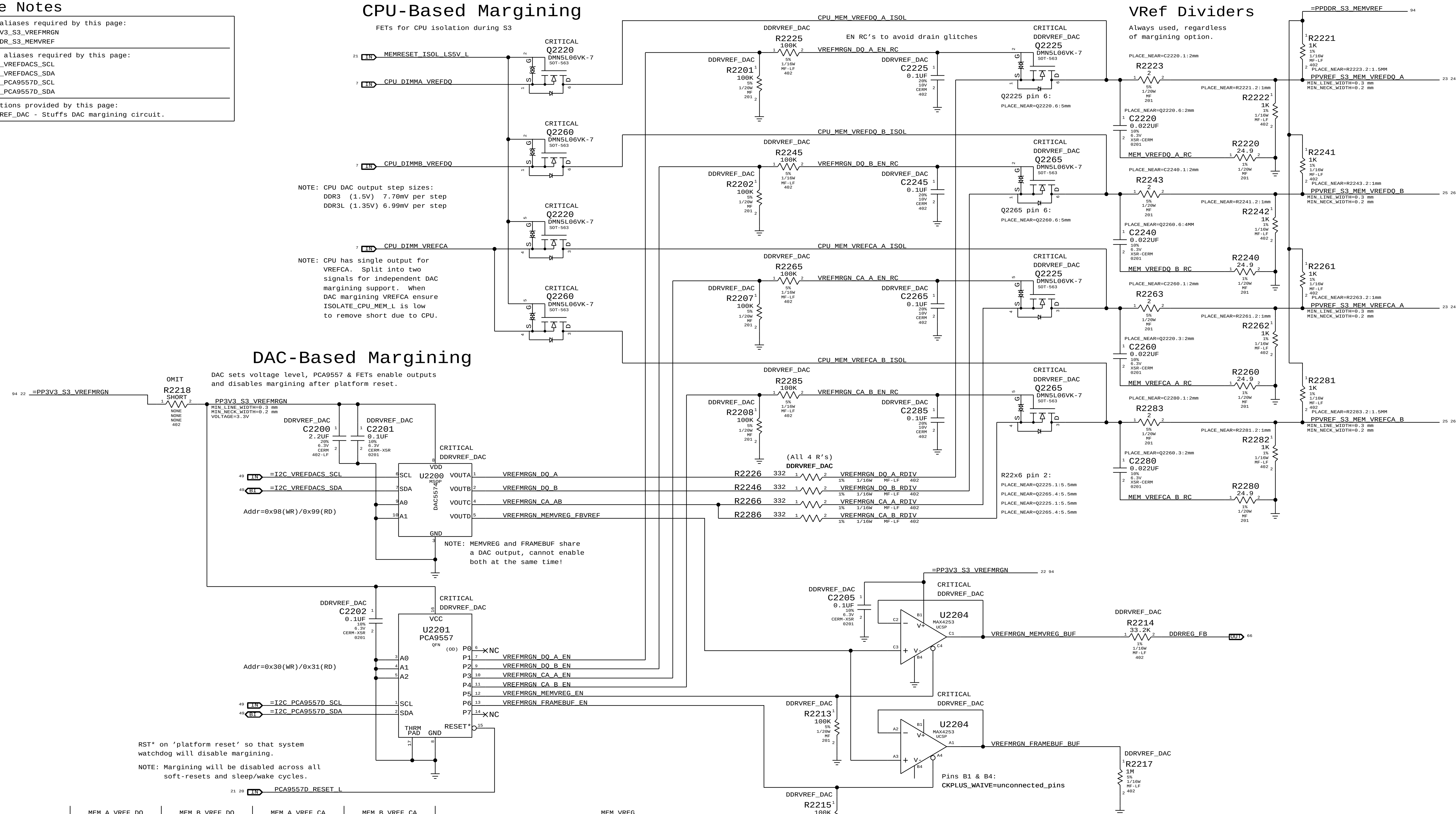
NOTE: In the event of a S3->S5 transition ISOLATE_CPU_MEM_L will still be asserted on next S5->S0 transition. Rails will power-up as if from S3, but MEM_RESET_L will not properly assert. Software must de-assert ISOLATE_CPU_MEM_L and then generate a valid reset cycle on CPU_MEM_RESET_L.

Page Notes

Power aliases required by this page:
- =PP3V3_S3_VREFMRGN
- =PPDDR_S3_MEMVREF

Signal aliases required by this page:
- =I2C_VREFDACS_SCL
- =I2C_VREFDACS_SDA
- =I2C_PCA9557D_SCL
- =I2C_PCA9557D_SDA

BOM options provided by this page:
- DDRVREF_DAC - Stuffs DAC margining circuit.



	MEM A VREF DQ	MEM B VREF DQ	MEM A VREF CA	MEM B VREF CA	MEM VREG
DAC Channel:	A	B	C	C	D
PCA9557D Pin:	1	2	3	4	5

	DDR3 (1.5V)	DDR3L (1.35V)	DDR3 (1.5V)	DDR3L (1.35V)
Nominal value	0.750V (DAC: 0x3A = 0.747mV)	0.675V (DAC: 0x34 = 0.670mV)	1.500V (DAC: 0x74 = 1.495V)	1.343V (DAC: 0x68 = 1.341V)
Margined target:	0.300V - 1.200V (+/- 450mV)	0.275V - 1.075V (+/- 400mV)	1.200V - 1.800V (+/- 300mV)	0.950V - 1.750V (+/- 400mV)
DAC range:	0.000V - 1.508V (0x00 - 0x75)	0.000V - 1.354V (0x00 - 0x69)	0.000V - 3.004V (0x00 - 0xE9)	0.000V - 2.707V (0x00 - 0xD2)
Margined range:	0.299V - 1.206V (+/- 453mV)	0.269V - 1.083V (+/- 406mV)	1.199V - 1.801V (+/- 301mV)	0.932V - 1.760V (+/- 414mV)
VRef current:	+901uA - -911uA (- = sourced)	+811uA - -816uA (- = sourced)	+36uA - -36uA (- = sourced)	+28uA - -29uA (- = sourced)
DAC step size:	7.68mV / step @ output	7.67mV / step @ output	2.575mV / step @ output	3.923mV / step @ output

NOTE: DDR3 assumes TPS51916 supply with 10.0k/49.9k divider
DDR3L assumes TPS51916 supply with 19.6k/57.6k divider

SYNC_MASTER=J78_MLB SYNC_DATE=02/21/2014
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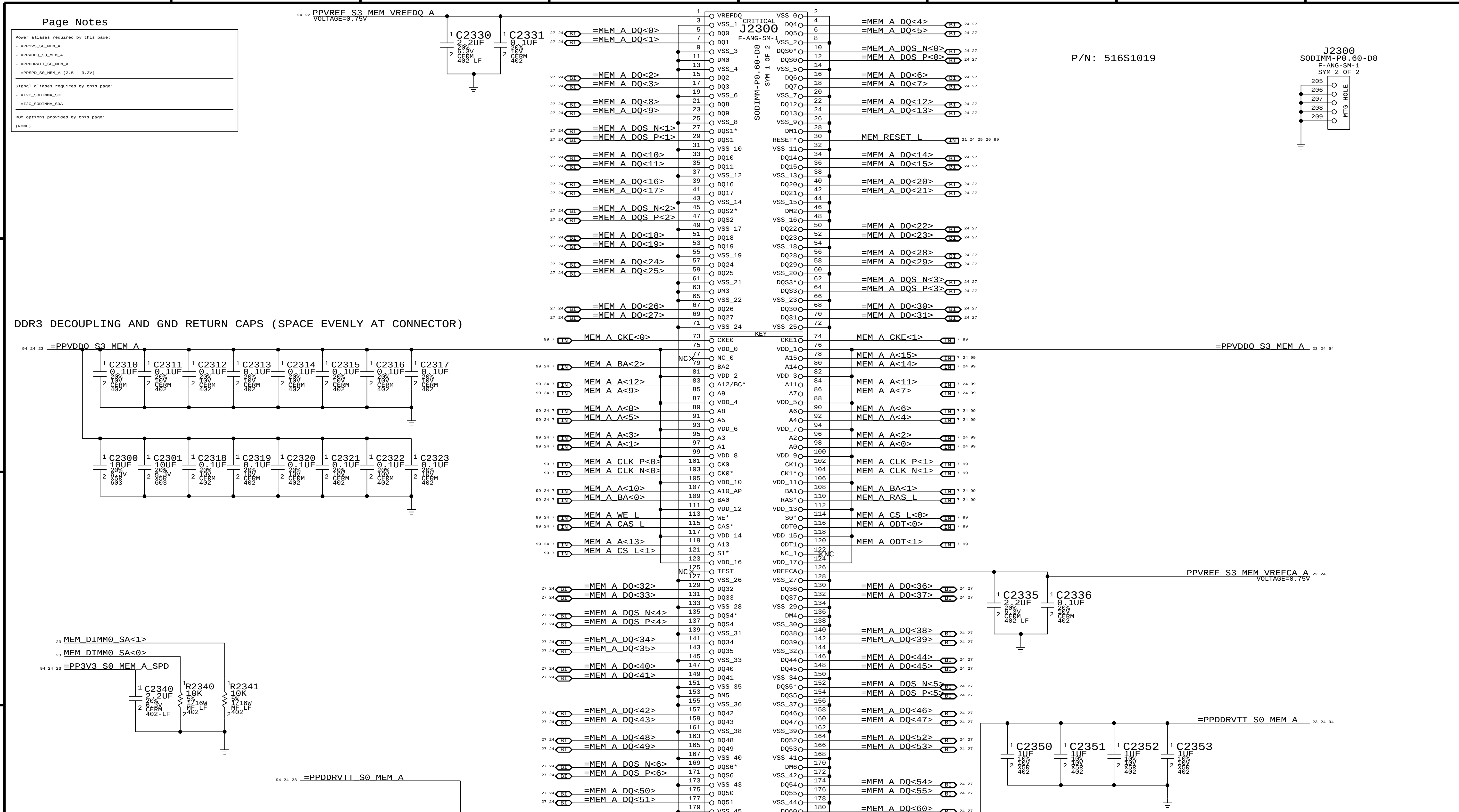

SYNC_MASTER=J78_MLB SYNC_DATE=02/21/2014

PAGE TITLE
DDR3 SO-DIMM Connector A Slot0

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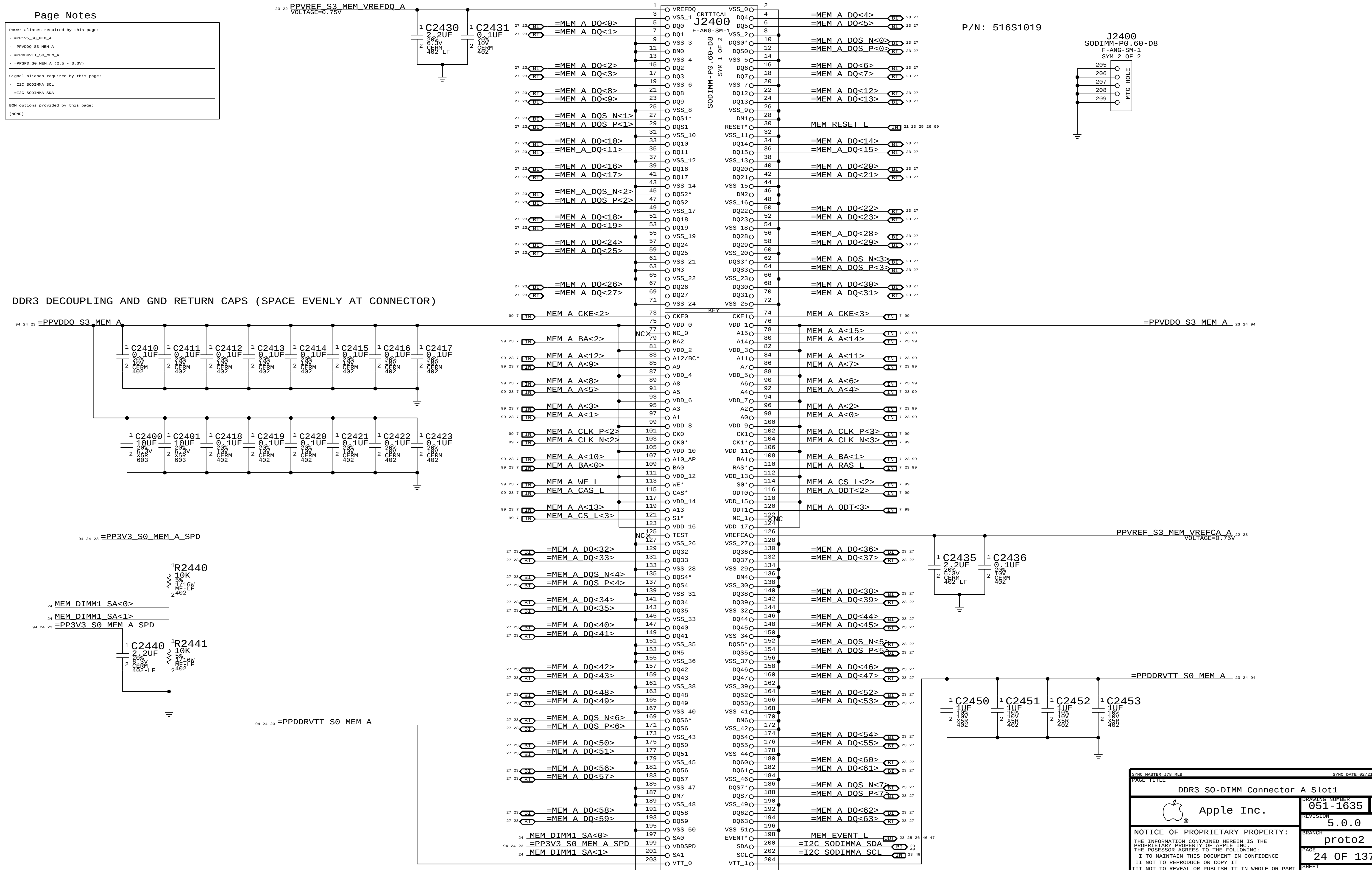
Page Notes

Power aliases required by this page:
- =PP1V5_S0_MEM_A
- =PPVDDQ_S3_MEM_A
- =PPDDRVTT_S0_MEM_A
- =PPSPD_S0_MEM_A (2.5 - 3.3V)

Signal aliases required by this page:
- =I2C_SODIMMA_SCL
- =I2C_SODIMMA_SDA

BOM options provided by this page:
(NONE)



SYNC_MASTER=J78_MLB SYNC_DATE=02/21/2014

PAGE TITLE: DDR3 SO-DIMM Connector A Slot1

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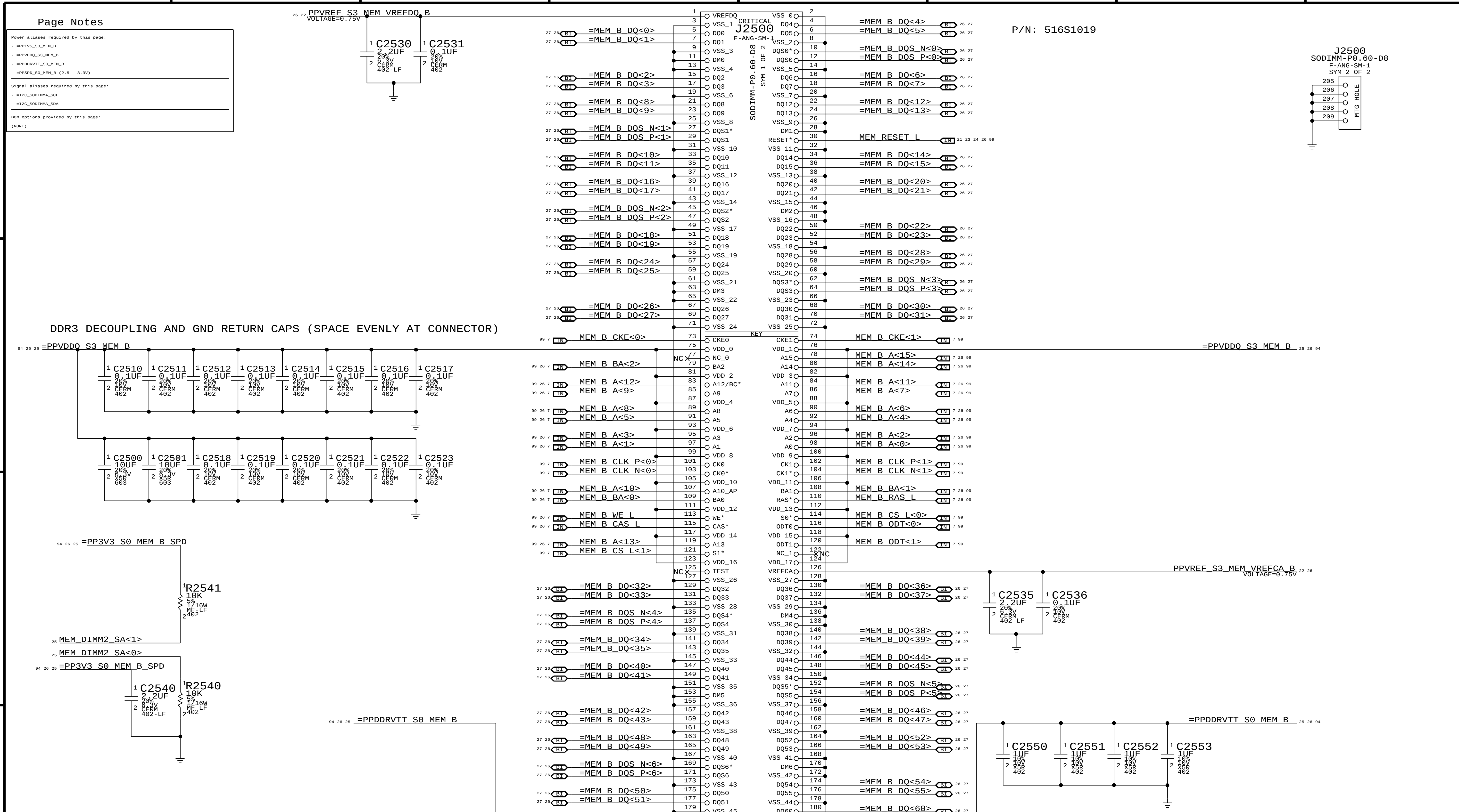

SYNC_MASTER=J78_MLB SYNC_DATE=02/21/2014

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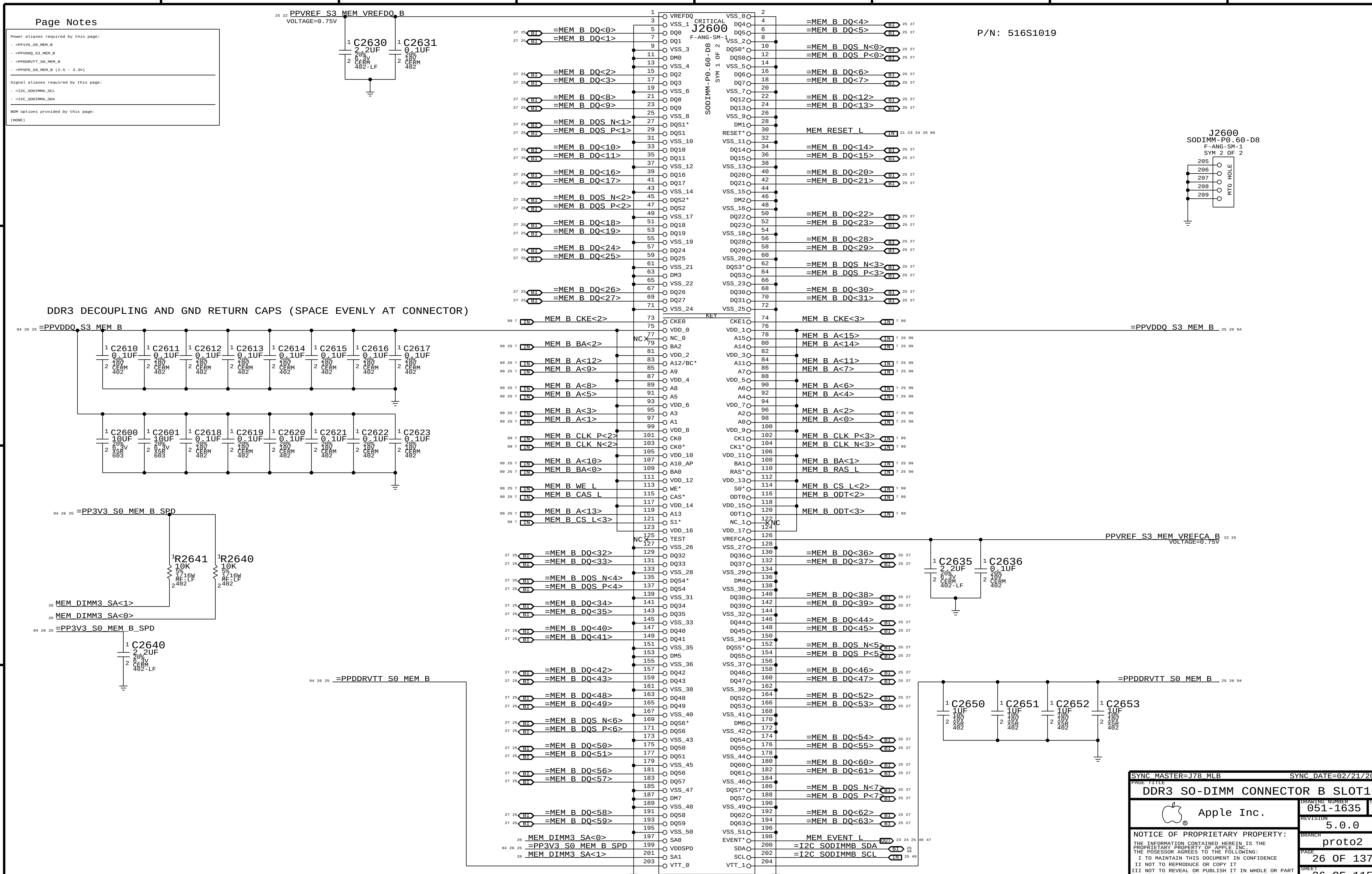

SYNC_MASTER=J78_MLB SYNC_DATE=02/21/2014

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DDR3 SO-DIMM CONNECTOR B SLOT1

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THERE ARE NO PIN SWAPS

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99 7	MEM_A_DQS_P<0>	MAKE_BASE=TRUE	=MEM_A_DQS_P<0>	23 24	99 7	MEM_B_DQS_P<0>	MAKE_BASE=TRUE	=MEM_B_DQS_P<0>	25 26
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PAGE TITLE

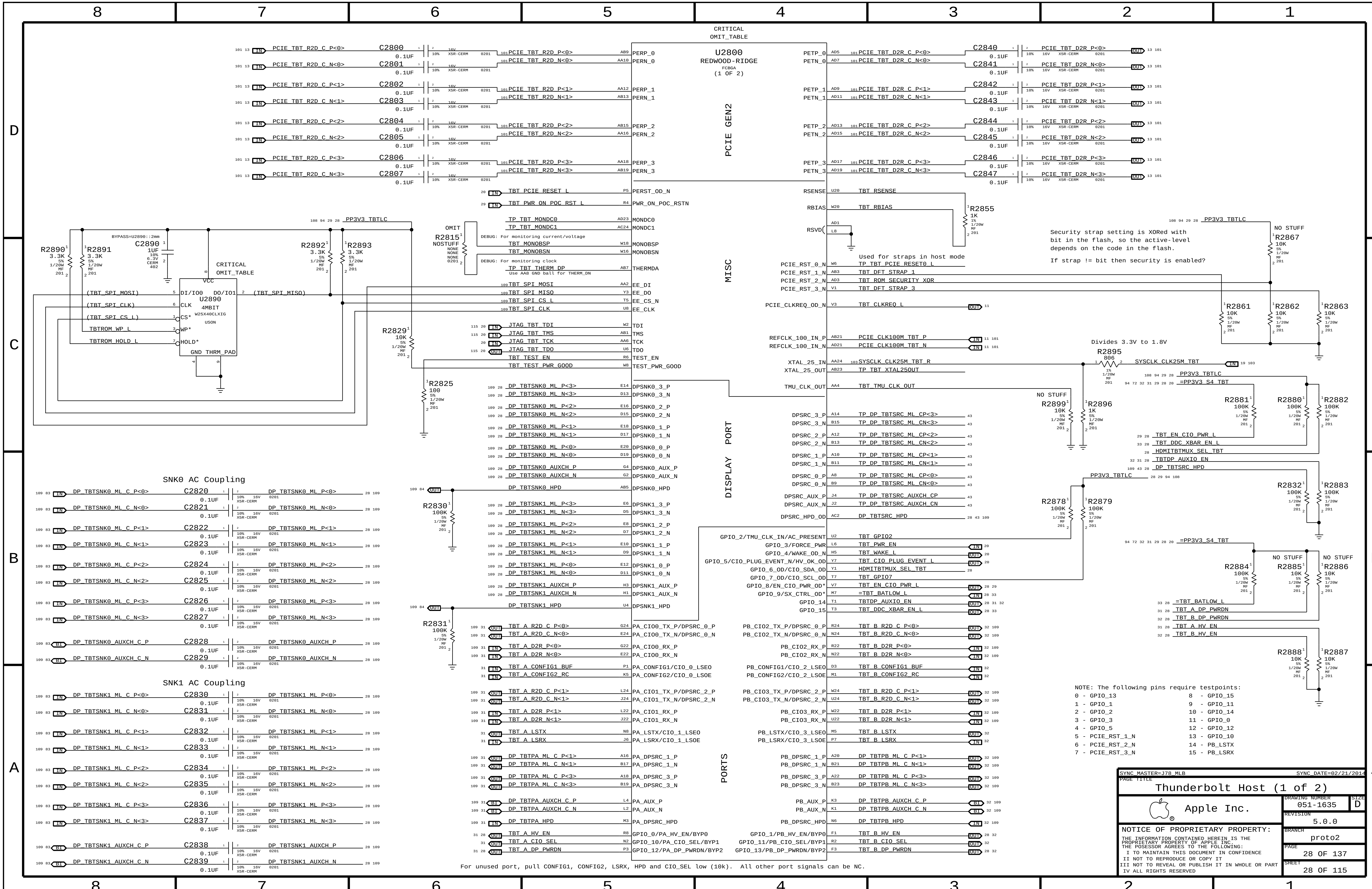
DDR3 ALIASES AND BITSWAPS

Apple Inc.

DRAWING NUMBER 051-1635 SIZE D

REVISION 5.0.0

BRANCH proto2



CRITICAL
OMIT_TABLE
U2800
REDWOOD-RIDGE
FCBGA
(1 OF 2)
PCIE GEN2
MISC
DISPLAY PORT
PORTS
PERP_0
PERN_0
PERP_1
PERN_1
PERP_2
PERN_2
PERP_3
PERN_3
PERST_OD_N
PWR_ON_POC_RSTN
MONDC0
MONDC1
MONOBSP
MONOBSN
THERMDA
EE_DI
EE_DO
EE_CS_N
EE_CLK
TDI
TMS
TCK
TDO
TEST_EN
TEST_PWR_GOOD
PETP_0
PETN_0
PETP_1
PETN_1
PETP_2
PETN_2
PETP_3
PETN_3
RSENSE
RBIAS
RSVD
PCIE_RST_0_N
PCIE_RST_1_N
PCIE_RST_2_N
PCIE_RST_3_N
PCIE_CLKREQ_OD_N
REFCLK_100_IN_P
REFCLK_100_IN_N
XTAL_25_IN
XTAL_25_OUT
TMU_CLK_OUT
DEBUG: For monitoring current/voltage
DEBUG: For monitoring clock
Use AA8 GND ball for THERM_DN
U2890
4MBIT
W25X40CLXIG
USON
CRITICAL
OMIT_TABLE
Used for straps in host mode
Security strap setting is XORed with
bit in the flash, so the active-level
depends on the code in the flash.
If strap != bit then security is enabled?
Divides 3.3V to 1.8V
SNK0 AC Coupling
SNK1 AC Coupling
For unused port, pull CONFIG1, CONFIG2, LSRX, HPD and CIO_SEL low (10k). All other port signals can be NC.
NOTE: The following pins require testpoints:
0 - GPIO_13
1 - GPIO_1
2 - GPIO_2
3 - GPIO_3
4 - GPIO_5
5 - PCIE_RST_1_N
6 - PCIE_RST_2_N
7 - PCIE_RST_3_N
8 - GPIO_15
9 - GPIO_11
10 - GPIO_14
11 - GPIO_0
12 - GPIO_12
13 - GPIO_10
14 - PB_LSTX
15 - PB_LSRX
SYNC_MASTER=J78_MLB
SYNC_DATE=02/21/2014
PAGE TITLE
Thunderbolt Host (1 of 2)
Apple Inc.
DRAWING NUMBER
051-1635
SIZE
D
REVISION
5.0.0
BRANCH
proto2
PAGE
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28 OF 115
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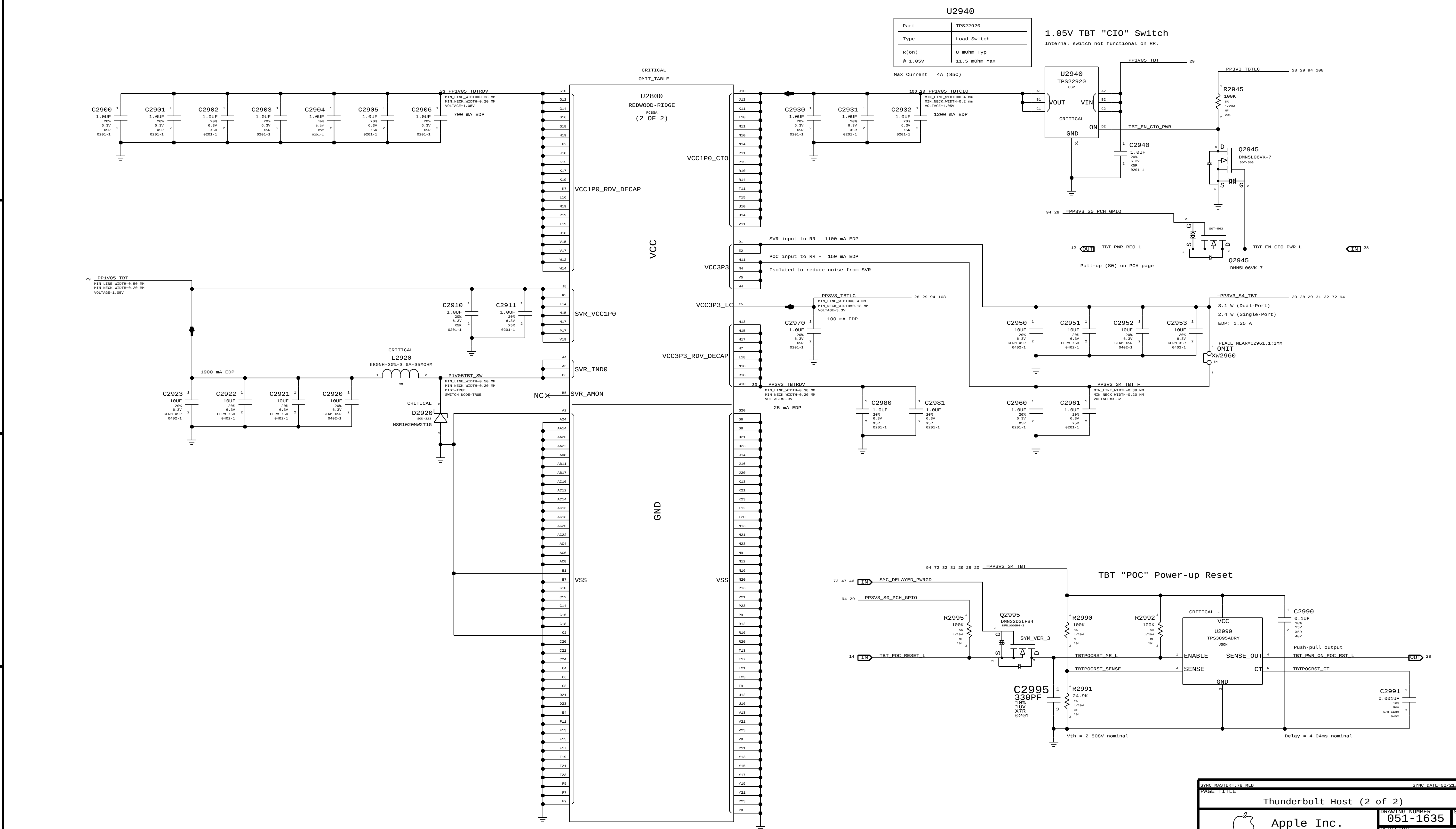

EDP current / power consumption figures copied from R68 schematic (Rev 2, dated October 28, 2012, not available on IBL).

SYNC_MASTER=J78_MLB SYNC_DATE=02/21/2014

PAGE TITLE: Thunderbolt Host (2 of 2)

Apple Inc.

DRAWING NUMBER: 051-1635 SIZE: D

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BRANCH: proto2

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Page Notes

Power aliases required by this page:
- =PPVIN_SW_TBTBST (8-13V Boost Input)
- =PP15V_TBT_REG (15V Boost Output)
- =PP3V3_TBT_P3V3TBTFET (3.3V FET Input)
- =PP3V3_TBT_FET (3.3V FET Output)
- =PP3V3_S0_TBTPWRCTL
- =PP1V05_TBT_P1V05TBTFET (1.05V FET Input)
- =PP1V05_TBT_FET (1.05V FET Output)

Signal aliases required by this page:
- =TBT_CLKREQ_L
- =TBT_RESET_L

BOM options provided by this page:
TBTBST:Y - Stuffs 15V boost circuitry.

SYNC_MASTER=J78_MLB SYNC_DATE=02/21/2014

PAGE TITLE
Thunderbolt Power Support

Apple Inc.

DRAWING NUMBER 051-1635 SIZE D

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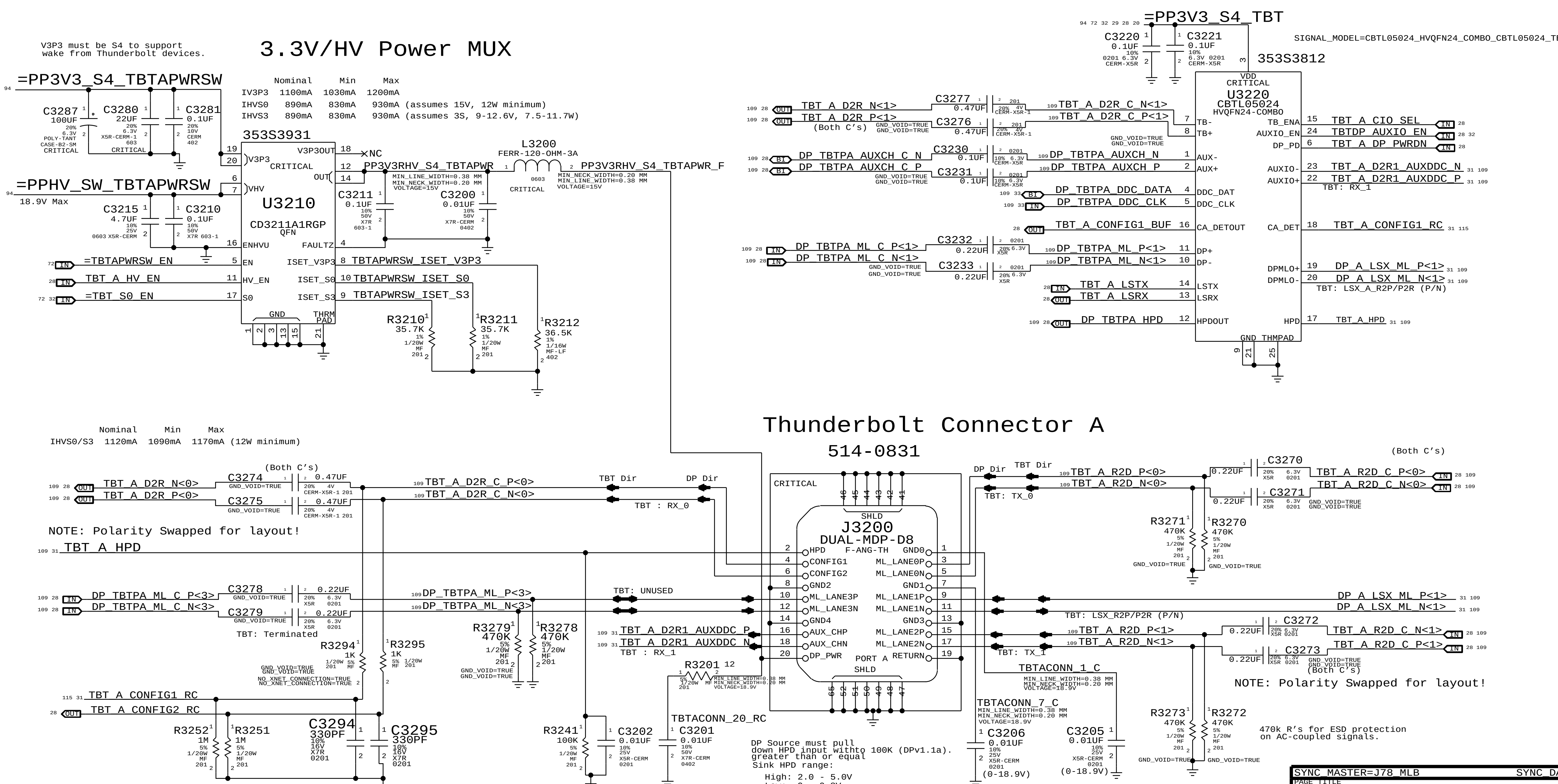

SYNC_MASTER=J78_MLB SYNC_DATE=02/21/2014

PAGE TITLE

Thunderbolt Connector A

Apple Inc.

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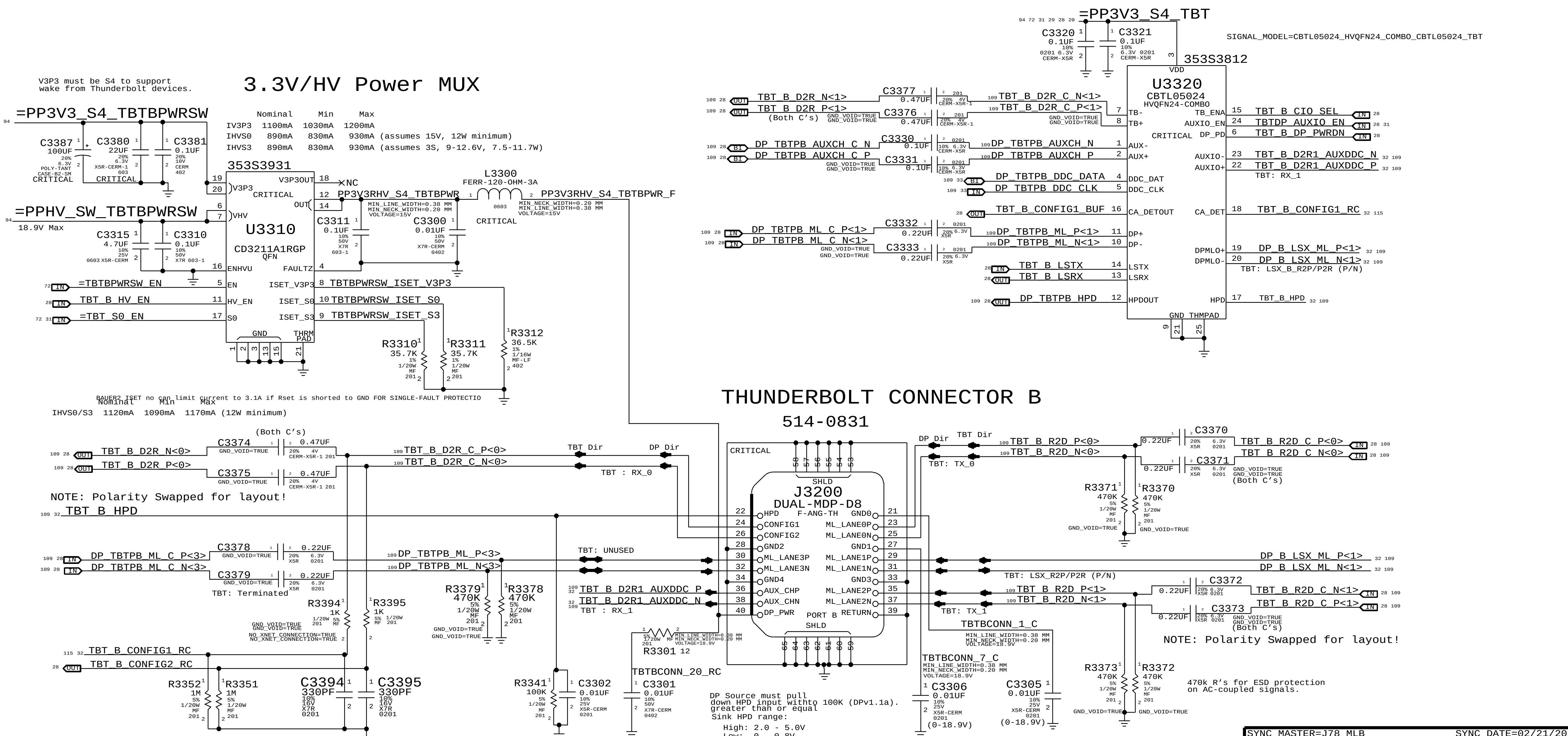

SYNC_MASTER=J78_MLB SYNC_DATE=02/21/2014

PAGE TITLE: Thunderbolt Connector B

Apple Inc.

DRAWING NUMBER: 051-1635 SIZE: D

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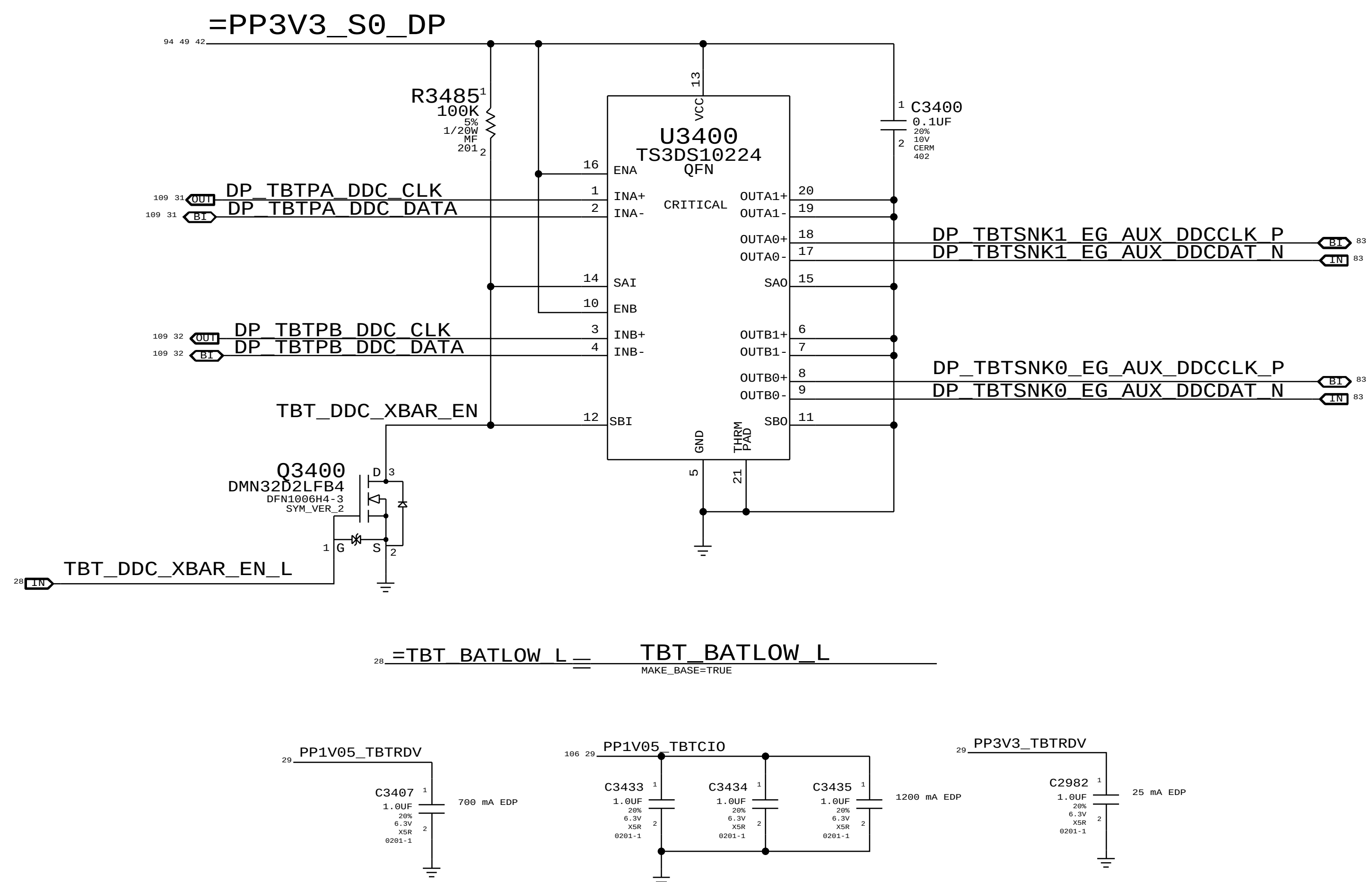
Dual-Port Host DDC Crossbar



Additional DeCaps added for CSA-29 (csa-29 is syncing from J44)

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PAGE TITLE

TBT DDC Crossbar

Apple Inc.

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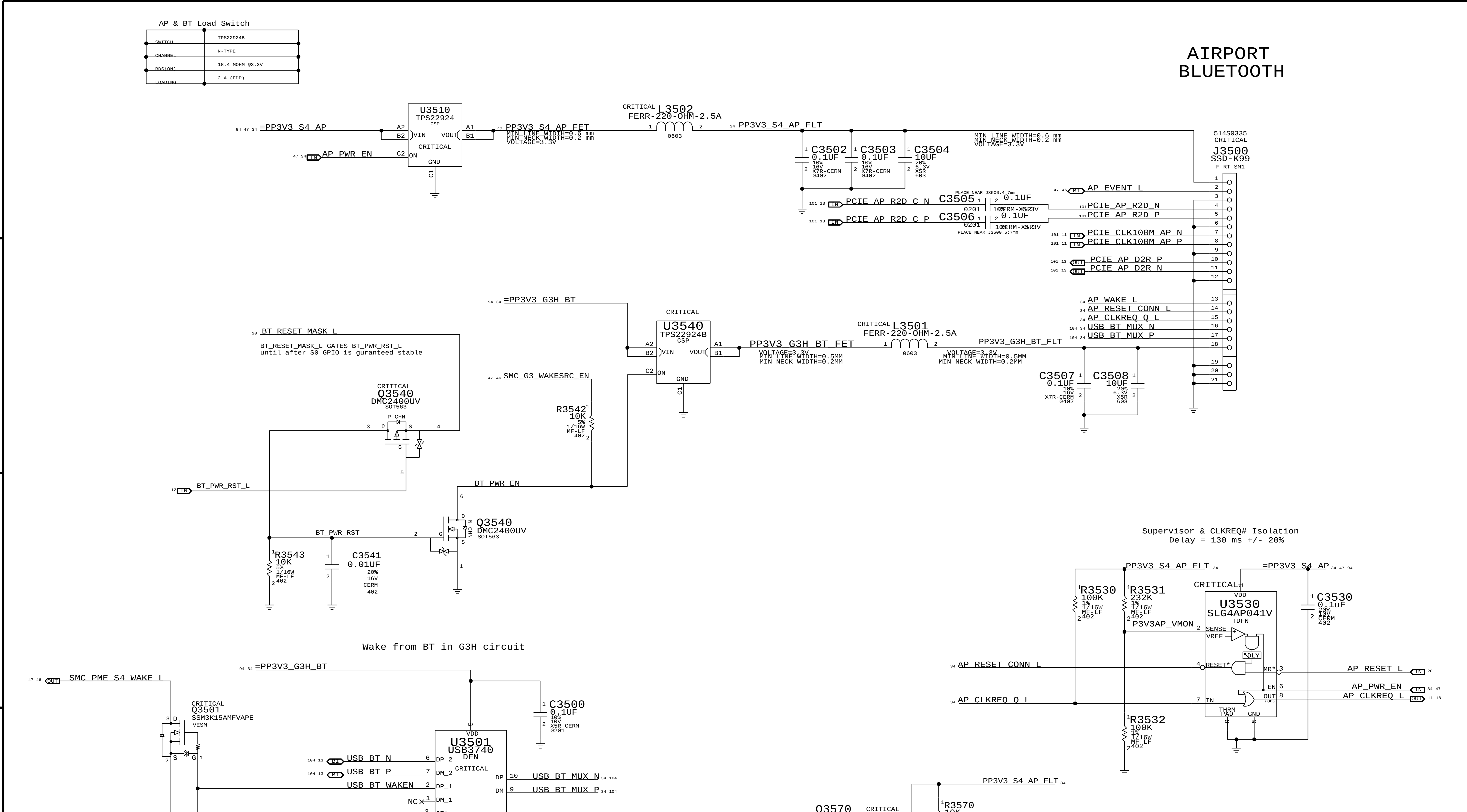

SYNC_MASTER=J78_MLB SYNC_DATE=02/21/2014

PAGE TITLE: AIRPORT/BT

Apple Inc.

DRAWING NUMBER: 051-1635 SIZE: D

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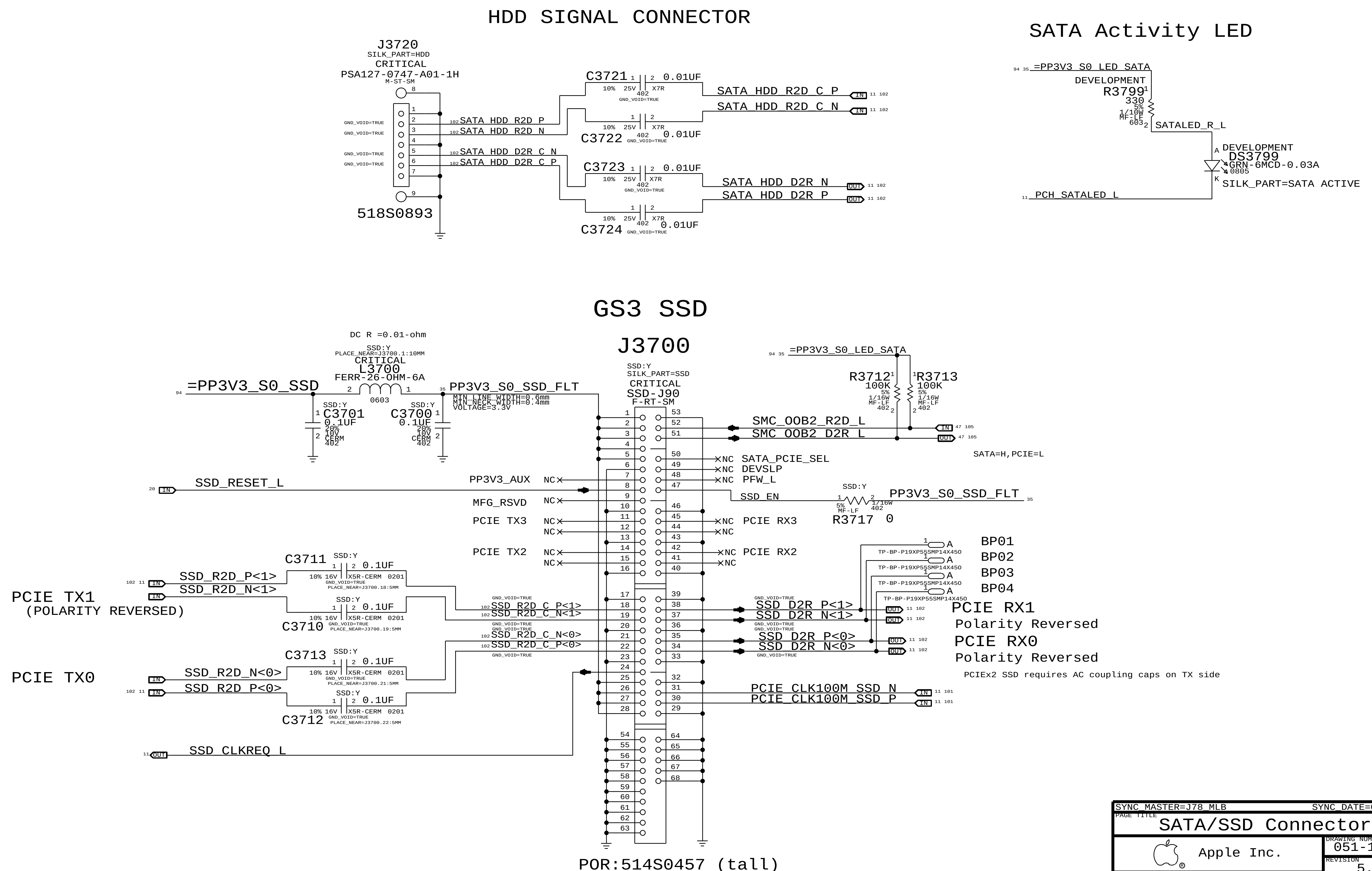

SYNC_MASTER=J78_MLB SYNC_DATE=02/21/2014

PAGE TITLE

SATA/SSD Connectors

Apple Inc.

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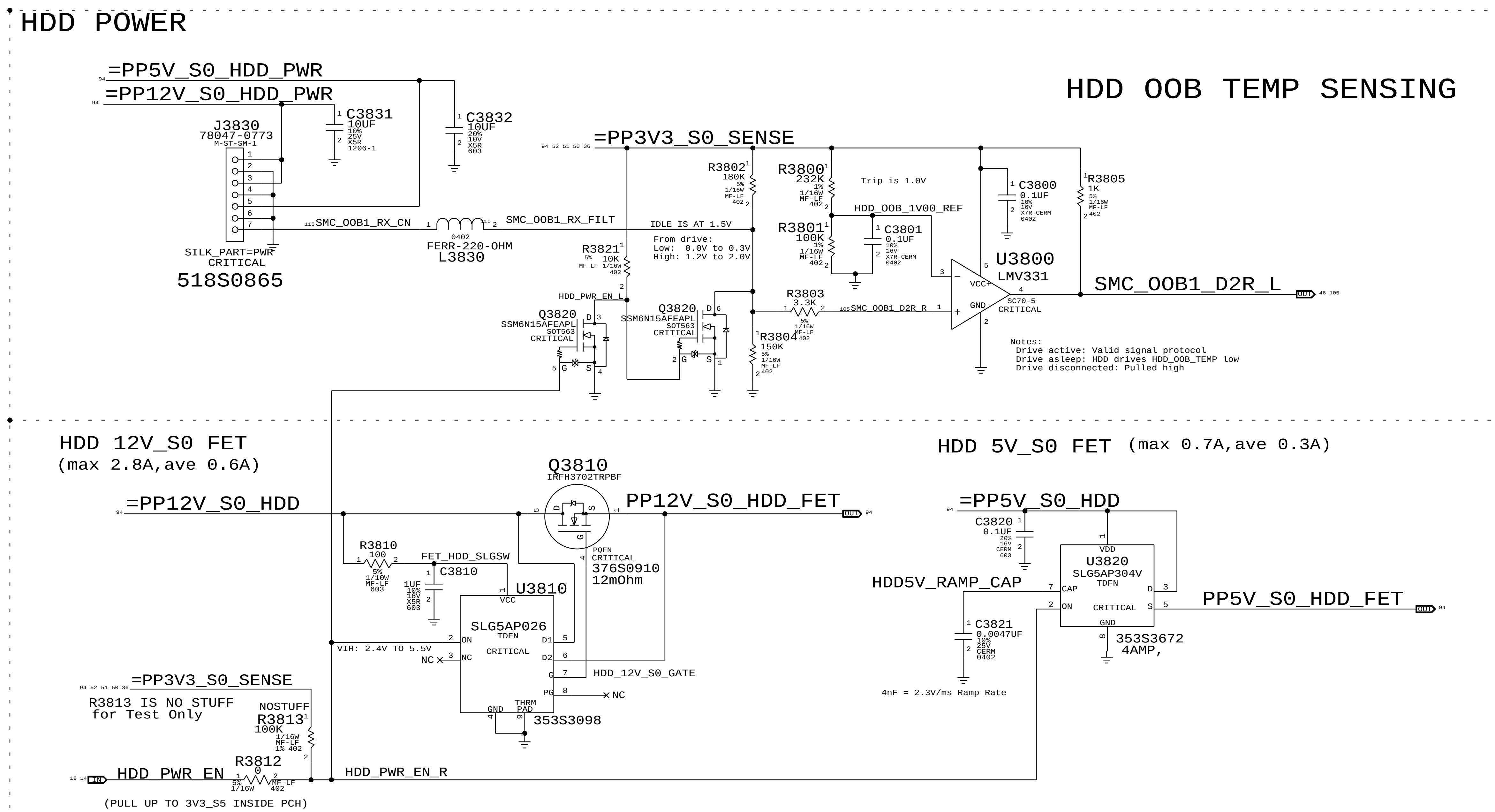

REMOVE R3813 AND SHORT R3812 AFTER HDD_PWR_EN WORKS

SYNC_MASTER=J78_MLB SYNC_DATE=02/21/2014

PAGE TITLE: HDD/SSD Temp Sense

Apple Inc.

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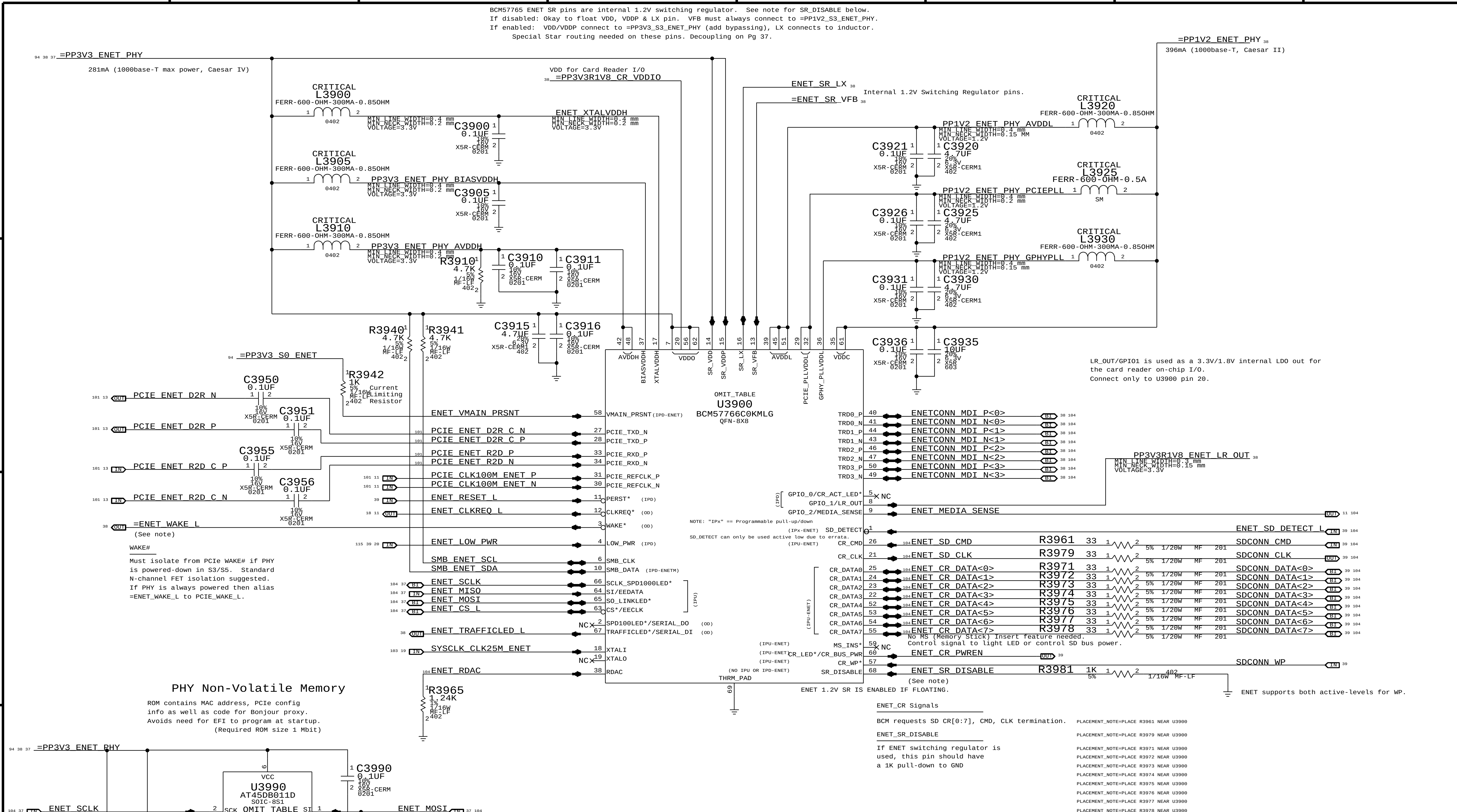

SYNC_MASTER=J78_MLB SYNC_DATE=02/21/2014

PAGE TITLE

ETHERNET PHY (CAESAR IV)

Apple Inc.

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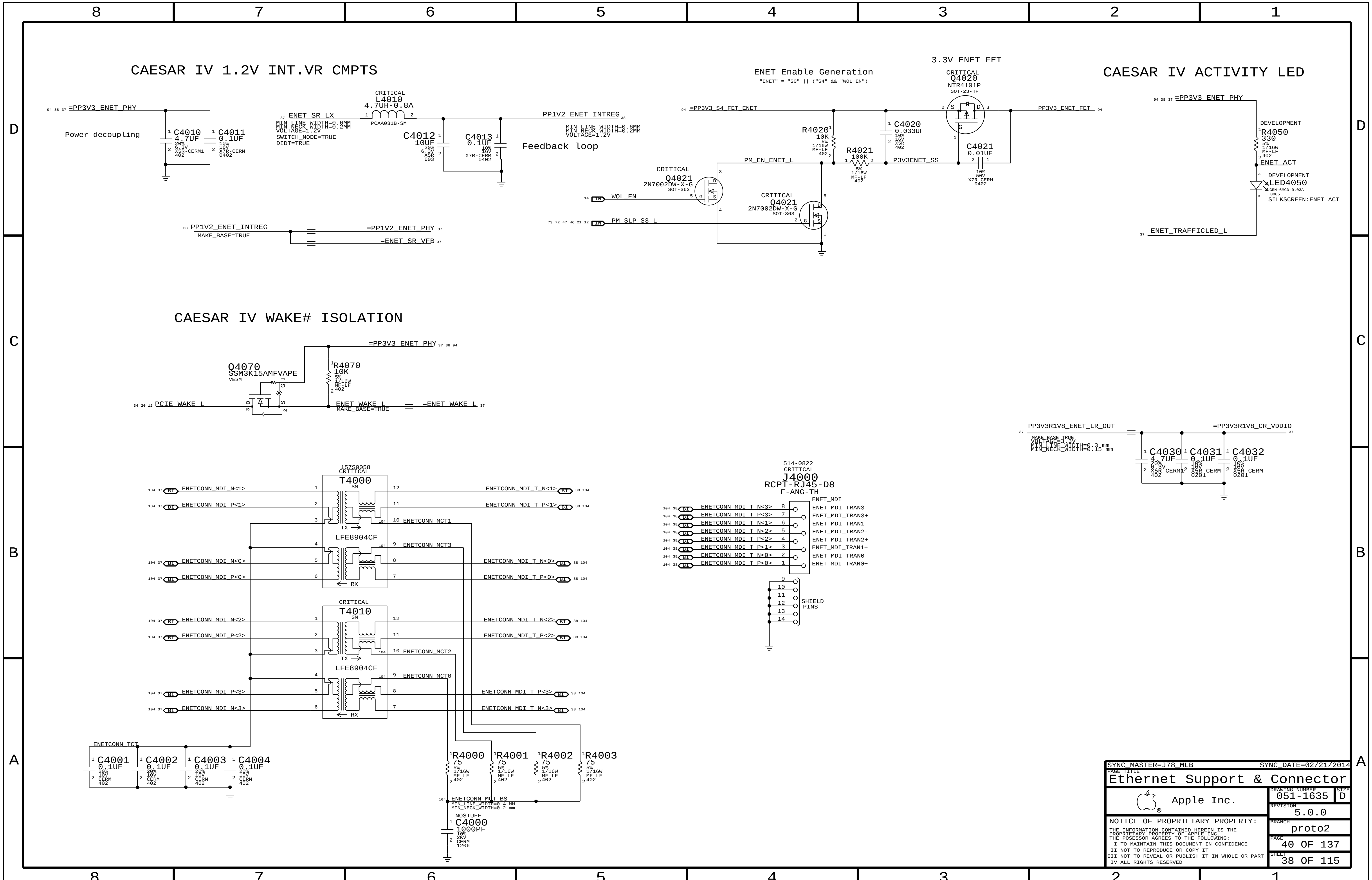

CAESAR IV 1.2V INT.VR CMPTS
=PP3V3_ENET_PHY
Power decoupling
C4010 4.7UF 20% 6.3V X5R-CERM1 402
C4011 0.1UF 10% 16V X7R-CERM 0402
CRITICAL
L4010 4.7UH-0.8A PCAA031B-SM
ENET_SR_LX
MIN_LINE_WIDTH=0.6MM
MIN_NECK_WIDTH=0.2MM
VOLTAGE=1.2V
SWITCH_NODE=TRUE
DIDT=TRUE
C4012 10UF 20% 6.3V X5R 603
C4013 0.1UF 10% 16V X7R-CERM 0402
PP1V2_ENET_INTREG
MIN_LINE_WIDTH=0.6MM
MIN_NECK_WIDTH=0.2MM
VOLTAGE=1.2V
Feedback loop
PP1V2_ENET_INTREG
MAKE_BASE=TRUE
=PP1V2_ENET_PHY
=ENET_SR_VFB
ENET Enable Generation
"ENET" = "S0" || ("S4" && "WOL_EN")
3.3V ENET FET
CRITICAL
Q4020 NTR4101P SOT-23-HF
=PP3V3_S4_FET_ENET
PP3V3_ENET_FET
R4020 10K 5% 1/16W MF-LF 402
C4020 0.033UF 10% 16V X5R 402
R4021 100K 5% 1/16W MF-LF 402
P3V3ENET_SS
C4021 0.01UF 10% 50V X7R-CERM 0402
PM_EN_ENET_L
CRITICAL
Q4021 2N7002DW-X-G SOT-363
WOL_EN
PM_SLP_S3_L
CAESAR IV ACTIVITY LED
=PP3V3_ENET_PHY
DEVELOPMENT
R4050 330 5% 1/16W MF-LF 402
ENET_ACT
LED4050 GRN-6MCD-0.03A 0805
SILKSCREEN:ENET ACT
ENET_TRAFFICLED_L
CAESAR IV WAKE# ISOLATION
=PP3V3_ENET_PHY
Q4070 SSM3K15AMFVAPE VESM
R4070 10K 5% 1/16W MF-LF 402
PCIE_WAKE_L
ENET_WAKE_L
MAKE_BASE=TRUE
=ENET_WAKE_L
PP3V3R1V8_ENET_LR_OUT
MAKE_BASE=TRUE
VOLTAGE=3.3V
MIN_LINE_WIDTH=0.3 mm
MIN_NECK_WIDTH=0.15 mm
=PP3V3R1V8_CR_VDDIO
C4030 4.7UF 20% 6.3V X5R-CERM1 402
C4031 0.1UF 10% 16V X5R-CERM 0201
C4032 0.1UF 10% 16V X5R-CERM 0201
157S0058
CRITICAL
T4000 SM
LFE8904CF
TX
RX
CRITICAL
T4010 SM
LFE8904CF
ENETCONN_MDI_N<1>
ENETCONN_MDI_P<1>
ENETCONN_MDI_N<0>
ENETCONN_MDI_P<0>
ENETCONN_MDI_N<2>
ENETCONN_MDI_P<2>
ENETCONN_MDI_P<3>
ENETCONN_MDI_N<3>
ENETCONN_MDI_T_N<1>
ENETCONN_MDI_T_P<1>
ENETCONN_MCT1
ENETCONN_MCT3
ENETCONN_MDI_T_N<0>
ENETCONN_MDI_T_P<0>
ENETCONN_MDI_T_N<2>
ENETCONN_MDI_T_P<2>
ENETCONN_MCT2
ENETCONN_MCT0
ENETCONN_MDI_T_P<3>
ENETCONN_MDI_T_N<3>
514-0822
CRITICAL
J4000 RCPT-RJ45-D8 F-ANG-TH
ENET_MDI
ENETCONN_MDI_T_N<3> 8 ENET_MDI_TRAN3-
ENETCONN_MDI_T_P<3> 7 ENET_MDI_TRAN3+
ENETCONN_MDI_T_N<1> 6 ENET_MDI_TRAN1-
ENETCONN_MDI_T_N<2> 5 ENET_MDI_TRAN2-
ENETCONN_MDI_T_P<2> 4 ENET_MDI_TRAN2+
ENETCONN_MDI_T_P<1> 3 ENET_MDI_TRAN1+
ENETCONN_MDI_T_N<0> 2 ENET_MDI_TRAN0-
ENETCONN_MDI_T_P<0> 1 ENET_MDI_TRAN0+
SHIELD PINS
ENETCONN_TCT
C4001 0.1UF 20% 10V CERM 402
C4002 0.1UF 20% 10V CERM 402
C4003 0.1UF 20% 10V CERM 402
C4004 0.1UF 20% 10V CERM 402
R4000 75 5% 1/16W MF-LF 402
R4001 75 5% 1/16W MF-LF 402
R4002 75 5% 1/16W MF-LF 402
R4003 75 5% 1/16W MF-LF 402
ENETCONN_MCT_BS
MIN_LINE_WIDTH=0.4 MM
MIN_NECK_WIDTH=0.2 MM
NOSTUFF
C4000 1000PF 10% 2KV CERM 1206
SYNC_MASTER=J78_MLB
SYNC_DATE=02/21/2014
PAGE TITLE
Ethernet Support & Connector
Apple Inc.
DRAWING NUMBER 051-1635
SIZE D
REVISION 5.0.0
BRANCH proto2
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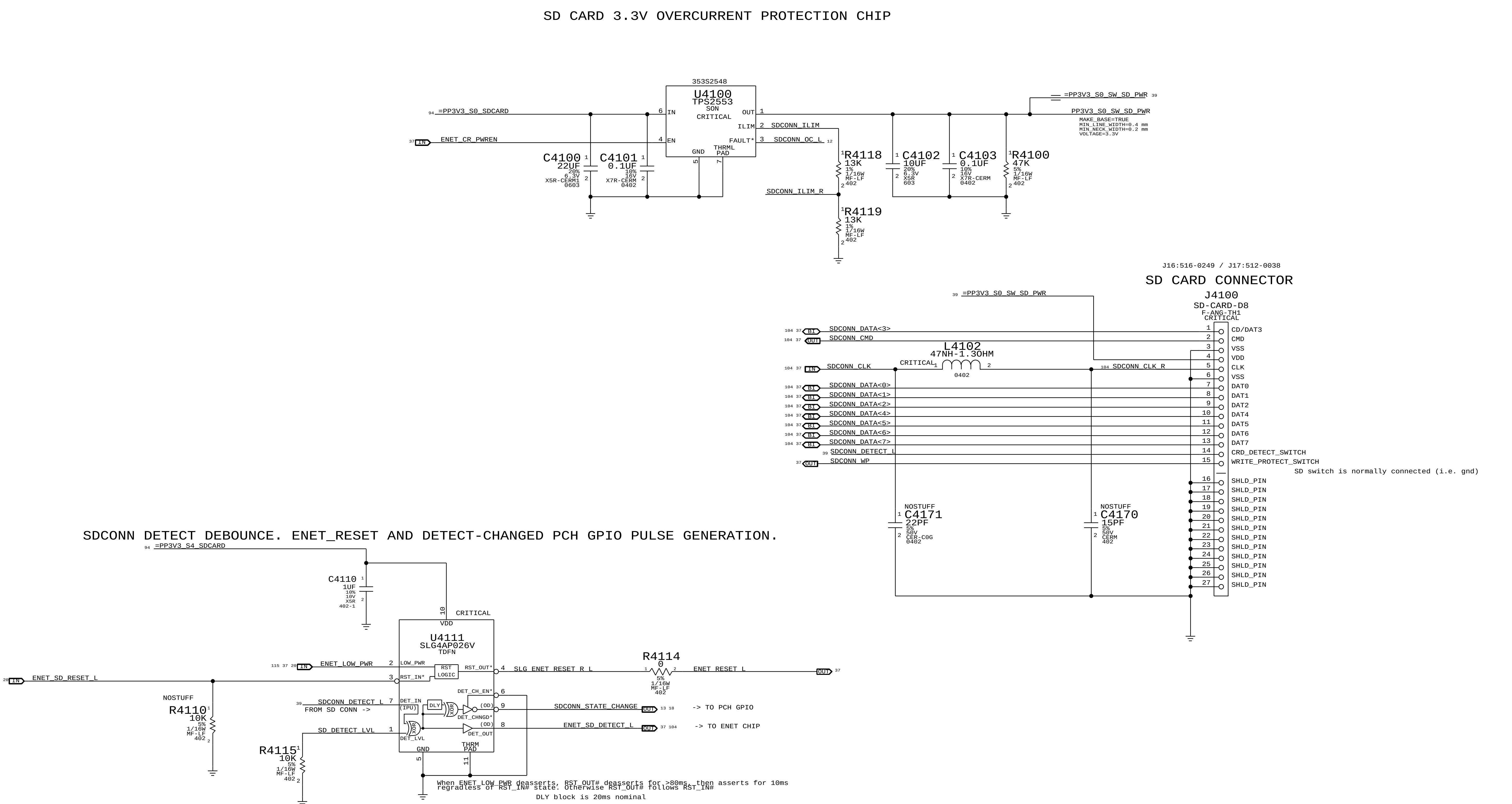

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PAGE TITLE

SD READER CONNECTOR

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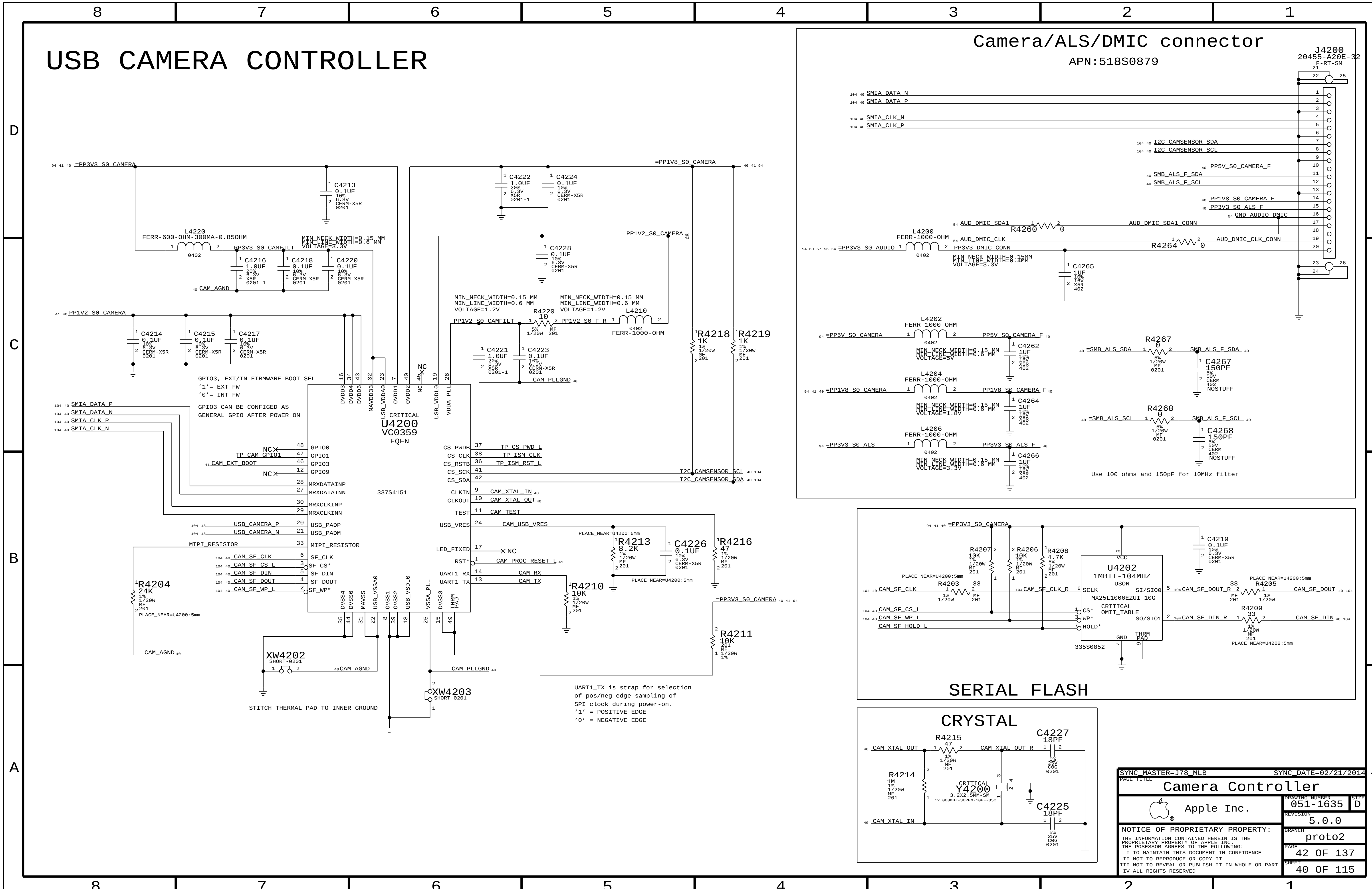

USB CAMERA CONTROLLER
Camera/ALS/DMIC connector
APN:518S0879
J4200
20455-A20E-32
F-RT-SM
U4200
VC0359
FQFN
CRITICAL
337S4151
GPIO3, EXT/IN FIRMWARE BOOT SEL
'1'= EXT FW
'0'= INT FW
GPIO3 CAN BE CONFIGED AS
GENERAL GPIO AFTER POWER ON
STITCH THERMAL PAD TO INNER GROUND
UART1_TX is strap for selection
of pos/neg edge sampling of
SPI clock during power-on.
'1' = POSITIVE EDGE
'0' = NEGATIVE EDGE
Use 100 ohms and 150pF for 10MHz filter
SERIAL FLASH
U4202
1MBIT-104MHZ
USON
MX25L1006EZUI-10G
CRITICAL
OMIT_TABLE
335S0852
CRYSTAL
Y4200
3.2X2.5MM-SM
12.000MHZ-30PPM-10PF-85C
SYNC_MASTER=J78_MLB
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PAGE TITLE
Camera Controller
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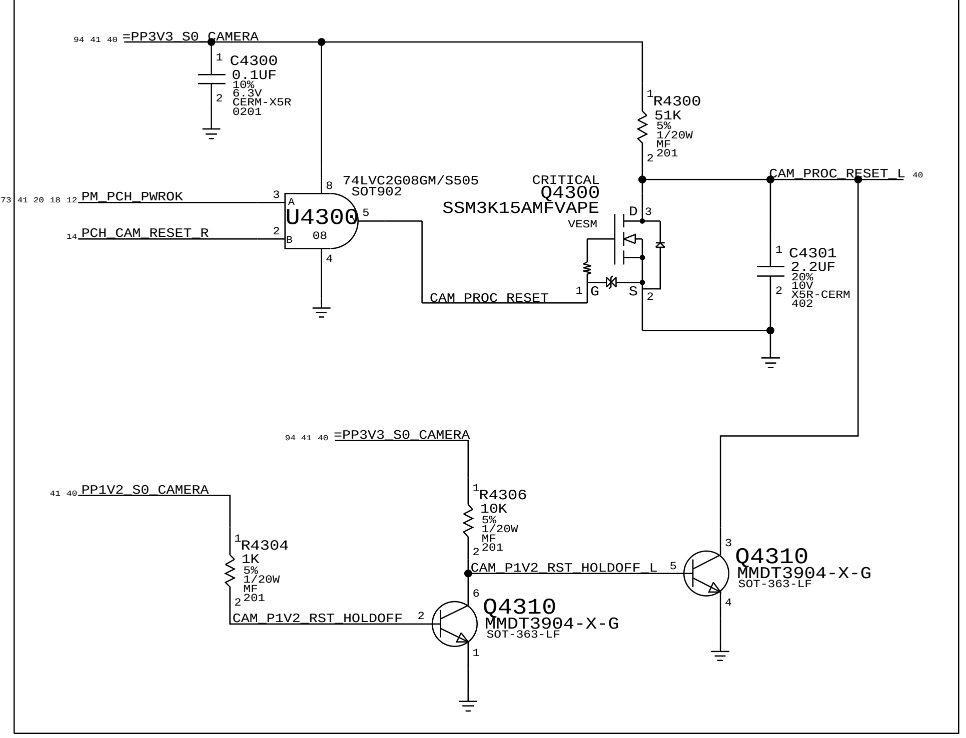
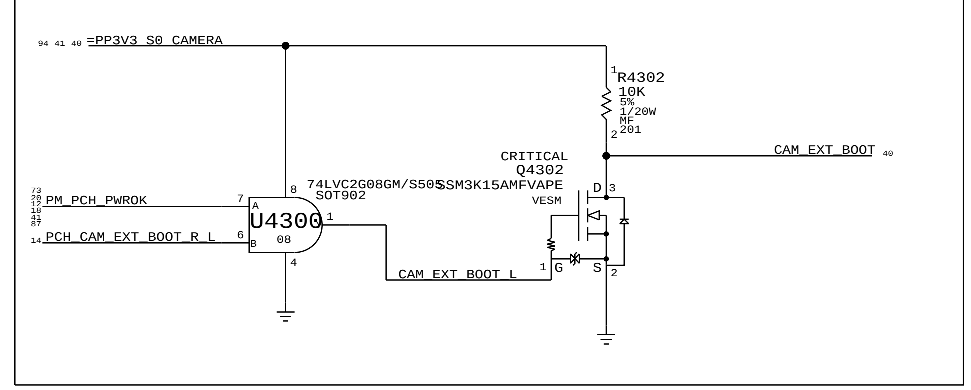
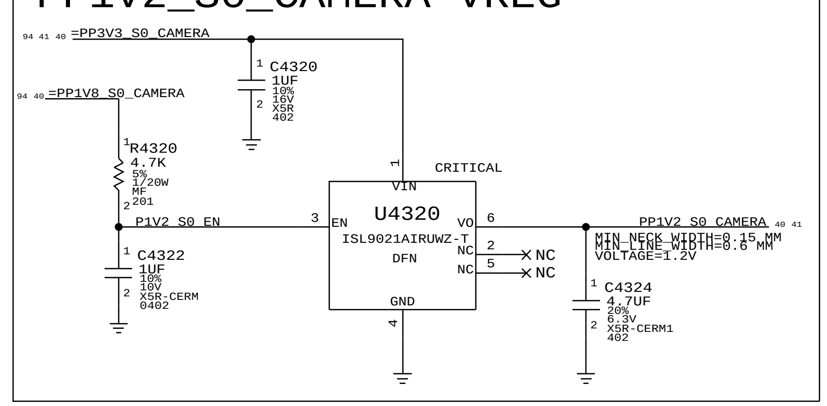
Camera Processor Reset



Camera Processor ExtBoot Cntl



PP1V2_S0_CAMERA VREG



SYNC_MASTER=J78_MLB SYNC_DATE=02/21/2014

PAGE TITLE
Camera Controller Support

Apple Inc.

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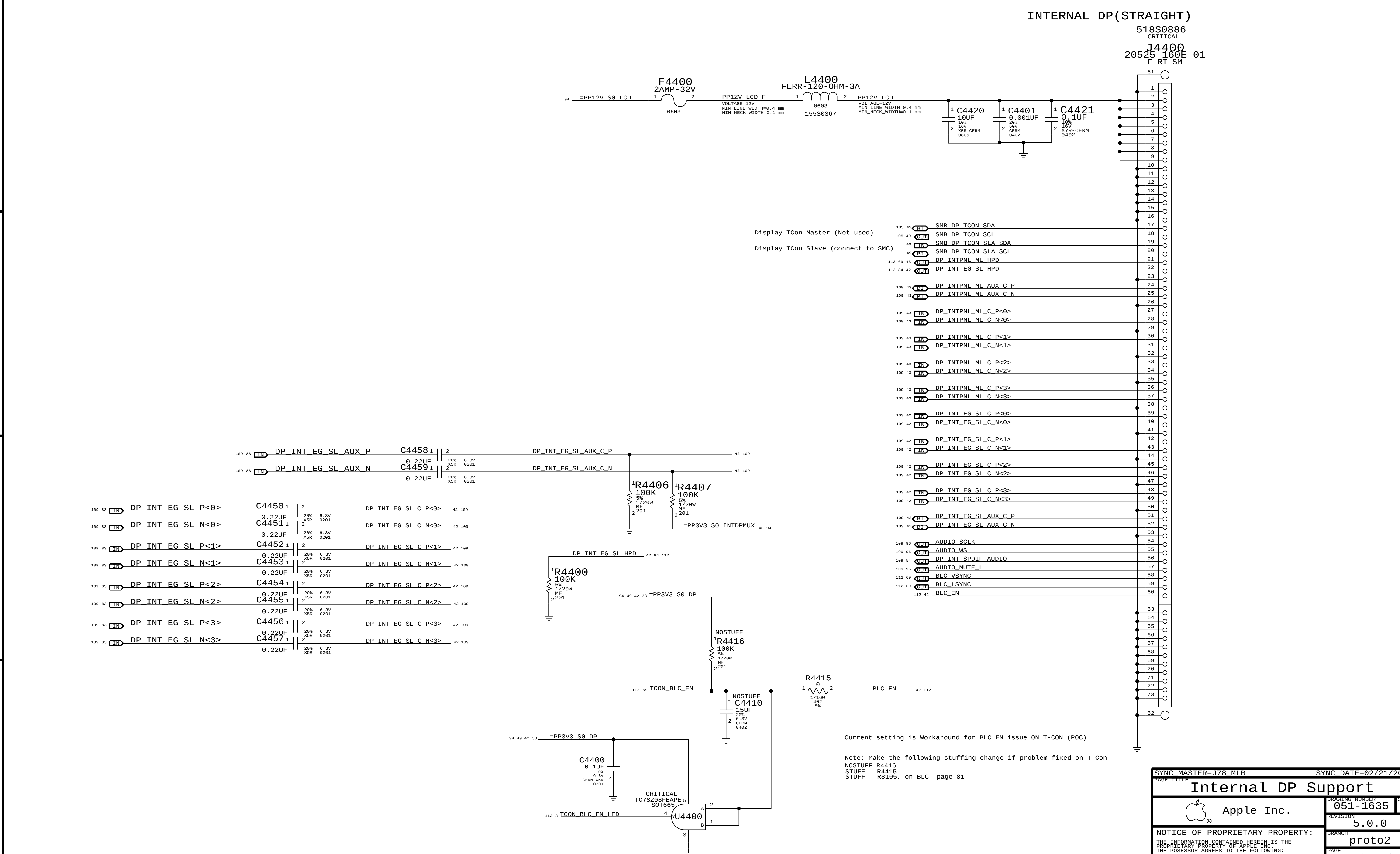

SYNC_MASTER=J78_MLB SYNC_DATE=02/21/2014

PAGE TITLE
Internal DP Support

Apple Inc.

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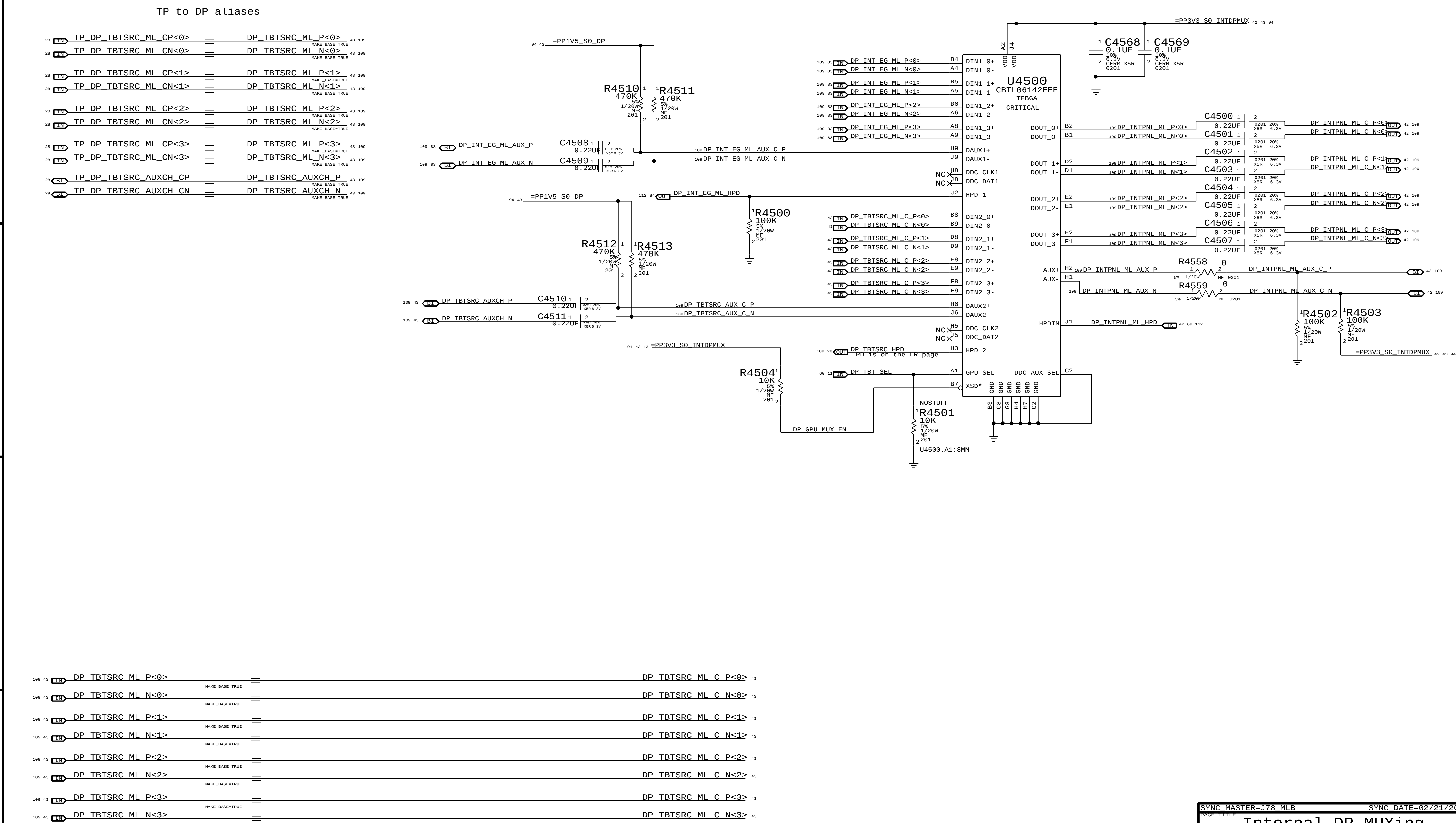

SYNC_MASTER=J78_MLB SYNC_DATE=02/21/2014

PAGE TITLE

Internal DP MUXing

Apple Inc.

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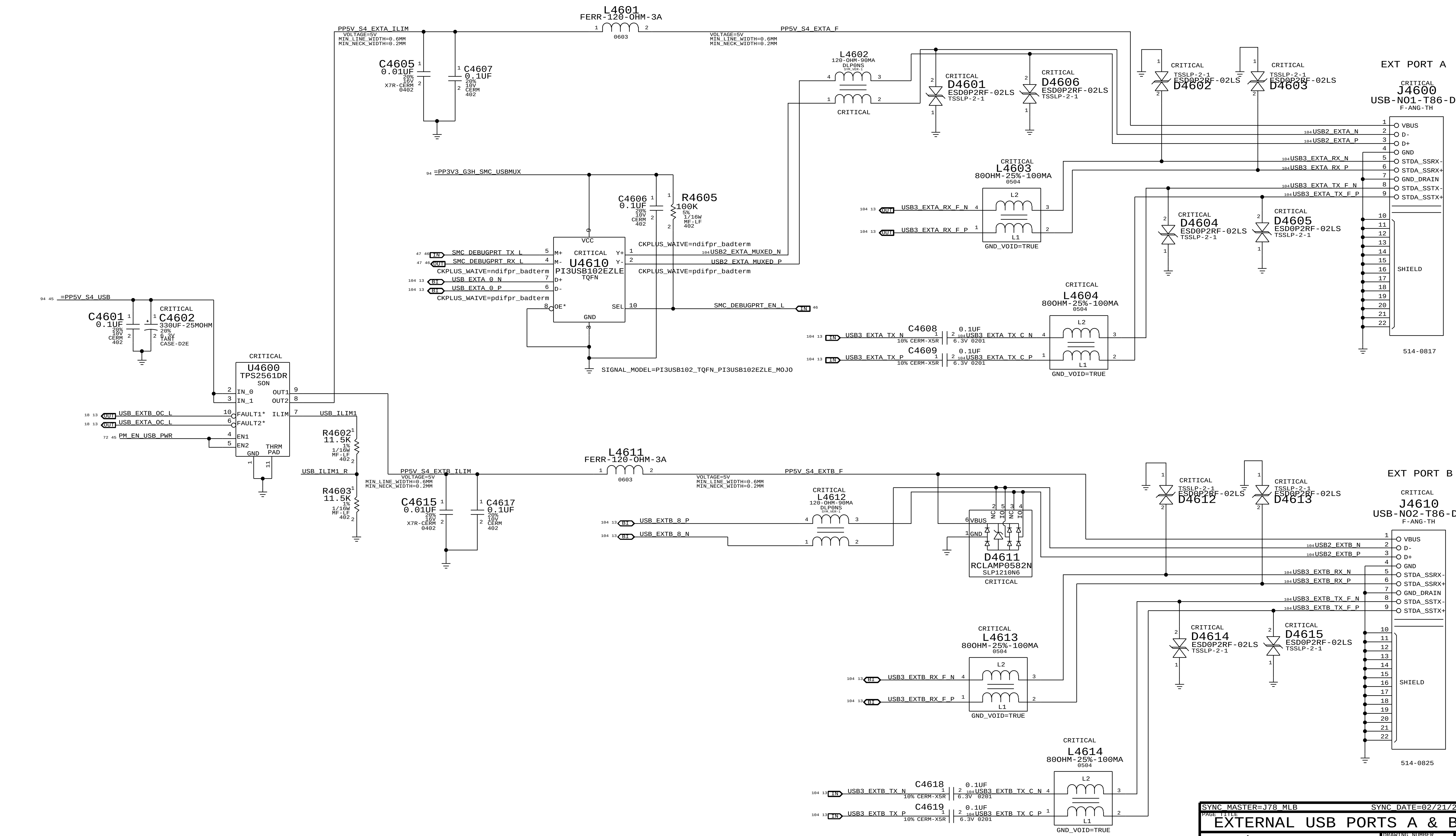

SYNC_MASTER=J78_MLB SYNC_DATE=02/21/2014

PAGE TITLE

EXTERNAL USB PORTS A & B

Apple Inc.

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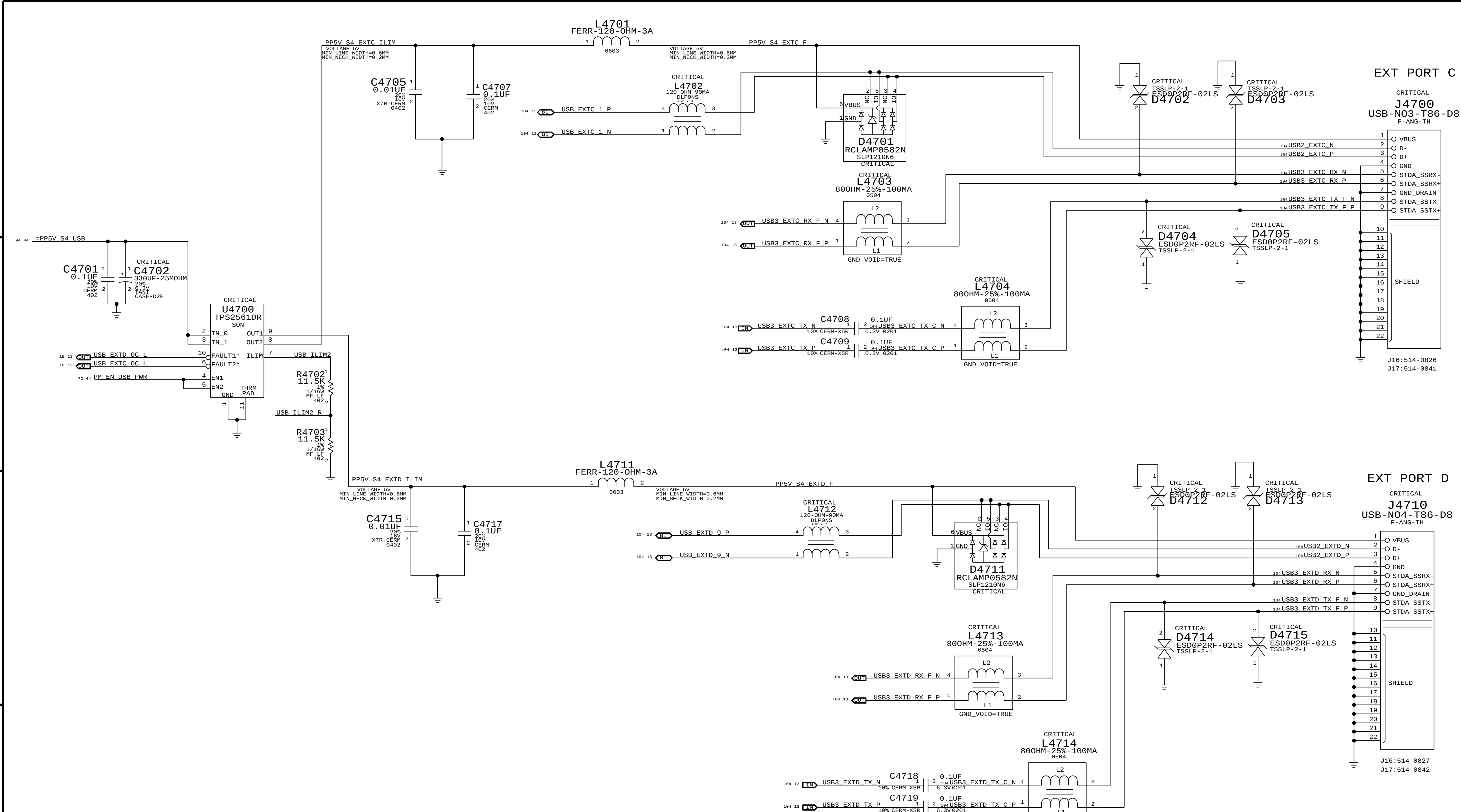

SYNC_MASTER=J78_MLB SYNC_DATE=02/21/2014

PAGE TITLE

EXTERNAL USB PORTS C & D

Apple Inc.

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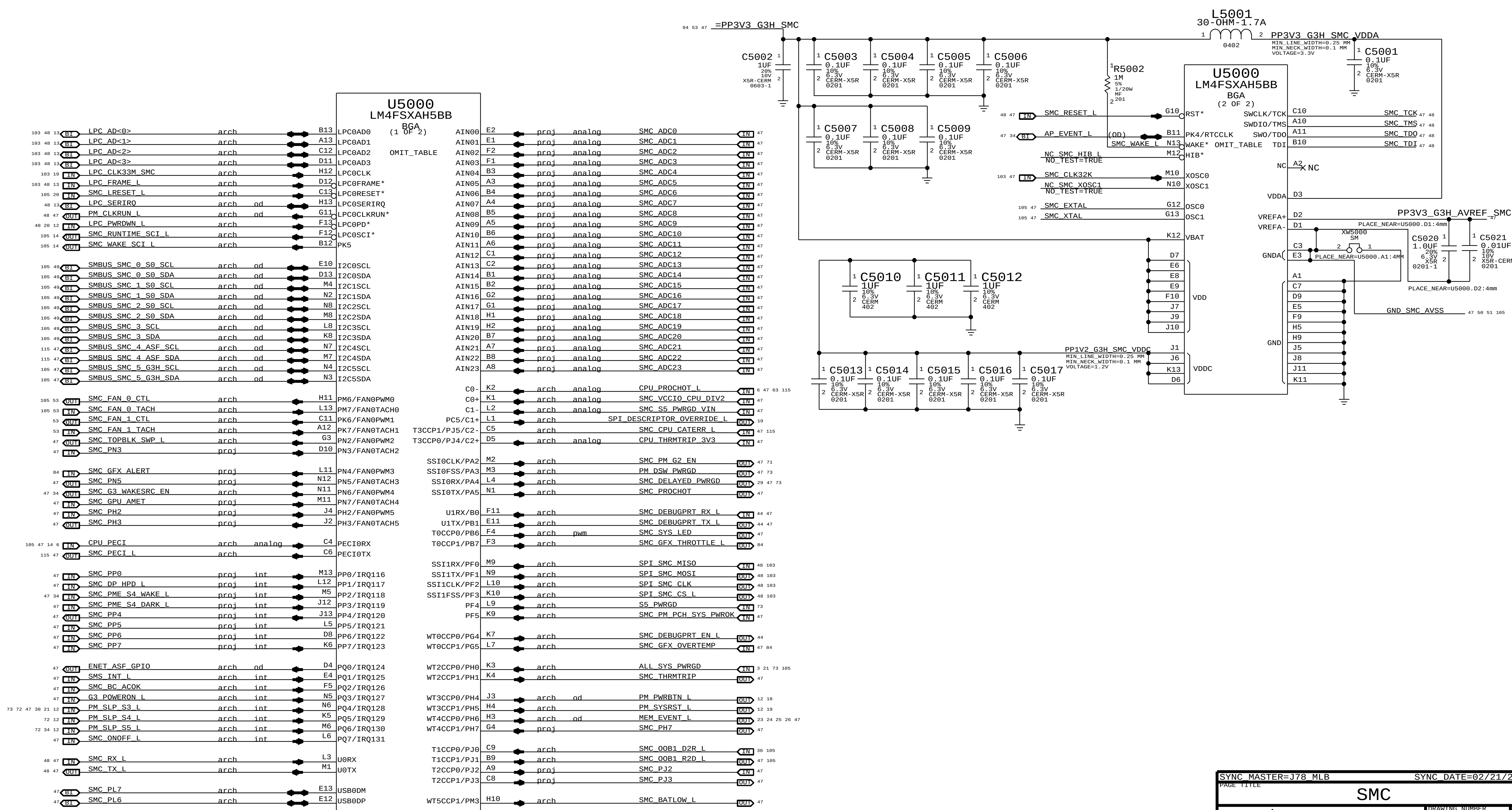
NOTE: Unused pins have "SMC_Pxx" names. Unused
pins designed as outputs can be left floating,
those designated as inputs require pull-ups.



SYNC_MASTER=J78_MLB		SYNC_DATE=02/21/2014
PAGE TITLE: SMC		
Apple Inc.	DRAWING NUMBER: 051-1635	SIZE: D
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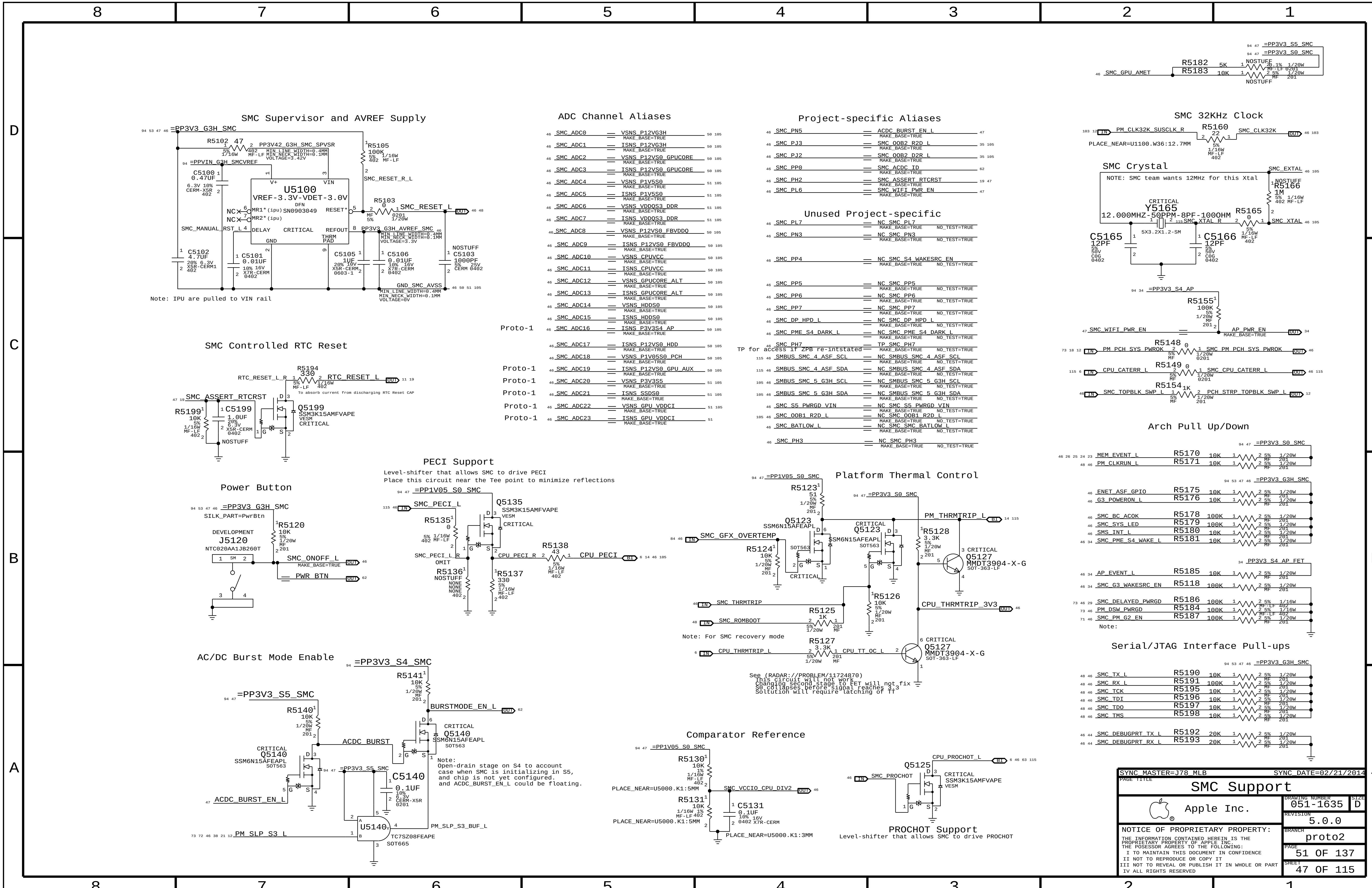
SMC Supervisor and AVREF Supply
=PP3V3_G3H_SMC
U5100
VREF-3.3V-VDET-3.0V
SN0903049
SMC_RESET_L
PP3V42_G3H_SMC_SPVSR
PP3V3_G3H_AVREF_SMC
GND_SMC_AVSS
Note: IPU are pulled to VIN rail
ADC Channel Aliases
SMC_ADC0 VSNS_P12VG3H
SMC_ADC1 ISNS_P12VG3H
SMC_ADC2 VSNS_P12VS0_GPUCORE
SMC_ADC3 ISNS_P12VS0_GPUCORE
SMC_ADC4 VSNS_P1V5S0
SMC_ADC5 ISNS_P1V5S0
SMC_ADC6 VSNS_VDDQS3_DDR
SMC_ADC7 ISNS_VDDQS3_DDR
SMC_ADC8 VSNS_P12VS0_FBVDDQ
SMC_ADC9 ISNS_P12VS0_FBVDDQ
SMC_ADC10 VSNS_CPUVCC
SMC_ADC11 ISNS_CPUVCC
SMC_ADC12 VSNS_GPUCORE_ALT
SMC_ADC13 ISNS_GPUCORE_ALT
SMC_ADC14 VSNS_HDDS0
SMC_ADC15 ISNS_HDDS0
Proto-1 SMC_ADC16 ISNS_P3V3S4_AP
SMC_ADC17 ISNS_P12VS0_HDD
SMC_ADC18 VSNS_P1V05S0_PCH
Proto-1 SMC_ADC19 ISNS_P12VS0_GPU_AUX
Proto-1 SMC_ADC20 VSNS_P3V3S5
Proto-1 SMC_ADC21 ISNS_SSDS0
Proto-1 SMC_ADC22 VSNS_GPU_VDDCI
Proto-1 SMC_ADC23 ISNS_GPU_VDDCI
Project-specific Aliases
Unused Project-specific
TP for access if ZPB re-intstated
SMC 32KHz Clock
SMC Crystal
NOTE: SMC team wants 12MHz for this Xtal
Y5165
12.000MHZ-50PPM-8PF-100OHM
SMC Controlled RTC Reset
To absorb current from discharging RTC Reset CAP
Power Button
PECI Support
Level-shifter that allows SMC to drive PECI
Place this circuit near the Tee point to minimize reflections
Platform Thermal Control
Note: For SMC recovery mode
See (RADAR://PROBLEM/11724870)
This circuit will not work
Changing second stage to FET will not fix
S0 collapses before signal reaches 3.3
Soltution will require latching of TT
Arch Pull Up/Down
Serial/JTAG Interface Pull-ups
AC/DC Burst Mode Enable
Note:
Open-drain stage on S4 to account case when SMC is initializing in S5, and chip is not yet configured, and ACDC_BURST_EN_L could be floating.
Comparator Reference
PROCHOT Support
Level-shifter that allows SMC to drive PROCHOT
SYNC_MASTER=J78_MLB
SYNC_DATE=02/21/2014
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SMC Support
Apple Inc.
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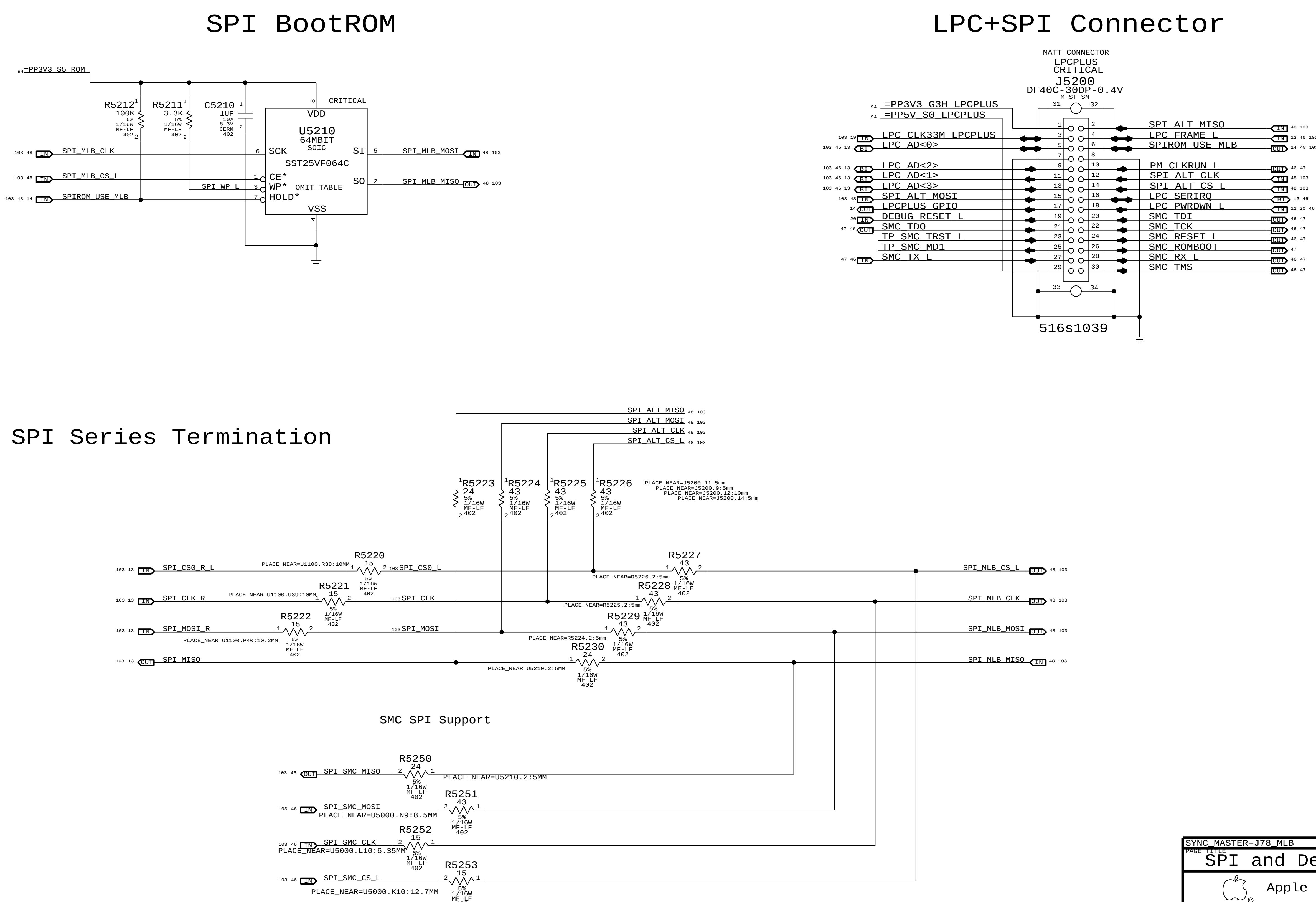

SYNC_MASTER=J78_MLB SYNC_DATE=02/21/2014

PAGE TITLE

SPI and Debug Connector

Apple Inc.

DRAWING NUMBER 051-1635 SIZE D

REVISION 5.0.0

BRANCH proto2

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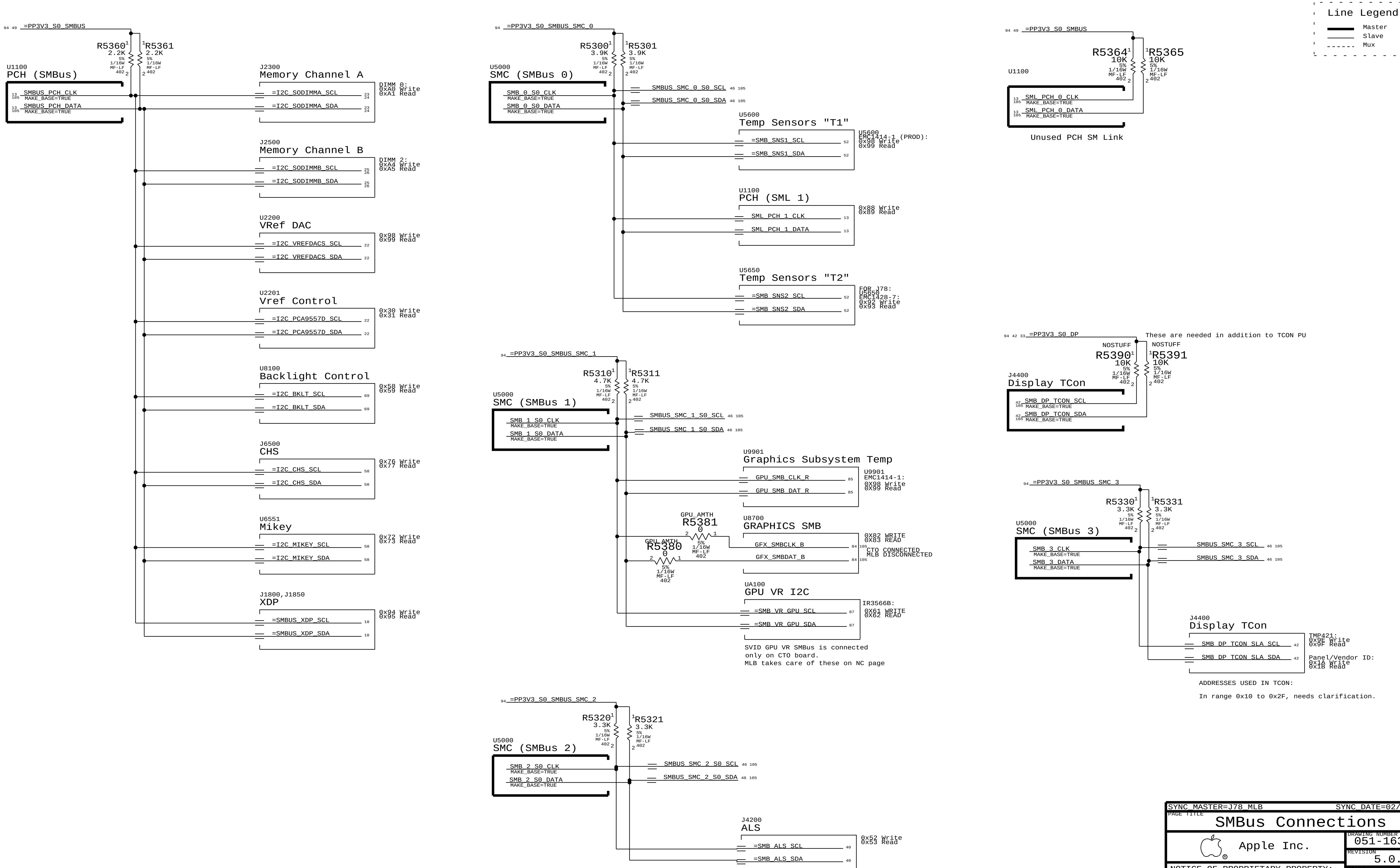
SHEET 48 OF 115



SYNC_MASTER=J78_MLB SYNC_DATE=02/21/2014

PAGE TITLE

SMBus Connections

Apple Inc.

DRAWING NUMBER 051-1635 SIZE D

REVISION 5.0.0

BRANCH proto2

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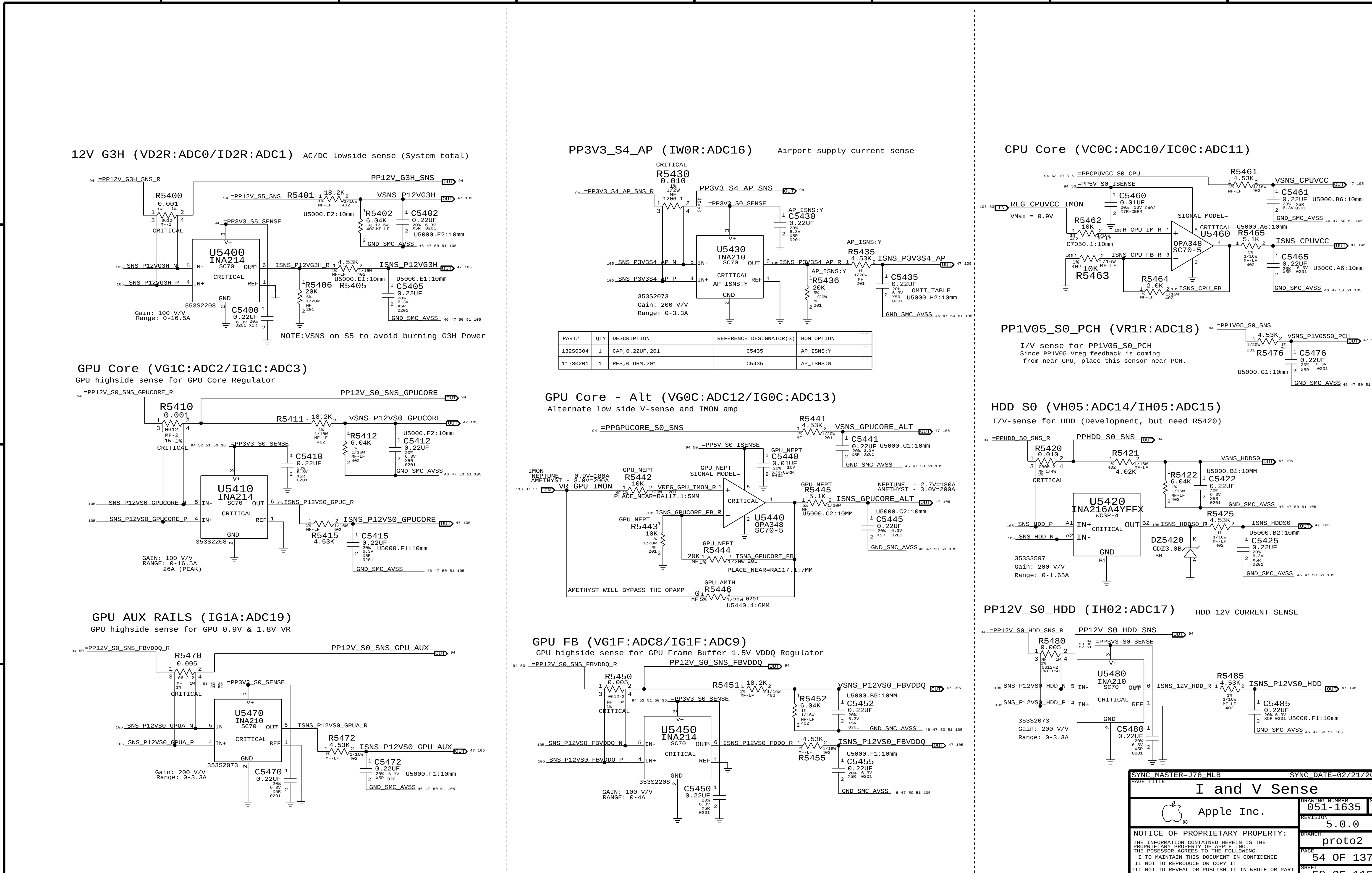

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	BOM OPTION
132S0304	1	CAP,0.22UF,201	C5435	AP_ISNS:Y
117S0201	1	RES,0 OHM,201	C5435	AP_ISNS:N

SYNC_MASTER=J78_MLB SYNC_DATE=02/21/2014

PAGE TITLE

I and V Sense

Apple Inc.

DRAWING NUMBER 051-1635 SIZE D

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BRANCH proto2

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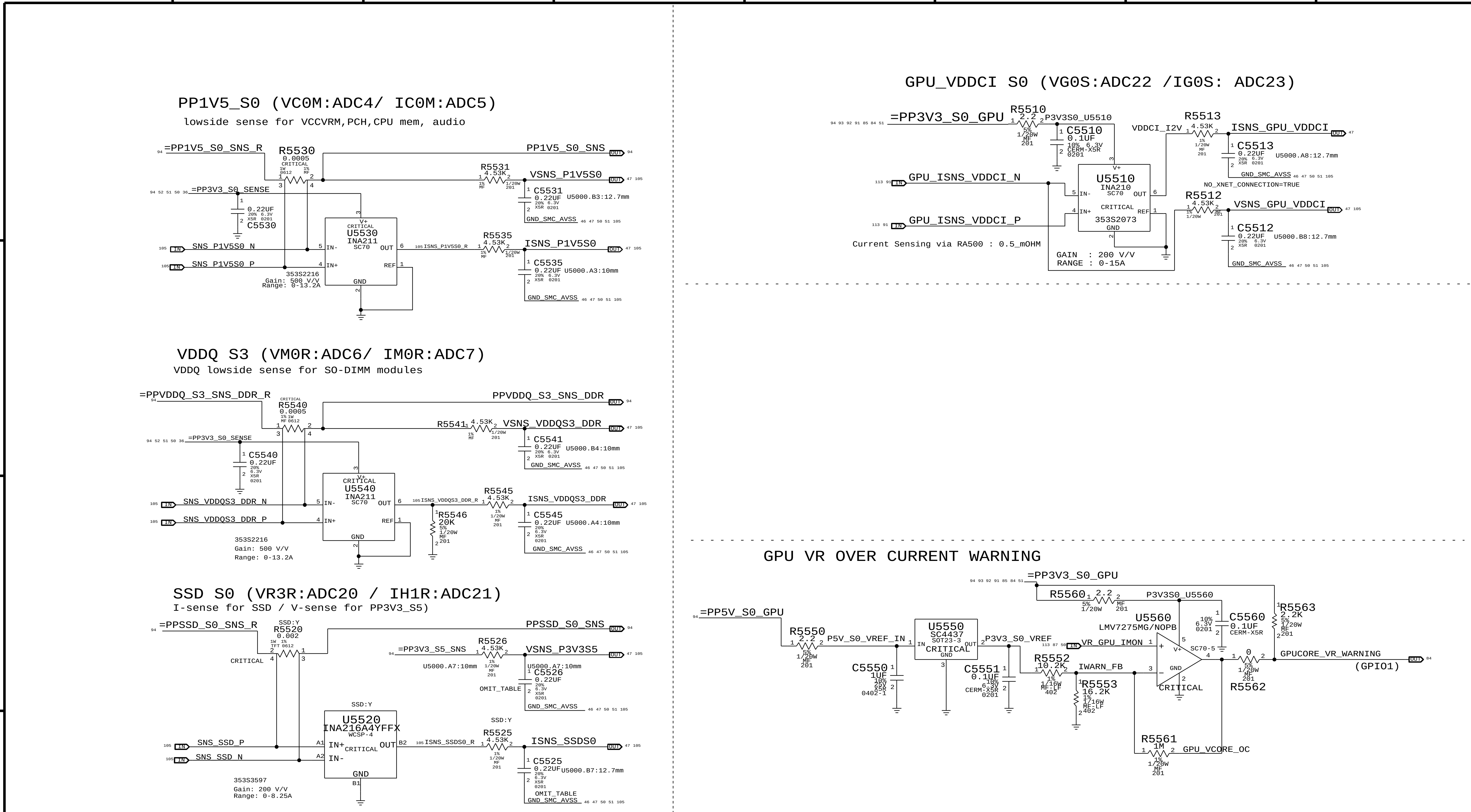

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	BOM OPTION
132S0304	2	CAP,0.22UF,201	C5525,C5526	SSD:Y
117S0201	2	RES,0 OHM,201	C5525,C5526	SSD:N

SYNC_MASTER=J78_NAT SYNC_DATE=12/09/2013

PAGE TITLE

I and V Sense(Continued)

Apple Inc.

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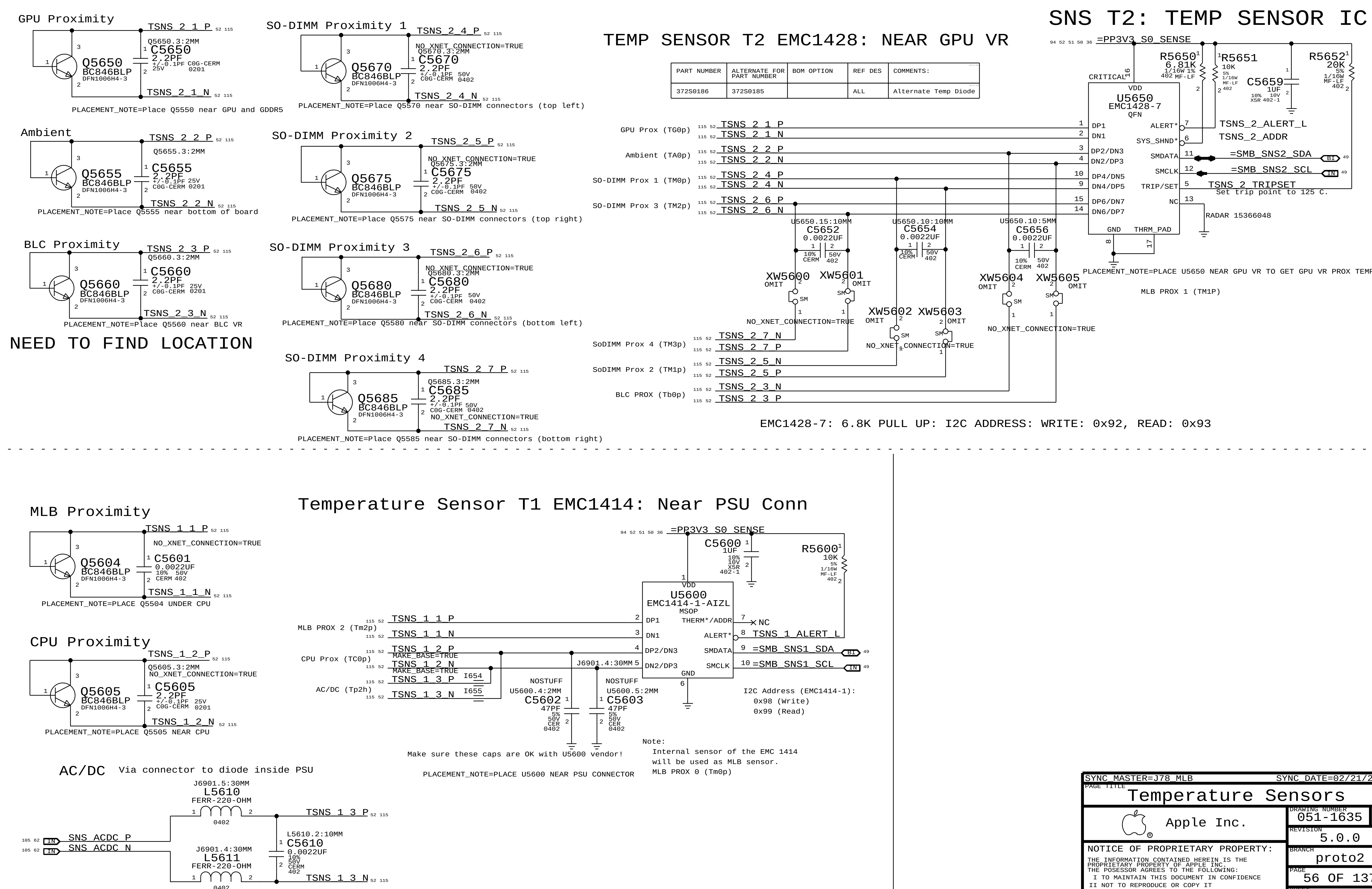
BRANCH proto2



PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
372S0186	372S0185		ALL	Alternate Temp Diode

SYNC_MASTER=J78_MLB SYNC_DATE=02/21/2014

PAGE TITLE: Temperature Sensors

Apple Inc.

DRAWING NUMBER: 051-1635 SIZE: D

REVISION: 5.0.0

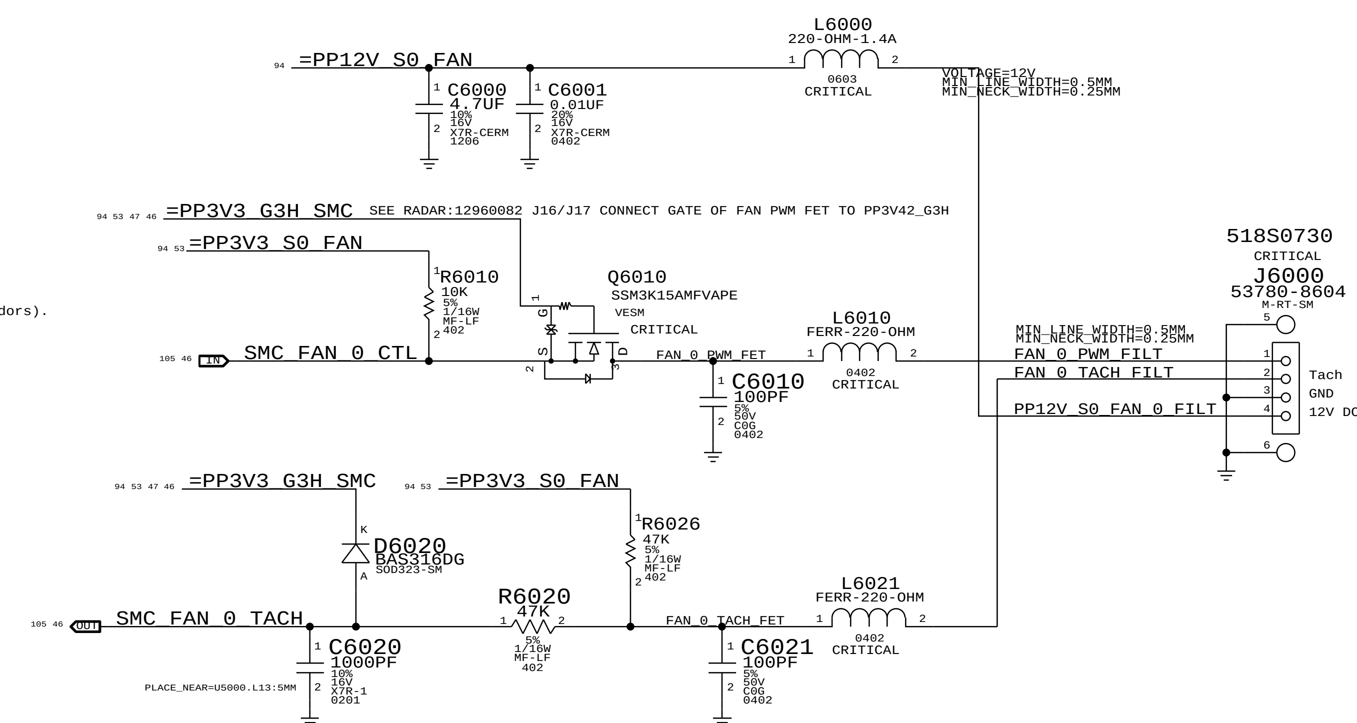
BRANCH: proto2

SMC Fan 0 (System)

Note:
The circuit for the PWM input to the fan acts as a non-inverting level-shifter to protect the SMC. It is assumed there is a pull-up to 5V/12V inside the fan, otherwise when the SMC PWM goes low and Q6010 turns on, there would be 5V/12V present on the SMC pin! Then by definition, the drain of Q6010 is at common and the SMC sinks current when Q6010 is on.

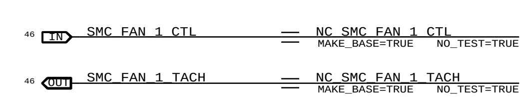
This resembles an open-drain if there is a pull-up, going to a Hi-Z FET input.

Otherwise, this is simply a pass-FET.
See RADAR: 10565825- D7: Need scematic and PCB file of fan(All Vendors).



Add C6020 1000pF Cap, Change R6020 to 47K -- Radar 11661918 D8 Proto1 Fan Tach instability.

SMC Fan 1 (Unused)



SYNC_MASTER=J78_MLB SYNC_DATE=02/21/2014

PAGE TITLE
System Fan

Apple Inc.

DRAWING NUMBER 051-1635 SIZE D

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BRANCH proto2

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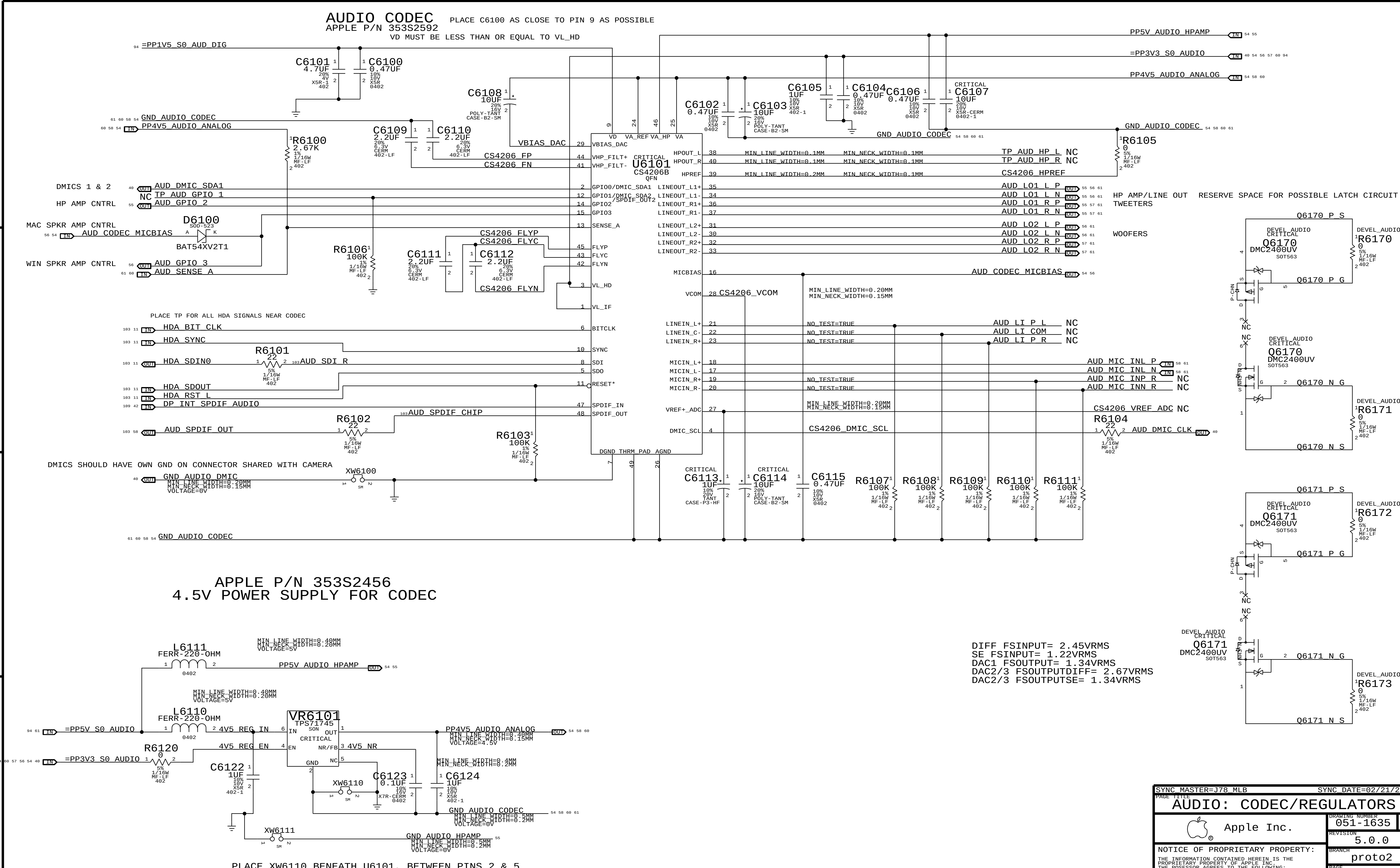

SYNC_MASTER=J78_MLB SYNC_DATE=02/21/2014

PAGE TITLE
AUDIO: CODEC/REGULATORS

Apple Inc.

DRAWING NUMBER	051-1635
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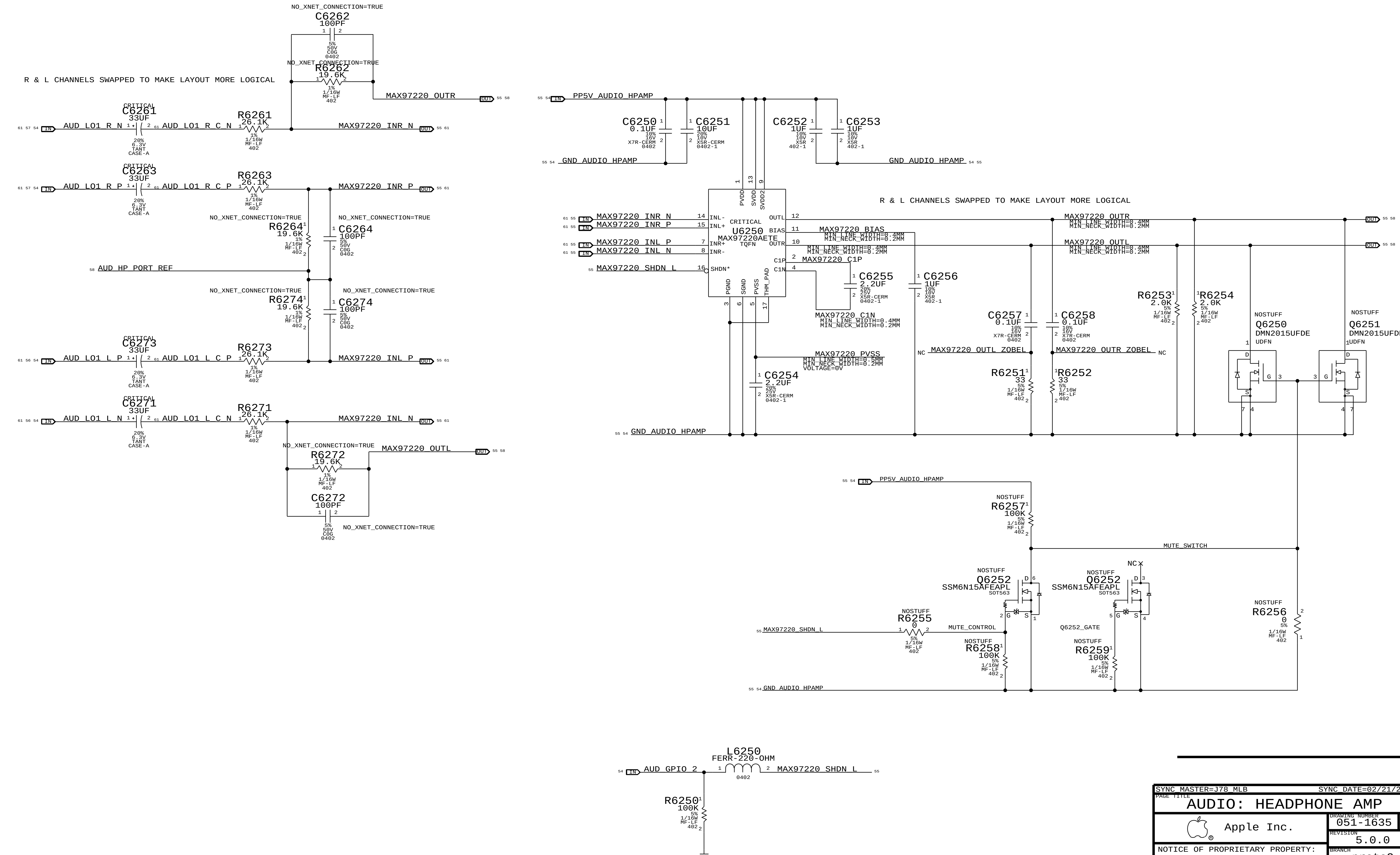

SYNC_MASTER=J78_MLB SYNC_DATE=02/21/2014

PAGE TITLE

AUDIO: HEADPHONE AMP

Apple Inc.

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BRANCH proto2

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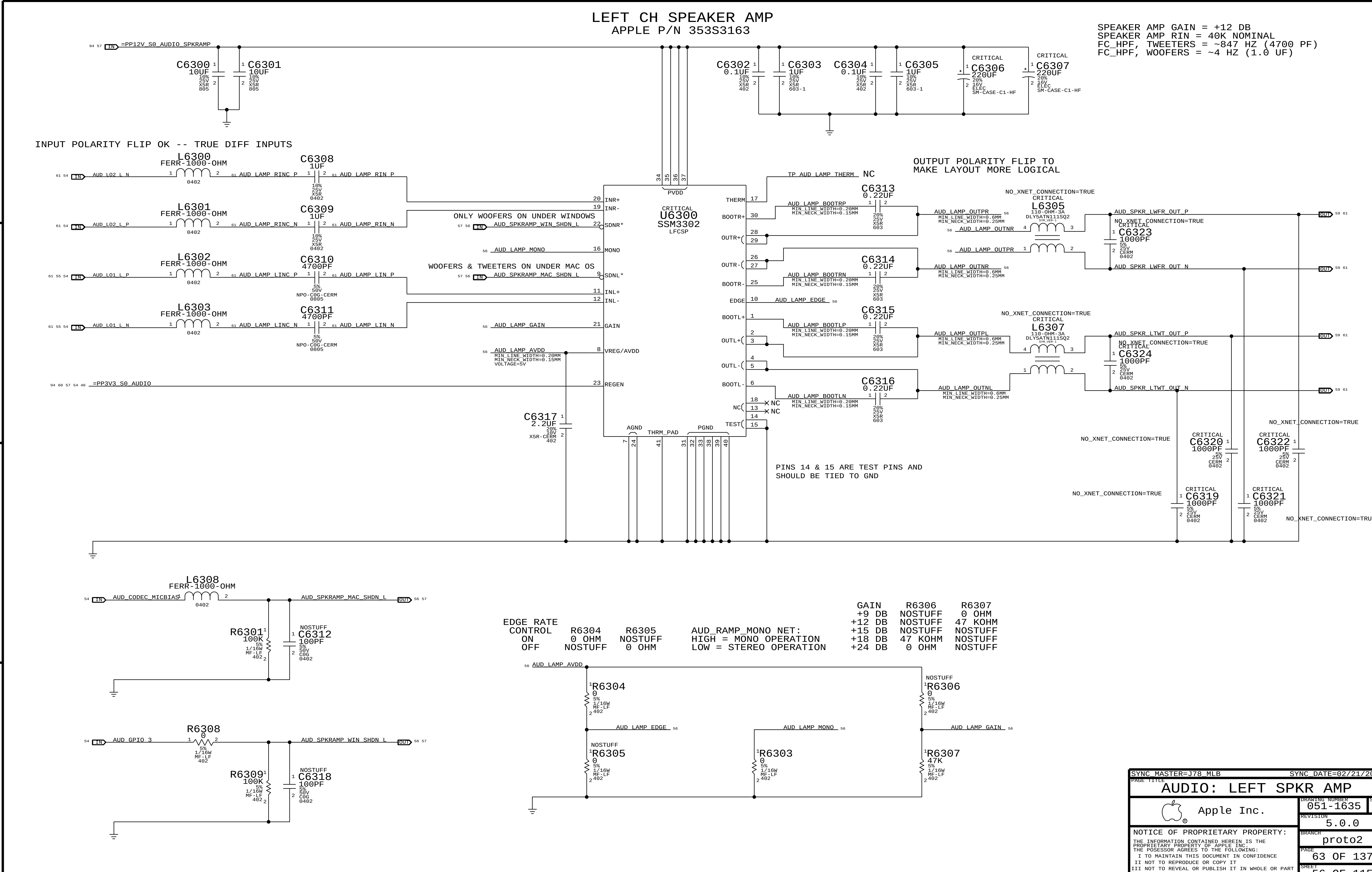
LEFT CH SPEAKER AMP

APPLE P/N 353S3163

SPEAKER AMP GAIN = +12 DB
SPEAKER AMP RIN = 40K NOMINAL
FC_HPF, TWEETERS = ~847 HZ (4700 PF)
FC_HPF, WOOFERS = ~4 HZ (1.0 UF)



EDGE RATE CONTROL	R6304	R6305
ON	0 OHM	NOSTUFF
OFF	NOSTUFF	0 OHM

AUD_RAMP_MONO NET:
HIGH = MONO OPERATION
LOW = STEREO OPERATION

GAIN	R6306	R6307
+9 DB	NOSTUFF	0 OHM
+12 DB	NOSTUFF	47 KOHM
+15 DB	NOSTUFF	NOSTUFF
+18 DB	47 KOHM	NOSTUFF
+24 DB	0 OHM	NOSTUFF

SYNC_MASTER=J78_MLB SYNC_DATE=02/21/2014

PAGE TITLE: AUDIO: LEFT SPKR AMP

Apple Inc.

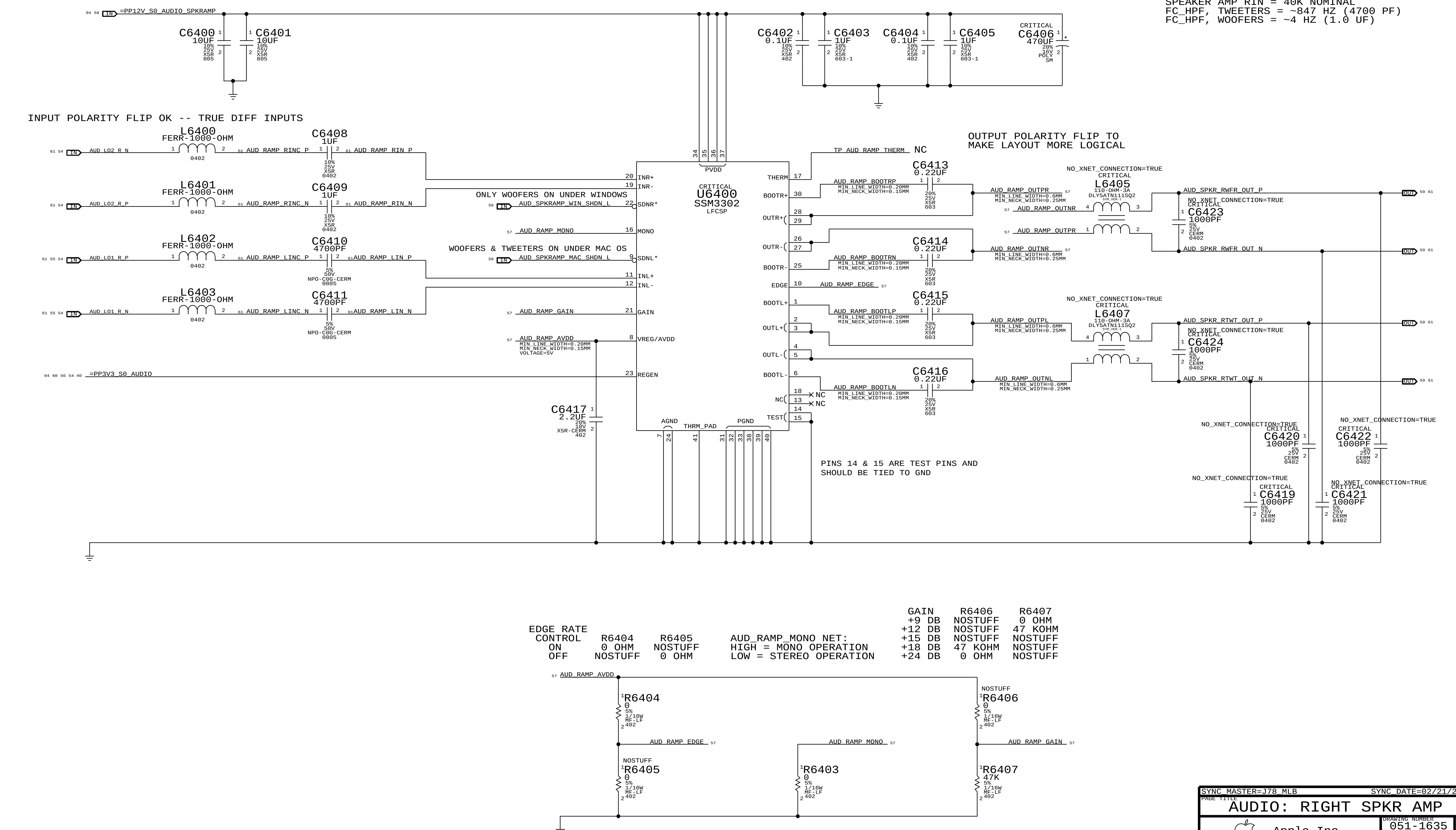
DRAWING NUMBER 051-1635 SIZE D

REVISION 5.0.0

BRANCH proto2

RIGHT CH SPEAKER AMP
APPLE P/N 353S3163

SPEAKER AMP GAIN = +12 DB
SPEAKER AMP RIN = 40K NOMINAL
FC_HPF, TWEETERS = ~847 HZ (4700 PF)
FC_HPF, WOOFERS = ~4 HZ (1.0 UF)



EDGE RATE

CONTROL	R6404	R6405
ON	0 OHM	NOSTUFF
OFF	NOSTUFF	0 OHM

AUD_RAMP_MONO NET:
HIGH = MONO OPERATION
LOW = STEREO OPERATION

GAIN	R6406	R6407
+9 DB	NOSTUFF	0 OHM
+12 DB	NOSTUFF	47 KOHM
+15 DB	NOSTUFF	NOSTUFF
+18 DB	47 KOHM	NOSTUFF
+24 DB	0 OHM	NOSTUFF

SYNC_MASTER=J78_MLB SYNC_DATE=02/21/2014

PAGE TITLE
AUDIO: RIGHT SPKR AMP

Apple Inc.

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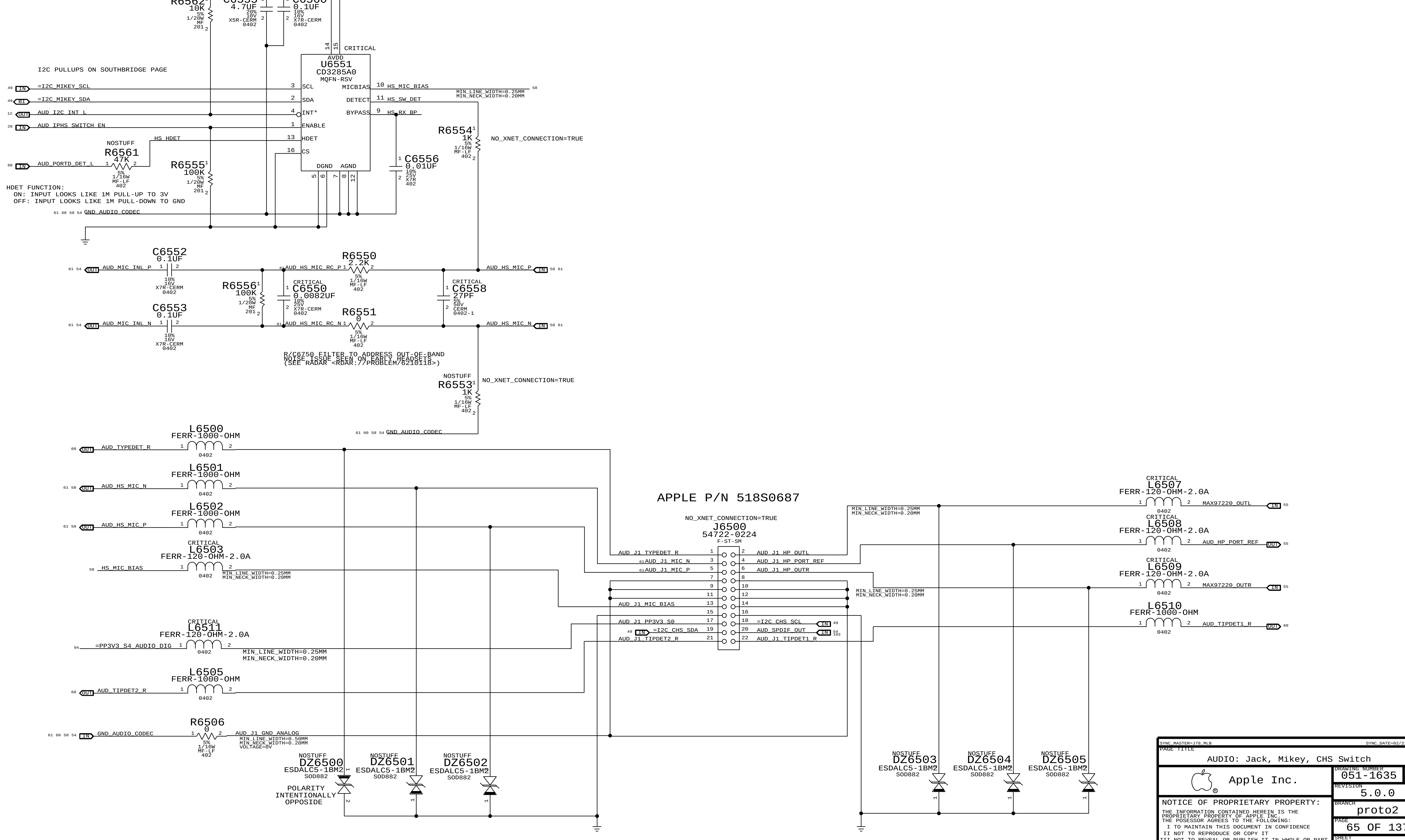
REVISION 5.0.0
BRANCH proto2

MIKEY RECEIVER CKT

WRITE: 0X72 READ: 0X73 APN 353S3231

I2C ADDRESSES

MIKEY	U6551	READ	0111 0011	0X73
MIKEY	U6551	WRITE	0111 0010	0X72



SYNC_MASTER=J78_MLB SYNC_DATE=02/21/2014

PAGE TITLE

AUDIO: Jack, Mikey, CHS Switch

Apple Inc.

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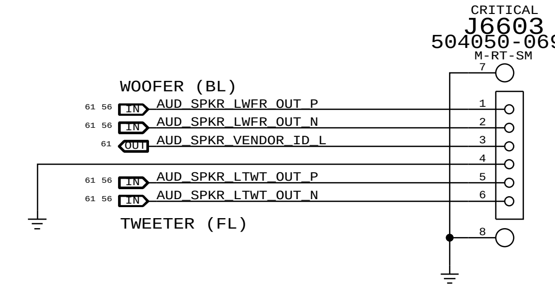
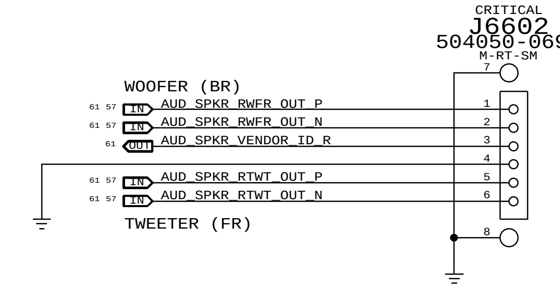
SPEAKER CABLE CONNECTORS

APPLE P/N 518S0862





SYNC_MASTER=J78_MLB	SYNC_DATE=02/21/2014
PAGE TITLE	Audio: Spkr/Mic Conn.
Apple Inc.	DRAWING NUMBER 051-1635 SIZE D
	REVISION 5.0.0
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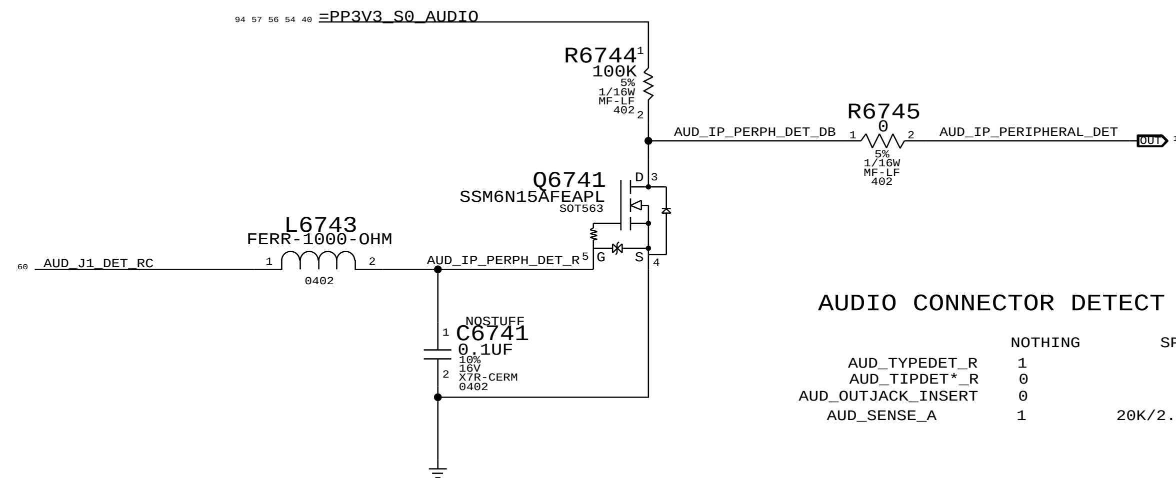
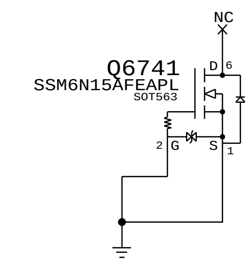
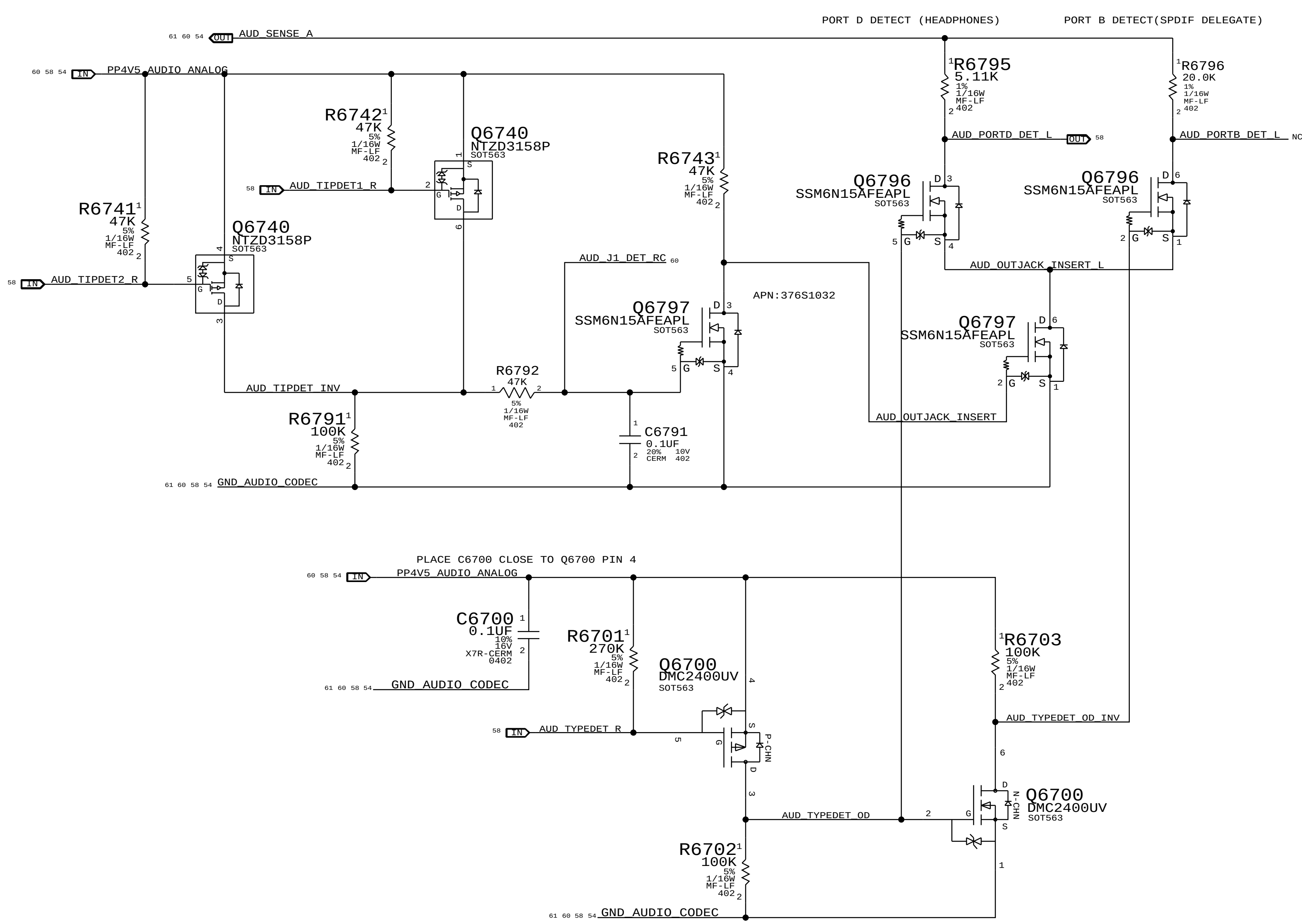
IPHS HS Detect Debounce CKT





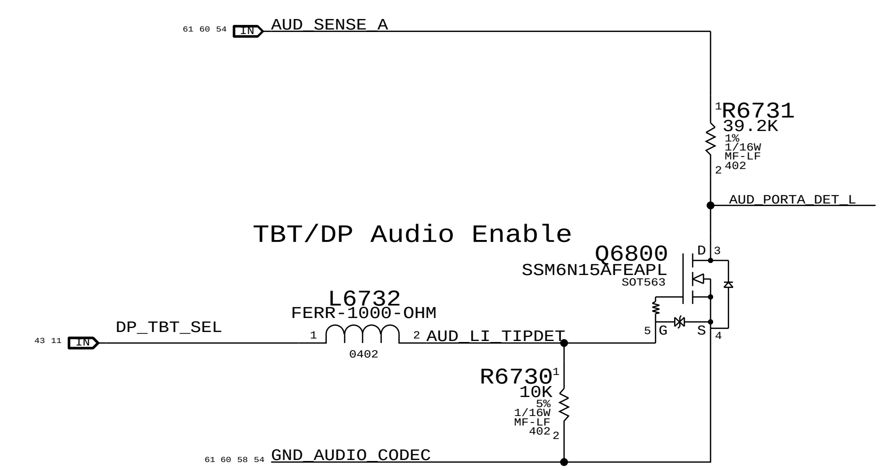
AUDIO CONNECTOR DETECT STATES

	NOTHING	SPDIF	HEADPHONE
AUD_TYPEDET_R	1	1	0
AUD_TIPDET*_R	0	1	1
AUD_OUTJACK_INSERT	0	1	1
AUD_SENSE_A	1	20K/2.67K RDIV	5.11K/2.67K RDIV



Target Display Mode Detect

TBT/DP Audio Enable



SYNC_MASTER=J78_MLB SYNC_DATE=02/21/2014

PAGE TITLE
AUDIO: Detects/Grounding

Apple Inc.

DRAWING NUMBER 051-1635 SIZE D

REVISION 5.0.0

BRANCH proto2

CODEC OUTPUT SIGNAL PATHS

FUNCTION	VOLUME/MUTE	CONVERTER	PIN COMPLEX	MAC SHDN	WIN SHDN	DET ASSIGNMENT
HP/LINE OUT	0X03 (3)	0X03 (3)	0X0A (10,D)	GPIO_2	GPIO_2	0X0A (DET D)
PRIMARY SPKRS (WFR)	0X04 (4)	0X04 (4)	0X0B (11)	MICBIAS	GPIO_3	N/A
SECONDARY SPKRS (TWT)	0X03 (3)	0X03 (3)	0X0A (10,V24)	MICBIAS	N/A	N/A
SPDIF OUT	N/A	0X08 (8)	0x10 (16)	N/A	N/A	0X0D (DET B)

CODEC INPUT SIGNAL PATHS

FUNCTION	CONVERTER	PIN COMPLEX	ENABLE/CONTROL	DET ASSIGNMENT
SPDIF IN	0X07 (7)	0X0F (15)	N/A	0X09 (DET A)
INTERNAL MIC ARRAY	0X06 (6) 0X05 (5)	0X0E (14,LEFT & RIGHT) 0X12 (18,LEFT)	N/A	N/A
EXTERNAL MIC	0X06 (6)	0X0D (13,V22,B,LEFT)	Lynx POINT GPIO 16	Lynx POINT GPIO 5 (RCVR INT) Lynx POINT GPIO 3 (PERIPH DET)

OTHER DETECT

FUNCTION	CONVERTER	PIN COMPLEX	ENABLE/CONTROL	DET ASSIGNMENT
MULTIPLE SPKR VENDORS	N/A	N/A	N/A	0X0C (DET C)

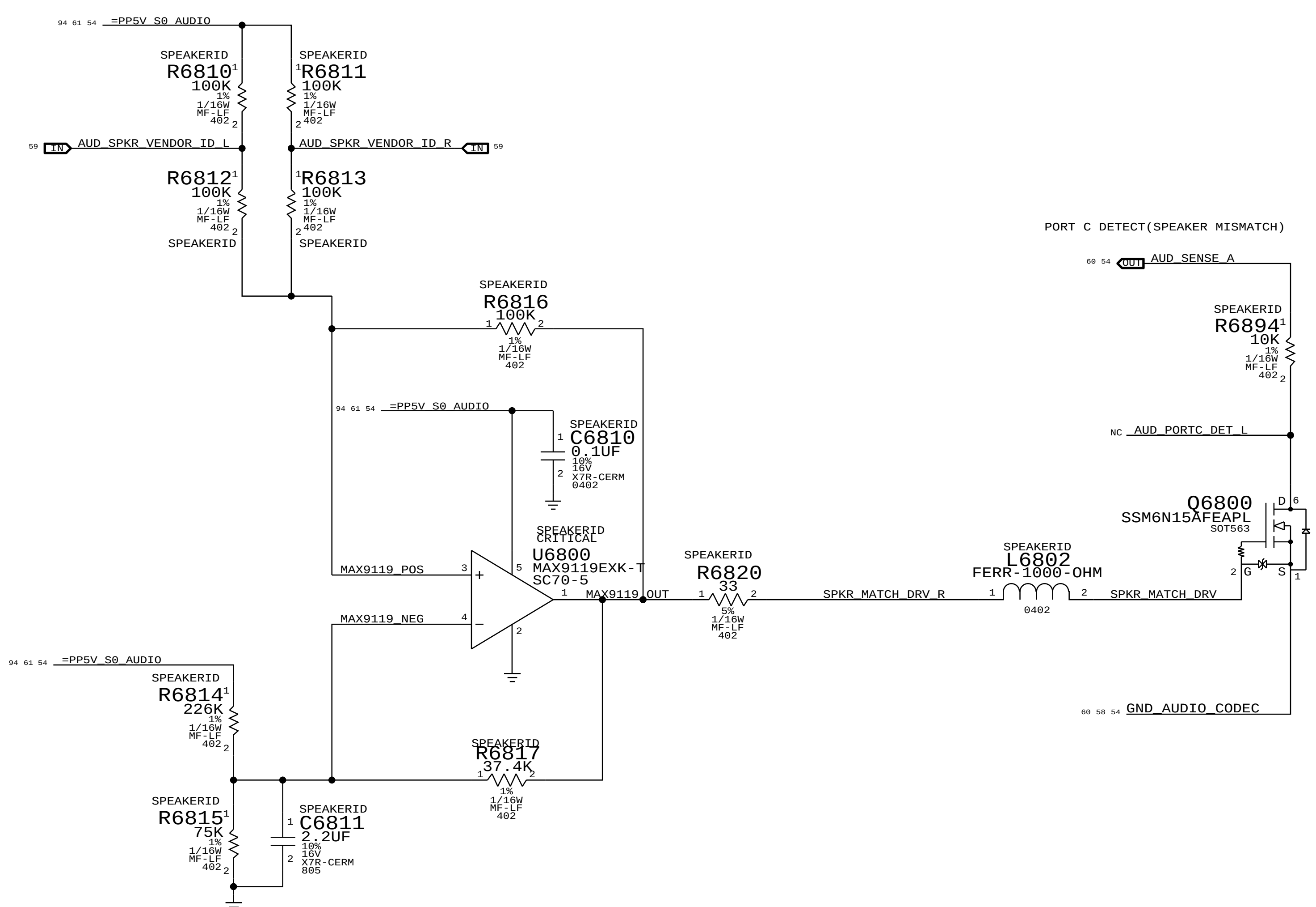
SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
AUDIO	*	0.1 MM	?
SPKROUT	*	0.2 MM	?

NET_PHYSICAL_TYPE	AREA_TYPE	PHYSICAL_RULE_SET
AUDIODIFF	*	AUDIODIFF
SPKROUTDIFF	*	SPKROUTDIFF

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
AUDIODIFF	*	Y	0.1 MM	0.1 MM	10 MM	0.1 MM	0.1 MM
SPKROUTDIFF	*	Y	0.6 MM	0.25 MM	10 MM	0.2 MM	0.2 MM

	ELECTRICAL_CONSTRAINT_SET	NET_TYPE PHYSICAL	NET_TYPE SPACING	NET	PAGES
I215	AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD_LO1_L_P	54 55 56
I216	AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD_LO1_L_N	54 55 56
I217	AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD_LO1_L_C_P	55
I218	AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD_LO1_L_C_N	55
I211	AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD_LO1_R_P	54 55 57
I212	AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD_LO1_R_N	54 55 57
I210	AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD_LO1_R_C_P	55
I209	AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD_LO1_R_C_N	55
I208	AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD_LO2_L_P	54 56
I206	AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD_LO2_L_N	54 56
I207	AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD_LO2_R_P	54 57
I204	AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD_LO2_R_N	54 57
I205	AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD_RAMP_LINC_P	57
I203	AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD_RAMP_LINC_N	57
I220	AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD_RAMP_RINC_P	57
I219	AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD_RAMP_RINC_N	57
I222	AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD_RAMP_LIN_P	57
I221	AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD_RAMP_LIN_N	57
I224	AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD_RAMP_RIN_P	57
I223	AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD_RAMP_RIN_N	57
I226	AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD_LAMP_LINC_P	56
I225	AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD_LAMP_LINC_N	56
I227	AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD_LAMP_RINC_P	56
I229	AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD_LAMP_RINC_N	56
I228	AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD_LAMP_LIN_P	56
I230	AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD_LAMP_LIN_N	56
I264	AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD_LAMP_RIN_P	56
I263	AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD_LAMP_RIN_N	56
I236	AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	MAX97220_INL_P	55
I238	AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	MAX97220_INL_N	55
I237	AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	MAX97220_INR_P	55
I239	AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	MAX97220_INR_N	55
I240	SPKROUT_DIFFPAIR	SPKROUTDIFF	SPKROUT	AUD_SPKR_RWFR_OUT_P	57 59
I241	SPKROUT_DIFFPAIR	SPKROUTDIFF	SPKROUT	AUD_SPKR_RWFR_OUT_N	57 59
I242	SPKROUT_DIFFPAIR	SPKROUTDIFF	SPKROUT	AUD_SPKR_RTWT_OUT_P	57 59
I243	SPKROUT_DIFFPAIR	SPKROUTDIFF	SPKROUT	AUD_SPKR_RTWT_OUT_N	57 59
I244	SPKROUT_DIFFPAIR	SPKROUTDIFF	SPKROUT	AUD_SPKR_LWFR_OUT_P	56 59
I246	SPKROUT_DIFFPAIR	SPKROUTDIFF	SPKROUT	AUD_SPKR_LWFR_OUT_N	56 59
I245	SPKROUT_DIFFPAIR	SPKROUTDIFF	SPKROUT	AUD_SPKR_LTWT_OUT_P	56 59
I247	SPKROUT_DIFFPAIR	SPKROUTDIFF	SPKROUT	AUD_SPKR_LTWT_OUT_N	56 59
I326	AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD_MIC_INL_P	54 58
I327	AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD_MIC_INL_N	54 58
I328	AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD_HS_MIC_RC_P	58
I329	AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD_HS_MIC_RC_N	58
I254	AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD_HS_MIC_P	58
I255	AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD_HS_MIC_N	58
I324	AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD_J1_MIC_P	58
I325	AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD_J1_MIC_N	58

CIRCUIT THEORY OF OPERATION AVAILABLE IN <RDAR://PROBLEM/9776522>



PAGE TITLE
AUDIO: Speaker ID

Apple Inc.

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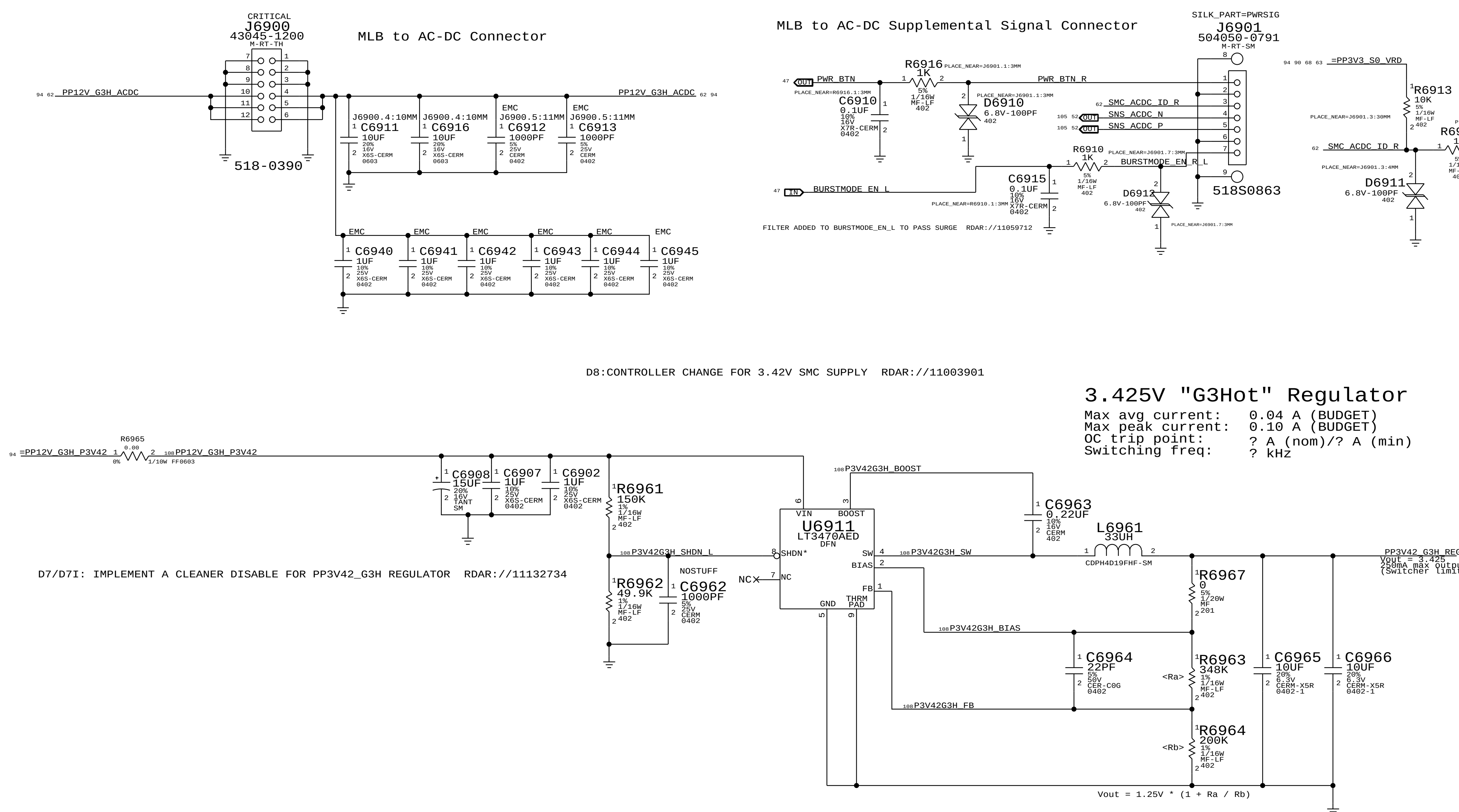

SYNC_MASTER=J78_MLB SYNC_DATE=02/21/2014

PAGE TITLE
Power Connectors / VReg G3Hot

Apple Inc.

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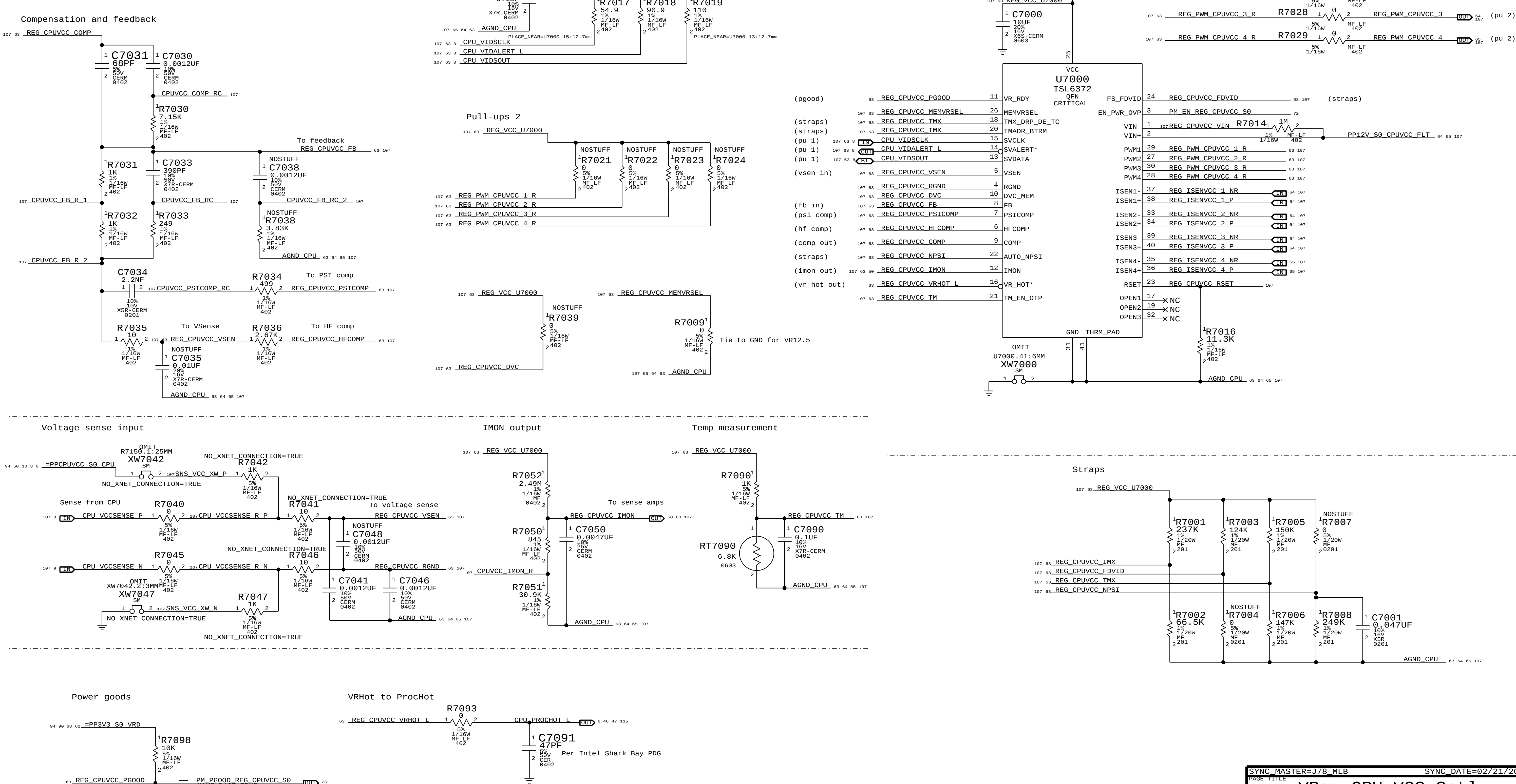
BRANCH proto2

CPU VCC S0 Regulator

OC trip point: 114 A

Switching freq: 403 kHz = 5 E10 / R7003



SYNC_MASTER=J78_MLB SYNC_DATE=02/21/2014

PAGE TITLE: VReg CPU VCC Cntl

Apple Inc.

DRAWING NUMBER: 051-1635 SIZE: D

REVISION: 5.0.0

BRANCH: proto2

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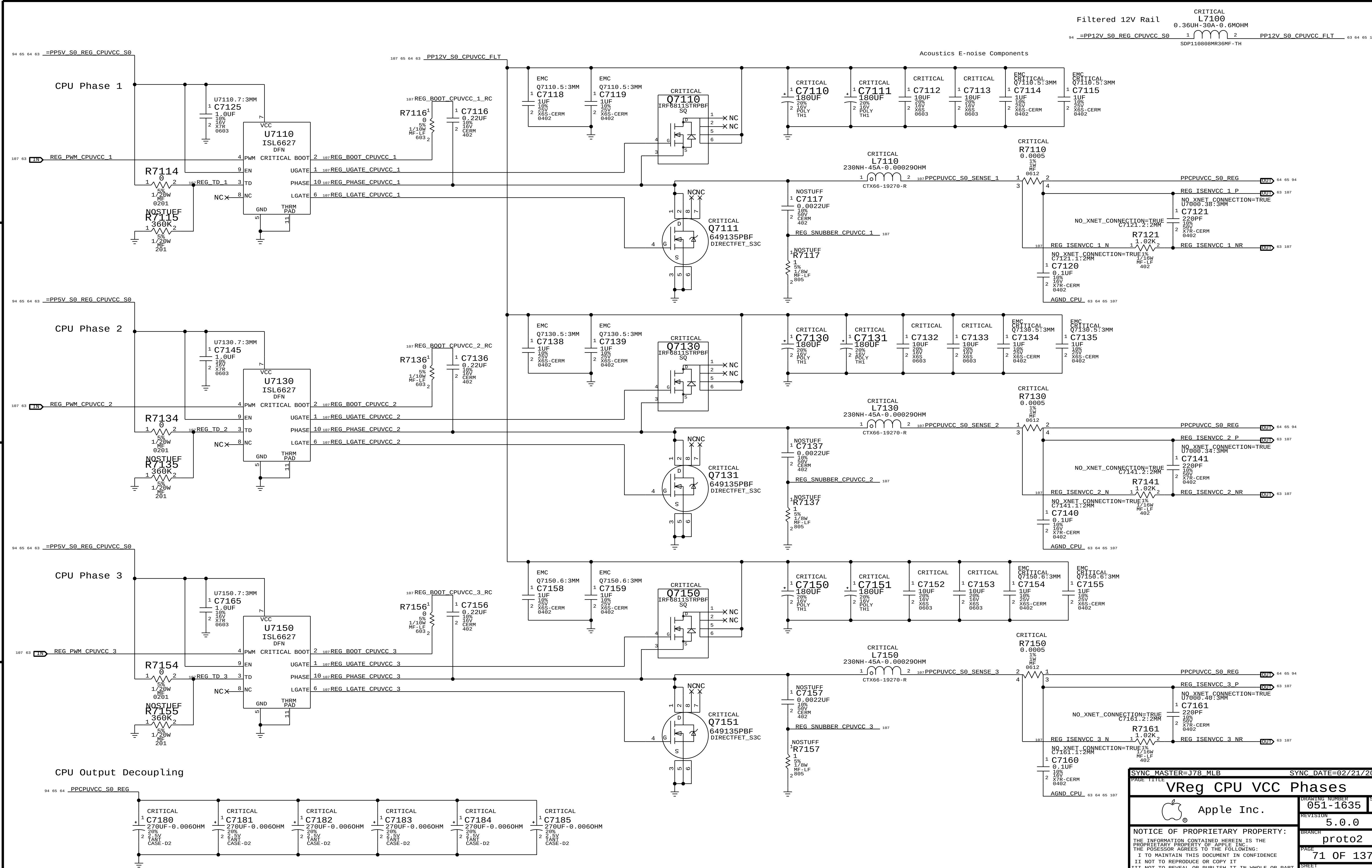

SYNC_MASTER=J78_MLB SYNC_DATE=02/21/2014

PAGE TITLE

VReg CPU VCC Phases

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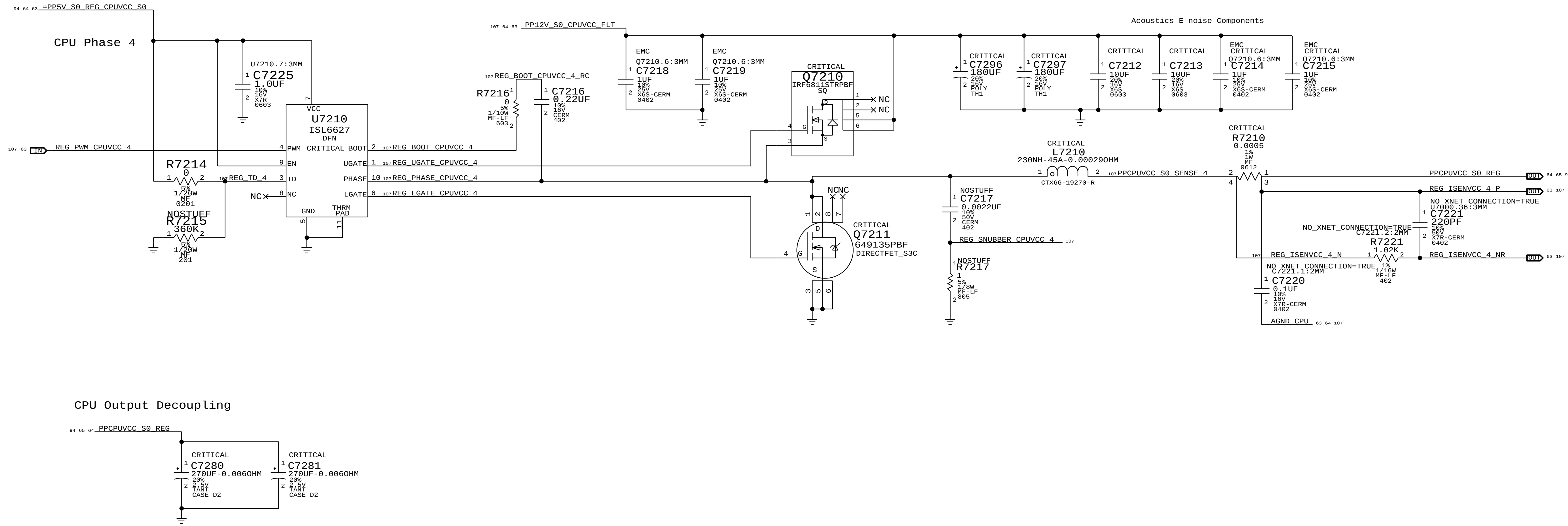

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PAGE TITLE

VReg CPU VCC Phase 4

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051-1635	D
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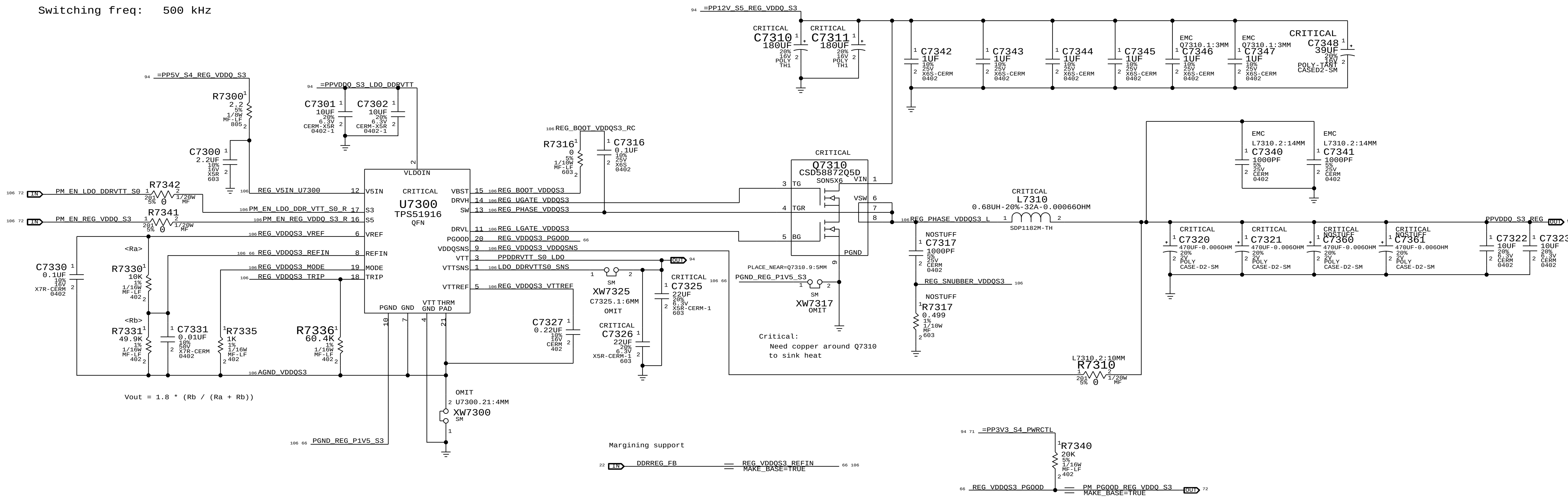
VDDQ (1.5V) S3 REGULATOR

OC trip point: 30.4 A VDDQ = $\frac{R7336}{8\ E5 * Rds(Q7310)} + \frac{0.65625}{L7310 * f(switch)}$

3 A VTT (FIXED)
10 mA VTTREF (FIXED)

Switching freq: 500 kHz



SYNC_MASTER=J78_MLB		SYNC_DATE=02/21/2014
PAGE TITLE: VReg VDDQ S3		
Apple Inc.	DRAWING NUMBER: 051-1635	SIZE: D
	REVISION: 5.0.0	
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	SHEET: 66 OF 115	

3.3V S5 Regulator

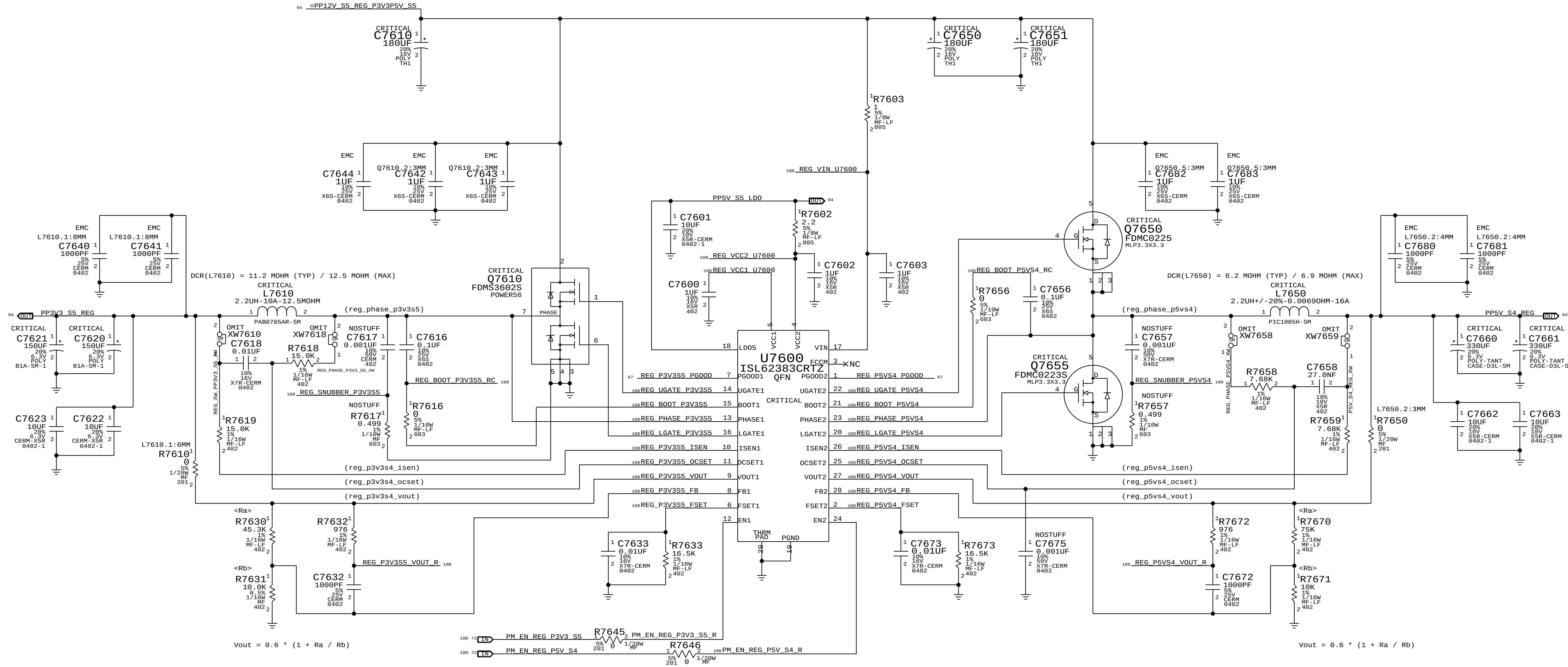
OC trip point: 12.5 A = $\frac{R7618 * 10 E-6}{DCR(L7610)}$

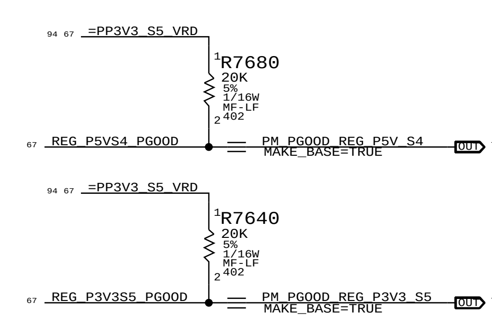
Switching freq: 356 kHz = $\frac{1}{170 E-12 * R7633}$

5V S4 Regulator

OC trip point: 14.1 A = $\frac{R7658 * 10 E-6}{DCR(L7650)}$

Switching freq: 356 kHz = $\frac{1}{170 E-12 * R7673}$





SYNC_MASTER=J78_MLB SYNC_DATE=02/21/2014

PAGE TITLE

VReg 3.3V S5/5V S4

Apple Inc.

DRAWING NUMBER 051-1635 SIZE D

REVISION 5.0.0

BRANCH proto2

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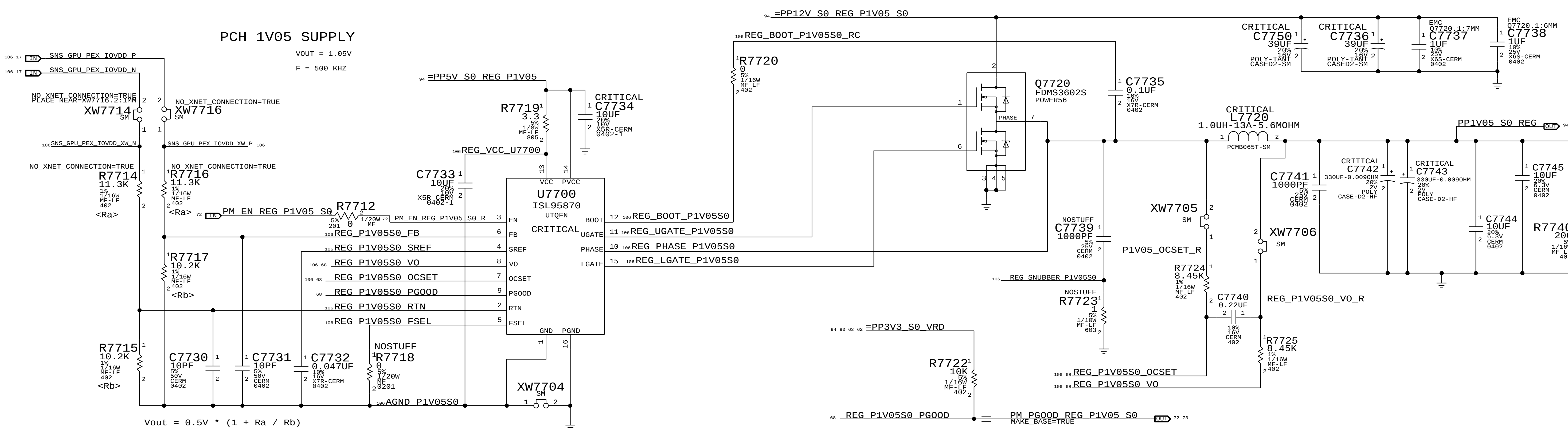

SYNC_MASTER=J78_MLB SYNC_DATE=02/21/2014

PAGE TITLE

1V05 PCH VREG

Apple Inc.

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REVISION 5.0.0

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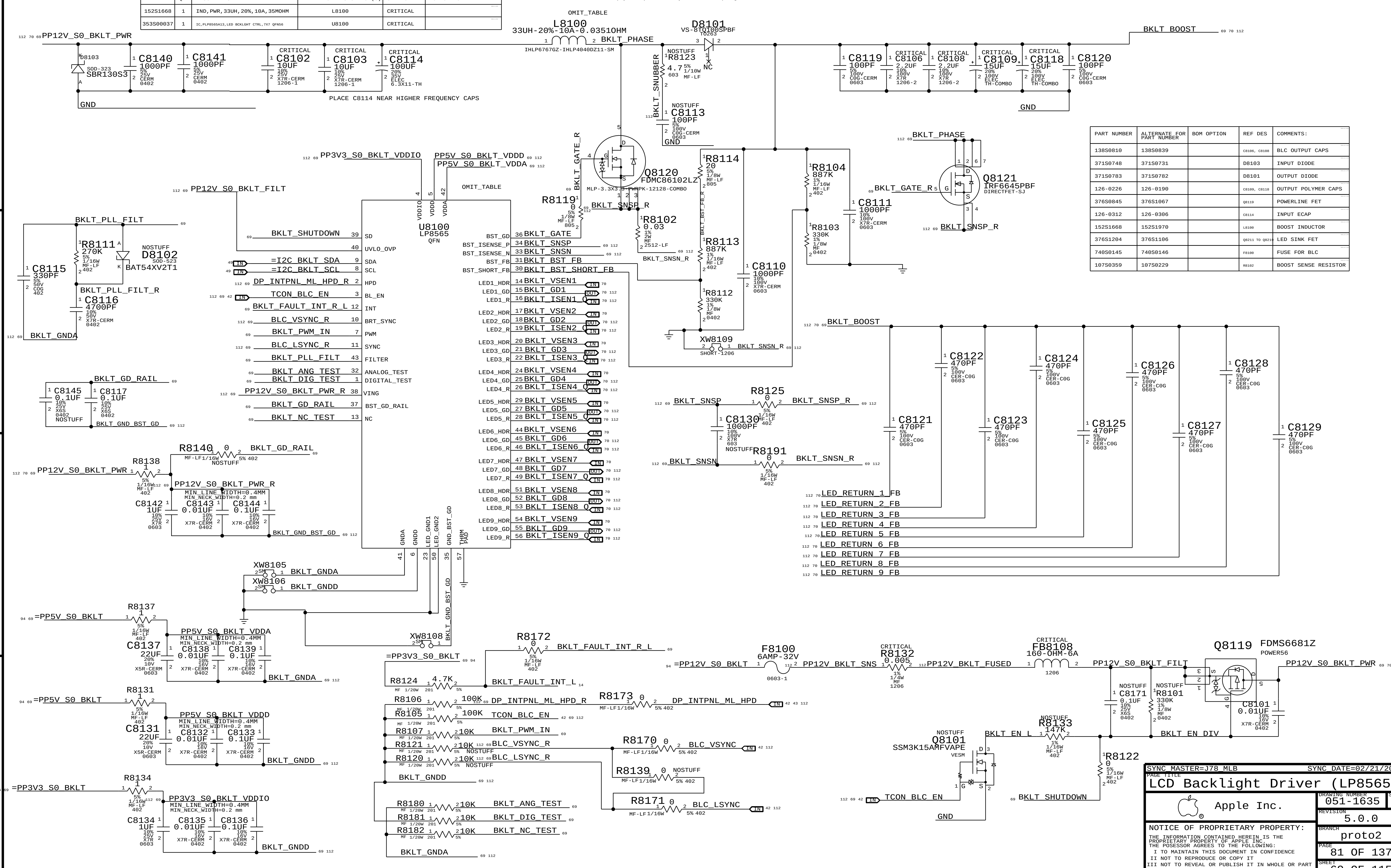
PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
152S1668	1	IND,PWR,33UH,20%,10A,35MOHM	L8100	CRITICAL	
353S00037	1	IC,PLP8565A13,LED BCKLGHT CTRL,7X7 QFN56	U8100	CRITICAL	

OMIT/table L8100 due to late sourcing change, did not want footprint change
Add 152S1668 (Cyntec) directly for next program



PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
138S0810	138S0839		C8106, C8108	BLC OUTPUT CAPS
371S0748	371S0731		D8103	INPUT DIODE
371S0783	371S0782		D8101	OUTPUT DIODE
126-0226	126-0190		C8109, C8118	OUTPUT POLYMER CAPS
376S0845	376S1067		Q8119	POWERLINE FET
126-0312	126-0306		C8114	INPUT ECAP
152S1668	152S1970		L8100	BOOST INDUCTOR
376S1204	376S1106		Q8211 TO Q8219	LED SINK FET
740S0145	740S0146		F8100	FUSE FOR BLC
107S0359	107S0229		R8102	BOOST SENSE RESISTOR

SYNC_MASTER=J78_MLB SYNC_DATE=02/21/2014

PAGE TITLE
LCD Backlight Driver (LP8565)

Apple Inc.

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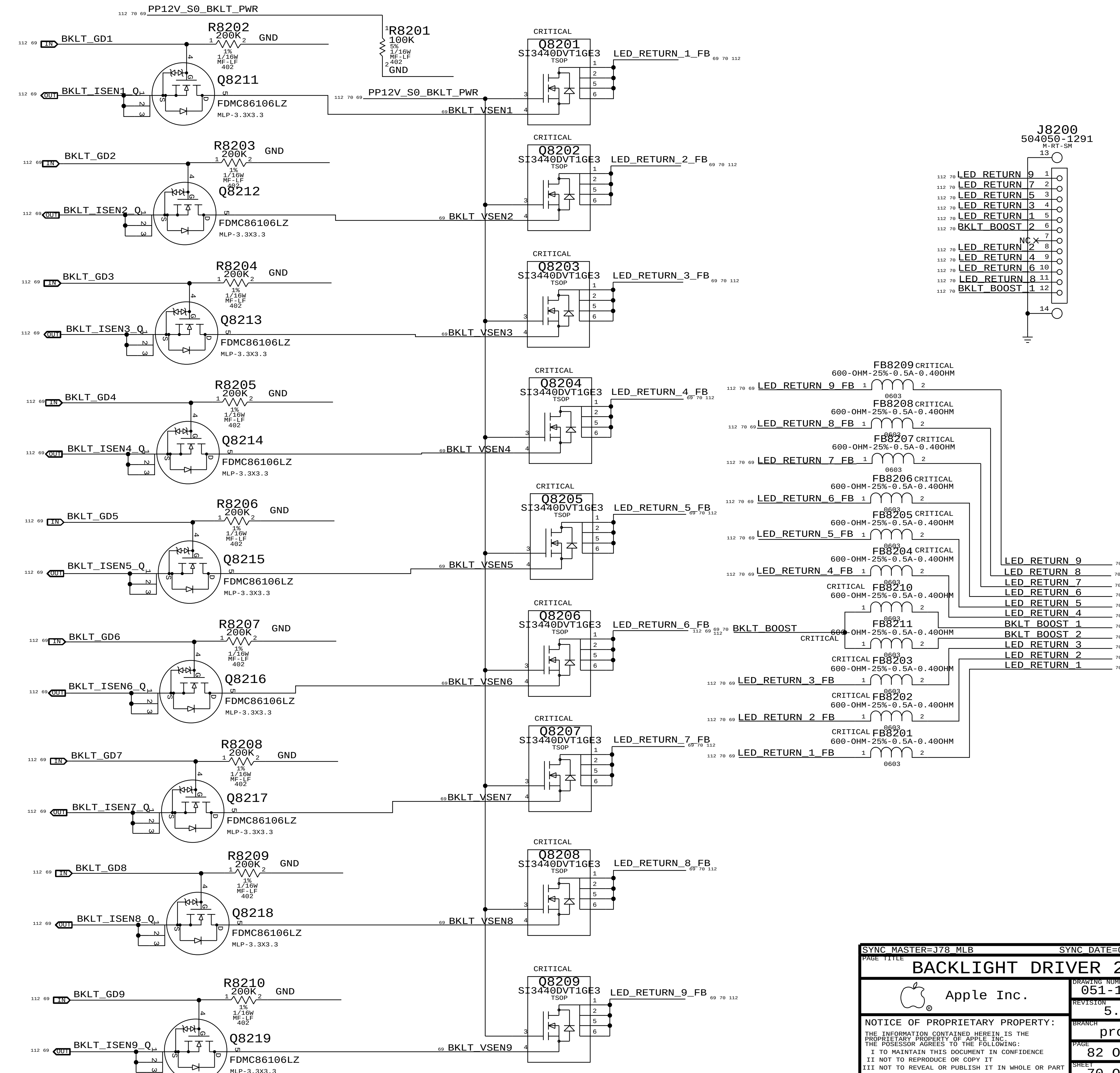
PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
376S1073	376S1256		ALL	Short Protection FET
155S0831	155S0797		ALL	FB8201 TO FB8211



SYNC_MASTER=J78_MLB	SYNC_DATE=02/21/2014
PAGE TITLE	BACKLIGHT DRIVER 2
DRAWING NUMBER	051-1635
SIZE	D
REVISION	5.0.0
BRANCH	proto2
PAGE	82 OF 137
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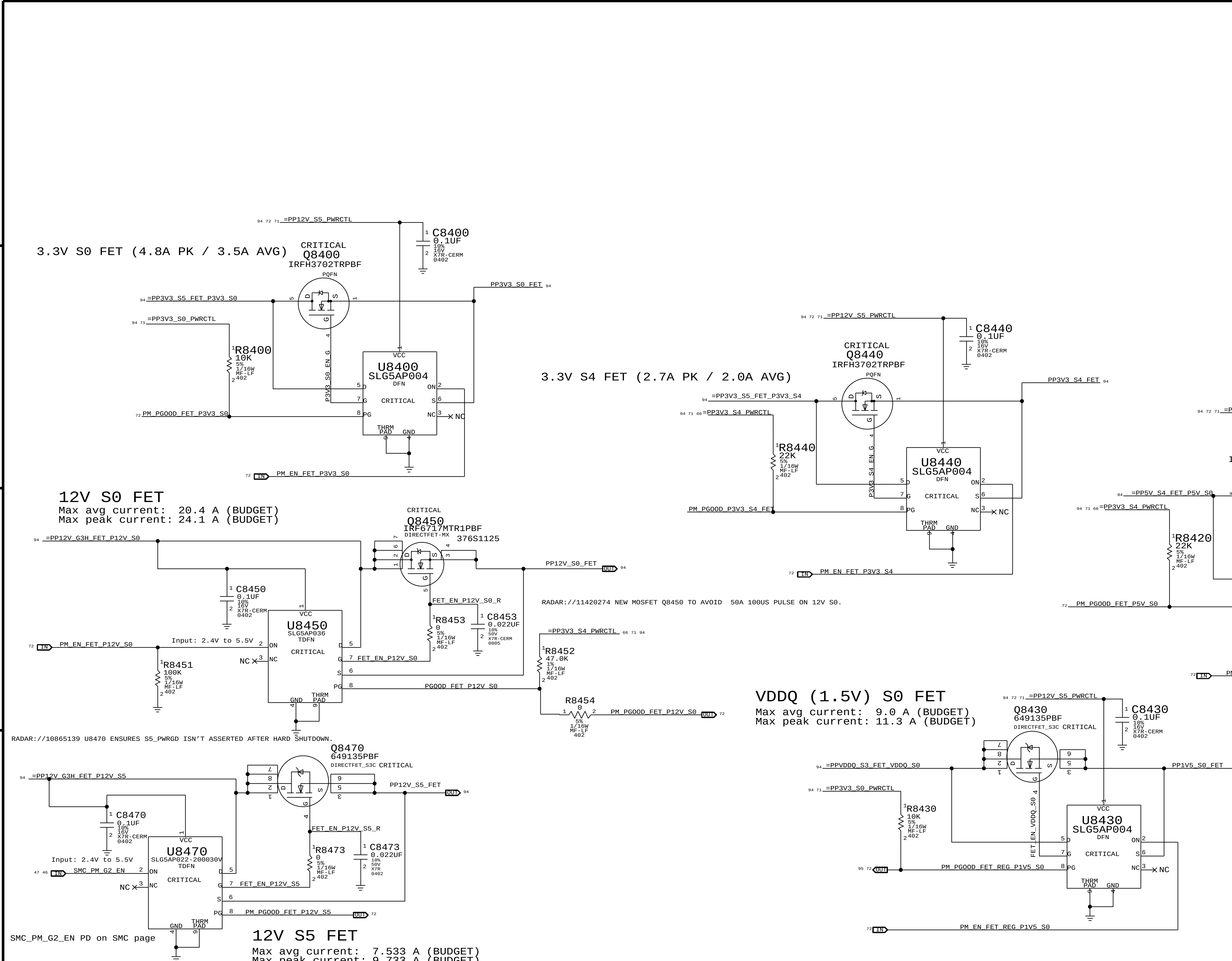
Apple Inc.



SYNC_MASTER=J78_MLB SYNC_DATE=02/21/2014

PAGE TITLE

FET-Controlled S0 and S4

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DRAWING NUMBER 051-1635 SIZE D

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BRANCH proto2

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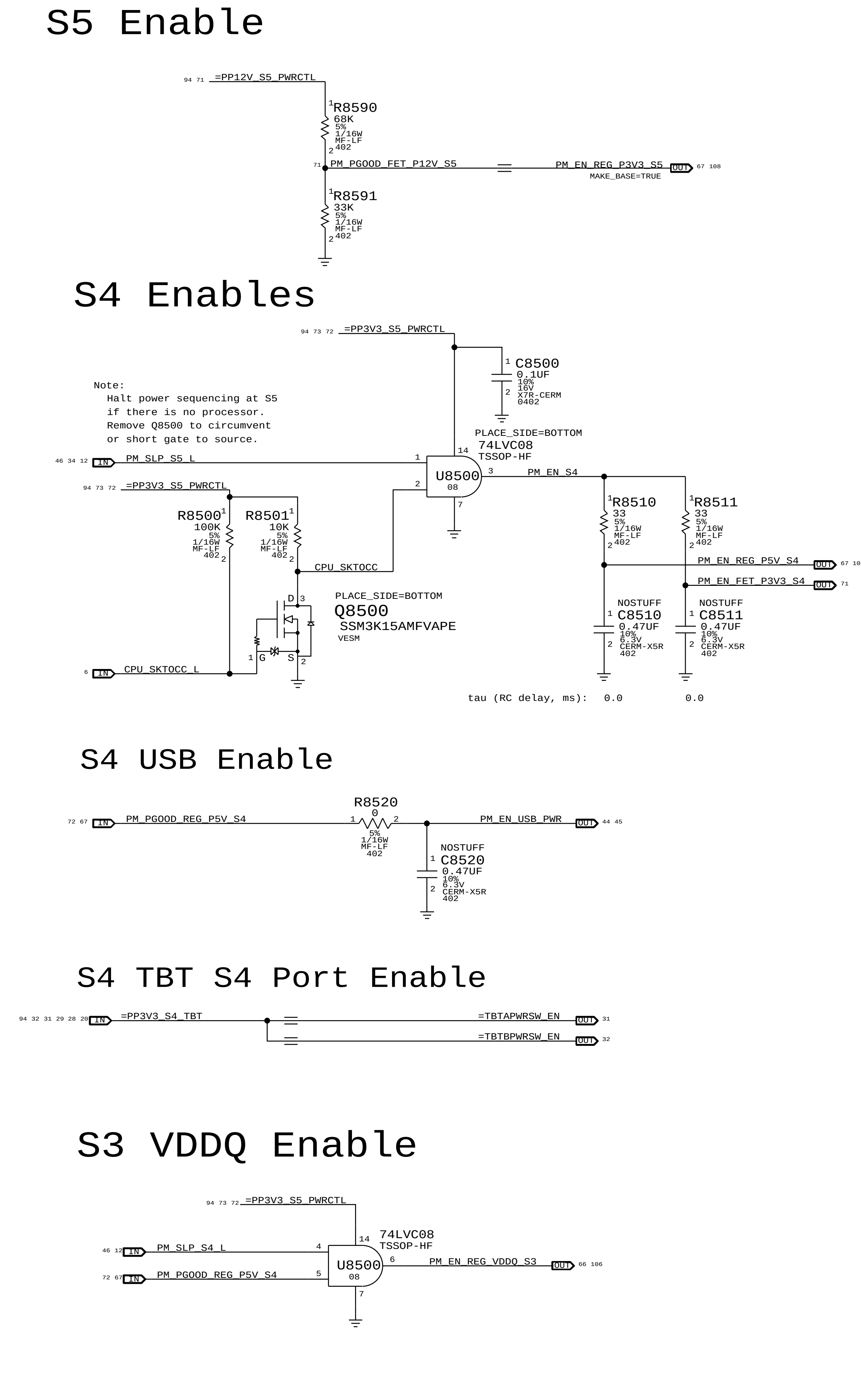
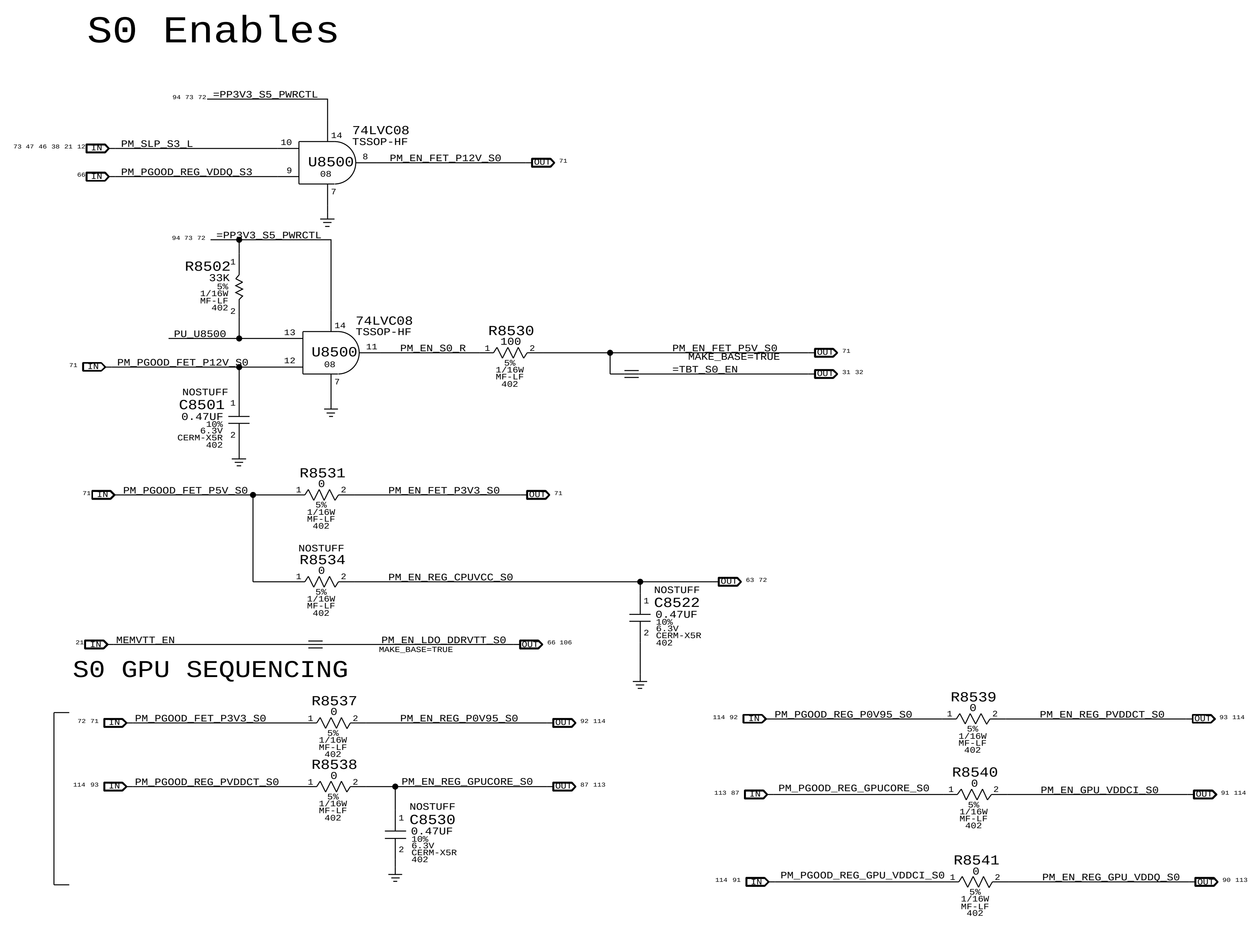
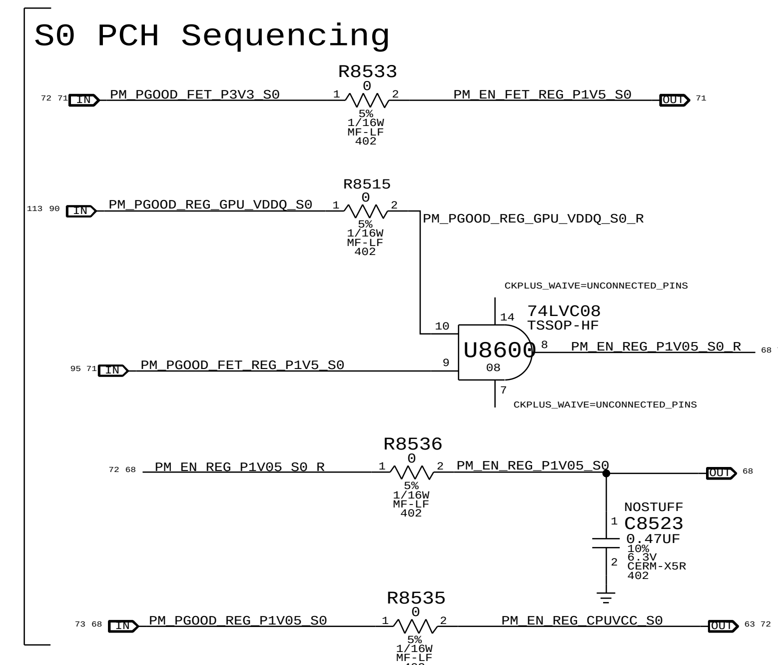






Rail definitions

Platform: ALL PROCESSOR NON-CORE AND NON-GRAPHICS (5 V, 3.3 V, 1.5 V, 1.05V FOR PCH/TBT)
Uncore: VDDQ

Notes on sequencing requirements

Intel:
1. No hard specification on platform rails
2. SMC guarantees timing on PCH DPWROK and PWROK
3. VCC3_3 may power up before VCC, VCC must ramp to 0.6V within 25ms of VCC3V3 ramping to 2.6V
4. VCC1_5 may power up before VCC, VCC must ramp to 0.6V within 25ms of VCC1V5 ramping to 1.35V
5. VCC may power down before VCC3_3, VCC3_3 must ramp down to 2.6V within 35ms
6. VCC may power down before VCC1_5, VCC1_5 must ramp down to 1.35V within 35ms

AMD GPU:
1. 5V_S0 -> 3V3_S0 -> 0V95 -> 1V8 (VDD_CT) -> GPUCORE (VDDC) -> VDDCI -> MVDD

SYNC_MASTER=J78_MLB SYNC_DATE=02/21/2014

PAGE TITLE: PM Regulator Enables

Apple Inc.

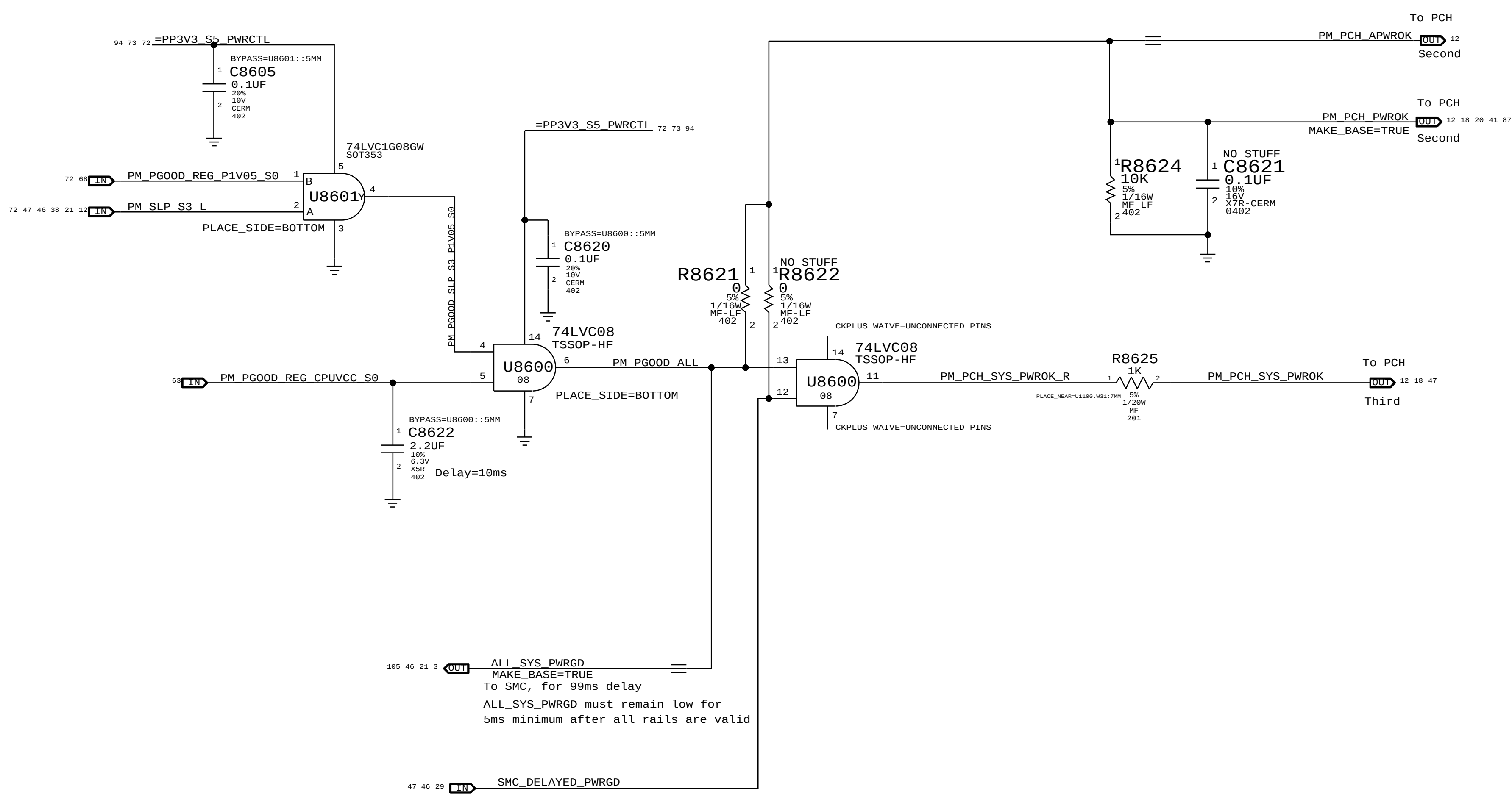
DRAWING NUMBER: 051-1635 SIZE: D

REVISION: 5.0.0

BRANCH: proto2

ALL_SYS_PWRGD,PCH_PWROK & SYS_PWROK Generation

PCH Power Goods



Resume Reset

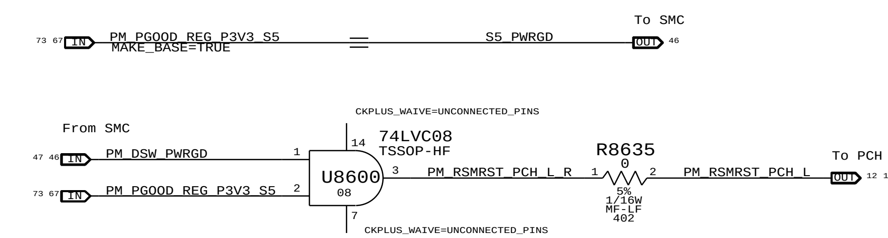
Intel Doc# 29517 Maho Bay PDG, Section 22.13
Intel Doc# 29562 Panther Point EDS, Section 8.7 and 8.8

Note:
THE IMAC J78 DESIGNS DOES NOT SUPPORT DEEP SX MODES SO BOTH DPWROK AND RSMRST# signals are shorted together

Requirements:
- Power on:
  - Asserted at least 10 ms after all suspend well power is valid
- Power off or loss of AC:
  - Transition to 0.8V or less before VccSUS3_3 drops to 2.90 V to allow PCH to switch suspend well to battery without excessive loading

Method:
The SMC guarantees proper assertion and de-assertion of RSMRST# for normal operation via PM_DSW_PWRGD.

RSMRST# is asserted when power good from regulator is de-asserted in the event AC is lost. Power good de-assertion should happen quickly enough to meet Intel spec.



SYNC_MASTER=J78_MLB SYNC_DATE=02/21/2014

PAGE TITLE: PM Power Good

Apple Inc.

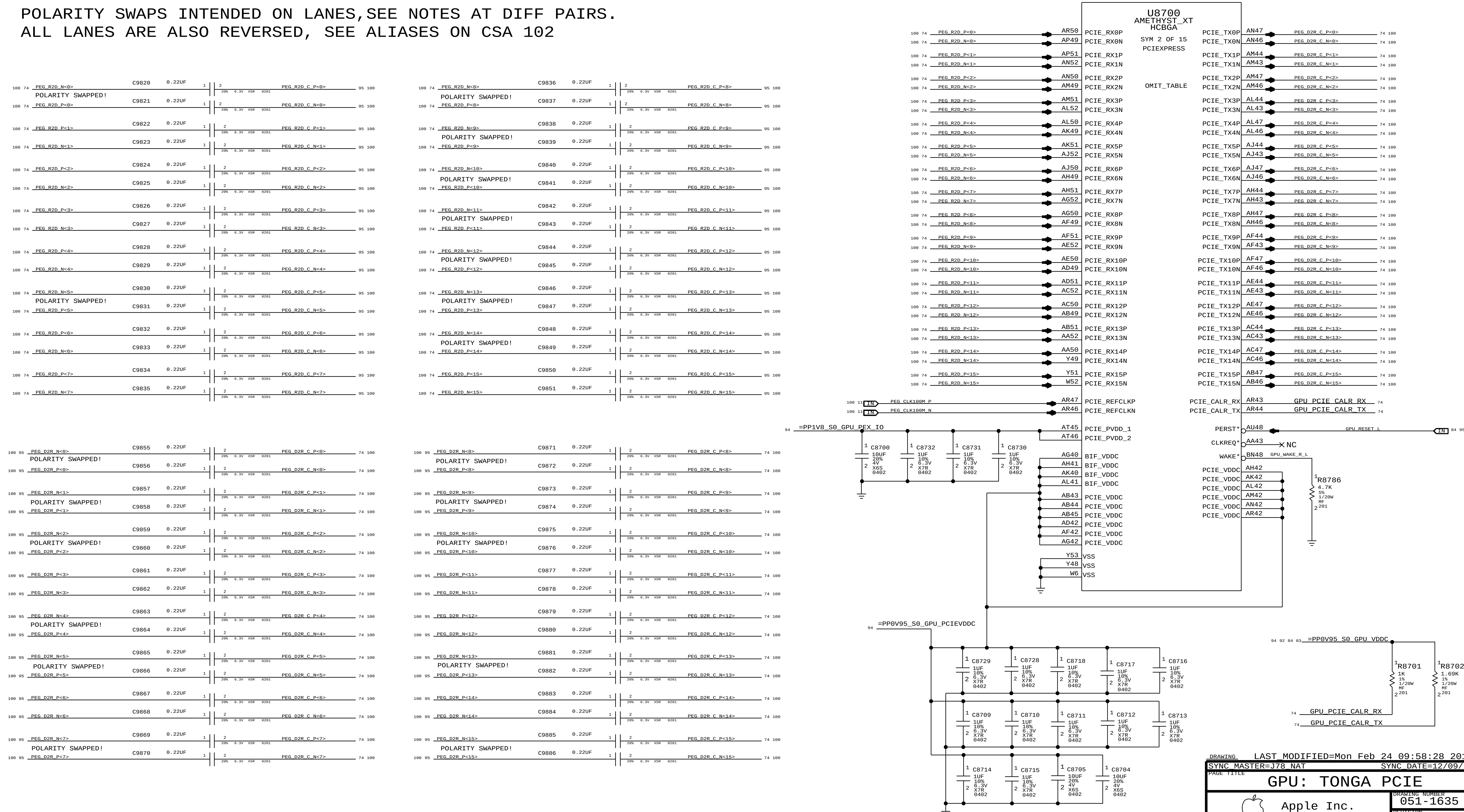
DRAWING NUMBER: 051-1635 SIZE: D

REVISION: 5.0.0

BRANCH: proto2

NOTE:
TONGA DOES NOT GENERATE PCIE CLKREQ.

POLARITY SWAPS INTENDED ON LANES,SEE NOTES AT DIFF PAIRS.
ALL LANES ARE ALSO REVERSED, SEE ALIASES ON CSA 102



DRAWING
LAST_MODIFIED=Mon Feb 24 09:58:28 2014

SYNC_MASTER=J78_NAT
SYNC_DATE=12/09/2013

PAGE TITLE
GPU: TONGA PCIE

Apple Inc.

DRAWING NUMBER
051-1635

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D

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proto2

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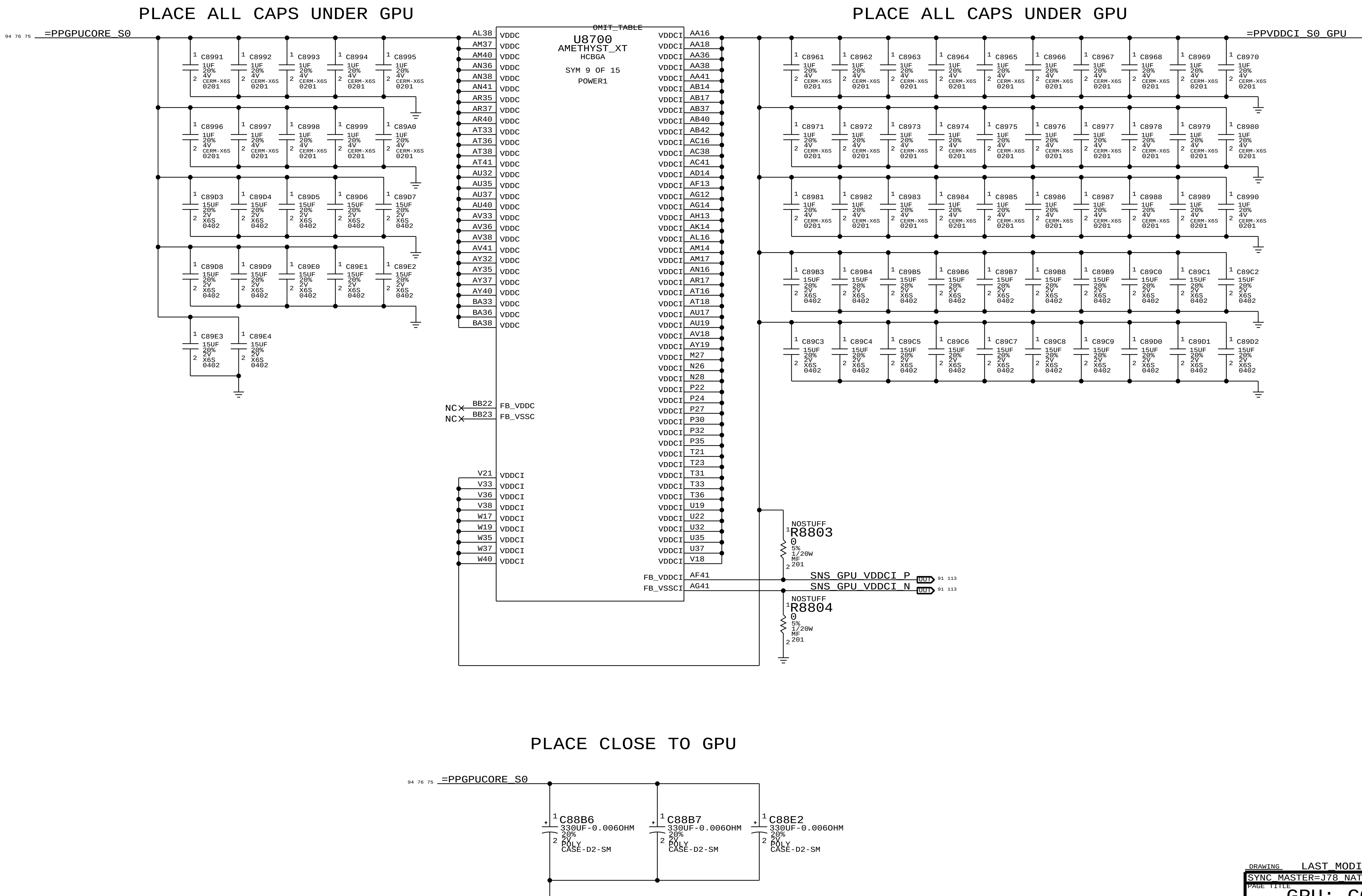
GPU VDDC/VDDCI DECOUPLING



DRAWING LAST_MODIFIED=Mon Feb 24 09:58:28 2014

SYNC_MASTER=J78_NAT SYNC_DATE=12/09/2013

PAGE TITLE

GPU: CORE VDDC/VDDCI

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BRANCH proto2

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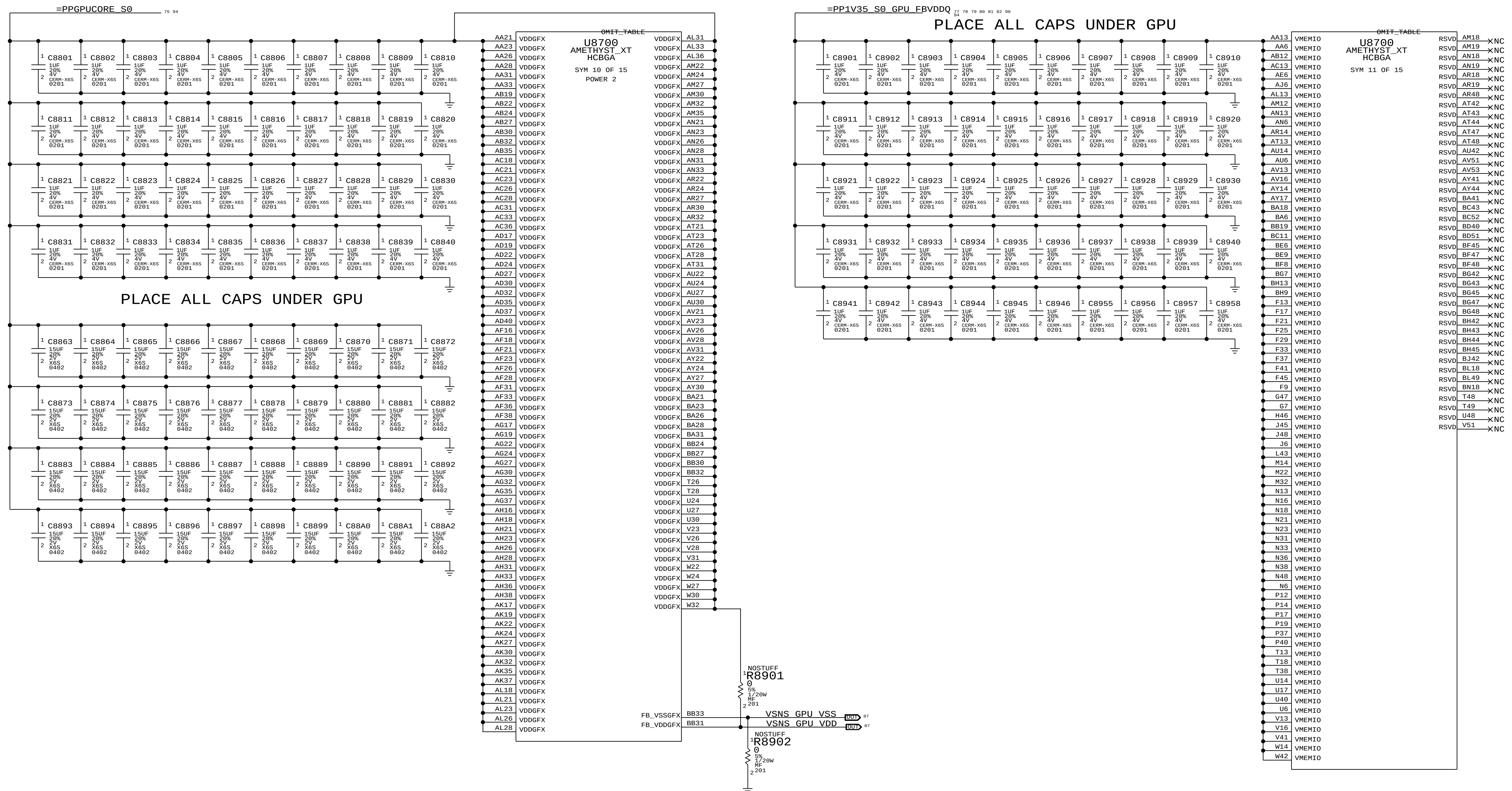
GPU MVDD/VDDGFX DECOUPLING



DRAWING LAST_MODIFIED=Mon Feb 24 09:58:29 2014

SYNC_MASTER=J78_NAT SYNC_DATE=12/09/2013

PAGE TITLE

GPU: MVDD/VDDGFX

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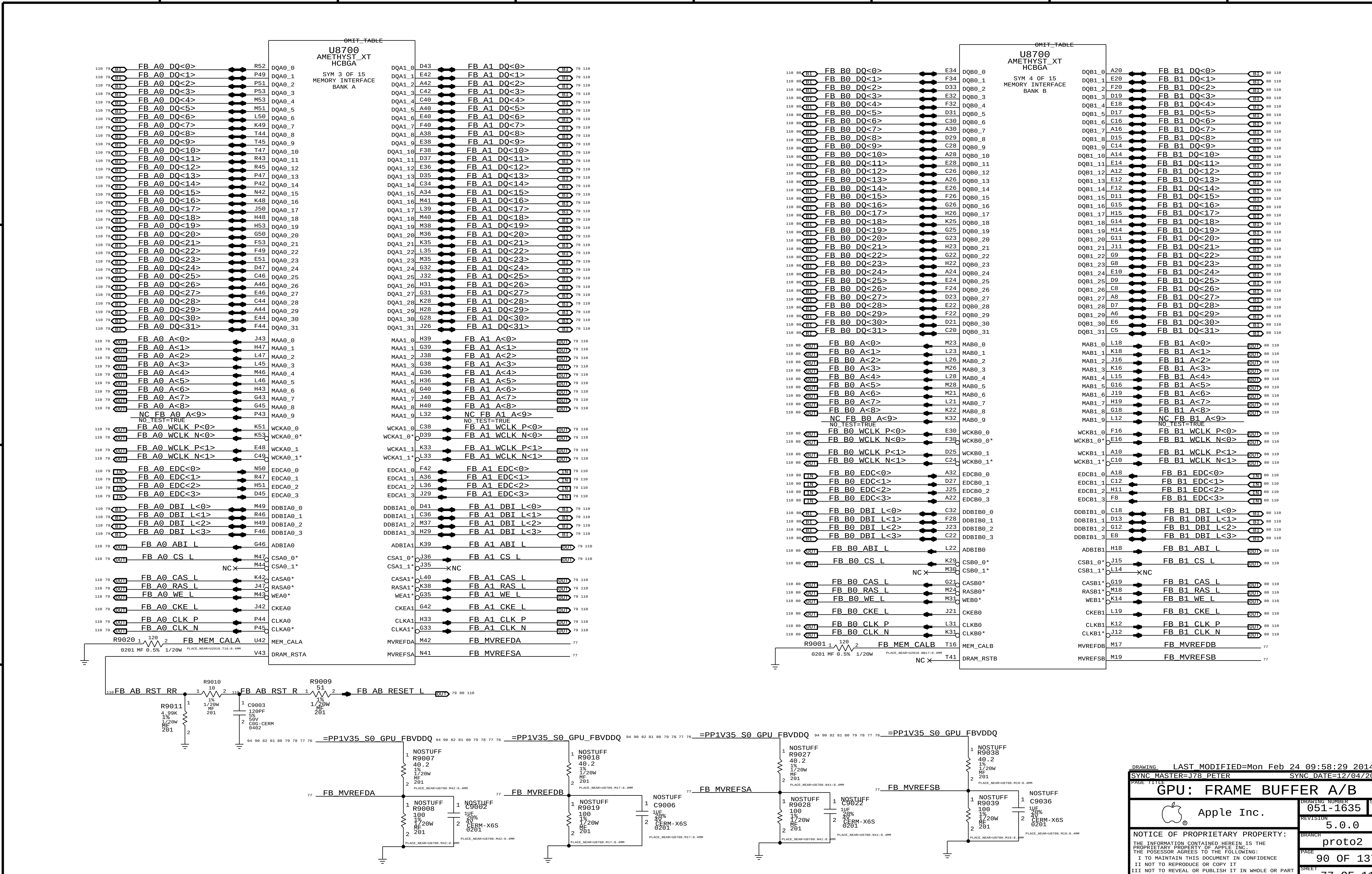

DRAWING LAST_MODIFIED=Mon Feb 24 09:58:29 2014

SYNC_MASTER=J78_PETER SYNC_DATE=12/04/2013

PAGE TITLE

GPU: FRAME BUFFER A/B

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BRANCH proto2

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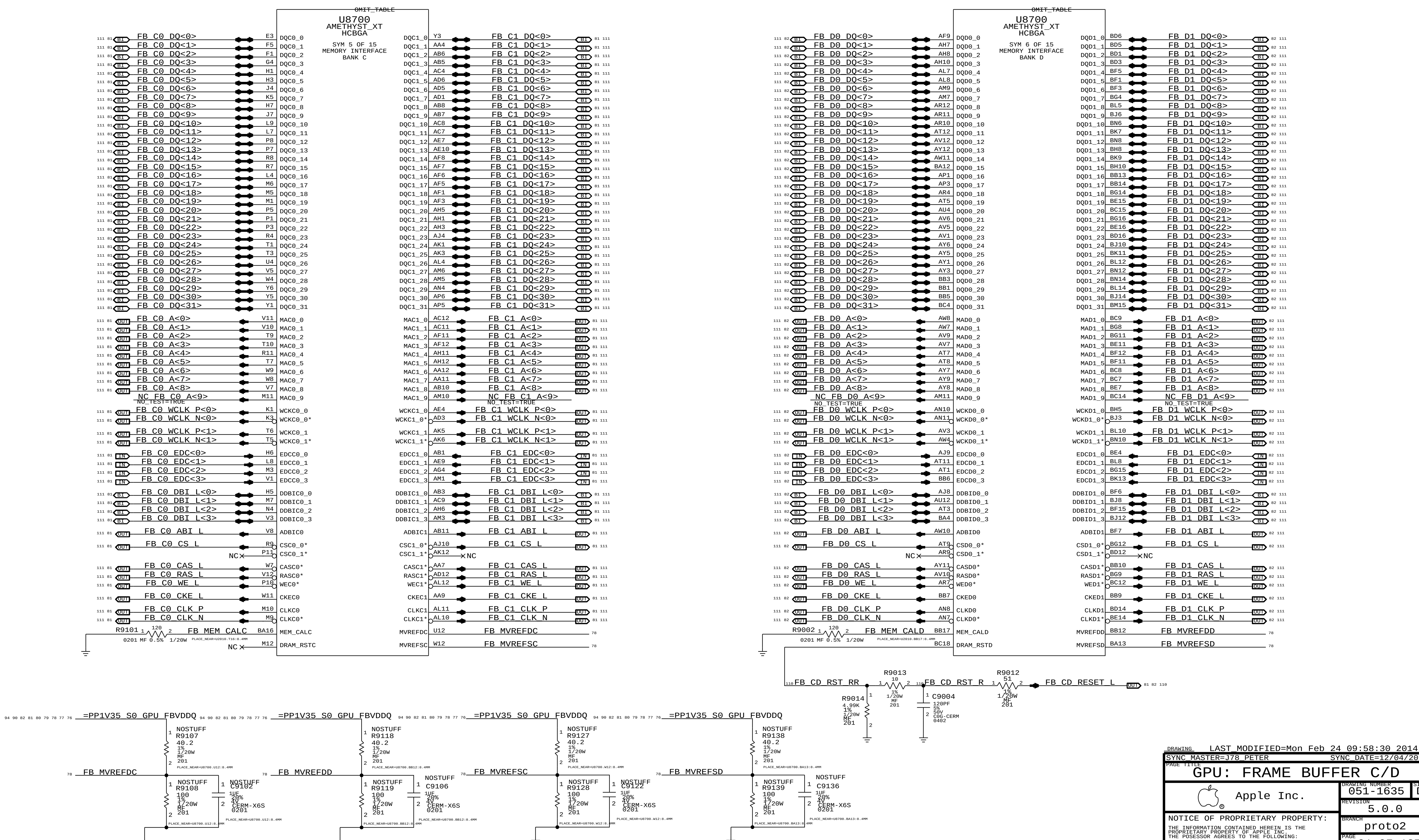

DRAWING LAST_MODIFIED=Mon Feb 24 09:58:30 2014

SYNC_MASTER=J78_PETER SYNC_DATE=12/04/2013

PAGE TITLE

GPU: FRAME BUFFER C/D

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BRANCH proto2

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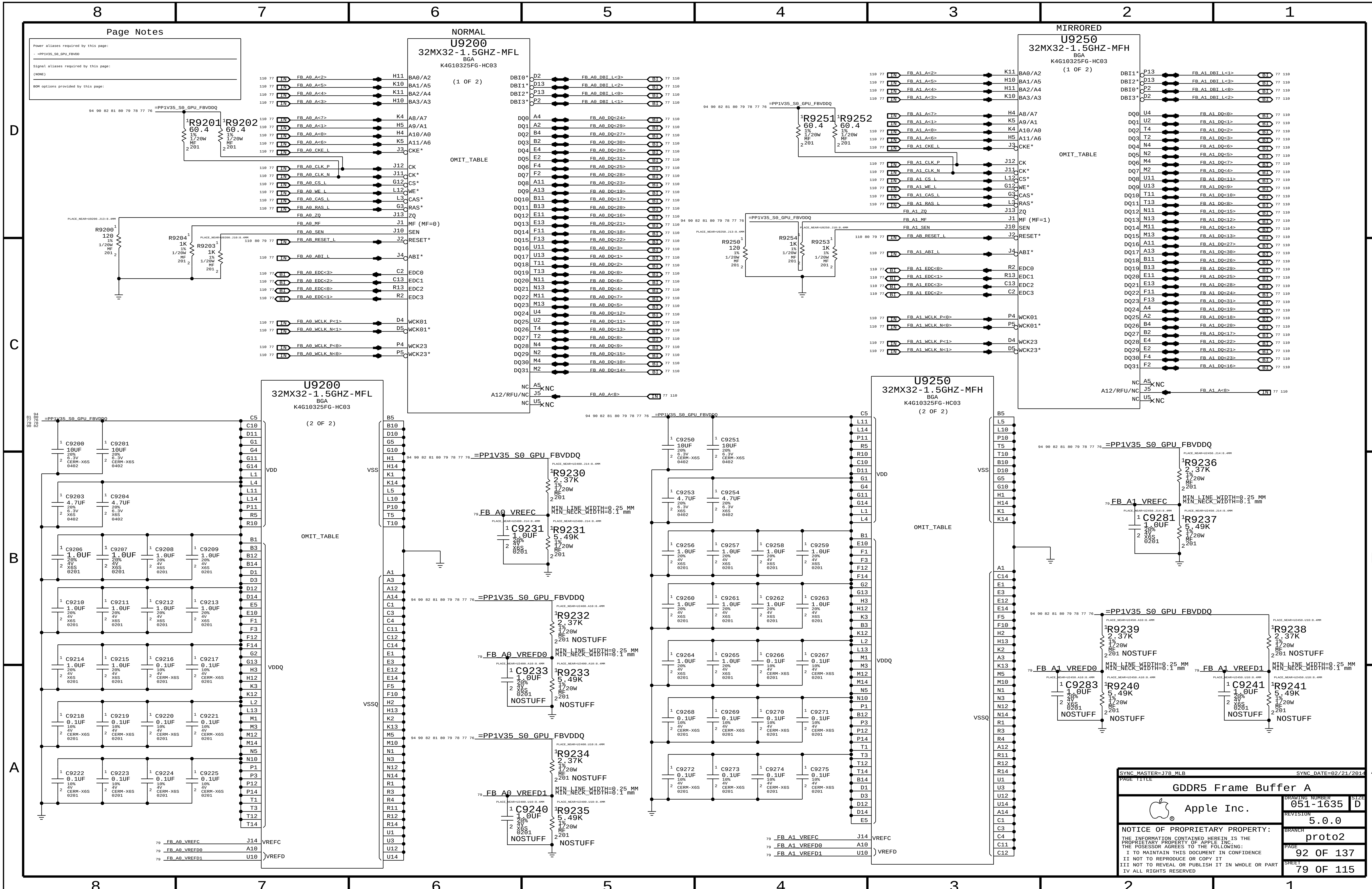

SYNC_MASTER=J78_MLB SYNC_DATE=02/21/2014

PAGE TITLE

GDDR5 Frame Buffer A

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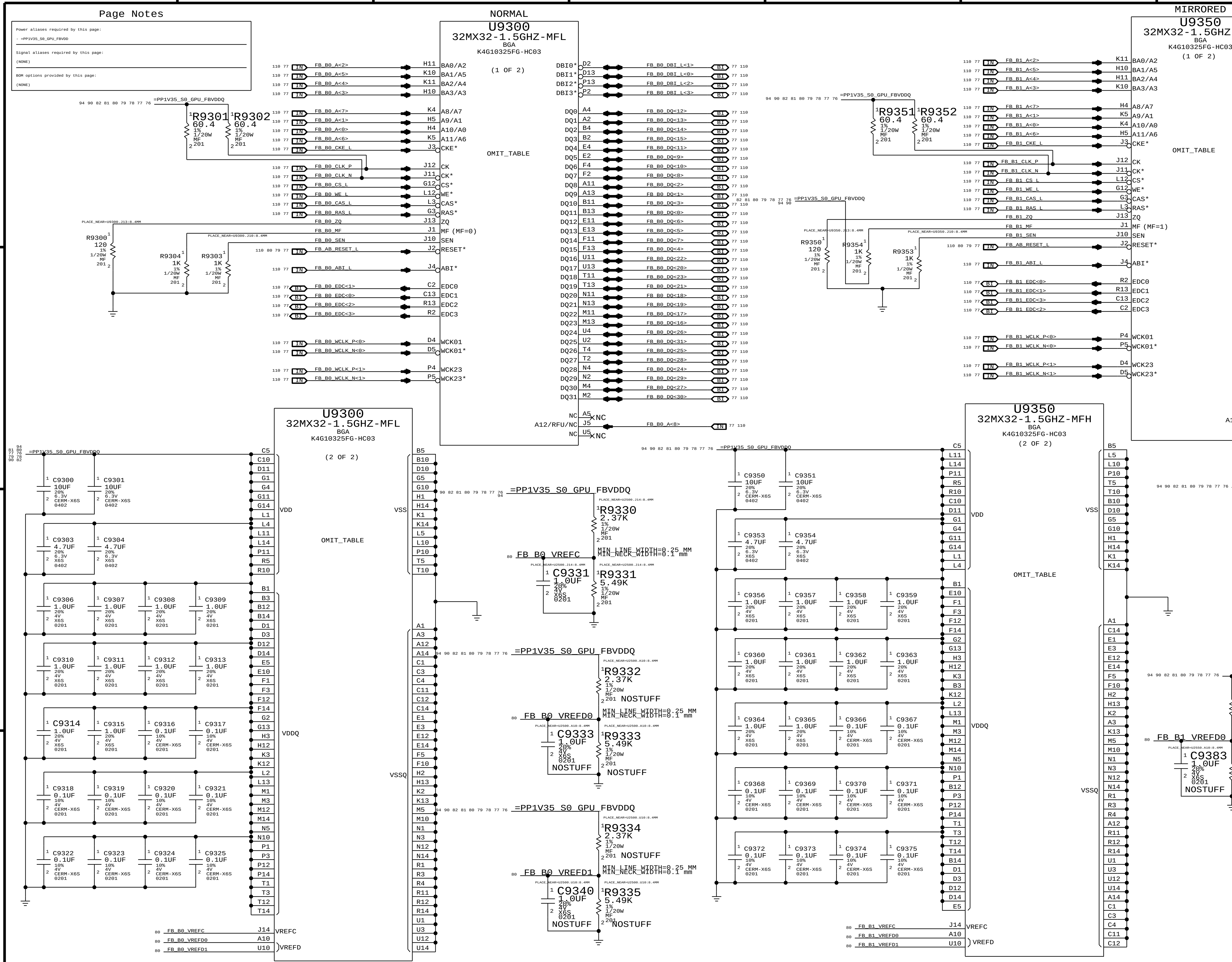

SYNC_MASTER=J78_MLB SYNC_DATE=02/21/2014

PAGE TITLE

GDDR5 Frame Buffer B

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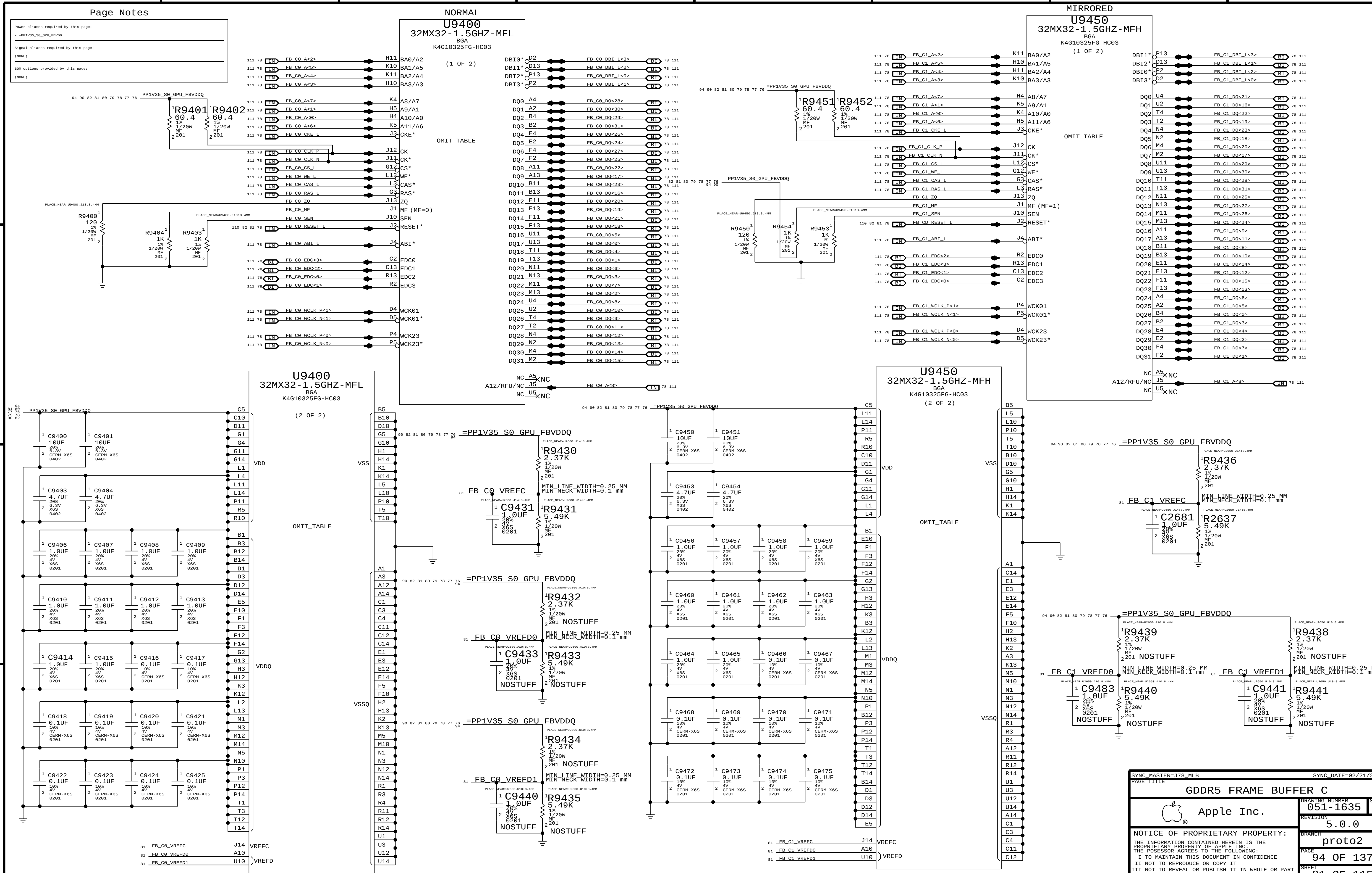

SYNC_MASTER=J78_MLB SYNC_DATE=02/21/2014

PAGE TITLE

GDDR5 FRAME BUFFER C

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Page Notes

Power aliases required by this page:

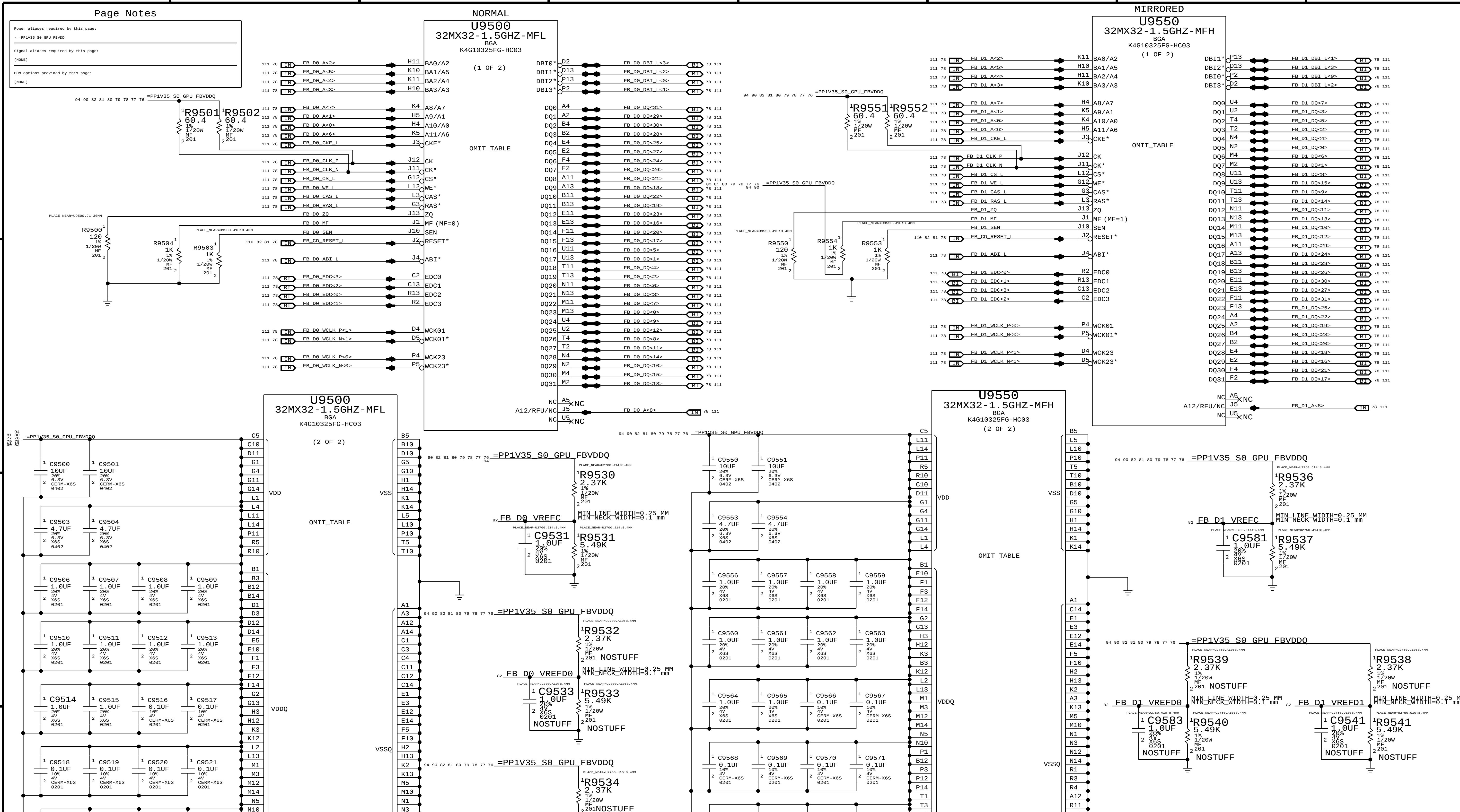
- =PP1V35_S0_GPU_FBVDD

Signal aliases required by this page:

(NONE)

BOM options provided by this page:

(NONE)



SYNC_MASTER=J78_MLB SYNC_DATE=02/21/2014

PAGE TITLE

GDDR5 FRAME BUFFER D

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DRAWING NUMBER 051-1635 SIZE D

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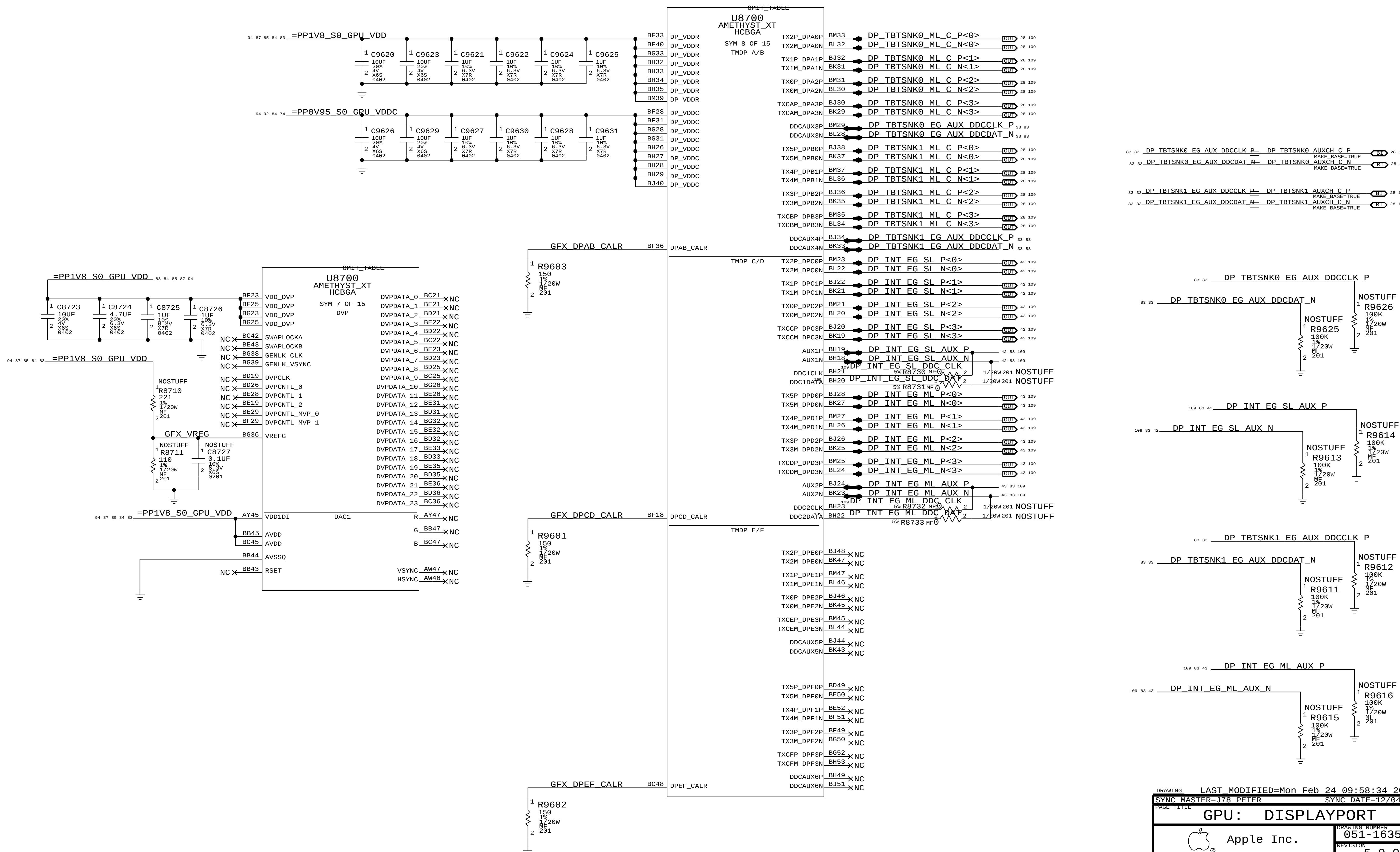

LAST_MODIFIED=Mon Feb 24 09:58:34 2014

SYNC_MASTER=J78_PETER SYNC_DATE=12/04/2013

PAGE TITLE

GPU: DISPLAYPORT

Apple Inc.

DRAWING NUMBER 051-1635 SIZE D

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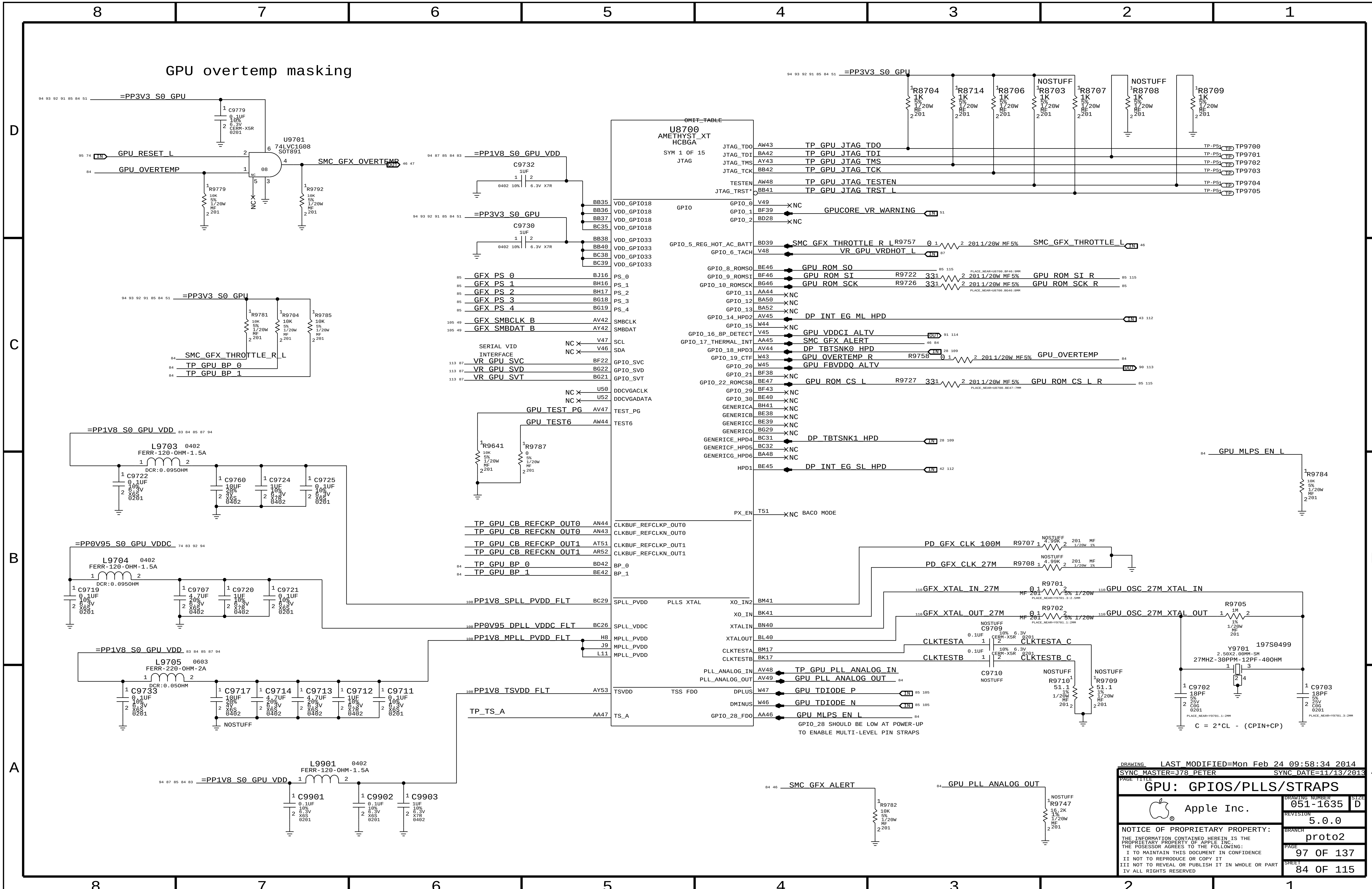

DRAWING LAST_MODIFIED=Mon Feb 24 09:58:34 2014

SYNC_MASTER=J78_PETER SYNC_DATE=11/13/2013

PAGE TITLE

GPU: GPIOS/PLLS/STRAPS

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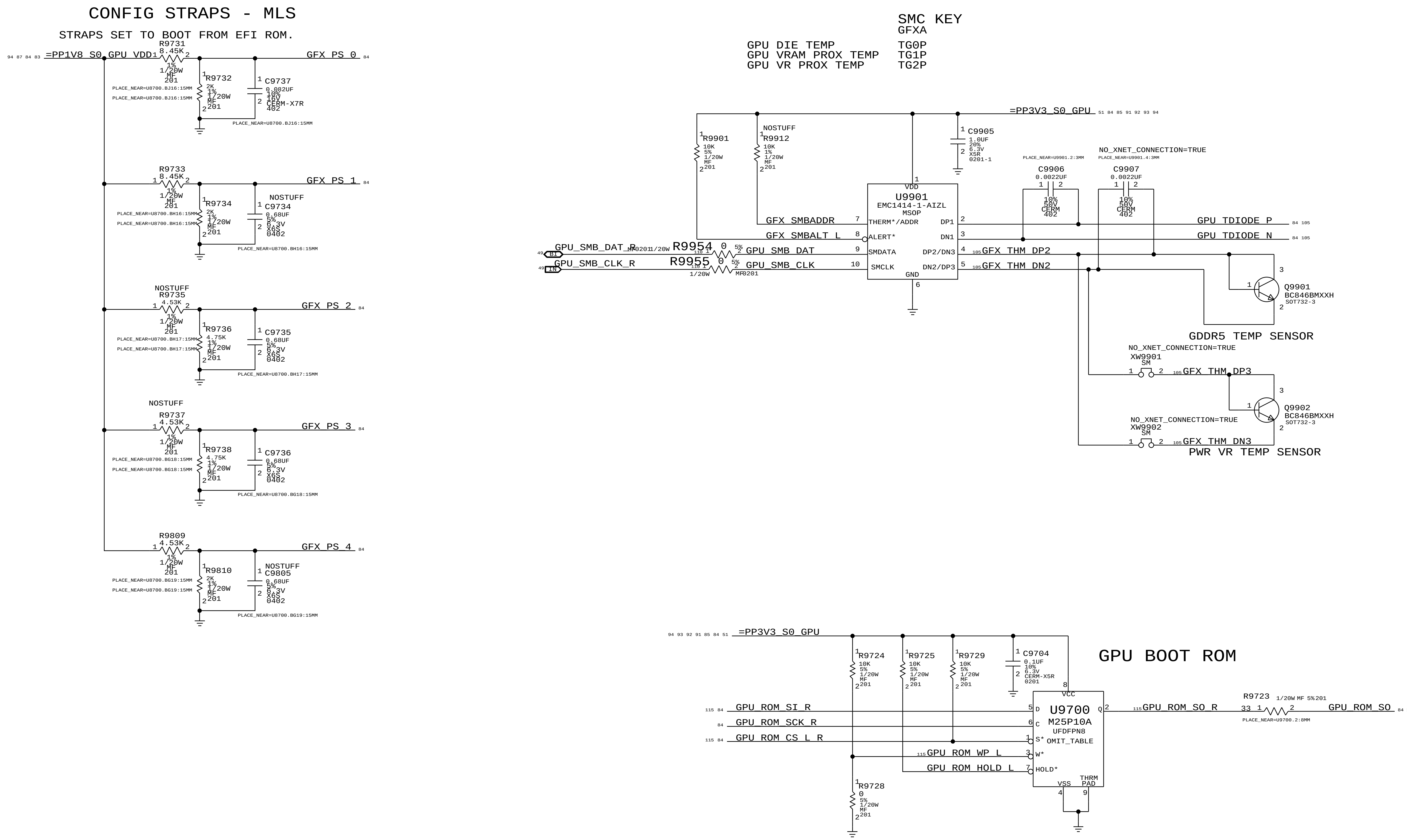

Page Notes

SYNC_MASTER=J78_PETER SYNC_DATE=11/13/2013

PAGE TITLE
GPU: STRAPS,SENSORS,ROM

Apple Inc.

DRAWING NUMBER 051-1635 SIZE D

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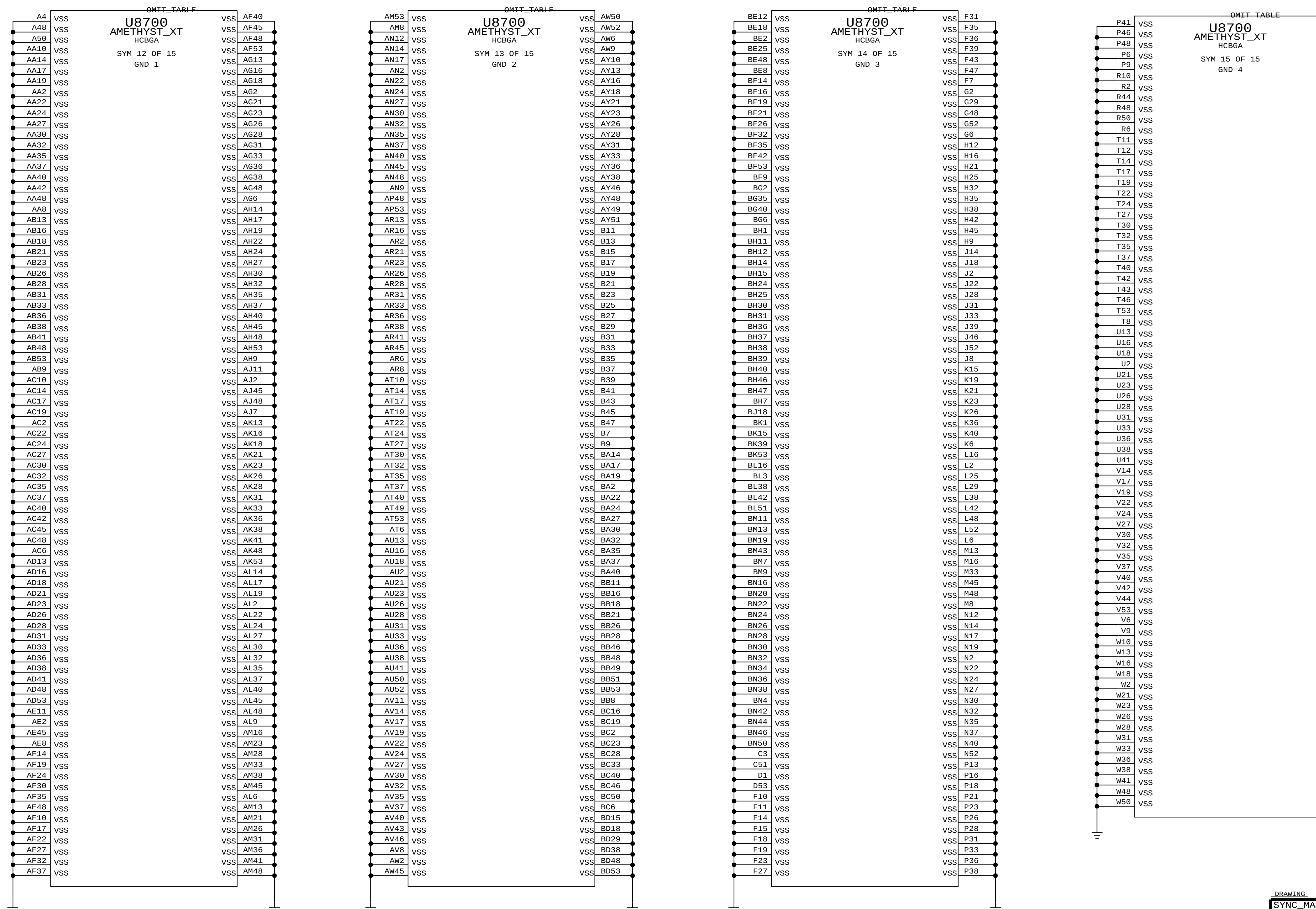

DRAWING LAST_MODIFIED=Mon Feb 24 09:58:35 2014

SYNC_MASTER=J78_PETER SYNC_DATE=11/13/2013

PAGE TITLE

GPU: GNDS & UNUSED

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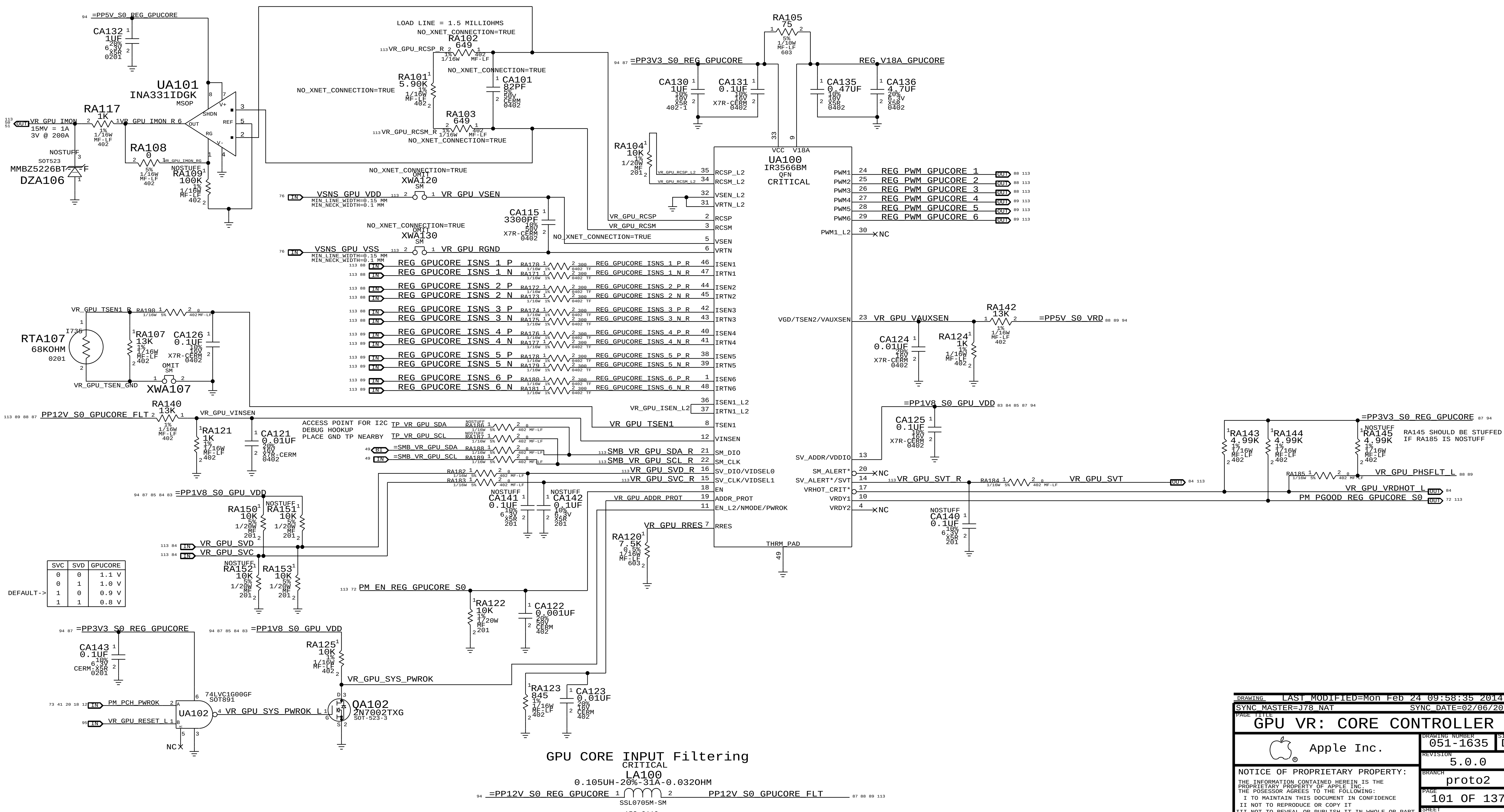
O/P= PPGPUCORE_S0_REG

GPUCORE
VOUT = VCORE
PEAK = 180A
AVG = 150A



SVC	SVD	GPUCORE
0	0	1.1 V
0	1	1.0 V
1	0	0.9 V
1	1	0.8 V

DRAWING LAST_MODIFIED=Mon Feb 24 09:58:35 2014

SYNC_MASTER=J78_NAT SYNC_DATE=02/06/2014

PAGE TITLE

GPU VR: CORE CONTROLLER

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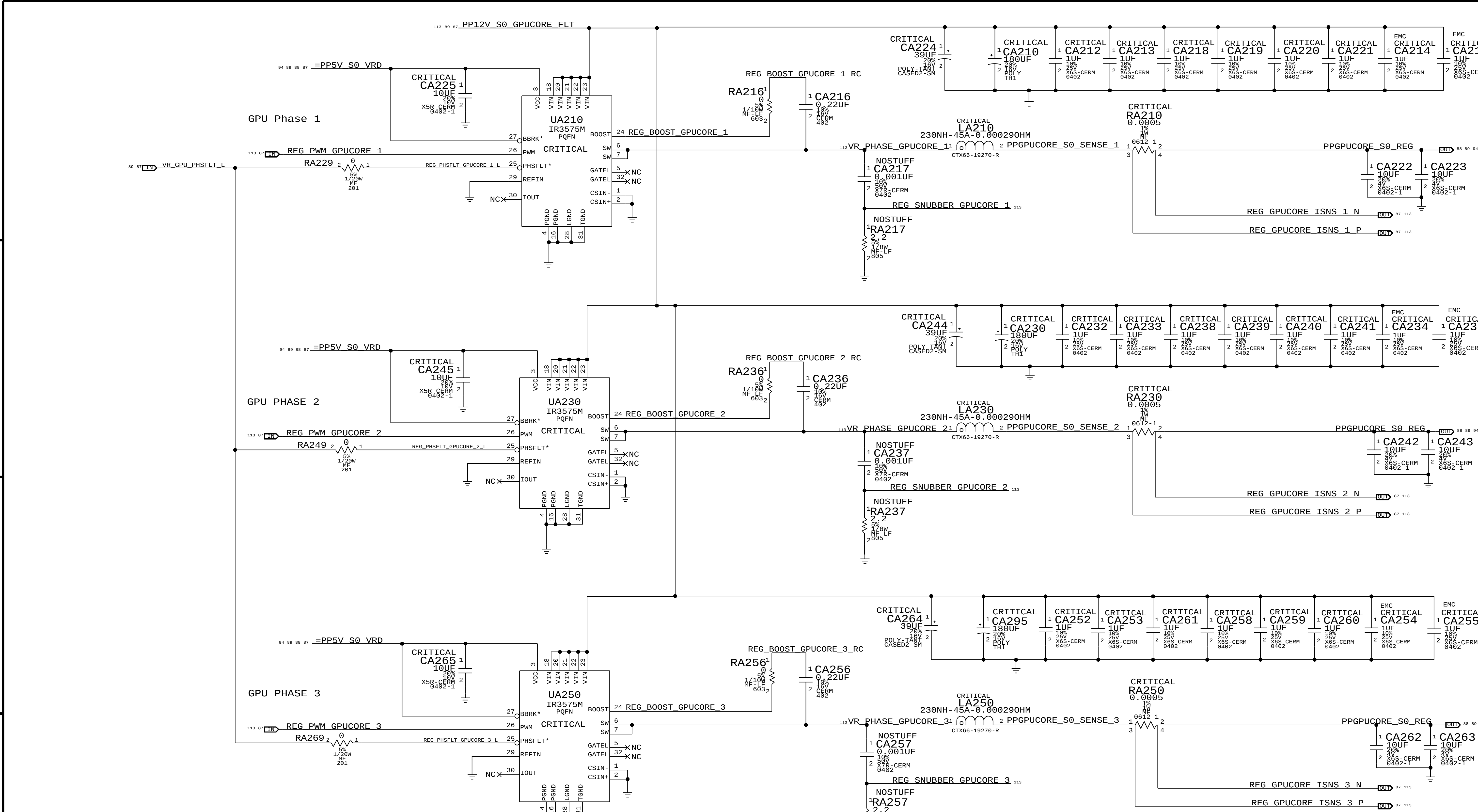

DRAWING LAST_MODIFIED=Mon Feb 24 09:58:35 2014

SYNC_MASTER=J78_NAT SYNC_DATE=02/13/2014

PAGE TITLE

GPU VR: CORE PHASES 1-3

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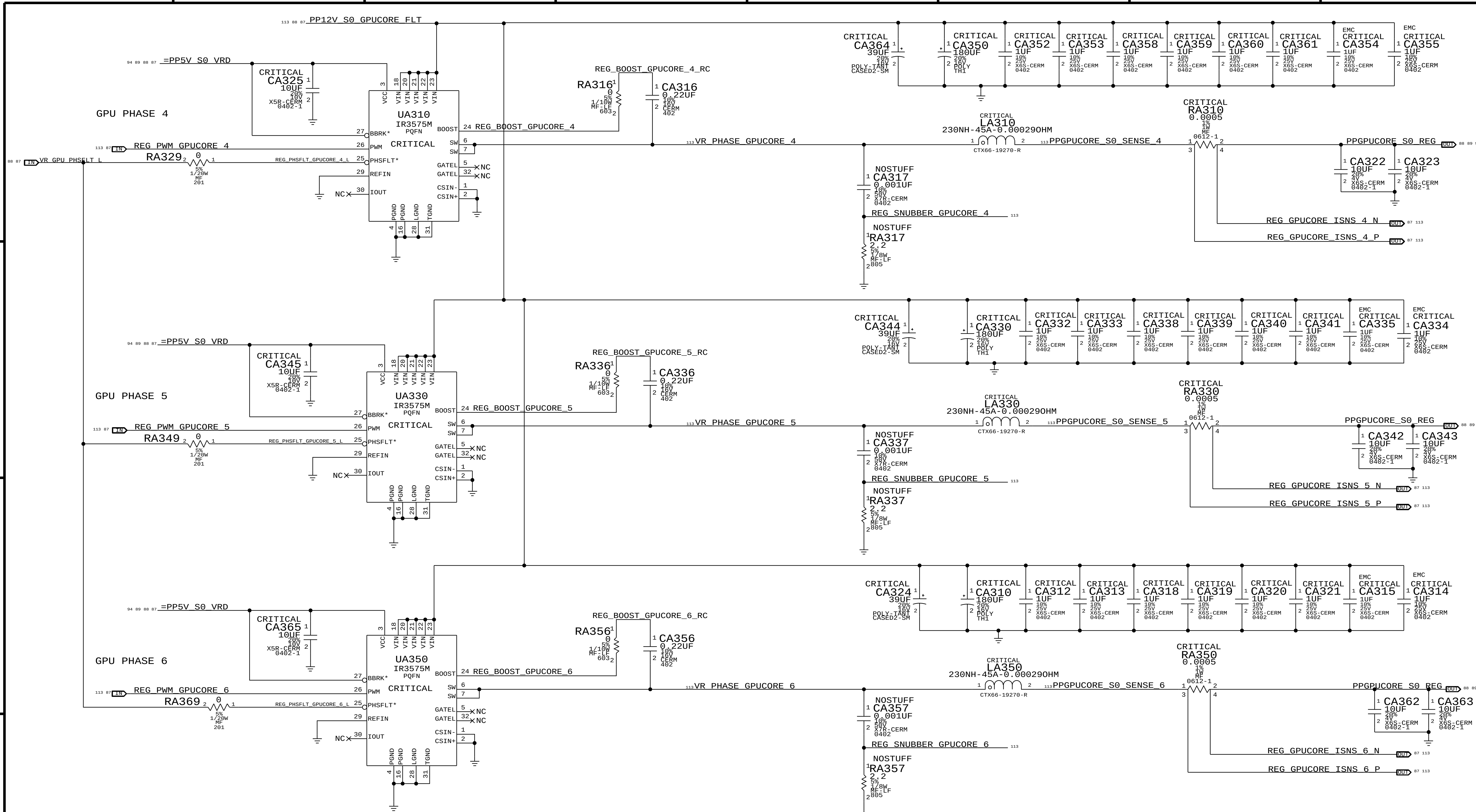

DRAWING LAST_MODIFIED=Mon Feb 24 09:58:36 2014

SYNC_MASTER=J78_NAT SYNC_DATE=02/13/2014

PAGE TITLE

GPU VR: CORE PHASES CONT

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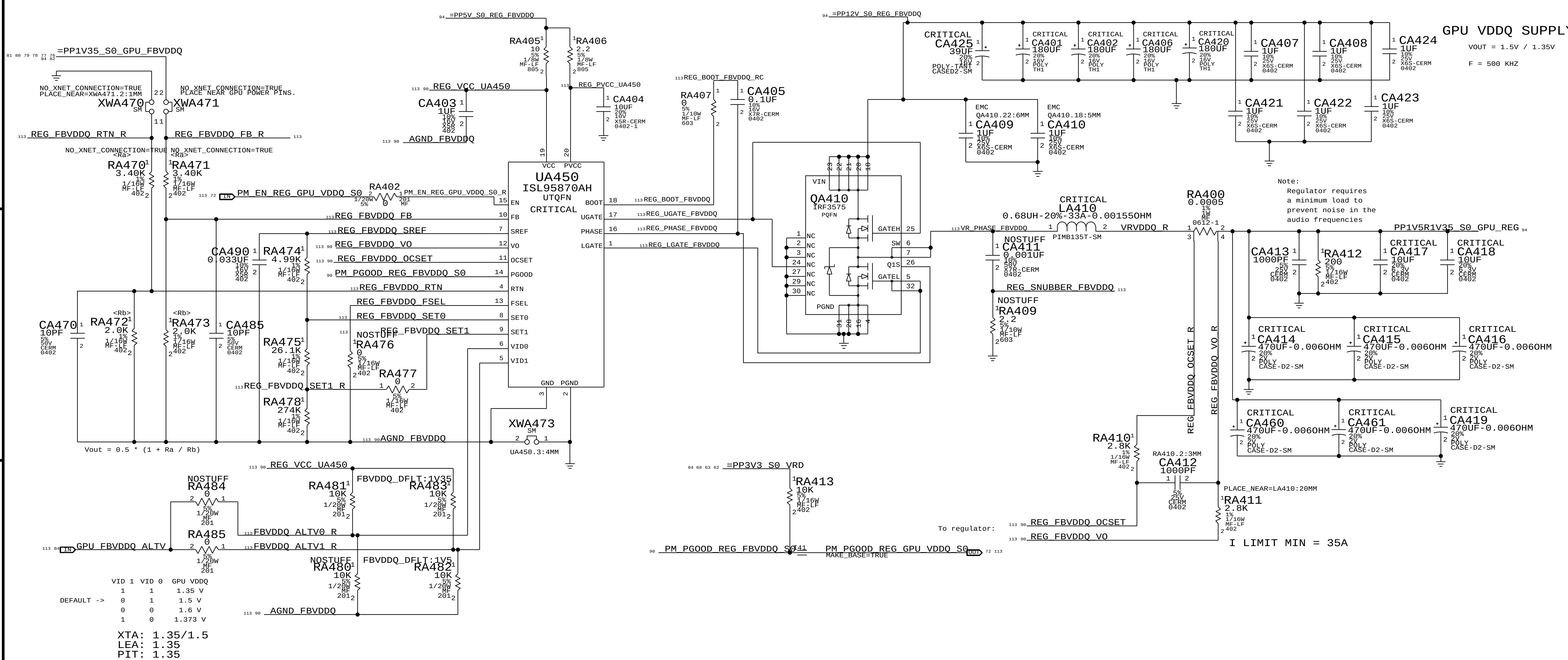

DRAWING	LAST_MODIFIED=Mon Feb 24 09:58:36 2014
SYNC_MASTER=J78_MLB	SYNC_DATE=02/20/2014
PAGE TITLE	GPU VR: VDDQ
Apple Inc.	DRAWING NUMBER 051-1635 SIZE D
	REVISION 5.0.0
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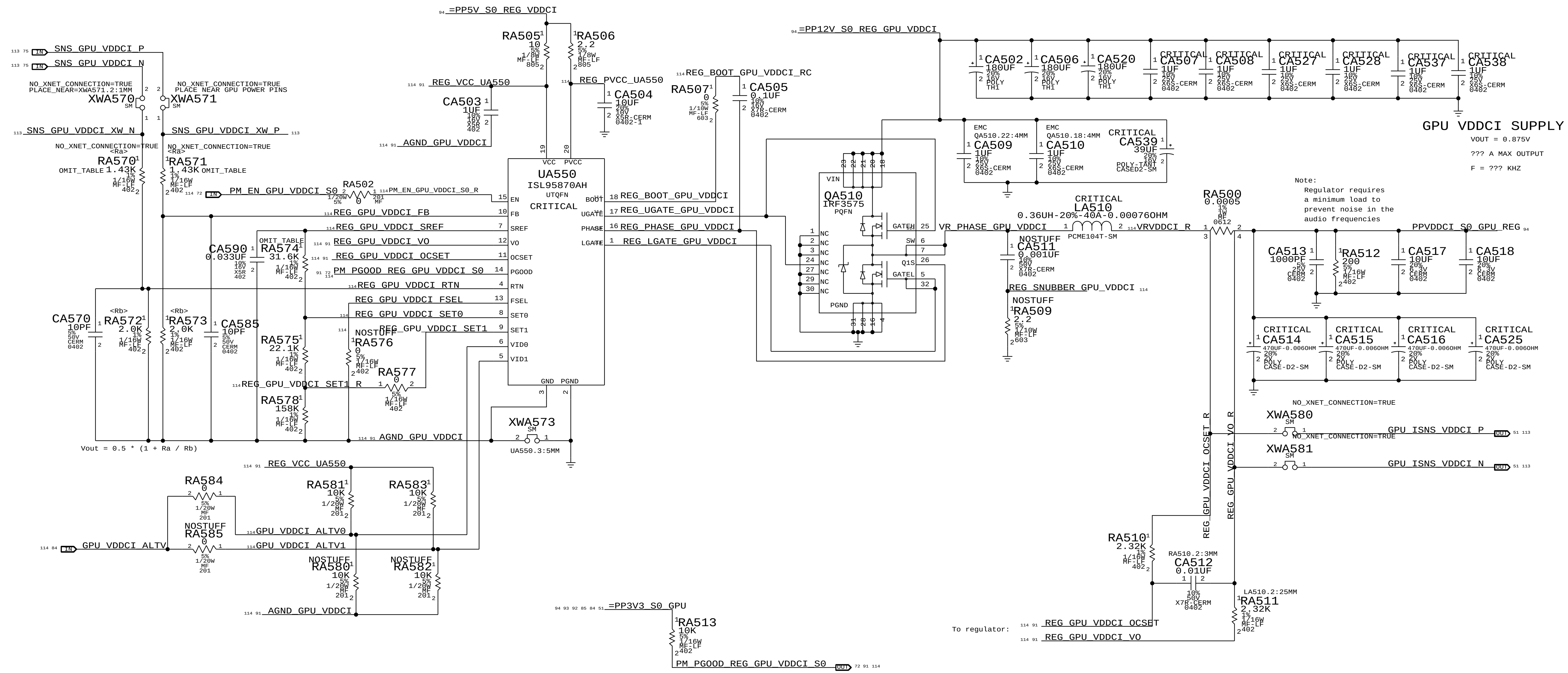

NEPTUNE

	VID 1	VID 0	GPU VDDCI
DEFAULT ->	1	1	0.850 V
	1	0	1.000 V

AMETHYST

	VID 1	VID 0	GPU VDDCI
DEFAULT ->	1	1	0.900 V
	1	0	1.000 V

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	BOM OPTION
114S0363	1	RES,31.6K,0402	RA574	GPU_NEPT
114S0232	2	RES,1.43K,0402	RA570,RA571	GPU_NEPT
114S0343	1	RES,20.0K,0402	RA574	GPU_AMTH
114S0568	2	RES,1.6K,0402	RA570,RA571	GPU_AMTH

SYNC_MASTER=J78_MLB SYNC_DATE=02/21/2014

PAGE TITLE

GPU VR: VDDCI

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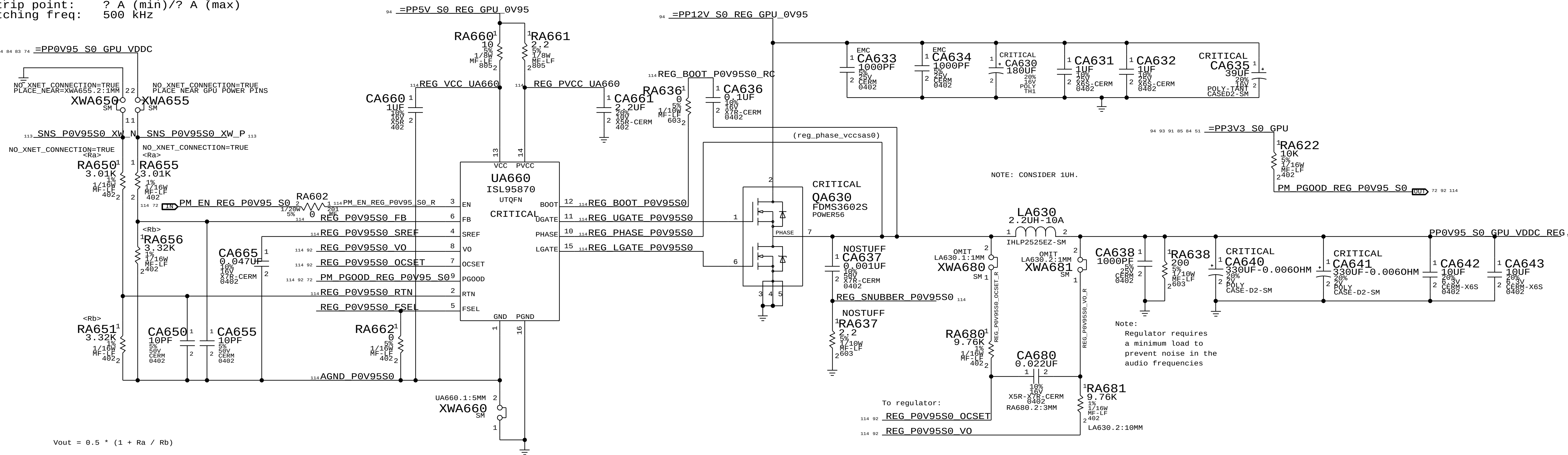
GPU VDDC (0.95V) S0 REGULATOR

Max avg current: 8.10 A (BUDGET)
Max peak current: 8.50 A (BUDGET)
OC trip point: ? A (min)/? A (max)
Switching freq: 500 kHz



DRAWING	LAST_MODIFIED=Mon Feb 24 09:58:37 2014
SYNC_MASTER=J78_MLB	SYNC_DATE=02/21/2014
PAGE TITLE	GPU VR: 0V95
Apple Inc.	DRAWING NUMBER 051-1635, SIZE D
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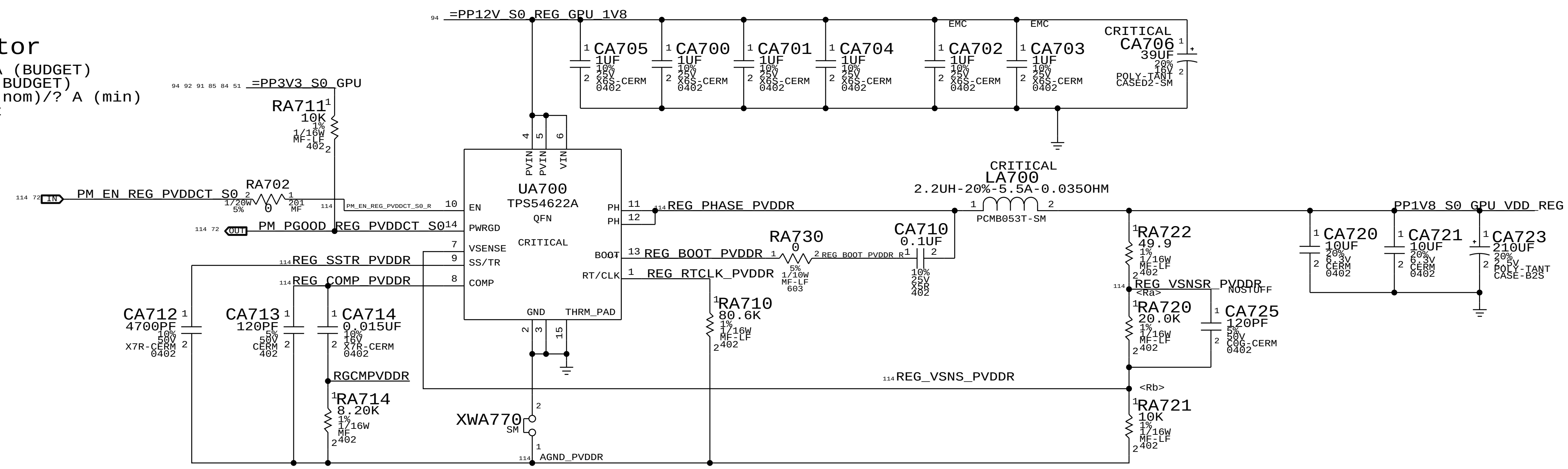
NOTICE OF PROPRIETARY PROPERTY:

1.8V S0 Regulator

Max avg current: 2.4 A (BUDGET)
Max peak current: 3 A (BUDGET)
OC trip point: ? A (nom)/? A (min)
Switching freq: ? kHz



VOUT = 0.6 * (1 + RA / RB)

DRAWING LAST_MODIFIED=Mon Feb 24 09:58:37 2014
SYNC_MASTER=J78_MLB SYNC_DATE=02/21/2014

PAGE TITLE
GPU VR: 1V8

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G3 Rails
Always on: Keeps the PCH RTC alive

PP3V3_G3 (MAKE_BASE=TRUE)
- PP3V3_G3_RTC 19
- =PP3V3_G3_PCH_RTC 15
- =PP3V3_G3_PCH 11 12

G3H Rails
On with AC/DC plugged in

PP12V_ACDC (MAKE_BASE=TRUE)
- PP12V_G3H_ACDC 62
- =PP12V_G3H_SNS_R 50

PP12V_G3H (MAKE_BASE=TRUE)
- PP12V_G3H_SNS 50
- =PP12V_G3H_P3V42 62
- =PP12V_G3H_FET_P12V_S0 71
- =PP12V_G3H_FET_P12V_S5 71

PP3V42_G3H (MAKE_BASE=TRUE)
- PP3V42_G3H_REG 62
- =PP3V3_G3H_BT 34
- =PP3V3_G3H_RTC_D 19
- =PP3V3_G3H_SMC 46 47 53
- =PP3V3_G3H_SMC_USBMUX 44
- =PPVIN_G3H_SMCVREF 47
- =PP3V3_G3H_LPCPLUS 48

Ground/Common

GND (MAKE_BASE=TRUE)

S5 Rails
On when in S5

PP12V_S5 (MAKE_BASE=TRUE)
- PP12V_S5_FET 71
- =PP12V_S5_REG_P3V3P5V_S5 67
- =PP12V_S5_REG_VDDQ_S3 66
- =PP12V_S5_PWRCTL 71 72
- =PPHV_SW_TBTAPWRSW 31
- =PPHV_SW_TBTBPWRSW 32
- =PP12V_S5_SNS 50

PP5V_S5 (MAKE_BASE=TRUE)
- PP5V_S5_LDO 67

PP3V3_S5 (MAKE_BASE=TRUE)
- PP3V3_S5_REG 67
- =PP3V3_S5_FET_P3V3_S0 71
- =PP3V3_S5_FET_P3V3_S4 71
- =PP3V3_S5_SMC 47
- =PP3V3_S5_LED 3
- =PP3V3_S5_PCH_VCCSUS_GPIO 11 12 13 14 15 17
- =PP3V3_S5_PCH_VCCDSW 15 17
- =PP3V3_S5_PCH_VCC_SPI 15 17
- =PP3V3_S5_PCH_VCCSUS_HDA 15 17
- =PP3V3_S5_PCH_VCCSUS_USB 15 17
- =PP3V3_S5_PCH_VCCSUS_RTC 15 17
- =PP3V3_S5_PCH_VCCSUS_USB3 15
- =PP3V3_S5_PCH 19 20
- =PP3V3_S5_PWRCTL 72 73
- =PP3V3_S5_ROM 48
- =PP3V3_S5_VRD 67
- =PP3V3_S5_XDP 18
- =PP3V3_S5_SENSE 50
- =PP3V3_S5_SNS 51

S4 Rails
Enabled when system has AC and is in run or sleep

PP5V_S4 (MAKE_BASE=TRUE)
- PP5V_S4_REG 67
- =PP5V_S4_REG_VDDQ_S3 66
- =PP5V_S4_FET_P5V_S0 71
- =PP5V_S4_MEMRESET 21
- =PP5V_S4_USB 44 45

PP3V3_S4 (MAKE_BASE=TRUE)
- PP3V3_S4_FET 71
- =PP3V3_S4_FET_ENET 38
- =PP3V3_S4_PWRCTL 66 71
- =PP3V3_S4_LED 3
- =PP3V3_S4_MEMRESET 21
- =PP3V3_S4_PCH 13
- =PP3V3_S4_PM 21
- =PP3V3_S4_SDCARD 39
- =PP3V3_S4_SMC 47
- =PP3V3_S4_TBT 20 28 29 31 32 72
- =PP3V3_S4_AUDIO_DIG 58
- =PP3V3_S4_AP_SNS_R 50
- =PP3V3_S3_VREFMRGN 22
- =PP3V3_S4_TBTAPWRSW 31
- =PP3V3_S4_TBTBPWRSW 32

PP3V3_S4_AP (MAKE_BASE=TRUE)
- PP3V3_S4_AP_SNS 50
- =PP3V3_S4_AP 34 47

S3 Rails
Enabled when system is in run or sleep

PP3V3_ENET (MAKE_BASE=TRUE)
- PP3V3_ENET_FET 38
- =PP3V3_ENET_PHY 37 38
- =PP3V3_ENET_SYSCLK 19
- =PPVDDIO_ENET_CLK 19

PPVDDQ_S3 (MAKE_BASE=TRUE)
THIS IS 1.5V RAIL
- PPVDDQ_S3_REG 66
- =PPVDDQ_S3_SNS_DDR_R 51
- =PPVDDQ_S3_FET_VDDQ_S0 71

PPVDDQ_S3_DDR (MAKE_BASE=TRUE)
THIS IS 1.5V RAIL
- PPVDDQ_S3_SNS_DDR 51
- =PPDDR_S3_MEMVREF 22
- =PPVDDQ_S3_MEM_A 23 24
- =PPVDDQ_S3_MEM_B 25 26
- =PPVDDQ_S3_MEMRESET 21
- =PPVDDQ_S3_LDO_DDRVTT 66

S0 Rails
Enabled when system is in run

PP12V_S0 (MAKE_BASE=TRUE)
- PP12V_S0_FET 71
- =PP12V_S0_AUDIO_SPKRAMP 56 57
- =PP12V_S0_FAN 53
- =PP12V_S0_SNS_GPUCORE_R 50
- =PP12V_S0_LCD 42
- =PP12V_S0_REG_CPUVCC_S0 64
- =PP12V_S0_REG_P1V05_S0 68
- =PP12V_S0_SNS_FBVDDQ_R 50
- =PP12V_S0_HDD 36
- =PP12V_S0_BKLT 69

PP12V_S0_GPU_AUX (MAKE_BASE=TRUE)
- PP12V_S0_SNS_GPU_AUX 50
- =PP12V_S0_REG_GPU_0V95 92
- =PP12V_S0_REG_GPU_1V8 93

PP12V_S0_GPUCORE (MAKE_BASE=TRUE)
- PP12V_S0_SNS_GPUCORE 50
- =PP12V_S0_REG_GPUCORE 87
- =PP12V_S0_REG_GPU_VDDCI 91

PPHDD12_S0 (MAKE_BASE=TRUE)
- PP12V_S0_HDD_FET 36
- =PP12V_S0_HDD_SNS_R 50

PP12V_S0_HDD (MAKE_BASE=TRUE)
- PP12V_S0_HDD_SNS 50
- =PP12V_S0_HDD_PWR 36

PP12V_S0_FBVDDQ (MAKE_BASE=TRUE)
- PP12V_S0_SNS_FBVDDQ 50
- =PP12V_S0_REG_FBVDDQ 90

PP5V_S0 (MAKE_BASE=TRUE)
- PP5V_S0_FET 71
- =PP5V_S0_AUDIO 54 61
- =PP5V_S0_BKLT 69
- =PP5V_S0_ISENSE 50
- =PP5V_S0_LPCPLUS 48
- =PP5V_S0_REG_CPUVCC_S0 63 64 65
- =PP5V_S0_REG_FBVDDQ 90
- =PP5V_S0_REG_P1V05 68
- =PP5V_S0_HDD 36
- =PP5V_S0_VRD 87 88 89
- =PP5V_S0_GPU 51
- =PP5V_S0_REG_VDDCI 91
- =PP5V_S0_REG_GPU_0V95 92
- =PP5V_S0_CAMERA 40
- =PP5V_S0_REG_GPUCORE 87

PPHDD_S0 (MAKE_BASE=TRUE)
- PP5V_S0_HDD_FET 36
- =PPHDD_S0_SNS_R 50

PP5V_S0_HDD (MAKE_BASE=TRUE)
- PPHDD_S0_SNS 50
- =PP5V_S0_HDD_PWR 36

PP3V3_S0 (MAKE_BASE=TRUE)
- PP3V3_S0_FET 71
- =PP3V3_S0_AUDIO 40 54 56 57 60
- =PP3V3_S0_AUDIO_DIG 58
- =PP3V3_S0_BKLT 69
- =PP3V3_S0_DP 33 42 49
- =PP3V3_S0_ENET 37
- =PP3V3_S0_FAN 53
- =PP3V3_S0_GPU 51 84 85 91 92 93
- =PP3V3_S0_INTDPMUX 42 43
- =PP3V3_S0_LED 3
- =PP3V3_S0_LED_SATA 35
- =PP3V3_S0_MEM_A_SPD 23 24
- =PP3V3_S0_MEM_B_SPD 25 26
- =PP3V3_S0_PCH 19
- =PP3V3_S0_PCH_VCC3_3_GPIO 11 12 13 14 15 17
- =PP3V3_S0_PCH_VCC3_3_THRM 15 17
- =PP3V3_S0_PCH_VCCCLK3_3 15 17
- =PP3V3_S0_PCH_VCC3_3_HVCMOS 15 17
- =PP3V3_S0_PCH_VCC_FUSE 15 17
- =PP3V3_S0_PWRCTL 71
- =PP3V3_S0_RSTBUF 20
- =PP3V3_S0_SDCARD 39
- =PP3V3_S0_SENSE 36 50 51 52
- =PP3V3_S0_SMBUS 49
- =PP3V3_S0_SMBUS_SMC_0 49
- =PP3V3_S0_SMBUS_SMC_1 49
- =PP3V3_S0_SMBUS_SMC_3 49
- =PP3V3_S0_SMC 47
- =PP3V3_S0_VRD 62 63 68 90
- =PP3V3_S0_PCH_GPIO 29
- =PP3V3_S0_PCH_VCC3_3_USB 15 17
- =PPSSD_S0_SNS_R 51
- =PP3V3_TBT_PCH_GPIO 20
- =PP3V3_S0_CAMERA 40 41
- =PP3V3_S0_ALS 40
- =PP3V3_S0_SMBUS_SMC_2 49
- =PP3V3_S0_REG_GPUCORE 87

PPSSD_S0 (MAKE_BASE=TRUE)
- PPSSD_S0_SNS 51
- =PP3V3_S0_SSD 35

PP1V5_S0 (MAKE_BASE=TRUE)
- PP1V5_S0_FET 71
- =PP1V5_S0_SNS_R 51

PP1V5_S0_PCH (MAKE_BASE=TRUE)
- PP1V5_S0_SNS 51
- =PP1V5_S0_CPU_MEM 6 8 10
- =PP1V5_S0_AUD_DIG 54
- =PP1V5_S0_PCH_VCCVRM 17
- =PP1V5_S0_PCH_RCOMP 12 13
- =PP1V5_S0_PCH_SATA 11
- =PP1V5_S0_PCH_CLK 11 19
- =PP1V5_S0_DP 43

PPDDRVTT_S0 (MAKE_BASE=TRUE)
- PPDDRVTT_S0_LDO 66
- =PPDDRVTT_S0_CLAMP 21
- =PPDDRVTT_S0_MEM_A 23 24
- =PPDDRVTT_S0_MEM_B 25 26

PPCPUVCC_S0_CPU (MAKE_BASE=TRUE)
- PPCPUVCC_S0_REG 64 65
- =PPCPUVCC_S0_CPU 6 8 10 50 63

GPU Rails (S0)
Enabled when system is in run

PPGPUCORE_S0 (MAKE_BASE=TRUE)
- PPGPUCORE_S0_REG 88 89
- =PPGPUCORE_S0 75 76
- =PPGPUCORE_S0_SNS 50

PPFBVDDQ_S0_GPU (MAKE_BASE=TRUE)
- PP1V5R1V35_S0_GPU_REG 90
- =PP1V35_S0_GPU_FBVDDQ 76 77 78 79 80 81 82 90

PP1V05_S0 (MAKE_BASE=TRUE)
- PP1V05_S0_REG 68
- =PP1V05_S0_PCH_VCC 15 17
- =PP1V05_S0_PCH_VCCCLK_CLK100 15 17
- =PP1V05_S0_PCH_VCCASW 15 17
- =PP1V05_S0_PCH_VCC_CLK 17
- =PP1V05_S0_PCH_VCCIO_FDI 15 17
- =PP1V05_S0_PCH_VCCCLK_SSC 15 17
- =PP1V05_S0_PCH_VCCUSBPLL 15 17
- =PP1V05_S0_PCH_VCCIO 15 17
- =PP1V05_S0_PCH_VCCCLK_SSC100 15 17
- =PP1V05_S0_PCH_VCCCLK_CLK135 15 17
- =PP1V05_S0_PCH_VCCIO_USB2 15 17
- =PP1V05_S0_PCH_V_PROC_IO 14 15 17
- =PP1V05_S0_SNS 50
- =PP1V05_S0_SMC 47
- =PP1V05_S0_XDP 18

PPVDDCI_S0_GPU (MAKE_BASE=TRUE)
- PPVDDCI_S0_GPU_REG 91
- =PPVDDCI_S0_GPU 75

PP0V95_S0_GPU_VDDC (MAKE_BASE=TRUE)
- PP0V95_S0_GPU_VDDC_REG 92
- =PP0V95_S0_GPU_VDDC 74 83 84 92
- =PP0V95_S0_GPU_PCIEVDDC 74

PP1V8_S0_GPU_VDD (MAKE_BASE=TRUE)
- PP1V8_S0_GPU_VDD_REG 93
- =PP1V8_S0_GPU_VDD 83 84 85 87
- =PP1V8_S0_CAMERA 40 41
- =PP1V8_S0_GPU_PEX_IO 74

Thunderbolt Rails (S0)
Enabled when Thunderbolt cable is plugged in

PP3V3_TBTLC (MAKE_BASE=TRUE)
- =PPVDDIO_TBT_CLK 19

SYNC_MASTER=J78_MLB SYNC_DATE=11/20/2013

PAGE TITLE: Power Connectors/Aliases

Apple Inc.

DRAWING NUMBER: 051-1635 SIZE: D

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PEG aliases

100 74 PEG_D2R_N<0..15> MAKE_BASE=TRUE =PEG_D2R_N<15..0> 5
100 74 PEG_D2R_P<0..15> MAKE_BASE=TRUE =PEG_D2R_P<15..0> 5
100 74 PEG_R2D_C_N<0..15> MAKE_BASE=TRUE =PEG_R2D_C_N<15..0> 5
100 74 PEG_R2D_C_P<0..15> MAKE_BASE=TRUE =PEG_R2D_C_P<15..0> 5

DDI/FDI ALIASES

I64
5 TP_DP_IG_B_MLN<0..3> MAKE_BASE=TRUE NO_TEST=TRUE NC_DP_IG_B_MLN<0..3>
5 TP_DP_IG_B_MLP<0..3> MAKE_BASE=TRUE NO_TEST=TRUE NC_DP_IG_B_MLP<0..3>
5 TP_DP_IG_C_MLN<0..3> MAKE_BASE=TRUE NO_TEST=TRUE NC_DP_IG_C_MLN<0..3>
5 TP_DP_IG_C_MLP<0..3> MAKE_BASE=TRUE NO_TEST=TRUE NC_DP_IG_C_MLP<0..3>
5 TP_DP_IG_D_MLN<0..3> MAKE_BASE=TRUE NO_TEST=TRUE NC_DP_IG_D_MLN<0..3>
5 TP_DP_IG_D_MLP<0..3> MAKE_BASE=TRUE NO_TEST=TRUE NC_DP_IG_D_MLP<0..3>

GPU ALIASES

84 74 GPU_RESET_L MAKE_BASE=TRUE TP_GPU_RESET_L 20
87 VR_GPU_RESET_L MAKE_BASE=TRUE TP_VR_GPU_RESET_L 20

MEMORY PGOOD

21 =PM_PGOOD_MEM_S0 PM_PGOOD_FET_REG_P1V5_S0 71 72 MAKE_BASE=TRUE

SYNC_MASTER=J78_MLB SYNC_DATE=12/05/2013

PAGE TITLE
Signal Aliases

Apple Inc.

DRAWING NUMBER 051-1635 SIZE D
REVISION 5.0.0
BRANCH proto2

CPU Reserved

102 18 6	CPU_CFG<15..12>	TP_CPU_CFG<15..12>	MAKE_BASE=TRUE

CPU Memory

7	TP_MEM_A_DQS_N<8>	NC_MEM_A_DQSNX<8>	MAKE_BASE=TRUE NO_TEST=TRUE
7	TP_MEM_A_DQS_P<8>	NC_MEM_A_DQSPX<8>	MAKE_BASE=TRUE NO_TEST=TRUE
7	TP_MEM_B_DQS_N<8>	NC_MEM_B_DQSNX<8>	MAKE_BASE=TRUE NO_TEST=TRUE
7	TP_MEM_B_DQS_P<8>	NC_MEM_B_DQSPX<8>	MAKE_BASE=TRUE NO_TEST=TRUE

PCH PCIe

11	TP_PCIE_CLK100M_PE4N	NC_PCIE_CLK100M_PE4NX	MAKE_BASE=TRUE NO_TEST=TRUE
11	TP_PCIE_CLK100M_PE4P	NC_PCIE_CLK100M_PE4PX	MAKE_BASE=TRUE NO_TEST=TRUE
11	TP_PCIE_CLK100M_PE5N	NC_PCIE_CLK100M_PE5NX	MAKE_BASE=TRUE NO_TEST=TRUE
11	TP_PCIE_CLK100M_PE5P	NC_PCIE_CLK100M_PE5PX	MAKE_BASE=TRUE NO_TEST=TRUE
11	TP_PCIE_CLK100M_PE7N	NC_PCIE_CLK100M_PE7NX	MAKE_BASE=TRUE NO_TEST=TRUE
11	TP_PCIE_CLK100M_PE7P	NC_PCIE_CLK100M_PE7PX	MAKE_BASE=TRUE NO_TEST=TRUE

PCH Clocks

11	TP_PCH_GPIO64_CLKOUTFLEX0	NC_PCH_GPIO64_CLKOUTFLEX0	MAKE_BASE=TRUE NO_TEST=TRUE	
11	TP_PCH_GPIO65_CLKOUTFLEX1	NC_PCH_GPIO65_CLKOUTFLEX1	MAKE_BASE=TRUE NO_TEST=TRUE	
11	TP_PCH_GPIO66_CLKOUTFLEX2	NC_PCH_GPIO66_CLKOUTFLEX2	MAKE_BASE=TRUE NO_TEST=TRUE	
11	TP_PCH_GPIO67_CLKOUTFLEX3	NC_PCH_GPIO67_CLKOUTFLEX3	MAKE_BASE=TRUE NO_TEST=TRUE	
	TP_ITPXDP_CLK100MP	ITPXDP_CLK100M_P	MAKE_BASE=TRUE NO_TEST=TRUE	11 102
	TP_ITPXDP_CLK100MN	ITPXDP_CLK100M_N	MAKE_BASE=TRUE NO_TEST=TRUE	11 102

PCH PCI

13	TP_LPC_DREQ0_L	NC_LPC_DREQ0_L	MAKE_BASE=TRUE NO_TEST=TRUE

PCH Miscellaneous

11	TP_HDA_SDIN1	NC_HDA_SDIN1	MAKE_BASE=TRUE NO_TEST=TRUE
11	TP_HDA_SDIN2	NC_HDA_SDIN2	MAKE_BASE=TRUE NO_TEST=TRUE
11	TP_HDA_SDIN3	NC_HDA_SDIN3	MAKE_BASE=TRUE NO_TEST=TRUE
11	TP_PCI_CLK33M_OUT2	NC_PCI_CLK33M_OUT2	MAKE_BASE=TRUE NO_TEST=TRUE
11	TP_PCI_CLK33M_OUT3	NC_PCI_CLK33M_OUT3	MAKE_BASE=TRUE NO_TEST=TRUE
11	PEG_CLKREQ_L	NC_PEG_CLKREQ_L	MAKE_BASE=TRUE NO_TEST=TRUE

AUDIO MISC

Pages	Signal	Alias	Properties
109 42	AUDIO_MUTE_L	NC_AUDIO_MUTE_L	MAKE_BASE=TRUE NO_TEST=TRUE
109 42	AUDIO_SCLK	NC_AUDIO_SCLK	MAKE_BASE=TRUE NO_TEST=TRUE
109 42	AUDIO_WS	NC_AUDIO_WS	MAKE_BASE=TRUE NO_TEST=TRUE

SYNC_MASTER=J78_MLB SYNC_DATE=12/05/2013

PAGE TITLE

Unused Signal Aliases

Apple Inc.

DRAWING NUMBER 051-1635 SIZE D

REVISION 5.0.0

BRANCH proto2

SYNC_MASTER=J78_MLB SYNC_DATE=02/21/2014

PAGE TITLE

Functional / ICT Test

Apple Inc.

DRAWING NUMBER	SIZE
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J17 BOARD SPECIFIC PHYSICAL AND SPACING CONSTRAINTS

BOARD LAYERS	BOARD AREAS	BOARD UNITS (MIL or MM)	ALLEGRO VERSION
TOP,ISL2,ISL3,ISL4,ISL5,ISL6,ISL7,ISL8,ISL9,ISL10,ISL11,BOTTOM	NO_TYPE,BGA,BGA_TBT,BGA_VRAM	MM	16.2

General Physical Rule Definitions

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
DEFAULT	*	Y	0.1 MM	0.070 MM	10 MM	0 MM	0 MM
STANDARD	*	Y	=DEFAULT	=DEFAULT	10 MM	=DEFAULT	=DEFAULT

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
34_OHM_SE	*	Y	0.185 MM	0.085 MM	=STANDARD	=STANDARD	=STANDARD
34_OHM_SE	ISL5,ISL8	Y	0.205 MM	0.085 MM	=STANDARD	=STANDARD	=STANDARD
34_OHM_SE	TOP,BOTTOM	Y	0.220 MM	0.085 MM	=STANDARD	=STANDARD	=STANDARD

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
39_OHM_SE	*	Y	0.150 MM	0.085 MM	=STANDARD	=STANDARD	=STANDARD
39_OHM_SE	ISL5,ISL8	Y	0.165 MM	0.085 MM	=STANDARD	=STANDARD	=STANDARD
39_OHM_SE	TOP,BOTTOM	Y	0.175 MM	0.085 MM	=STANDARD	=STANDARD	=STANDARD

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
42_OHM_SE	*	Y	0.130 MM	0.085 MM	=STANDARD	=STANDARD	=STANDARD
42_OHM_SE	ISL5,ISL8	Y	0.145 MM	0.085 MM	=STANDARD	=STANDARD	=STANDARD
42_OHM_SE	TOP,BOTTOM	Y	0.155 MM	0.085 MM	=STANDARD	=STANDARD	=STANDARD

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
45_OHM_SE	*	Y	0.115 MM	0.085 MM	=STANDARD	=STANDARD	=STANDARD
45_OHM_SE	ISL5,ISL8	Y	0.126 MM	0.085 MM	=STANDARD	=STANDARD	=STANDARD
45_OHM_SE	TOP,BOTTOM	Y	0.135 MM	0.085 MM	=STANDARD	=STANDARD	=STANDARD

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
50_OHM_SE	*	Y	0.090 MM	0.085 MM	=STANDARD	=STANDARD	=STANDARD
50_OHM_SE	ISL5,ISL8	Y	0.100 MM	0.085 MM	=STANDARD	=STANDARD	=STANDARD
50_OHM_SE	TOP,BOTTOM	Y	0.105 MM	0.085 MM	=STANDARD	=STANDARD	=STANDARD

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
55_OHM_SE	*	Y	0.075 MM	0.075 MM	=STANDARD	=STANDARD	=STANDARD
55_OHM_SE	ISL5,ISL8	Y	0.080 MM	0.080 MM	=STANDARD	=STANDARD	=STANDARD
55_OHM_SE	TOP,BOTTOM	Y	0.085 MM	0.085 MM	=STANDARD	=STANDARD	=STANDARD

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
68_OHM_DIFF	*	Y	0.171 MM	0.085 MM	=STANDARD	0.130 MM	0.1 MM
68_OHM_DIFF	TOP,BOTTOM	Y	0.185 MM	0.085 MM	=STANDARD	0.150 MM	0.1 MM

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
80_OHM_DIFF	*	Y	0.136 MM	0.085 MM	=STANDARD	0.190 MM	0.1 MM
80_OHM_DIFF	TOP,BOTTOM	Y	0.141 MM	0.085 MM	=STANDARD	0.185 MM	0.1 MM

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
85_OHM_DIFF	*	Y	0.121 MM	0.085 MM	=STANDARD	0.190 MM	0.1 MM
85_OHM_DIFF	TOP,BOTTOM	Y	0.125 MM	0.085 MM	=STANDARD	0.190 MM	0.1 MM

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
90_OHM_DIFF	*	Y	0.109 MM	0.085 MM	=STANDARD	0.200 MM	0.1 MM
90_OHM_DIFF	TOP,BOTTOM	Y	0.111 MM	0.085 MM	=STANDARD	0.200 MM	0.1 MM

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
100_OHM_DIFF	*	Y	0.086 MM	0.085 MM	=STANDARD	0.200 MM	0.1 MM
100_OHM_DIFF	TOP,BOTTOM	Y	0.090 MM	0.085 MM	=STANDARD	0.230 MM	0.1 MM

BGA

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
BGA_P1MM	*	=STANDARD	?

Power and Common

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
GND_ISO	*	=STANDARD	8000
GND_P2MM	*	=2:1_SPACING	1000
PWR_P2MM	*	=2:1_SPACING	1100

GENERIC

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
GENERIC_ISO	*	=1:1_SPACING	?

BGA Area Constraints

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
*	*	BGA	BGA_P1MM

General Spacing Definitions

Default

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
DEFAULT	*	0.1 MM	?
STANDARD	*	=DEFAULT	?

Fixed and Dielectric

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
1:1_SPACING	*	0.1 MM	?
1X_DIELECTRIC	TOP,BOTTOM	0.071 MM	?
1X_DIELECTRIC	ISL3,ISL10	0.101 MM	?
1X_DIELECTRIC	*	0.076 MM	?

Board Stack-up

FINISHED BOARD THICKNESS: 1.94 MM

Top	Signal	0.5 oz (Cu plated)
	Prepreg	0.071 MM
2	Plane	1 oz
	Core	0.101 MM
3	Signal	0.5 oz
	Prepreg	0.115 MM
4	Plane	1 oz
	Core	0.076 MM
5	Signal	0.5 oz
	Prepreg	0.380 MM
6	Plane	1 oz
	Core	0.076 MM
7	Plane	1 oz
	Prepreg	0.380 MM
8	Signal	0.5 oz
	Core	0.076 MM
9	Plane	1 oz
	Prepreg	0.115 MM
10	Signal	0.5 oz
	Core	0.101 MM
11	Plane	1 oz
	Prepreg	0.071 MM
Btm	Signal	0.5 oz (Cu plated)

SYNC_MASTER=J78_MLB SYNC_DATE=02/21/2014

PAGE TITLE: J17 RULE DEFINITIONS

Apple Inc.

DRAWING NUMBER: 051-1635 SIZE: D

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8	7	6	5	4	3	2	1

DDR3

DDR3-specific Physical Rules

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
DDR_34S	*	=34_OHM_SE	=34_OHM_SE	=34_OHM_SE	=34_OHM_SE	=STANDARD	=STANDARD
DDR_39S	*	=39_OHM_SE	=39_OHM_SE	=39_OHM_SE	=39_OHM_SE	=STANDARD	=STANDARD
DDR_42S	*	=42_OHM_SE	=42_OHM_SE	=42_OHM_SE	=42_OHM_SE	=STANDARD	=STANDARD
DDR_42S_D	*	=42_OHM_SE	=42_OHM_SE	=42_OHM_SE	=42_OHM_SE	0.1016 MM	0.1016 MM
DDR_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=STANDARD	=STANDARD
DDR_68D	*	=68_OHM_DIFF	=68_OHM_DIFF	=68_OHM_DIFF	=68_OHM_DIFF	=68_OHM_DIFF	=68_OHM_DIFF
DDR_COMP	*	Y	0.305 MM	0.105 MM	=STANDARD	=STANDARD	=STANDARD

Minimum diff spacing is 4 mil
Table 4-5, Intel Doc# 486712

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
POWER_DDR_P4MM	*	Y	0.400 MM	0.100 MM	3.0 MM	=STANDARD	=STANDARD

Physical Net Type to Rule Map

NET_PHYSICAL_TYPE	AREA_TYPE	PHYSICAL_RULE_SET
POWER_DDR	*	POWER_DDR_P4MM
DDR_CLK_PHY	*	DDR_68D
DDR_CTRL_PHY	*	DDR_39S
DDR_CMD_PHY	*	DDR_34S
DDR_DQ_PHY	*	DDR_42S
DDR_DQS_PHY	*	DDR_42S_D
DDR_COMP_PHY	*	DDR_COMP

DDR3 Power-specific Spacing Definitions

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
POWER_DDR	*	=2:1_SPACING	?

DDR3-specific Spacing Definitions

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
DDR_CLK_ISO	*	=5:1_SPACING	?
DDR_CTRL_ISO	*	=3.5:1_SPACING	?
DDR_CTRL2CTRL	*	=2.5:1_SPACING	?
DDR_CMD_ISO	*	=3.5:1_SPACING	?
DDR_CMD2CMD	*	=2:1_SPACING	?
DDR_DATA_ISO	*	=3:1_SPACING	?
DDR_DQ2DQ	*	=2:1_SPACING	900
DDR_DQ2DQS	*	=3:1_SPACING	?
DDR_BL2BL	*	=3:1_SPACING	?
DDR_CH2CH	*	=6.5:1_SPACING	?
DDR_COMP_ISO	*	0.381 MM	?

Main Segment Min Spacing Rules (mils) (Shark Bay PDG, Intel Doc# 486712)

Table	Trace	Design (diff)	Iso	Design	Comments
4-2	4		15	19.69	CLK trace spacing controlled by =68_OHM_DIFF
4-3	8	9.84	12	13.78	
4-4	6	7.87	12	13.78	
4-5	8.5	7.87	12	11.81	DQ or DQS to other signals not in the same bytelane (but not ch)
					DQ to DQ in the same bytelane of the same channel
			10	11.81	DQ to DQS in the same bytelane of the same channel
			12	11.81	DQ or DQS in different bytelanes of the same channel
			25	25.59	DQ or DQS in different channels
			-	25.59	DDR3 to any other signal not DDR3

Constraints

Clocks: CK[3:0], CK#[3:0]

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
DDR_CLK	*	*	DDR_CLK_ISO

Control: CS#[3:0], CKE[3:0], ODT[3:0]

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
DDR_CTRL	*	*	DDR_CTRL_ISO
DDR_CTRL	DDR_CTRL	*	DDR_CTRL2CTRL

Command: MA[15:0], RAS#, CAS#, WE# BS[2:0]

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
DDR_CMD	*	*	DDR_CMD_ISO
DDR_CMD	DDR_CMD	*	DDR_CMD2CMD

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
DDR_COMP	*	*	DDR_COMP_ISO

Data: DQS[7:0], DQS#[7:0], DQ[63:0]

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET	
DDR_A_DQ_BYTE*	*	*	DDR_DATA_ISO	
DDR_A_DQS*	*	*	DDR_DATA_ISO	
DDR_B_DQ_BYTE*	*	*	DDR_DATA_ISO	
DDR_B_DQS*	*	*	DDR_DATA_ISO	
DDR_*_DQ_BYTE*	=SAME	*	DDR_DQ2DQ	See Note (3)
DDR_A_DQ_BYTE*	DDR_A_DQS*	*	DDR_DQ2DQS	See Note (1)
DDR_A_DQ_BYTE*	DDR_A_DQ_BYTE*	*	DDR_BL2BL	See Note (3)
DDR_B_DQ_BYTE*	DDR_B_DQS*	*	DDR_DQ2DQS	See Note (1)
DDR_B_DQ_BYTE*	DDR_B_DQ_BYTE*	*	DDR_BL2BL	
DDR_A_*	DDR_B_*	*	DDR_CH2CH	See Note (2)

Note (1):
Deliberately set DQ to DQS spacing to 3:1 to avoid adding complexity to contraints, even though it can be less. Only one rule per channel is needed by trading off a little space.

Note (2):
Intel suggests 25 mil (0.65 mm) spacing for via to channel, and via to pad to two different channels. DDR3 draws about 20 mA per trace with edge rates in the 100s of ps. The main coupling mechanism is capacitive. A 0.65 mm spacing is used for power nets, which draw far more current (inductive coupling however). These rules are far too conservative. To meet these rules, the spacing must be applied to the net.

Note (3):
In order for the constraints DDR_*_DQ_BYTE* to =SAME to win out over DDR_{A,B}_DQ_BYTE* to DDR_{A,B}_DQ_BYTE* so that the small intra-bytelane spacing is used, the spacing rule DDR_DQ2DQ must have a weight greater than DDR_BL2BL.

DDR3

	Electrical Contraint Set	Physical	Spacing	Net	
Channel A					
I178	DDR_A_CLK0	DDR_CLK_PHY	DDR_CLK	MEM_A_CLK_P<1..0>	7 23
I179	DDR_A_CLK0	DDR_CLK_PHY	DDR_CLK	MEM_A_CLK_N<1..0>	7 23
I255	DDR_A_CLK1	DDR_CLK_PHY	DDR_CLK	MEM_A_CLK_P<3..2>	7 24
I256	DDR_A_CLK1	DDR_CLK_PHY	DDR_CLK	MEM_A_CLK_N<3..2>	7 24
I180	DDR_A_CTRL0	DDR_CTRL_PHY	DDR_CTRL	MEM_A_CKE<1..0>	7 23
I181	DDR_A_CTRL0	DDR_CTRL_PHY	DDR_CTRL	MEM_A_CS_L<1..0>	7 23
I182	DDR_A_CTRL0	DDR_CTRL_PHY	DDR_CTRL	MEM_A_ODT<1..0>	7 23
I247	DDR_A_CTRL1	DDR_CTRL_PHY	DDR_CTRL	MEM_A_CKE<3..2>	7 24
I248	DDR_A_CTRL1	DDR_CTRL_PHY	DDR_CTRL	MEM_A_CS_L<3..2>	7 24
I249	DDR_A_CTRL1	DDR_CTRL_PHY	DDR_CTRL	MEM_A_ODT<3..2>	7 24
I183	DDR_A_CMD	DDR_CMD_PHY	DDR_CMD	MEM_A_A<15..0>	7 23 24
I184	DDR_A_CMD	DDR_CMD_PHY	DDR_CMD	MEM_A_BA<2..0>	7 23 24
I185	DDR_A_CMD	DDR_CMD_PHY	DDR_CMD	MEM_A_RAS_L	7 23 24
I186	DDR_A_CMD	DDR_CMD_PHY	DDR_CMD	MEM_A_CAS_L	7 23 24
I187	DDR_A_CMD	DDR_CMD_PHY	DDR_CMD	MEM_A_WE_L	7 23 24
I188	DDR_A_DQ_BYTE0	DDR_DQ_PHY	DDR_A_DQ_BYTE0	MEM_A_DQ<7..0>	7 27
I189	DDR_A_DQ_BYTE1	DDR_DQ_PHY	DDR_A_DQ_BYTE1	MEM_A_DQ<15..8>	7 27
I191	DDR_A_DQ_BYTE2	DDR_DQ_PHY	DDR_A_DQ_BYTE2	MEM_A_DQ<23..16>	7 27
I190	DDR_A_DQ_BYTE3	DDR_DQ_PHY	DDR_A_DQ_BYTE3	MEM_A_DQ<31..24>	7 27
I192	DDR_A_DQ_BYTE4	DDR_DQ_PHY	DDR_A_DQ_BYTE4	MEM_A_DQ<39..32>	7 27
I193	DDR_A_DQ_BYTE5	DDR_DQ_PHY	DDR_A_DQ_BYTE5	MEM_A_DQ<47..40>	7 27
I194	DDR_A_DQ_BYTE6	DDR_DQ_PHY	DDR_A_DQ_BYTE6	MEM_A_DQ<55..48>	7 27
I195	DDR_A_DQ_BYTE7	DDR_DQ_PHY	DDR_A_DQ_BYTE7	MEM_A_DQ<63..56>	7 27
I196	DDR_A_DQS0	DDR_DQS_PHY	DDR_A_DQS0	MEM_A_DQS_P<0>	7 27
I197	DDR_A_DQS0	DDR_DQS_PHY	DDR_A_DQS0	MEM_A_DQS_N<0>	7 27
I198	DDR_A_DQS1	DDR_DQS_PHY	DDR_A_DQS1	MEM_A_DQS_P<1>	7 27
I199	DDR_A_DQS1	DDR_DQS_PHY	DDR_A_DQS1	MEM_A_DQS_N<1>	7 27
I200	DDR_A_DQS2	DDR_DQS_PHY	DDR_A_DQS2	MEM_A_DQS_P<2>	7 27
I201	DDR_A_DQS2	DDR_DQS_PHY	DDR_A_DQS2	MEM_A_DQS_N<2>	7 27
I203	DDR_A_DQS3	DDR_DQS_PHY	DDR_A_DQS3	MEM_A_DQS_P<3>	7 27
I202	DDR_A_DQS3	DDR_DQS_PHY	DDR_A_DQS3	MEM_A_DQS_N<3>	7 27
I204	DDR_A_DQS4	DDR_DQS_PHY	DDR_A_DQS4	MEM_A_DQS_P<4>	7 27
I205	DDR_A_DQS4	DDR_DQS_PHY	DDR_A_DQS4	MEM_A_DQS_N<4>	7 27
I206	DDR_A_DQS5	DDR_DQS_PHY	DDR_A_DQS5	MEM_A_DQS_P<5>	7 27
I207	DDR_A_DQS5	DDR_DQS_PHY	DDR_A_DQS5	MEM_A_DQS_N<5>	7 27
I208	DDR_A_DQS6	DDR_DQS_PHY	DDR_A_DQS6	MEM_A_DQS_P<6>	7 27
I209	DDR_A_DQS6	DDR_DQS_PHY	DDR_A_DQS6	MEM_A_DQS_N<6>	7 27
I210	DDR_A_DQS7	DDR_DQS_PHY	DDR_A_DQS7	MEM_A_DQS_P<7>	7 27
I211	DDR_A_DQS7	DDR_DQS_PHY	DDR_A_DQS7	MEM_A_DQS_N<7>	7 27
Channel B					
I212	DDR_B_CLK0	DDR_CLK_PHY	DDR_CLK	MEM_B_CLK_P<1..0>	7 25
I213	DDR_B_CLK0	DDR_CLK_PHY	DDR_CLK	MEM_B_CLK_N<1..0>	7 25
I254	DDR_B_CLK1	DDR_CLK_PHY	DDR_CLK	MEM_B_CLK_P<3..2>	7 26
I253	DDR_B_CLK1	DDR_CLK_PHY	DDR_CLK	MEM_B_CLK_N<3..2>	7 26
I214	DDR_B_CTRL0	DDR_CTRL_PHY	DDR_CTRL	MEM_B_CKE<1..0>	7 25
I215	DDR_B_CTRL0	DDR_CTRL_PHY	DDR_CTRL	MEM_B_CS_L<1..0>	7 25
I216	DDR_B_CTRL0	DDR_CTRL_PHY	DDR_CTRL	MEM_B_ODT<1..0>	7 25
I250	DDR_B_CTRL1	DDR_CTRL_PHY	DDR_CTRL	MEM_B_CKE<3..2>	7 26
I251	DDR_B_CTRL1	DDR_CTRL_PHY	DDR_CTRL	MEM_B_CS_L<3..2>	7 26
I252	DDR_B_CTRL1	DDR_CTRL_PHY	DDR_CTRL	MEM_B_ODT<3..2>	7 26
I217	DDR_B_CMD	DDR_CMD_PHY	DDR_CMD	MEM_B_A<15..0>	7 25 26
I218	DDR_B_CMD	DDR_CMD_PHY	DDR_CMD	MEM_B_BA<2..0>	7 25 26
I219	DDR_B_CMD	DDR_CMD_PHY	DDR_CMD	MEM_B_RAS_L	7 25 26
I220	DDR_B_CMD	DDR_CMD_PHY	DDR_CMD	MEM_B_CAS_L	7 25 26
I221	DDR_B_CMD	DDR_CMD_PHY	DDR_CMD	MEM_B_WE_L	7 25 26
I222	DDR_B_DQ_BYTE0	DDR_DQ_PHY	DDR_B_DQ_BYTE0	MEM_B_DQ<7..0>	7 27
I223	DDR_B_DQ_BYTE1	DDR_DQ_PHY	DDR_B_DQ_BYTE1	MEM_B_DQ<15..8>	7 27
I225	DDR_B_DQ_BYTE2	DDR_DQ_PHY	DDR_B_DQ_BYTE2	MEM_B_DQ<23..16>	7 27
I224	DDR_B_DQ_BYTE3	DDR_DQ_PHY	DDR_B_DQ_BYTE3	MEM_B_DQ<31..24>	7 27
I226	DDR_B_DQ_BYTE4	DDR_DQ_PHY	DDR_B_DQ_BYTE4	MEM_B_DQ<39..32>	7 27
I227	DDR_B_DQ_BYTE5	DDR_DQ_PHY	DDR_B_DQ_BYTE5	MEM_B_DQ<47..40>	7 27
I228	DDR_B_DQ_BYTE6	DDR_DQ_PHY	DDR_B_DQ_BYTE6	MEM_B_DQ<55..48>	7 27
I229	DDR_B_DQ_BYTE7	DDR_DQ_PHY	DDR_B_DQ_BYTE7	MEM_B_DQ<63..56>	7 27
I230	DDR_B_DQS0	DDR_DQS_PHY	DDR_B_DQS0	MEM_B_DQS_P<0>	7 27
I231	DDR_B_DQS0	DDR_DQS_PHY	DDR_B_DQS0	MEM_B_DQS_N<0>	7 27
I232	DDR_B_DQS1	DDR_DQS_PHY	DDR_B_DQS1	MEM_B_DQS_P<1>	7 27
I233	DDR_B_DQS1	DDR_DQS_PHY	DDR_B_DQS1	MEM_B_DQS_N<1>	7 27
I234	DDR_B_DQS2	DDR_DQS_PHY	DDR_B_DQS2	MEM_B_DQS_P<2>	7 27
I235	DDR_B_DQS2	DDR_DQS_PHY	DDR_B_DQS2	MEM_B_DQS_N<2>	7 27
I237	DDR_B_DQS3	DDR_DQS_PHY	DDR_B_DQS3	MEM_B_DQS_P<3>	7 27
I236	DDR_B_DQS3	DDR_DQS_PHY	DDR_B_DQS3	MEM_B_DQS_N<3>	7 27
I238	DDR_B_DQS4	DDR_DQS_PHY	DDR_B_DQS4	MEM_B_DQS_P<4>	7 27
I239	DDR_B_DQS4	DDR_DQS_PHY	DDR_B_DQS4	MEM_B_DQS_N<4>	7 27
I240	DDR_B_DQS5	DDR_DQS_PHY	DDR_B_DQS5	MEM_B_DQS_P<5>	7 27
I241	DDR_B_DQS5	DDR_DQS_PHY	DDR_B_DQS5	MEM_B_DQS_N<5>	7 27
I242	DDR_B_DQS6	DDR_DQS_PHY	DDR_B_DQS6	MEM_B_DQS_P<6>	7 27
I243	DDR_B_DQS6	DDR_DQS_PHY	DDR_B_DQS6	MEM_B_DQS_N<6>	7 27
I244	DDR_B_DQS7	DDR_DQS_PHY	DDR_B_DQS7	MEM_B_DQS_P<7>	7 27
I245	DDR_B_DQS7	DDR_DQS_PHY	DDR_B_DQS7	MEM_B_DQS_N<7>	7 27
Reset					
I246		DDR_50S		MEM_RESET_L	21 23 24 25 26
SM COMP					
I257		DDR_COMP_PHY	DDR_COMP	CPU_SM_RCOMP<0..2>	6

SYNC_MASTER=J78_MLB SYNC_DATE=02/21/2014

PAGE TITLE
DDR3 Constraints

Apple Inc.

DRAWING NUMBER 051-1635 SIZE D

REVISION 5.0.0

BRANCH proto2

PCI Express/DMI

PCIe-specific Physical Rules

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
PCIE_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=STANDARD	=STANDARD
PCIE_80D	*	=80_OHM_DIFF	=80_OHM_DIFF	=80_OHM_DIFF	=80_OHM_DIFF	=80_OHM_DIFF	=80_OHM_DIFF
PCIE_R2D_80D	*	=80_OHM_DIFF	0.12 MM	=80_OHM_DIFF	=80_OHM_DIFF	0.12 MM	=80_OHM_DIFF
PCIE_85D	*	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF
PCIE_90D	*	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF
PCIE_COMP	*	Y	0.305 MM	0.105 MM	=STANDARD	=STANDARD	=STANDARD

Physical Net Type to Rule Map

NET_PHYSICAL_TYPE	AREA_TYPE	PHYSICAL_RULE_SET
PCIE3_PHY	*	PCIE_80D
PCIE3_R2D_PHY	*	PCIE_R2D_80D
CLK_PCIE_PHY	*	PCIE_90D
COMP_PCIE_PHY	*	PCIE_COMP

PCIe and DMI Compensation Rules (mils)

Table	Imp	Design	Iso	Design	Comments
4-5	50	50	15	15.75	PCIe. Impedance inferred from Table 4-7.
4-7	50	50	8	15.75	DMI. Numbers based on Intel stack-up.

PCIe-specific Spacing Definitions

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
CLK_PCIE_ISO	*	=5:1_SPACING	?
COMP_PCIE_ISO	*	=4:1_SPACING	?

Spacing Constraints

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
CLK_PCIE	*	*	CLK_PCIE_ISO
COMP_PCIE	*	*	COMP_PCIE_ISO
PEG_R2D	PEG_R2D	*	PEG_SAME_DIR
PEG_D2R	PEG_D2R	*	PEG_SAME_DIR
PEG_D2R	PEG_R2D	*	PEG_ALT_DIR
PEG_D2R	POWER	*	PEG2PWR_ISO
PEG_R2D	POWER	*	PEG2PWR_ISO
PEG_D2R	VR_SWITCH	*	PEG2VR_ISO
PEG_R2D	VR_SWITCH	*	PEG2VR_ISO
PEG_D2R	*	*	PEG_ISO
PEG_R2D	*	*	PEG_ISO

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
PEG_SAME_DIR	TOP,BOTTOM	=5X_DIELECTRIC	?
PEG_SAME_DIR	*	=3.5X_DIELECTRIC	?
PEG_ALT_DIR	*	=7X_DIELECTRIC	?
PEG2PWR_ISO	*	1 MM	?
PEG2VR_ISO	*	1 MM	?
PEG_ISO	*	=4:1_SPACING	?
PEG_GDDR_ISO	*	=5.6X_DIELECTRIC	?

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
PEG*	GDDR*	*	PEG_GDDR_ISO

PEG Min Spacing Rules (mils) (Maho Bay PDG, Intel Doc# 473718)

Section	Imp	Design	Iso	Design	Comments
4.2.1	80	80	16	15.75	PCIe Gen3. Allow looser spacing for same direction on stripline per Anil

PCIe (CPU)

x16 Graphics

	Electrical Contraint Set	Physical	Spacing		
I440	PCIE_GEN3_R2D	PCIE3_R2D_PHY	PEG_R2D	PEG_R2D_P<15>	74
I441	PCIE_GEN3_R2D	PCIE3_R2D_PHY	PEG_R2D	PEG_R2D_N<15>	74
I442		PCIE3_R2D_PHY	PEG_R2D	PEG_R2D_C_P<15>	74 95
I443		PCIE3_R2D_PHY	PEG_R2D	PEG_R2D_C_N<15>	74 95
I444	PCIE_GEN3_D2R_RVSD	PCIE3_PHY	PEG_D2R	PEG_D2R_P<15>	74 95
I445	PCIE_GEN3_D2R_RVSD	PCIE3_PHY	PEG_D2R	PEG_D2R_N<15>	74 95
I446		PCIE3_PHY	PEG_D2R	PEG_D2R_C_P<15>	74
I447		PCIE3_PHY	PEG_D2R	PEG_D2R_C_N<15>	74
I449	PCIE_GEN3_R2D_RVSD	PCIE3_R2D_PHY	PEG_R2D	PEG_R2D_P<14>	74
I448	PCIE_GEN3_R2D_RVSD	PCIE3_R2D_PHY	PEG_R2D	PEG_R2D_N<14>	74
I450		PCIE3_R2D_PHY	PEG_R2D	PEG_R2D_C_P<14>	74 95
I452		PCIE3_R2D_PHY	PEG_R2D	PEG_R2D_C_N<14>	74 95
I451	PCIE_GEN3_D2R	PCIE3_PHY	PEG_D2R	PEG_D2R_P<14>	74 95
I453	PCIE_GEN3_D2R	PCIE3_PHY	PEG_D2R	PEG_D2R_N<14>	74 95
I455		PCIE3_PHY	PEG_D2R	PEG_D2R_C_P<14>	74
I454		PCIE3_PHY	PEG_D2R	PEG_D2R_C_N<14>	74
I457	PCIE_GEN3_R2D_RVSD	PCIE3_R2D_PHY	PEG_R2D	PEG_R2D_P<13>	74
I456	PCIE_GEN3_R2D_RVSD	PCIE3_R2D_PHY	PEG_R2D	PEG_R2D_N<13>	74
I458		PCIE3_R2D_PHY	PEG_R2D	PEG_R2D_C_P<13>	74 95
I460		PCIE3_R2D_PHY	PEG_R2D	PEG_R2D_C_N<13>	74 95
I459	PCIE_GEN3_D2R_RVSD	PCIE3_PHY	PEG_D2R	PEG_D2R_P<13>	74 95
I461	PCIE_GEN3_D2R_RVSD	PCIE3_PHY	PEG_D2R	PEG_D2R_N<13>	74 95
I463		PCIE3_PHY	PEG_D2R	PEG_D2R_C_P<13>	74
I462		PCIE3_PHY	PEG_D2R	PEG_D2R_C_N<13>	74
I464	PCIE_GEN3_R2D_RVSD	PCIE3_R2D_PHY	PEG_R2D	PEG_R2D_P<12>	74
I465	PCIE_GEN3_R2D_RVSD	PCIE3_R2D_PHY	PEG_R2D	PEG_R2D_N<12>	74
I467		PCIE3_R2D_PHY	PEG_R2D	PEG_R2D_C_P<12>	74 95
I466		PCIE3_R2D_PHY	PEG_R2D	PEG_R2D_C_N<12>	74 95
I468	PCIE_GEN3_D2R	PCIE3_PHY	PEG_D2R	PEG_D2R_P<12>	74 95
I470	PCIE_GEN3_D2R	PCIE3_PHY	PEG_D2R	PEG_D2R_N<12>	74 95
I469		PCIE3_PHY	PEG_D2R	PEG_D2R_C_P<12>	74
I471		PCIE3_PHY	PEG_D2R	PEG_D2R_C_N<12>	74
I472	PCIE_GEN3_R2D_RVSD	PCIE3_R2D_PHY	PEG_R2D	PEG_R2D_P<11>	74
I474	PCIE_GEN3_R2D_RVSD	PCIE3_R2D_PHY	PEG_R2D	PEG_R2D_N<11>	74
I473		PCIE3_R2D_PHY	PEG_R2D	PEG_R2D_C_P<11>	74 95
I475		PCIE3_R2D_PHY	PEG_R2D	PEG_R2D_C_N<11>	74 95
I477	PCIE_GEN3_D2R	PCIE3_PHY	PEG_D2R	PEG_D2R_P<11>	74 95
I476	PCIE_GEN3_D2R	PCIE3_PHY	PEG_D2R	PEG_D2R_N<11>	74 95
I478		PCIE3_PHY	PEG_D2R	PEG_D2R_C_P<11>	74
I479		PCIE3_PHY	PEG_D2R	PEG_D2R_C_N<11>	74
I480	PCIE_GEN3_R2D_RVSD	PCIE3_R2D_PHY	PEG_R2D	PEG_R2D_P<10>	74
I482	PCIE_GEN3_R2D_RVSD	PCIE3_R2D_PHY	PEG_R2D	PEG_R2D_N<10>	74
I481		PCIE3_R2D_PHY	PEG_R2D	PEG_R2D_C_P<10>	74 95
I484		PCIE3_R2D_PHY	PEG_R2D	PEG_R2D_C_N<10>	74 95
I483	PCIE_GEN3_D2R_RVSD	PCIE3_PHY	PEG_D2R	PEG_D2R_P<10>	74 95
I485	PCIE_GEN3_D2R_RVSD	PCIE3_PHY	PEG_D2R	PEG_D2R_N<10>	74 95
I487		PCIE3_PHY	PEG_D2R	PEG_D2R_C_P<10>	74
I486		PCIE3_PHY	PEG_D2R	PEG_D2R_C_N<10>	74
I489	PCIE_GEN3_R2D_RVSD	PCIE3_R2D_PHY	PEG_R2D	PEG_R2D_P<9>	74
I488	PCIE_GEN3_R2D_RVSD	PCIE3_R2D_PHY	PEG_R2D	PEG_R2D_N<9>	74
I491		PCIE3_R2D_PHY	PEG_R2D	PEG_R2D_C_P<9>	74 95
I490		PCIE3_R2D_PHY	PEG_R2D	PEG_R2D_C_N<9>	74 95
I492	PCIE_GEN3_D2R_RVSD	PCIE3_PHY	PEG_D2R	PEG_D2R_P<9>	74 95
I494	PCIE_GEN3_D2R_RVSD	PCIE3_PHY	PEG_D2R	PEG_D2R_N<9>	74 95
I493		PCIE3_PHY	PEG_D2R	PEG_D2R_C_P<9>	74
I496		PCIE3_PHY	PEG_D2R	PEG_D2R_C_N<9>	74
I495	PCIE_GEN3_R2D_RVSD	PCIE3_R2D_PHY	PEG_R2D	PEG_R2D_P<8>	74
I497	PCIE_GEN3_R2D_RVSD	PCIE3_R2D_PHY	PEG_R2D	PEG_R2D_N<8>	74
I498		PCIE3_R2D_PHY	PEG_R2D	PEG_R2D_C_P<8>	74 95
I499		PCIE3_R2D_PHY	PEG_R2D	PEG_R2D_C_N<8>	74 95
I501	PCIE_GEN3_D2R_RVSD	PCIE3_PHY	PEG_D2R	PEG_D2R_P<8>	74 95
I500	PCIE_GEN3_D2R_RVSD	PCIE3_PHY	PEG_D2R	PEG_D2R_N<8>	74 95
I502		PCIE3_PHY	PEG_D2R	PEG_D2R_C_P<8>	74
I503		PCIE3_PHY	PEG_D2R	PEG_D2R_C_N<8>	74

PCIe (CPU)

x16 Graphics

	Electrical Contraint Set	Physical	Spacing		
I504	PCIE_GEN3_R2D	PCIE3_R2D_PHY	PEG_R2D	PEG_R2D_P<7>	74
I506	PCIE_GEN3_R2D	PCIE3_R2D_PHY	PEG_R2D	PEG_R2D_N<7>	74
I505		PCIE3_R2D_PHY	PEG_R2D	PEG_R2D_C_P<7>	74 95
I508		PCIE3_R2D_PHY	PEG_R2D	PEG_R2D_C_N<7>	74 95
I507	PCIE_GEN3_D2R_RVSD	PCIE3_PHY	PEG_D2R	PEG_D2R_P<7>	74 95
I509	PCIE_GEN3_D2R_RVSD	PCIE3_PHY	PEG_D2R	PEG_D2R_N<7>	74 95
I511		PCIE3_PHY	PEG_D2R	PEG_D2R_C_P<7>	74
I510		PCIE3_PHY	PEG_D2R	PEG_D2R_C_N<7>	74
I513	PCIE_GEN3_R2D	PCIE3_R2D_PHY	PEG_R2D	PEG_R2D_P<6>	74
I512	PCIE_GEN3_R2D	PCIE3_R2D_PHY	PEG_R2D	PEG_R2D_N<6>	74
I515		PCIE3_R2D_PHY	PEG_R2D	PEG_R2D_C_P<6>	74 95
I514		PCIE3_R2D_PHY	PEG_R2D	PEG_R2D_C_N<6>	74 95
I516	PCIE_GEN3_D2R	PCIE3_PHY	PEG_D2R	PEG_D2R_P<6>	74 95
I518	PCIE_GEN3_D2R	PCIE3_PHY	PEG_D2R	PEG_D2R_N<6>	74 95
I517		PCIE3_PHY	PEG_D2R	PEG_D2R_C_P<6>	74
I519		PCIE3_PHY	PEG_D2R	PEG_D2R_C_N<6>	74
I521	PCIE_GEN3_R2D_RVSD	PCIE3_R2D_PHY	PEG_R2D	PEG_R2D_P<5>	74
I520	PCIE_GEN3_R2D_RVSD	PCIE3_R2D_PHY	PEG_R2D	PEG_R2D_N<5>	74
I522		PCIE3_R2D_PHY	PEG_R2D	PEG_R2D_C_P<5>	74 95
I523		PCIE3_R2D_PHY	PEG_R2D	PEG_R2D_C_N<5>	74 95
I524	PCIE_GEN3_D2R_RVSD	PCIE3_PHY	PEG_D2R	PEG_D2R_P<5>	74 95
I525	PCIE_GEN3_D2R_RVSD	PCIE3_PHY	PEG_D2R	PEG_D2R_N<5>	74 95
I526		PCIE3_PHY	PEG_D2R	PEG_D2R_C_P<5>	74
I527		PCIE3_PHY	PEG_D2R	PEG_D2R_C_N<5>	74
I528	PCIE_GEN3_R2D	PCIE3_R2D_PHY	PEG_R2D	PEG_R2D_P<4>	74
I529	PCIE_GEN3_R2D	PCIE3_R2D_PHY	PEG_R2D	PEG_R2D_N<4>	74
I530		PCIE3_R2D_PHY	PEG_R2D	PEG_R2D_C_P<4>	74 95
I531		PCIE3_R2D_PHY	PEG_R2D	PEG_R2D_C_N<4>	74 95
I532	PCIE_GEN3_D2R_RVSD	PCIE3_PHY	PEG_D2R	PEG_D2R_P<4>	74 95
I533	PCIE_GEN3_D2R_RVSD	PCIE3_PHY	PEG_D2R	PEG_D2R_N<4>	74 95
I534		PCIE3_PHY	PEG_D2R	PEG_D2R_C_P<4>	74
I535		PCIE3_PHY	PEG_D2R	PEG_D2R_C_N<4>	74
I536	PCIE_GEN3_R2D	PCIE3_R2D_PHY	PEG_R2D	PEG_R2D_P<3>	74
I538	PCIE_GEN3_R2D	PCIE3_R2D_PHY	PEG_R2D	PEG_R2D_N<3>	74
I537		PCIE3_R2D_PHY	PEG_R2D	PEG_R2D_C_P<3>	74 95
I539		PCIE3_R2D_PHY	PEG_R2D	PEG_R2D_C_N<3>	74 95
I541	PCIE_GEN3_D2R	PCIE3_PHY	PEG_D2R	PEG_D2R_P<3>	74 95
I540	PCIE_GEN3_D2R	PCIE3_PHY	PEG_D2R	PEG_D2R_N<3>	74 95
I543		PCIE3_PHY	PEG_D2R	PEG_D2R_C_P<3>	74
I542		PCIE3_PHY	PEG_D2R	PEG_D2R_C_N<3>	74
I544	PCIE_GEN3_R2D	PCIE3_R2D_PHY	PEG_R2D	PEG_R2D_P<2>	74
I546	PCIE_GEN3_R2D	PCIE3_R2D_PHY	PEG_R2D	PEG_R2D_N<2>	74
I545		PCIE3_R2D_PHY	PEG_R2D	PEG_R2D_C_P<2>	74 95
I548		PCIE3_R2D_PHY	PEG_R2D	PEG_R2D_C_N<2>	74 95
I547	PCIE_GEN3_D2R_RVSD	PCIE3_PHY	PEG_D2R	PEG_D2R_P<2>	74 95
I549	PCIE_GEN3_D2R_RVSD	PCIE3_PHY	PEG_D2R	PEG_D2R_N<2>	74 95
I551		PCIE3_PHY	PEG_D2R	PEG_D2R_C_P<2>	74
I550		PCIE3_PHY	PEG_D2R	PEG_D2R_C_N<2>	74
I552	PCIE_GEN3_R2D	PCIE3_R2D_PHY	PEG_R2D	PEG_R2D_P<1>	74
I554	PCIE_GEN3_R2D	PCIE3_R2D_PHY	PEG_R2D	PEG_R2D_N<1>	74
I553		PCIE3_R2D_PHY	PEG_R2D	PEG_R2D_C_P<1>	74 95
I555		PCIE3_R2D_PHY	PEG_R2D	PEG_R2D_C_N<1>	74 95
I557	PCIE_GEN3_D2R_RVSD	PCIE3_PHY	PEG_D2R	PEG_D2R_P<1>	74 95
I556	PCIE_GEN3_D2R_RVSD	PCIE3_PHY	PEG_D2R	PEG_D2R_N<1>	74 95
I559		PCIE3_PHY	PEG_D2R	PEG_D2R_C_P<1>	74
I558		PCIE3_PHY	PEG_D2R	PEG_D2R_C_N<1>	74
I560	PCIE_GEN3_R2D_RVSD	PCIE3_R2D_PHY	PEG_R2D	PEG_R2D_P<0>	74
I562	PCIE_GEN3_R2D_RVSD	PCIE3_R2D_PHY	PEG_R2D	PEG_R2D_N<0>	74
I561		PCIE3_R2D_PHY	PEG_R2D	PEG_R2D_C_P<0>	74 95
I564		PCIE3_R2D_PHY	PEG_R2D	PEG_R2D_C_N<0>	74 95
I563	PCIE_GEN3_D2R_RVSD	PCIE3_PHY	PEG_D2R	PEG_D2R_P<0>	74 95
I565	PCIE_GEN3_D2R_RVSD	PCIE3_PHY	PEG_D2R	PEG_D2R_N<0>	74 95
I567		PCIE3_PHY	PEG_D2R	PEG_D2R_C_P<0>	74
I566		PCIE3_PHY	PEG_D2R	PEG_D2R_C_N<0>	74

CPU PCIe Clocks

I568	PEG_REF_CLK	CLK_PCIE_PHY	CLK_PCIE	PEG_CLK100M_P	11 74
I569	PEG_REF_CLK	CLK_PCIE_PHY	CLK_PCIE	PEG_CLK100M_N	11 74

CPU PCIe Compensation

I570		COMP_PCIE_PHY	COMP_PCIE	CPU_PEG_RCOMP	5

CPU eDP Compensation

I571		COMP_PCIE_PHY	COMP_PCIE	CPU_EDP_RCOMP	5

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CPU PCIe Constraints

Apple Inc.

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REVISION 5.0.0

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Physical Net Type to Rule Map

NET_PHYSICAL_TYPE	AREA_TYPE	PHYSICAL_RULE_SET
PCIE_PHY	*	PCIE_85D
COMP_DMI_PHY	*	DMI_COMP

PCIe-specific Spacing Definitions

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
PCIE_SAME_DIR	TOP,BOTTOM	=5X_DIELECTRIC	?
PCIE_SAME_DIR	*	=3.5X_DIELECTRIC	?
PCIE_ALT_DIR	*	=7X_DIELECTRIC	?
PCIE_ISO	*	=4:1_SPACING	?

TBT x4 PCIE Spacing Constraints

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
PCIE_TBT_R2D	PCIE_TBT_R2D	*	PCIE_SAME_DIR
PCIE_TBT_D2R	PCIE_TBT_D2R	*	PCIE_SAME_DIR
PCIE_TBT_D2R	PCIE_TBT_R2D	*	PCIE_ALT_DIR
PCIE_TBT_D2R	*	*	PCIE_ISO
PCIE_TBT_R2D	*	*	PCIE_ISO

PCH x1 PCIE Constraints

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
PCIE	*	*	PCIE_ISO

DMI-specific Spacing Definitions

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
DMI_SAME_DIR	TOP,BOTTOM	=5X_DIELECTRIC	?
DMI_SAME_DIR	*	=4X_DIELECTRIC	?
DMI_ALT_DIR	*	=5X_DIELECTRIC	?
DMI_ISO	*	=4X_DIELECTRIC	?
DMI2PWR_ISO	*	0.5 MM	?
DMI2VR_ISO	*	1 MM	?

DMI x4 PCIE Spacing Constraints

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
DMI_N2S	DMI_N2S	*	DMI_SAME_DIR
DMI_S2N	DMI_S2N	*	DMI_SAME_DIR
DMI_N2S	DMI_S2N	*	DMI_ALT_DIR
DMI_N2S	POWER	*	DMI2PWR_ISO
DMI_S2N	POWER	*	DMI2PWR_ISO
DMI_N2S	VR_SWITCH	*	DMI2VR_ISO
DMI_S2N	VR_SWITCH	*	DMI2VR_ISO
DMI_N2S	*	*	DMI_ISO
DMI_S2N	*	*	DMI_ISO

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
DMI_COMP	*	Y	0.2032 MM	0.2032 MM	3 MM	=STANDARD	=STANDARD

PCIe (PCH)

	Electrical Contraint Set	Physical	Spacing		
	x4 Thunderbolt				
I132	PCIE_GEN2_R2D	PCIE_PHY	PCIE_TBT_R2D	PCIE_TBT_R2D_P<3..0>	28
I133	PCIE_GEN2_R2D	PCIE_PHY	PCIE_TBT_R2D	PCIE_TBT_R2D_N<3..0>	28
I134		PCIE_PHY	PCIE_TBT_R2D	PCIE_TBT_R2D_C_P<3..0>	13 28
I135		PCIE_PHY	PCIE_TBT_R2D	PCIE_TBT_R2D_C_N<3..0>	13 28
I136	PCIE_GEN2_D2R	PCIE_PHY	PCIE_TBT_D2R	PCIE_TBT_D2R_P<3..0>	13 28
I137	PCIE_GEN2_D2R	PCIE_PHY	PCIE_TBT_D2R	PCIE_TBT_D2R_N<3..0>	13 28
I138		PCIE_PHY	PCIE_TBT_D2R	PCIE_TBT_D2R_C_P<3..0>	28
I139		PCIE_PHY	PCIE_TBT_D2R	PCIE_TBT_D2R_C_N<3..0>	28
I140	PCIE_REF_CLK	CLK_PCIE_PHY	CLK_PCIE	PCIE_CLK100M_TBT_P	11 28
I141	PCIE_REF_CLK	CLK_PCIE_PHY	CLK_PCIE	PCIE_CLK100M_TBT_N	11 28
	x1 AirPort				
I142	PCIE_GEN2_R2D_CONN_AP	PCIE_PHY	PCIE	PCIE_AP_R2D_P	34
I143	PCIE_GEN2_R2D_CONN_AP	PCIE_PHY	PCIE	PCIE_AP_R2D_N	34
I144		PCIE_PHY	PCIE	PCIE_AP_R2D_C_P	13 34
I146		PCIE_PHY	PCIE	PCIE_AP_R2D_C_N	13 34
I145	PCIE_GEN2_D2R_CONN_AP	PCIE_PHY	PCIE	PCIE_AP_D2R_P	13 34
I147	PCIE_GEN2_D2R_CONN_AP	PCIE_PHY	PCIE	PCIE_AP_D2R_N	13 34
I148	PCIE_REF_CLK_CONN	CLK_PCIE_PHY	CLK_PCIE	PCIE_CLK100M_AP_P	11 34
I149	PCIE_REF_CLK_CONN	CLK_PCIE_PHY	CLK_PCIE	PCIE_CLK100M_AP_N	11 34
	x1 Caesar IV				
I151	PCIE_GEN2_R2D	PCIE_PHY	PCIE	PCIE_ENET_R2D_P	37
I150	PCIE_GEN2_R2D	PCIE_PHY	PCIE	PCIE_ENET_R2D_N	37
I152		PCIE_PHY	PCIE	PCIE_ENET_R2D_C_P	13 37
I154		PCIE_PHY	PCIE	PCIE_ENET_R2D_C_N	13 37
I153	PCIE_GEN2_D2R	PCIE_PHY	PCIE	PCIE_ENET_D2R_P	13 37
I155	PCIE_GEN2_D2R	PCIE_PHY	PCIE	PCIE_ENET_D2R_N	13 37
I156		PCIE_PHY	PCIE	PCIE_ENET_D2R_C_P	37
I157		PCIE_PHY	PCIE	PCIE_ENET_D2R_C_N	37
I158	PCIE_REF_CLK	CLK_PCIE_PHY	CLK_PCIE	PCIE_CLK100M_ENET_P	11 37
I159	PCIE_REF_CLK	CLK_PCIE_PHY	CLK_PCIE	PCIE_CLK100M_ENET_N	11 37
	x2 SSD				
I172	PCIE_REF_CLK_CONN	CLK_PCIE_PHY	CLK_PCIE	PCIE_CLK100M_SSD_P	11 35
I173	PCIE_REF_CLK_CONN	CLK_PCIE_PHY	CLK_PCIE	PCIE_CLK100M_SSD_N	11 35
	PCH PCIE Compensation				
I178		COMP_DMI_PHY	COMP_PCIE	PCH_PCIE_RCOMP	13

DMI

	Electrical Contraint Set	Physical	Spacing		
	DMI				
I166	DMI_N2S	PCIE_PHY	DMI_N2S	DMI_N2S_P<3..0>	5 12
I165	DMI_N2S	PCIE_PHY	DMI_N2S	DMI_N2S_N<3..0>	5 12
I167	DMI_S2N	PCIE_PHY	DMI_S2N	DMI_S2N_P<3..0>	5 12
I168	DMI_S2N	PCIE_PHY	DMI_S2N	DMI_S2N_N<3..0>	5 12
I169	PCIE_REF_CLK	CLK_PCIE_PHY	CLK_PCIE	DMI_CLK100M_CPU_P	6 11
I170	PCIE_REF_CLK	CLK_PCIE_PHY	CLK_PCIE	DMI_CLK100M_CPU_N	6 11
	DMI Compensation				
I171		COMP_DMI_PHY	COMP_PCIE	PCH_DMI_RCOMP	12

CPU DP REF CLK

	Electrical Contraint Set	Physical	Spacing		
	CPU DP REF CLK				
I175	CPU_CLK135_PLL	PCIE_PHY	CLK_PCIE	CPU_CLK135M_DPLLREF_N	6 11
I174	CPU_CLK135_PLL	PCIE_PHY	CLK_PCIE	CPU_CLK135M_DPLLREF_P	6 11
I176	CPU_CLK135_PLL	PCIE_PHY	CLK_PCIE	CPU_CLK135M_DPLLSS_N	6 11
I177	CPU_CLK135_PLL	PCIE_PHY	CLK_PCIE	CPU_CLK135M_DPLLSS_P	6 11

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PAGE TITLE

PCH PCIe/DMI Constaints

Apple Inc.

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SATA

SATA-specific Physical Rules

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
SATA_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=STANDARD	=STANDARD
SATA_85D	*	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF
SATA_90D	*	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF

Physical Net Type to Rule Map

NET_PHYSICAL_TYPE	AREA_TYPE	PHYSICAL_RULE_SET
SATA_PHY	*	SATA_85D
COMP_SATA_PHY	*	SATA_50S
SATA_PHY_90	*	SATA_90D

SATA-specific Spacing Definitions

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
SATA_ISO	*	=6:1_SPACING	?
SATA2PWR_ISO	*	0.5 MM	?
SATA2VR_ISO	*	1 MM	?
COMP_SATA_ISO	*	=4:1_SPACING	?

Constraints

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
SATA	*	*	SATA_ISO
SATA	POWER	*	SATA2PWR_ISO
SATA	VR_SWITCH	*	SATA2VR_ISO
COMP_SATA	*	*	COMP_SATA_ISO

SATA Min Spacing Rules (mils) (Maho Bay PDG, Intel Doc# 473718)

Section	Imp	Design	Iso	Design	Comments
15.2.1	90	95	20	23.62	SATA Gen2, SATA Gen3

SATA Compensation Rules (mils)

Table	Imp	Design	Iso	Design	Comments
15-3	50	50	15	15.75	SATA Gen2, SATA Gen3

FDI

FDI-specific Physical Rules

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
FDI_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=STANDARD	=STANDARD
COMP_FDI	*	Y	0.25 MM	0.25 MM	3 MM	=STANDARD	=STANDARD

Physical Net Type to Rule Map

NET_PHYSICAL_TYPE	AREA_TYPE	PHYSICAL_RULE_SET
FDI_SE_PHY	*	FDI_50S
COMP_FDI_PHY	*	COMP_FDI

FDI-specific Spacing Definitions

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
FDI_ISO	*	=3:1_SPACING	?
COMP_FDI_ISO	*	=4:1_SPACING	?

Constraints

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
FDI	*	*	FDI_ISO
COMP_FDI	*	*	COMP_FDI_ISO

FDI Min Spacing Rules (mils) (Maho Bay PDG, Intel Doc# 473718)

Table	Imp	Design	Iso	Design	Comments
6-1/6-2	85	85	12	11.81	FDI main length

FDI Compensation Rules (mils)

Table	Trace	Design	Iso	Design	Comments
6-4	10	11.81	-	15.75	Using PCIe guidelines

XDP

XDP-specific Physical Rules

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
XDP_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD

Physical Net Type to Rule Map

NET_PHYSICAL_TYPE	AREA_TYPE	PHYSICAL_RULE_SET
XDP_PHY	*	XDP_55S

XDP-specific Spacing Definitions

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
XDP_ISO	*	=2:1_SPACING	?
CLK_JTAG_ISO	*	=4:1_SPACING	?

Constraints

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
XDP	*	*	XDP_ISO
CLK_JTAG	*	*	CLK_JTAG_ISO

Desktop Debug Design Guide (Intel Doc# 430883)

Section	Imp	Design	Iso	Design	Comments
1.5	45-65	55	-	15.75	Isolation is for JTAG clocks. All signals default are 50 Ohm SE.

SATA

	Electrical Contraint Set	Physical	Spacing			
	PCH SATA Port 0 (HDD)					
I90	SATA_R2D	SATA_PHY_90	SATA	SATA_HDD_R2D_P	NO_TEST=TRUE	35
I92	SATA_R2D	SATA_PHY_90	SATA	SATA_HDD_R2D_N	NO_TEST=TRUE	35
I88		SATA_PHY_90	SATA	SATA_HDD_R2D_C_P		11 35
I89		SATA_PHY_90	SATA	SATA_HDD_R2D_C_N		11 35
I94	SATA_D2R	SATA_PHY_90	SATA	SATA_HDD_D2R_P		11 35
I95	SATA_D2R	SATA_PHY_90	SATA	SATA_HDD_D2R_N		11 35
I91		SATA_PHY_90	SATA	SATA_HDD_D2R_C_P	NO_TEST=TRUE	35
I93		SATA_PHY_90	SATA	SATA_HDD_D2R_C_N	NO_TEST=TRUE	35
	PCH SATA Port 1 (SSD)					
I96	SATA_SSD_R2D	SATA_PHY	SATA	SSD_R2D_P<0..1>		11 35
I98	SATA_SSD_R2D	SATA_PHY	SATA	SSD_R2D_N<0..1>		11 35
I115	SATA_SSD_R2D	SATA_PHY	SATA	SSD_R2D_C_P<0..1>		35
I116	SATA_SSD_R2D	SATA_PHY	SATA	SSD_R2D_C_N<0..1>		35
I103	SATA_SSD_D2R	SATA_PHY	SATA	SSD_D2R_P<0..1>		11 35
I102	SATA_SSD_D2R	SATA_PHY	SATA	SSD_D2R_N<0..1>		11 35
	PCH SATA Compensation					
I104		COMP_SATA_PHY	COMP_SATA	PCH_SATA_RCOMP		11

FDI

	Electrical Contraint Set	Physical	Spacing		
	FDI				
I112		FDI_SE_PHY	FDI	FDI_CSYNC	5 12
I113		FDI_SE_PHY	FDI	FDI_INT	5 12
	FDI Compensation				
I114		COMP_FDI_PHY	COMP_FDI	PCH_FDI_RCOMP	12

XDP

	Electrical Contraint Set	Physical	Spacing		
	CPU XDP				
I37		XDP_PHY	XDP	XDP_BPM_L<7..0>	6 18
I38		XDP_PHY	XDP	CPU_CFG<17..0>	6 18 96
I39	ITP_CLK_CONN	CLK_PCIE_PHY	CLK_PCIE	ITPCPU_CLK100M_P	
I40	ITP_CLK_CONN	CLK_PCIE_PHY	CLK_PCIE	ITPCPU_CLK100M_N	
I41	ITP_CLK_CONN	CLK_PCIE_PHY	CLK_PCIE	ITPXDP_CLK100M_P	11 96
I43	ITP_CLK_CONN	CLK_PCIE_PHY	CLK_PCIE	ITPXDP_CLK100M_N	11 96
I42		CLK_PCIE_PHY	CLK_PCIE	XDP_CPU_CLK100M_P	
I44		CLK_PCIE_PHY	CLK_PCIE	XDP_CPU_CLK100M_N	
I45		XDP_PHY	CLK_JTAG	XDP_CPU_TCK	6 18
I46		XDP_PHY	XDP	XDP_CPU_TMS	6 18
I47		XDP_PHY	XDP	XDP_CPU_TDI	6 18
I48		XDP_PHY	XDP	XDP_CPU_TDO	6 18
	PCH XDP				
I49		XDP_PHY	CLK_JTAG	XDP_PCH_TCK	11 18
I50		XDP_PHY	XDP	XDP_PCH_TMS	11 18
I51		XDP_PHY	XDP	XDP_PCH_TDI	11 18
I52		XDP_PHY	XDP	XDP_PCH_TDO	11 18

PCH

PCH-specific Physical Rules

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
PCH_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD
CLK_PCH_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD

PCH-specific Spacing Definitions

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
CLK_PCH	*	=4:1_SPACING	?
COMP_PCH	*	=2:1_SPACING	?

PCI

PCI-specific Physical Rules

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
CLK_PCI_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD

PCI-specific Spacing Definitions

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
CLK_PCI	*	=2:1_SPACING	?

LPC

LPC-specific Physical Rules

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
LPC_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD
CLK_LPC_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD

LPC-specific Spacing Definitions

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
LPC	*	=1.5:1_SPACING	?
CLK_LPC	*	=2:1_SPACING	?

HDA

HDA-specific Physical Rules

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
HDA_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD

HDA-specific Spacing Definitions

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
HDA	*	=2x_DIELECTRIC	?

Crystal

Crystal-specific Physical Rules

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
CLK_XTAL	*	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF

Crystal-specific Spacing Definitions

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
XTAL	*	=4X_DIELECTRIC	?

SPI

SPI-specific Physical Rules

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
SPI_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=STANDARD	=STANDARD
SPI_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD

SPI-specific Spacing Definitions

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
SPI	*	=2:1_SPACING	?

PCI

Electrical Contraint Set	Physical	Spacing	Net	Pages
PCI Clock				
I333	CLK_PCI_55S	CLK_PCI	PCH_CLK33M_PCIIN	11 19
I334	CLK_PCI_55S	CLK_PCI	PCH_CLK33M_PCIOUT	11 19

LPC

Electrical Contraint Set	Physical	Spacing	Net	Pages
LPC				
I336	LPC_55S	LPC	LPC_AD<3..0>	13 46 48
I335	LPC_55S	LPC	LPC_AD_R<3..0>	13
I338	LPC_55S	LPC	LPC_FRAME_L	13 46 48
I337	LPC_55S	LPC	LPC_FRAME_R_L	13
LPC Clocks				
I339	CLK_LPC_55S	CLK_LPC	LPC_CLK33M_LPCPLUS	19 48
I340	CLK_LPC_55S	CLK_LPC	LPC_CLK33M_LPCPLUS_R	11 19
I341	CLK_LPC_55S	CLK_LPC	LPC_CLK33M_SMC	19 46
I342	CLK_LPC_55S	CLK_LPC	LPC_CLK33M_SMC_R	11 19

PCH Clocks

Electrical Contraint Set	Physical	Spacing	Net	Pages
PCH Reference Clock				
I385	CLK_PCH_55S	CLK_PCH	SYSCLK_CLK25M_SB	11 19
I406	CLK_PCH_55S	CLK_PCH	SYSCLK_CLK25M_SB_R	11
PCH RTC 32K				
I347	CLK_XTAL	XTAL	PCH_CLK32K_RTCX1	11 19
I348	CLK_XTAL	XTAL	PCH_CLK32K_RTCX2	11 19
I352	CLK_XTAL	XTAL	PCH_CLK32K_RTCX2_R	19
SMC 32K				
I349	CLK_PCH_55S	CLK_PCH	PM_CLK32K_SUSCLK_R	12 47
I350	CLK_PCH_55S	CLK_PCH	SMC_CLK32K	46 47

25 MHz Reference Clocks

Electrical Contraint Set	Physical	Spacing	Net	Pages
25M Reference Crystal				
I379	CLK_XTAL	XTAL	SYSCLK_CLK25M_X1	19
I380	CLK_XTAL	XTAL	SYSCLK_CLK25M_X2	19
I381	CLK_XTAL	XTAL	SYSCLK_CLK25M_X2_R	19
25M Reference Clocks				
I384	CLK_PCH_55S	CLK_PCH	SYSCLK_CLK25M_ENET	19 37
I386	CLK_PCH_55S	CLK_PCH	SYSCLK_CLK25M_ENET_R	19
I388	CLK_PCH_55S	CLK_PCH	SYSCLK_CLK25M_TBT	19 28
I387	CLK_PCH_55S	CLK_PCH	SYSCLK_CLK25M_TBT_R	28

HDA

Electrical Contraint Set	Physical	Spacing	Net	Pages
HDA				
I353	HDA_55S	HDA	HDA_BIT_CLK	11 54
I361	HDA_55S	HDA	HDA_BIT_CLK_R	11
I363	HDA_55S	HDA	HDA_RST_L	11 54
I362	HDA_55S	HDA	HDA_RST_R_L	11
I364	HDA_55S	HDA	HDA_SDOUT	11 54
I367	HDA_55S	HDA	HDA_SDOUT_R	11 19
I366	HDA_55S	HDA	HDA_SYNC	11 54
I368	HDA_55S	HDA	HDA_SYNC_R	11
I365	HDA_55S	HDA	HDA_SDIN0	11 54
I369	HDA_55S	HDA	AUD_SDI_R	54
SPDIF				
I370		HDA	AUD_SPDIF_CHIP	54
I371		HDA	AUD_SPDIF_OUT	54 58

SPI Bootrom

Electrical Contraint Set	Physical	Spacing	Net	Pages
SPI ROM				
I391	SPI_50S	SPI	SPI_CLK_R	13 48
I392 SPI_CLK	SPI_50S	SPI	SPI_CLK	48
I393	SPI_50S	SPI	SPI_ALT_CLK	48
I408	SPI_50S	SPI	SPI_SMC_CLK	46 48
I407	SPI_50S	SPI	SPI_MLB_CLK	48
I394	SPI_50S	SPI	SPI_CS0_R_L	13 48
I395 SPI_CS_L	SPI_50S	SPI	SPI_CS0_L	48
I396	SPI_50S	SPI	SPI_ALT_CS_L	48
I409	SPI_50S	SPI	SPI_SMC_CS_L	46 48
I397	SPI_50S	SPI	SPI_MLB_CS_L	48
I399	SPI_50S	SPI	SPI_MOSI_R	13 48
I400 SPI_MOSI	SPI_50S	SPI	SPI_MOSI	48
I401	SPI_50S	SPI	SPI_ALT_MOSI	48
I410	SPI_50S	SPI	SPI_SMC_MOSI	46 48
I411	SPI_50S	SPI	SPI_MLB_MOSI	48
I404 SPI_MISO	SPI_50S	SPI	SPI_MISO	13 48
I403	SPI_50S	SPI	SPI_ALT_MISO	48
I412	SPI_50S	SPI	SPI_SMC_MISO	46 48
I413	SPI_50S	SPI	SPI_MLB_MISO	48
I405	SPI_50S	SPI	SPIROM_USE_MLB	14 48

SYNC_MASTER=J78_MLB SYNC_DATE=02/21/2014

PAGE TITLE

PCH and BR Constraints

Apple Inc.

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USB

USB-specific Physical Rules

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
USB_85D	*	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF
USB_90D	*	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF

Physical Net Type to Rule Map

NET_PHYSICAL_TYPE	AREA_TYPE	PHYSICAL_RULE_SET
USB2_PHY	*	USB_90D
USB3_PHY	*	USB_85D

USB-specific Spacing Definitions

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
USB2_ISO	*	=3:1_SPACING	?
USB2_ISO	TOP,BOTTOM	=3:1_SPACING	?
USB3_ISO	*	=5.5:1_SPACING	?
USB3_ISO	TOP,BOTTOM	=5.5:1_SPACING	?
USB32PWR_ISO	*	0.5 MM	?
USB32VR_ISO	*	1 MM	?

USB Min Spacing Rules (mils) (Maho Bay PDG, Intel Doc# 473718)

Section	Imp	Design	Iso	Design	Comments
12.2.1	90	90	12	11.81	USB 2.0
13.3.1	85	85	20	21.65	USB 3.0

Constraints

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
USB2	*	*	USB2_ISO
USB3	*	*	USB3_ISO
USB3	POWER	*	USB32PWR_ISO
USB3	VR_SWITCH	*	USB32VR_ISO

Caesar IV (Ethernet/SD)

CIV-specific Physical Rules

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
ENET_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=STANDARD	=STANDARD
ENET_100D	*	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF
SD_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=STANDARD	=STANDARD

Physical Net Type to Rule Map

NET_PHYSICAL_TYPE	AREA_TYPE	PHYSICAL_RULE_SET
ENET_COMP_PHY	*	ENET_50S
ENET_DIFF_PHY	*	ENET_100D
SD_PHY	*	SD_50S
CIV_SPI	*	SPI_55S

CIV-specific Spacing Definitions
Ethernet

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
ENET_DIFF_ISO	*	=6:1_SPACING	?
ENET_DIFF2DIFF	*	=3:1_SPACING	?
ENET_TRANS_ISO	*	1.27 MM	?
COMP_ENET_ISO	*	=4:1_SPACING	?

Constraints
Ethernet

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
ENET_DIFF	*	*	ENET_DIFF_ISO
ENET_DIFF	ENET_DIFF	*	ENET_DIFF2DIFF
ENET_TRANS	*	*	ENET_TRANS_ISO
COMP_ENET	*	*	COMP_ENET_ISO
ENET_TRANS	ENET_TRANS	*	ENET_DIFF2DIFF

2 kV isolation

SD

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
SD_ISO	*	=3:1_SPACING	?

SD

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
SD	*	*	SD_ISO

Camera Processor-to-Camera Sensor I/F (SMIA/MIPI)

Camera Processor's SMIA Interface Physical Rules

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
SMIA_100D	*	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF

NET_PHYSICAL_TYPE	AREA_TYPE	PHYSICAL_RULE_SET
SMIA_DIFF_PHY	*	SMIA_100D

Camera Processor's SMIA Interface Spacing Definitions

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
SMIA_DIFF_ISO	*	=6:1_SPACING	?
SMIA_DIFF2DIFF	*	=3:1_SPACING	?

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
SMIA_DIFF	*	*	SMIA_DIFF_ISO
SMIA_DIFF	SMIA_DIFF	*	SMIA_DIFF2DIFF

USB 3.0 and USB 2.0 Trixies Muxing

	Electrical Contraint Set	Physical	Spacing	Net	
External Port A (J4600)					
I320	USB3_RX_CONN	USB3_PHY	USB3	USB3_EXTA_RX_P	44
I321	USB3_RX_CONN	USB3_PHY	USB3	USB3_EXTA_RX_N	44
I391		USB3_PHY	USB3	USB3_EXTA_RX_F_P	13 44
I392		USB3_PHY	USB3	USB3_EXTA_RX_F_N	13 44
I413	USB3_TX_CONN	USB3_PHY	USB3	USB3_EXTA_TX_P	13 44
I414	USB3_TX_CONN	USB3_PHY	USB3	USB3_EXTA_TX_N	13 44
I415		USB3_PHY	USB3	USB3_EXTA_TX_F_P	44
I416		USB3_PHY	USB3	USB3_EXTA_TX_F_N	44
I493		USB3_PHY	USB3	USB3_EXTA_TX_C_P	44
I494		USB3_PHY	USB3	USB3_EXTA_TX_C_N	44
I417	USB2_MUXED_MOJO_CONN	USB2_PHY	USB2	USB_EXTA_0_P	13 44
I418	USB2_MUXED_MOJO_CONN	USB2_PHY	USB2	USB_EXTA_0_N	13 44
I419		USB2_PHY	USB2	USB2_EXTA_MUXED_P	44
I420		USB2_PHY	USB2	USB2_EXTA_MUXED_N	44
I393		USB2_PHY	USB2	USB2_EXTA_P	44
I394		USB2_PHY	USB2	USB2_EXTA_N	44
External Port B (J4610)					
I421	USB3_RX_CONN	USB3_PHY	USB3	USB3_EXTB_RX_P	44
I422	USB3_RX_CONN	USB3_PHY	USB3	USB3_EXTB_RX_N	44
I423		USB3_PHY	USB3	USB3_EXTB_RX_F_P	13 44
I424		USB3_PHY	USB3	USB3_EXTB_RX_F_N	13 44
I425	USB3_TX_CONN	USB3_PHY	USB3	USB3_EXTB_TX_P	13 44
I427	USB3_TX_CONN	USB3_PHY	USB3	USB3_EXTB_TX_N	13 44
I426		USB3_PHY	USB3	USB3_EXTB_TX_F_P	44
I428		USB3_PHY	USB3	USB3_EXTB_TX_F_N	44
I495		USB3_PHY	USB3	USB3_EXTB_TX_C_P	44
I496		USB3_PHY	USB3	USB3_EXTB_TX_C_N	44
I429	USB2_CONN	USB2_PHY	USB2	USB_EXTB_8_P	13 44
I430	USB2_CONN	USB2_PHY	USB2	USB_EXTB_8_N	13 44
I434		USB2_PHY	USB2	USB2_EXTB_P	44
I433		USB2_PHY	USB2	USB2_EXTB_N	44
External Port C (J4700)					
I437	USB3_RX_CONN	USB3_PHY	USB3	USB3_EXTC_RX_P	45
I439	USB3_RX_CONN	USB3_PHY	USB3	USB3_EXTC_RX_N	45
I438		USB3_PHY	USB3	USB3_EXTC_RX_F_P	13 45
I440		USB3_PHY	USB3	USB3_EXTC_RX_F_N	13 45
I441	USB3_TX_CONN	USB3_PHY	USB3	USB3_EXTC_TX_P	13 45
I442	USB3_TX_CONN	USB3_PHY	USB3	USB3_EXTC_TX_N	13 45
I444		USB3_PHY	USB3	USB3_EXTC_TX_F_P	45
I443		USB3_PHY	USB3	USB3_EXTC_TX_F_N	45
I498		USB3_PHY	USB3	USB3_EXTC_TX_C_P	45
I497		USB3_PHY	USB3	USB3_EXTC_TX_C_N	45
I446	USB2_CONN	USB2_PHY	USB2	USB_EXTC_1_P	13 45
I445	USB2_CONN	USB2_PHY	USB2	USB_EXTC_1_N	13 45
I450		USB2_PHY	USB2	USB2_EXTC_P	45
I452		USB2_PHY	USB2	USB2_EXTC_N	45
External Port D (J4710)					
I453	USB3_RX_CONN	USB3_PHY	USB3	USB3_EXTD_RX_P	45
I455	USB3_RX_CONN	USB3_PHY	USB3	USB3_EXTD_RX_N	45
I454		USB3_PHY	USB3	USB3_EXTD_RX_F_P	13 45
I456		USB3_PHY	USB3	USB3_EXTD_RX_F_N	13 45
I457	USB3_TX_CONN	USB3_PHY	USB3	USB3_EXTD_TX_P	13 45
I458	USB3_TX_CONN	USB3_PHY	USB3	USB3_EXTD_TX_N	13 45
I460		USB3_PHY	USB3	USB3_EXTD_TX_F_P	45
I459		USB3_PHY	USB3	USB3_EXTD_TX_F_N	45
I499		USB3_PHY	USB3	USB3_EXTD_TX_C_P	45
I500		USB3_PHY	USB3	USB3_EXTD_TX_C_N	45
I462	USB2_CONN	USB2_PHY	USB2	USB_EXTD_9_P	13 45
I461	USB2_CONN	USB2_PHY	USB2	USB_EXTD_9_N	13 45
I466		USB2_PHY	USB2	USB2_EXTD_P	45
I468		USB2_PHY	USB2	USB2_EXTD_N	45
Camera (J3510)					
I324	USB2_CONN_INT	USB2_PHY	USB2	USB_CAMERA_P	13 40
I326	USB2_CONN_INT	USB2_PHY	USB2	USB_CAMERA_N	13 40
PCH USB Compensation					
I384		PCH_55S	COMP_PCH	PCH_USB_RBIAS	13

RMH Love

	Electrical Contraint Set	Physical	Spacing	Net	
I409	USB2_MUXED_BT	USB2_PHY	USB2	USB_BT_P	13 34
I410	USB2_MUXED_BT	USB2_PHY	USB2	USB_BT_N	13 34
I505	USB2_MUXED_BT	USB2_PHY	USB2	USB_BT_MUX_P	34
I506	USB2_MUXED_BT	USB2_PHY	USB2	USB_BT_MUX_N	34

Et tu Brute?

	Electrical Contraint Set	Physical	Spacing	Net	
Ethernet					
I474	ENET_MDI	ENET_DIFF_PHY	ENET_DIFF	ENETCONN_MDI_P<3..0>	37 38
I475	ENET_MDI	ENET_DIFF_PHY	ENET_DIFF	ENETCONN_MDI_N<3..0>	37 38
I476		ENET_DIFF_PHY	ENET_TRANS	ENETCONN_MDI_T_P<3..0>	38
I477		ENET_DIFF_PHY	ENET_TRANS	ENETCONN_MDI_T_N<3..0>	38
I519			ENET_TRANS	ENETCONN_MCT0	38
I520			ENET_TRANS	ENETCONN_MCT1	38
I521			ENET_TRANS	ENETCONN_MCT2	38
I522			ENET_TRANS	ENETCONN_MCT3	38
I523			ENET_TRANS	ENETCONN_MCT_BS	38
I478		ENET_COMP_PHY	COMP_ENET	ENET_RDAC	37
SD					
I479	SD_DATA	SD_PHY	SD	ENET_CR_DATA<7..0>	37
I480		SD_PHY	SD	SDCONN_DATA<7..0>	37 39
I515		SD_PHY	SD	SDCONN_DATA_R_<7..0>	
I481	SD_CMD	SD_PHY	SD	ENET_SD_CMD	37
I482		SD_PHY	SD	SDCONN_CMD	37 39
I516		SD_PHY	SD	SDCONN_CMD_R	
I483	SD_CLK	SD_PHY	SD	ENET_SD_CLK	37
I484		SD_PHY	SD	SDCONN_CLK	37 39
I486		SD_PHY	SD	SDCONN_CLK_R	39
I485		SD_PHY	SD	ENET_MEDIA_SENSE	11 37
I488		SD_PHY	SD	ENET_SD_DETECT_L	37 39
CIV SPI					
I490		CIV_SPI	SPI	ENET_SCLK	37
I489		CIV_SPI	SPI	ENET_MISO	37
I492		CIV_SPI	SPI	ENET_MOSI	37
I491		CIV_SPI	SPI	ENET_CS_L	37

Camera Processor-Camera Sensor I/F

	Electrical Contraint Set	Physical	Spacing	Net	
I501	SMIA_DP	SMIA_DIFF_PHY	SMIA_DIFF	SMIA_DATA_P	40
I502	SMIA_DP	SMIA_DIFF_PHY	SMIA_DIFF	SMIA_DATA_N	40
I504	SMIA_DP	SMIA_DIFF_PHY	SMIA_DIFF	SMIA_CLK_P	40
I503	SMIA_DP	SMIA_DIFF_PHY	SMIA_DIFF	SMIA_CLK_N	40
I507		SPI_50S	SPI	CAM_SF_CLK	40
I508		SPI_50S	SPI	CAM_SF_CLK_R	40
I509		SPI_50S	SPI	CAM_SF_DIN	40
I510		SPI_50S	SPI	CAM_SF_DIN_R	40
I511		SPI_50S	SPI	CAM_SF_CS_L	40
I513		SPI_50S	SPI	CAM_SF_WP_L	40
I512		SPI_50S	SPI	CAM_SF_DOUT	40
I514		SPI_50S	SPI	CAM_SF_DOUT_R	40
I517		SMB_PHY	SMB	I2C_CAMSENSOR_SDA	40
I518		SMB_PHY	SMB	I2C_CAMSENSOR_SCL	40

SMBus

SMBus-specific Physical Rules

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
SMB_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD

Physical Net Type to Rule Map

NET_PHYSICAL_TYPE	AREA_TYPE	PHYSICAL_RULE_SET
SMB_PHY	*	SMB_55S

SMBus-specific Spacing Definitions

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
SMB_ISO	*	=2x_DIELECTRIC	?

Constraints

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
SMB	*	*	SMB_ISO

Sensor

Sensor-specific Physical Rules

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
1:1_DIFFPAIR	*	Y	=STANDARD	=STANDARD	=STANDARD	0.1 MM	0.085 MM
SNS_DIFF_J90	*	Y	0.200 MM	0.100 MM	200 MM	0.150 MM	0.100 MM

Physical Net Type to Rule Map

NET_PHYSICAL_TYPE	AREA_TYPE	PHYSICAL_RULE_SET
SNS_DIFF_PHY	*	1:1_DIFFPAIR

Sensor-specific Spacing Definitions

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
SENSE_ISO	*	=1.5:1_SPACING	?

Constraints

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
SENSE	*	*	SENSE_ISO
SENSE	POWER	*	PWR_P2MM
SENSE	GND	*	GND_P2MM

SMC Generic Control Line Spacing Definitions

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
SMC_ISO	*	=1:1_SPACING	?
SMC_3xISO	*	=3:1_SPACING	?

Constraints

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
SMC_CTRL	*	*	SMC_ISO
SMC_3XCTRL	*	*	SMC_3xISO

SMC Generic Control Line Physical Rules

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
SMC_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=STANDARD	=STANDARD

Physical Net Type to Rule Map

NET_PHYSICAL_TYPE	AREA_TYPE	PHYSICAL_RULE_SET
SMC_GEN	*	SMC_50S

Current/Voltage Sense

	Electrical Contraint Set	Physical	Spacing	Net	Pages
Common					
I3			SENSE	GND_SMC_AVSS	46 47 50 51
12V S5 (System Total)					
I4	SNS_CURRENT	SNS_DIFF_PHY	SENSE	SNS_P12VG3H_P	50
I5	SNS_CURRENT	SNS_DIFF_PHY	SENSE	SNS_P12VG3H_N	50
I6			SENSE	ISNS_P12VG3H_R	50
I7			SENSE	ISNS_P12VG3H	47 50
I8			SENSE	VSNS_P12VG3H	47 50
12V S0 (GPU Core)					
I22	SNS_CURRENT	SNS_DIFF_PHY	SENSE	SNS_P12VS0_GPUCORE_P	50
I23	SNS_CURRENT	SNS_DIFF_PHY	SENSE	SNS_P12VS0_GPUCORE_N	50
I24			SENSE	ISNS_P12VS0_GPUC_R	50
I25			SENSE	ISNS_P12VS0_GPUCORE	47 50
I26			SENSE	VSNS_P12VS0_GPUCORE	47 50
I140	SNS_CURRENT	SNS_DIFF_PHY	SENSE	SNS_P12VS0_GPUA_P	50
I141	SNS_CURRENT	SNS_DIFF_PHY	SENSE	SNS_P12VS0_GPUA_N	50
HDD					
I80	SNS_CURRENT	SNS_DIFF_PHY	SENSE	SNS_HDD_P	50
I82	SNS_CURRENT	SNS_DIFF_PHY	SENSE	SNS_HDD_N	50
I81			SENSE	ISNS_HDDS0_R	50
I83			SENSE	ISNS_HDDS0	47 50
I84			SENSE	VSNS_HDDS0	47 50
I125	SNS_CURRENT	SNS_DIFF_PHY	SENSE	SNS_P12VS0_HDD_P	50
I126	SNS_CURRENT	SNS_DIFF_PHY	SENSE	SNS_P12VS0_HDD_N	50
I127			SENSE	ISNS_12V_HDD_R	50
I128			SENSE	ISNS_P12VS0_HDD	47 50
SSD					
I85	SNS_CURRENT	SNS_DIFF_PHY	SENSE	SNS_SSD_P	51
I87	SNS_CURRENT	SNS_DIFF_PHY	SENSE	SNS_SSD_N	51
I86			SENSE	ISNS_SSDS0_R	51
I88			SENSE	ISNS_SSDS0	47 51
I89			SENSE	VSNS_P3V3S5	47 51
GPU Core (alternate low side sense)					
I118			SENSE	ISNS_GPUCORE_ALT	47 50
I129			SENSE	ISNS_GPUCORE_FB	50
I145			SENSE	ISNS_GPUCORE_FB_R	50
I122			SENSE	VSNS_GPUCORE_ALT	47 50
I146			SENSE	VREG_GPU_IMON_R	50
GPU VDDCI					
I137			SENSE	VSNS_GPU_VDDCI	47 51
GPU AUX					
I144			SENSE	ISNS_P12VS0_GPU_AUX	47 50
I147			SENSE	ISNS_P12VS0_GPUA_R	50
VDDQ S3 (DDR)					
I33	SNS_CURRENT	SNS_DIFF_PHY	SENSE	SNS_VDDQS3_DDR_P	51
I32	SNS_CURRENT	SNS_DIFF_PHY	SENSE	SNS_VDDQS3_DDR_N	51
I35			SENSE	ISNS_VDDQS3_DDR_R	51
I34			SENSE	ISNS_VDDQS3_DDR	47 51
I36			SENSE	VSNS_VDDQS3_DDR	47 51
VDDQ S0 (GPU FBVDDQ)					
I37	SNS_CURRENT	SNS_DIFF_PHY	SENSE	SNS_P12VS0_FBVDDQ_P	50
I38	SNS_CURRENT	SNS_DIFF_PHY	SENSE	SNS_P12VS0_FBVDDQ_N	50
I39			SENSE	ISNS_P12VS0_FDDQ_R	50
I40			SENSE	ISNS_P12VS0_FBVDDQ	47 50
I143			SENSE	VSNS_P12VS0_FBVDDQ	47 50
CPU Core					
I42			SENSE	R_CPU_IM_R	50
I43			SENSE	ISNS_CPU_FB_R	50
I44			SENSE	ISNS_CPU_FB	50
I45			SENSE	ISNS_CPUVCC	47 50
I46			SENSE	VSNS_CPUVCC	47 50
PP1V05_S0_PCH					
I110			SENSE	VSNS_P1V05S0_PCH	47 50
PP1V5_S0					
I111	SNS_CURRENT	SNS_DIFF_PHY	SENSE	SNS_P1V5S0_P	51
I113	SNS_CURRENT	SNS_DIFF_PHY	SENSE	SNS_P1V5S0_N	51
I112			SENSE	ISNS_P1V5S0_R	51
I114			SENSE	ISNS_P1V5S0	47 51
I115			SENSE	VSNS_P1V5S0	47 51
Airport					
I102	SNS_CURRENT	SNS_DIFF_PHY	SENSE	SNS_P3V3S4_AP_P	50
I103	SNS_CURRENT	SNS_DIFF_PHY	SENSE	SNS_P3V3S4_AP_N	50
I104			SENSE	ISNS_P3V3S4_AP_R	50
I105			SENSE	ISNS_P3V3S4_AP	47 50

SMC

	Electrical Contraint Set	Physical	Spacing	Net	Pages
SMC					
I1		CLK_XTAL	XTAL	SMC_XTAL	46 47
I2		CLK_XTAL	XTAL	SMC_EXTAL	46 47
I91		SMC_GEN	SMC_CTRL	SMC_LRESET_L	20 46
I90		SMC_GEN	SMC_CTRL	SMC_RUNTIME_SCI_L	14 46
I92		SMC_GEN	SMC_CTRL	SMC_WAKE_SCI_L	14 46
I93		SMC_GEN	SMC_CTRL	SMC_FAN_0_CTL	46 53
I94		SMC_GEN	SMC_CTRL	SMC_FAN_0_TACH	46 53
I96		SMC_GEN	SMC_3XCTRL	CPU_PECI	6 14 46 47
I130		SMC_GEN	SMC_3XCTRL	ALL_SYS_PWRGD	3 21 46 73

SMBus

	Electrical Contraint Set	Physical	Spacing	Net	Pages
SMC					
I10		SMB_PHY	SMB	SMBUS_SMC_0_S0_SCL	46 49
I9		SMB_PHY	SMB	SMBUS_SMC_0_S0_SDA	46 49
I11		SMB_PHY	SMB	SMBUS_SMC_1_S0_SCL	46 49
I12		SMB_PHY	SMB	SMBUS_SMC_1_S0_SDA	46 49
I13		SMB_PHY	SMB	SMBUS_SMC_2_S0_SCL	46 49
I14		SMB_PHY	SMB	SMBUS_SMC_2_S0_SDA	46 49
I15		SMB_PHY	SMB	SMBUS_SMC_3_SCL	46 49
I16		SMB_PHY	SMB	SMBUS_SMC_3_SDA	46 49
I19		SMB_PHY	SMB	SMBUS_SMC_5_G3H_SCL	46 47
I20		SMB_PHY	SMB	SMBUS_SMC_5_G3H_SDA	46 47
GPU					
I152		SMB_PHY	SMB	GFX_SMBDAT_B	49 84
I153		SMB_PHY	SMB	GFX_SMBCLK_B	49 84
PCH					
I21		SMB_PHY	SMB	SMBUS_PCH_CLK	13 49
I52		SMB_PHY	SMB	SMBUS_PCH_DATA	13 49
I53		SMB_PHY	SMB	SML_PCH_0_CLK	13 49
I54		SMB_PHY	SMB	SML_PCH_0_DATA	13 49
Display TCon					
I55		SMB_PHY	SMB	SMB_DP_TCON_SCL	42 49
I56		SMB_PHY	SMB	SMB_DP_TCON_SDA	42 49

Temperature Sense

	Electrical Contraint Set	Physical	Spacing	Net	Pages
EMC1414-1 (Production)					
I65	SNS_TEMP	SNS_DIFF_PHY	SENSE	SNS_ACDC_P	52 62
I66	SNS_TEMP	SNS_DIFF_PHY	SENSE	SNS_ACDC_N	52 62
HDD Out-of-Band					
I117			SENSE	SMC_OOB1_D2R_L	36 46
I76			SENSE	SMC_OOB1_R2D_L	46 47
I116			SENSE	SMC_OOB1_D2R_R	36
SSD Out-of-Band					
I77			SENSE	SMC_OOB2_R2D_L	35 47
I78			SENSE	SMC_OOB2_D2R_L	35 47
GPU PCIE					
I131	SNS_TEMP	SNS_DIFF_PHY	SENSE	GPU_TDIODE_P	84 85
I132	SNS_TEMP	SNS_DIFF_PHY	SENSE	GPU_TDIODE_N	84 85
I148	SNS_TEMP	SNS_DIFF_PHY	SENSE	GFX_THM_DP2	85
I149	SNS_TEMP	SNS_DIFF_PHY	SENSE	GFX_THM_DN3	85
I150	SNS_TEMP	SNS_DIFF_PHY	SENSE	GFX_THM_DP3	85
I151	SNS_TEMP	SNS_DIFF_PHY	SENSE	GFX_THM_DN2	85

DC-DC

Power-specific Physical Rules

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
GND_P3MM	*	Y	0.300 MM	0.150 MM	12.7 MM	=STANDARD	=STANDARD
GND_P5MM	*	Y	0.500 MM	0.150 MM	12.7 MM	=STANDARD	=STANDARD
POWER_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=STANDARD	=STANDARD
POWER_P3MM	*	Y	0.300 MM	0.150 MM	12.7 MM	=STANDARD	=STANDARD
POWER_P6MM	*	Y	0.600 MM	0.150 MM	12.7 MM	=STANDARD	=STANDARD
VR_SENSE	*	Y	0.300 MM	0.200 MM	25 MM	0.400 MM	0.200 MM
POWER_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD
GPU_VSEN_PHY	*	Y	0.150 MM	0.100 MM	25 MM	0.400 MM	0.200 MM

Physical Net Type to Rule Map

NET_PHYSICAL_TYPE	AREA_TYPE	PHYSICAL_RULE_SET
GND	*	GND_P5MM
GND	BGA	GND_P3MM
POWER	*	POWER_P6MM
POWER	BGA	POWER_P3MM
VR_CTL_PHY	*	POWER_P3MM
VR_CTL_PHY	BGA	STANDARD
VR_VID_PHY	*	POWER_50S
VR_SVID_PHY	*	POWER_55S
VR_GPU_VSEN	*	GPU_VSEN_PHY

NET_PHYSICAL_TYPE	AREA_TYPE	PHYSICAL_RULE_SET
VR_DIDT_PHY	*	POWER_P6MM
VR_DIDT_PHY	BGA	STANDARD

Power-specific Spacing Definitions
Power and Common

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
POWER_ISO	*	=STANDARD	?
GND_ISO	*	=STANDARD	?

Constraints
Power and Common

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
POWER	*	*	POWER_ISO
GND	*	*	GND_ISO

DC-DC Baddies

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
SWNODE_ISO	*	=8:1_SPACING	1000
SWNODE_SW2SW	*	=1:1_SPACING	?
SWNODE_SW2PWR	*	=2:1_SPACING	?
SWNODE_SW2GND	*	=2:1_SPACING	?

DC-DC Baddies

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
VR_SWITCH	*	*	SWNODE_ISO
VR_SWITCH	*	BGA	BGA_P1MM
VR_SWITCH	VR_SWITCH	*	SWNODE_SW2SW
VR_SWITCH	POWER	*	SWNODE_SW2PWR
VR_SWITCH	GND	*	SWNODE_SW2GND

DC-DC Control

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
VR_CTL_ISO	*	=3:1_SPACING	?
VR_VID_ISO	*	=4X_DIELECTRIC	?
VR_SVID_ISO	*	=2.6:1_SPACING	?

DC-DC Control

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
VR_CTL	*	*	VR_CTL_ISO
VR_VID	*	*	VR_VID_ISO
VR_SVID	*	*	VR_SVID_ISO

PCH/GPU/TBT 1.05V S0

	Electrical Contraint Set	Physical	Spacing	Voltage	DIDT	NO_TEST		
	Input Bus							
I1		POWER	POWER	5V			REG_VCC_U7700	68
	Local Ground							
I3		GND	GND	0V			AGND_P1V05S0	68
	1.05V S0							
I5		VR_DIDT_PHY	VR_SWITCH	12V	TRUE		REG_PHASE_P1V05S0	68
I4		VR_DIDT_PHY	VR_SWITCH	12V	TRUE		REG_PHASE_P1V05S0_L	
I7		VR_DIDT_PHY	VR_SWITCH	12V	TRUE		REG_BOOT_P1V05S0	68
I6		VR_DIDT_PHY	VR_SWITCH	12V	TRUE	TRUE	REG_BOOT_P1V05S0_RC	68
I8		VR_DIDT_PHY	VR_SWITCH	12V	TRUE		REG_UGATE_P1V05S0	68
I10		VR_DIDT_PHY	VR_SWITCH	12V	TRUE		REG_LGATE_P1V05S0	68
I12		VR_DIDT_PHY	VR_SWITCH	12V	TRUE		REG_SNUBBER_P1V05S0	68
I11		VR_CTL_PHY	VR_CTL				REG_P1V05S0_OCSET	68
I14		VR_CTL_PHY	VR_CTL				REG_P1V05S0_VO	68
I72	VSNS_GPU_P1V05	SNS_DIFF_PHY	SENSE				SNS_GPU_PEX_IOVDD_P	17 68
I73	VSNS_GPU_P1V05	SNS_DIFF_PHY	SENSE				SNS_GPU_PEX_IOVDD_N	17 68
I16	VSNS_GPU_P1V05	SNS_DIFF_PHY	SENSE				SNS_GPU_PEX_IOVDD_XW_P	68
I17	VSNS_GPU_P1V05	SNS_DIFF_PHY	SENSE				SNS_GPU_PEX_IOVDD_XW_N	68
I19			SENSE				REG_P1V05S0_FB	68
I18			SENSE				REG_P1V05S0_RTN	68
I21		VR_CTL_PHY	VR_CTL				REG_P1V05S0_SREF	68
I20		VR_CTL_PHY	VR_CTL				REG_P1V05S0_FSEL	68
	Output Bus							
I22		POWER	POWER	1.05V			PP1V05_S0	94
	FET Switched							
I66		POWER	POWER	1.05V			PP1V05_TBTCIO	29 33

VDDQ S3 (1.5V)/VTT S0

	Physical	Spacing	Voltage	DIDT	NO_TEST		
Input Bus							
I43	POWER	POWER	5V			REG_V5IN_U7300	66
Local Ground							
I44	GND	GND	0V			AGND_VDDQS3	66
I74	GND	GND	0V			PGND_REG_P1V5_S3	66
VDDQ S3							
I45	VR_DIDT_PHY	VR_SWITCH	12V	TRUE		REG_PHASE_VDDQS3	66
I47	VR_DIDT_PHY	VR_SWITCH	12V	TRUE		REG_PHASE_VDDQS3_L	66
I46	VR_DIDT_PHY	VR_SWITCH	12V	TRUE		REG_BOOT_VDDQS3	66
I48	VR_DIDT_PHY	VR_SWITCH	12V	TRUE	TRUE	REG_BOOT_VDDQS3_RC	66
I49	VR_DIDT_PHY	VR_SWITCH	12V	TRUE		REG_UGATE_VDDQS3	66
I52	VR_DIDT_PHY	VR_SWITCH	12V	TRUE		REG_LGATE_VDDQS3	66
I51	VR_DIDT_PHY	VR_SWITCH	12V	TRUE		REG_SNUBBER_VDDQS3	66
I53	VR_CTL_PHY	VR_CTL				REG_VDDQS3_VDDQSNS	66
I55	VR_CTL_PHY	VR_CTL				REG_VDDQS3_VREF	66
I56	VR_CTL_PHY	VR_CTL				REG_VDDQS3_REFIN	66
I57	VR_CTL_PHY	VR_CTL				REG_VDDQS3_MODE	66
I58	VR_CTL_PHY	VR_CTL				REG_VDDQS3_TRIP	66
I59	VR_CTL_PHY	VR_CTL				LDO_DDRVTTS0_SNS	66
I61	VR_CTL_PHY	VR_CTL				REG_VDDQS3_VTTREF	66
I70	VR_CTL_PHY	VR_CTL				PM_EN_REG_VDDQ_S3	66 72
I71	VR_CTL_PHY	VR_CTL				PM_EN_REG_VDDQ_S3_R	66
I68	VR_CTL_PHY	VR_CTL				PM_EN_LDO_DDRVTT_S0	66 72
I69	VR_CTL_PHY	VR_CTL				PM_EN_LDO_DDR_VTT_S0_R	66
Output Bus							
I60	POWER	POWER	1.5V			PPVDDQ_S3	94
I62	POWER_DDR	POWER_DDR	0.75V			PPDDRVTT_S0	94
FET Switched							
I63	POWER	POWER	1.5V			PP1V5_S0	94
Sensed							
I64	POWER	POWER	1.5V			PPVDDQ_S3_DDR	94
I67	POWER	POWER	1.5V			PP1V5_S0_PCH	94

CPU VCC Phases

	Electrical Contraint Set	Physical	Spacing	Voltage	DIDT	NO_TEST	Net	Pages
Input Bus								
I836		POWER	POWER	12V			PP12V_S0_CPUVCC_FLT	63 64 65
I113		POWER	POWER	5V			REG_VCC_U7000	63
Local Ground								
I126		GND	GND	0V			AGND_CPU	63 64 65
Phase 1								
I883		VR_CTL_PHY	VR_CTL				REG_TD_1	64
I884		VR_CTL_PHY	VR_CTL				REG_PWM_CPUVCC_1	63 64
I885		VR_CTL_PHY	VR_CTL				REG_PWM_CPUVCC_1_R	63
I887		VR_DIDT_PHY	VR_SWITCH	12V	TRUE		REG_PHASE_CPUVCC_1	64
I888		VR_DIDT_PHY	VR_SWITCH	12V	TRUE		REG_BOOT_CPUVCC_1	64
I890		VR_DIDT_PHY	VR_SWITCH	12V	TRUE	TRUE	REG_BOOT_CPUVCC_1_RC	64
I889		VR_DIDT_PHY	VR_SWITCH	12V	TRUE		REG_UGATE_CPUVCC_1	64
I891		VR_DIDT_PHY	VR_SWITCH	12V	TRUE		REG_LGATE_CPUVCC_1	64
I892		VR_DIDT_PHY	VR_SWITCH	12V	TRUE		REG_SNUBBER_CPUVCC_1	64
I893		POWER	POWER	1.8V			PPCPUVCC_S0_SENSE_1	64
I894	ISNS_CPU_CORE	SNS_DIFF_PHY	SENSE				REG_ISENVCC_1_P	63 64
I895	ISNS_CPU_CORE	SNS_DIFF_PHY	SENSE				REG_ISENVCC_1_N	64
I896			SENSE				REG_ISENVCC_1_NR	63 64
Phase 2								
I113		VR_CTL_PHY	VR_CTL				REG_TD_2	64
I114		VR_CTL_PHY	VR_CTL				REG_PWM_CPUVCC_2	63 64
I114		VR_CTL_PHY	VR_CTL				REG_PWM_CPUVCC_2_R	63
I114		VR_DIDT_PHY	VR_SWITCH	12V	TRUE		REG_PHASE_CPUVCC_2	64
I114		VR_DIDT_PHY	VR_SWITCH	12V	TRUE		REG_BOOT_CPUVCC_2	64
I114		VR_DIDT_PHY	VR_SWITCH	12V	TRUE	TRUE	REG_BOOT_CPUVCC_2_RC	64
I114		VR_DIDT_PHY	VR_SWITCH	12V	TRUE		REG_UGATE_CPUVCC_2	64
I114		VR_DIDT_PHY	VR_SWITCH	12V	TRUE		REG_LGATE_CPUVCC_2	64
I114		VR_DIDT_PHY	VR_SWITCH	12V	TRUE		REG_SNUBBER_CPUVCC_2	64
I114		POWER	POWER	1.8V			PPCPUVCC_S0_SENSE_2	64
I114	ISNS_CPU_CORE	SNS_DIFF_PHY	SENSE				REG_ISENVCC_2_P	63 64
I115	ISNS_CPU_CORE	SNS_DIFF_PHY	SENSE				REG_ISENVCC_2_N	64
I115			SENSE				REG_ISENVCC_2_NR	63 64
Phase 3								
I115		VR_CTL_PHY	VR_CTL				REG_TD_3	64
I115		VR_CTL_PHY	VR_CTL				REG_PWM_CPUVCC_3	63 64
I115		VR_CTL_PHY	VR_CTL				REG_PWM_CPUVCC_3_R	63
I115		VR_DIDT_PHY	VR_SWITCH	12V	TRUE		REG_PHASE_CPUVCC_3	64
I115		VR_DIDT_PHY	VR_SWITCH	12V	TRUE		REG_BOOT_CPUVCC_3	64
I115		VR_DIDT_PHY	VR_SWITCH	12V	TRUE	TRUE	REG_BOOT_CPUVCC_3_RC	64
I116		VR_DIDT_PHY	VR_SWITCH	12V	TRUE		REG_UGATE_CPUVCC_3	64
I116		VR_DIDT_PHY	VR_SWITCH	12V	TRUE		REG_LGATE_CPUVCC_3	64
I116		VR_CTL_PHY	VR_SWITCH	12V	TRUE		REG_SNUBBER_CPUVCC_3	64
I116		POWER	POWER	1.8V			PPCPUVCC_S0_SENSE_3	64
I116	ISNS_CPU_CORE	SNS_DIFF_PHY	SENSE				REG_ISENVCC_3_P	63 64
I116	ISNS_CPU_CORE	SNS_DIFF_PHY	SENSE				REG_ISENVCC_3_N	64
I116			SENSE				REG_ISENVCC_3_NR	63 64
Phase 4								
I120		VR_CTL_PHY	VR_CTL				REG_TD_4	65
I120		VR_CTL_PHY	VR_CTL				REG_PWM_CPUVCC_4	63 65
I120		VR_CTL_PHY	VR_CTL				REG_PWM_CPUVCC_4_R	63
I120		VR_DIDT_PHY	VR_SWITCH	12V	TRUE		REG_PHASE_CPUVCC_4	65
I126		VR_DIDT_PHY	VR_SWITCH	12V	TRUE		REG_BOOT_CPUVCC_4	65
I127		VR_DIDT_PHY	VR_SWITCH	12V	TRUE	TRUE	REG_BOOT_CPUVCC_4_RC	65
I127		VR_DIDT_PHY	VR_SWITCH	12V	TRUE		REG_UGATE_CPUVCC_4	65
I127		VR_DIDT_PHY	VR_SWITCH	12V	TRUE		REG_LGATE_CPUVCC_4	65
I127		VR_CTL_PHY	VR_SWITCH	12V	TRUE		REG_SNUBBER_CPUVCC_4	65
I127		POWER	POWER	1.8V			PPCPUVCC_S0_SENSE_4	65
I127	ISNS_CPU_CORE	SNS_DIFF_PHY	SENSE				REG_ISENVCC_4_P	63 65
I127	ISNS_CPU_CORE	SNS_DIFF_PHY	SENSE				REG_ISENVCC_4_N	65
I127			SENSE				REG_ISENVCC_4_NR	63 65

CPU VCC Controller

	Electrical Contraint Set	Physical	Spacing	Voltage	DIDT	NO_TEST	Net	Pages
ISL6372								
I102		VR_CTL_PHY	VR_CTL				REG_CPUVCC_DVC	63
I102		VR_CTL_PHY	VR_CTL				CPUVCC_DVC_RC	
I102		VR_CTL_PHY	VR_CTL				CPUVCC_FB_RC_2	63
I102		VR_CTL_PHY	VR_CTL				REG_CPUVCC_COMP	63
I102		VR_CTL_PHY	VR_CTL				CPUVCC_COMP_RC	63
I102		VR_CTL_PHY	VR_CTL				REG_CPUVCC_FB	63
I102		VR_CTL_PHY	VR_CTL				CPUVCC_FB_RC	63
I103		VR_CTL_PHY	VR_CTL				CPUVCC_FB_R_1	63
I103		VR_CTL_PHY	VR_CTL				CPUVCC_FB_R_2	63
I103		VR_CTL_PHY	VR_CTL				CPUVCC_PSICOMP_RC	63
I104		VR_CTL_PHY	VR_CTL				REG_CPUVCC_PSICOMP	63
I104		VR_CTL_PHY	VR_CTL				REG_CPUVCC_HFCOMP	63
I103	VSNS_CPU_CORE	SNS_DIFF_PHY	SENSE				CPU_VCCSENSE_P	8 63
I104	VSNS_CPU_CORE	SNS_DIFF_PHY	SENSE				CPU_VCCSENSE_N	9 63
I103		SNS_DIFF_PHY	SENSE				CPU_VCCSENSE_R_P	63
I103		SNS_DIFF_PHY	SENSE				CPU_VCCSENSE_R_N	63
I103	VSNS_CPU_CORE	SNS_DIFF_PHY	SENSE				SNS_VCC_XW_P	63
I103	VSNS_CPU_CORE	SNS_DIFF_PHY	SENSE				SNS_VCC_XW_N	63
I103			SENSE				REG_CPUVCC_VSEN	63
I103			SENSE				REG_CPUVCC_RGND	63
I104			SENSE				REG_CPUVCC_VIN	63
I104		VR_CTL_PHY	VR_CTL				REG_CPUVCC_IMON	50 63
I104		VR_CTL_PHY	VR_CTL				CPUVCC_IMON_R	63
I105		VR_CTL_PHY	VR_CTL				REG_CPUVCC_TM	63
I105		VR_CTL_PHY	VR_CTL				REG_CPUVCC_IMX	63
I105		VR_CTL_PHY	VR_CTL				REG_CPUVCC_NPSI	63
I105		VR_CTL_PHY	VR_CTL				REG_CPUVCC_FDVID	63
I105		VR_CTL_PHY	VR_CTL				REG_CPUVCC_TMX	63
I105		VR_CTL_PHY	VR_CTL				REG_CPUVCC_MEMVRSEL	63
I106		VR_CTL_PHY	VR_CTL				REG_CPUVCC_RSET	63
I106	CPU_VIDSCLK	VR_VID_PHY	VR_VID				CPU_VIDSCLK	8 63
I106		VR_VID_PHY	VR_VID				CPU_VIDSCLK_R	8
I106	CPU_VIDALERT_L	VR_VID_PHY	VR_VID				CPU_VIDALERT_L	8 63
I106		VR_VID_PHY	VR_VID				CPU_VIDALERT_R_L	8
I106	CPU_VIDSOUT	VR_VID_PHY	VR_VID				CPU_VIDSOUT	8 63
I106		VR_VID_PHY	VR_VID				CPU_VIDSOUT_R	8
Output Bus								
I106		POWER	POWER	1.8V			PPCPUVCC_S0_CPU	94

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3.3V S5/5V S4

Tag	Physical	Spacing	Voltage	DIDT	NO_TEST	Net	Pages
Input Bus							
I843	POWER	POWER	12V			REG_VIN_U7600	67
I125	POWER	POWER	5V			REG_VCC1_U7600	67
I125	POWER	POWER	5V			REG_VCC2_U7600	67
3.3V S5							
I849	VR_DIDT_PHY	VR_SWITCH	12V	TRUE		REG_PHASE_P3V3S5	67
I847	VR_DIDT_PHY	VR_SWITCH	12V	TRUE		REG_BOOT_P3V3S5	67
I851	VR_DIDT_PHY	VR_SWITCH	12V	TRUE	TRUE	REG_BOOT_P3V3S5_RC	67
I848	VR_DIDT_PHY	VR_SWITCH	12V	TRUE		REG_UGATE_P3V3S5	67
I850	VR_DIDT_PHY	VR_SWITCH	12V	TRUE		REG_LGATE_P3V3S5	67
I852	VR_DIDT_PHY	VR_SWITCH	12V	TRUE		REG_SNUBBER_P3V3S5	67
I107	VR_CTL_PHY	VR_CTL				REG_P3V3S5_ISEN	67
I107	VR_CTL_PHY	VR_CTL				REG_P3V3S5_OCSET	67
I113	VR_CTL_PHY	VR_CTL				REG_P3V3S5_FSET	67
I108	VR_CTL_PHY	VR_CTL				REG_P3V3S5_VOUT	67
I107	VR_CTL_PHY	VR_CTL				REG_P3V3S5_VOUT_R	67
I107	VR_CTL_PHY	VR_CTL				REG_P3V3S5_FB	67
I150	VR_CTL_PHY	VR_CTL				PM_EN_REG_P3V3_S5	67 72
I150	VR_CTL_PHY	VR_CTL				PM_EN_REG_P3V3_S5_R	67
5V S3							
I113	VR_DIDT_PHY	VR_SWITCH	12V	TRUE		REG_PHASE_P5VS4	67
I113	VR_DIDT_PHY	VR_SWITCH	12V	TRUE		REG_BOOT_P5VS4	67
I113	VR_DIDT_PHY	VR_SWITCH	12V	TRUE	TRUE	REG_BOOT_P5VS4_RC	67
I114	VR_DIDT_PHY	VR_SWITCH	12V	TRUE		REG_UGATE_P5VS4	67
I114	VR_DIDT_PHY	VR_SWITCH	12V	TRUE		REG_LGATE_P5VS4	67
I114	VR_DIDT_PHY	VR_SWITCH	12V	TRUE		REG_SNUBBER_P5VS4	67
I114	VR_CTL_PHY	VR_CTL				REG_P5VS4_ISEN	67
I114	VR_CTL_PHY	VR_CTL				REG_P5VS4_OCSET	67
I114	VR_CTL_PHY	VR_CTL				REG_P5VS4_FSET	67
I114	VR_CTL_PHY	VR_CTL				REG_P5VS4_VOUT	67
I114	VR_CTL_PHY	VR_CTL				REG_P5VS4_VOUT_R	67
I114	VR_CTL_PHY	VR_CTL				REG_P5VS4_FB	67
I150	VR_CTL_PHY	VR_CTL				PM_EN_REG_P5V_S4	67 72
I150	VR_CTL_PHY	VR_CTL				PM_EN_REG_P5V_S4_R	67
Output Bus							
I115	POWER	POWER	5V			PP5V_S5	94
I115	POWER	POWER	5V			PP5V_S4	94
I114	POWER	POWER	3.3V			PP3V3_S5	94
I151	POWER	POWER	3.3V			PP3V3_S5_PCH_R	20
FET Switched							
I123	POWER	POWER	5V			PP5V_S0	94
I121	POWER	POWER	3.3V			PP3V3_S4	94
I123	POWER	POWER	3.3V			PP3V3_S0	94
I150	POWER	POWER	3.3V			PP3V3_TBT_PCH_GPIO	20
I121	POWER	POWER	3.3V			PP3V3_ENET	94
I138	POWER	POWER	3.3V			PP3V3_TBTLC	28 29 94
Sensed							
I145	POWER	POWER	3.3V			PPSSD_S0	94
I147	POWER	POWER	3.3V			PP3V3_S4_AP	94

J90 GPU VREG POWER NETS

Tag	Physical	Spacing	Voltage	DIDT	NO_TEST	Net	Pages
I148	POWER	POWER	1.8V			PP1V8_S0_GPU_VDD	94
I148	POWER	POWER	0.95V			PP0V95_S0_GPU_VDDC	94
I148	POWER	POWER	0.85V			PPVDDCI_S0_GPU	94
I149	POWER	POWER	1.8V			PP1V8_TSVDD_FLT	84
I149	POWER	POWER	1.8V			PP1V8_SPLL_PVDD_FLT	84
I149	POWER	POWER	1.8V			PP1V8_AVDD_FLT	
I149	POWER	POWER	1.8V			PP1V8_VDD1DI_FLT	
I149	POWER	POWER	1.8V			PP1V8_XTAL_VDDR_FLT	
I149	POWER	POWER	1.8V			PP1V8_MPLL_PVDD_FLT	84
I149	POWER	POWER	1.8V			PP1V8_DPLL_PVDD_FLT	
I149	POWER	POWER	0.95V			PP0V95_SPLL_VDDC_FLT	
I150	POWER	POWER	0.95V			PP0V95_DPLL_VDDC_FLT	84

3.42V G3H

Tag	Physical	Spacing	Voltage	DIDT	NO_TEST	Net	Pages
3.42V G3H							
I144	POWER	VR_SWITCH	12V	TRUE		P3V42G3H_BOOST	62
I150	POWER	VR_SWITCH	12V	TRUE		P3V42G3H_BOOST_R	
I144	POWER	VR_SWITCH	12V	TRUE		P3V42G3H_SW	62
I144	VR_CTL_PHY	VR_CTL				P3V42G3H_FB	62
I150	VR_CTL_PHY	VR_CTL				P3V42G3H_BIAS	62
I144	VR_CTL_PHY	VR_CTL				P3V42G3H_SHDN_L	62
Output Bus							
I144	POWER	POWER	3.425V			PP3V42_G3H	94

3.3V G3

Tag	Physical	Spacing	Voltage	DIDT	NO_TEST	Net	Pages
I147	POWER	POWER	3.3V			PP3V3_G3	94

HDD S0

Tag	Physical	Spacing	Voltage	DIDT	NO_TEST	Net	Pages
FET Switched							
I139	POWER	POWER	5V			PPHDD_S0	94
Sensed							
I146	POWER	POWER	5V			PP5V_S0_HDD	94

12V

Tag	Physical	Spacing	Voltage	DIDT	NO_TEST	Net	Pages
Input Bus							
I144	POWER	POWER	12V			PP12V_ACDC	94
FET Switched							
I144	POWER	POWER	12V			PP12V_S5	94
I145	POWER	POWER	12V			PP12V_S0	94
I148	POWER	POWER	12V			PPHDD12_S0	94
Sensed							
I145	POWER	POWER	12V			PP12V_G3H	94
I147	POWER	POWER	12V			PP12V_G3H_P3V42	62
I145	POWER	POWER	12V			PP12V_S0_GPUCORE	94
I145	POWER	POWER	12V			PP12V_S0_FBVDDQ	94
I147	POWER	POWER	12V			PP12V_S0_HDD	94
I150	POWER	POWER	12V			PP12V_S0_GPU_AUX	94

Ground/Common

Tag	Physical	Spacing	Voltage	DIDT	NO_TEST	Net
Common						
I139	GND	GND	0V			GND

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Thunderbolt

Thunderbolt-specific Physical Rules

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
TBT_I2C_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD
TBT_SPI_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD
TBTDP_90D	*	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF
TBT_MISC_55S	*	=55_OHM_SE	=55_OHM_SE	0.070 MM	=55_OHM_SE	=STANDARD	=STANDARD

Thunderbolt-specific Spacing Definitions

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
TBT_I2C	*	=2x_DIELECTRIC	?
TBT_SPI	*	=2x_DIELECTRIC	?
TBTDP_ISO	*	=5x_DIELECTRIC	?
TBTDP_ISO	TOP,BOTTOM	=7x_DIELECTRIC	?
TBT2PWR_ISO	*	0.5 MM	?
BGA_TBT_AREA	*	0.075MM	?
TBT_MISC	*	=3:1_SPACING	?
TBT2VR_ISO	*	1 MM	?

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
*	*	BGA_TBT	BGA_TBT_AREA
TBTDP	*	*	TBTDP_ISO
TBTDP	POWER	*	TBT2PWR_ISO
TBTDP	VR_SWITCH	*	TBT2VR_ISO

SOURCE: Bill Cornelius's T29 Routing Notes

DisplayPort

DP-specific Physical Rules

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
DP_85D	*	=85_OHM_DIFF	=85_OHM_DIFF	0.08MM	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF

DP-specific Spacing Definitions

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
DP2DP_ISO	*	=4X_DIELECTRIC	?
DP2DP_ISO	TOP,BOTTOM	=5X_DIELECTRIC	?
DP_ISO	*	=5X_DIELECTRIC	?
DP_ISO	TOP,BOTTOM	=7X_DIELECTRIC	?
DP2PWR_ISO	*	0.5 MM	?
DP2VR_ISO	*	1 MM	?

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
DISPLAYPORT	DISPLAYPORT	*	DP2DP_ISO
DISPLAYPORT	POWER	*	DP2PWR_ISO
DISPLAYPORT	VR_SWITCH	*	DP2VR_ISO
DISPLAYPORT	*	*	DP_ISO

Pairs should be within 100 mils of clock length.
Max length of DisplayPort traces: 12 inches

DisplayPort intra-pair matching should be 5 ps. Inter-pair matching should be within 150 ps.
DisplayPort AUX channel intra-pair matching should be 5 ps. No relationship to other signals.

TBT IC Net Properties

	Electrical Contraint Set	Physical	Spacing	Net		
I511		DP_85D	DISPLAYPORT	DP_TBTSNK0_ML_C_P<3..0>	NO_TEST=TRUE	28 83
I513		DP_85D	DISPLAYPORT	DP_TBTSNK0_ML_C_N<3..0>	NO_TEST=TRUE	28 83
I512	DP_TBTSNK0_ML	DP_85D	DISPLAYPORT	DP_TBTSNK0_ML_P<3..0>	NO_TEST=TRUE	28
I514	DP_TBTSNK0_ML	DP_85D	DISPLAYPORT	DP_TBTSNK0_ML_N<3..0>	NO_TEST=TRUE	28
I516		TBTDP_90D	DISPLAYPORT	DP_TBTSNK0_AUXCH_C_P		28 83
I515		TBTDP_90D	DISPLAYPORT	DP_TBTSNK0_AUXCH_C_N		28 83
I517	DP_TBTSNK0_AUX	TBTDP_90D	DISPLAYPORT	DP_TBTSNK0_AUXCH_P		28
I518	DP_TBTSNK0_AUX	TBTDP_90D	DISPLAYPORT	DP_TBTSNK0_AUXCH_N		28
I519		DP_85D	DISPLAYPORT	DP_TBTSNK1_ML_C_P<3..0>	NO_TEST=TRUE	28 83
I521		DP_85D	DISPLAYPORT	DP_TBTSNK1_ML_C_N<3..0>	NO_TEST=TRUE	28 83
I520	DP_TBTSNK1_ML	DP_85D	DISPLAYPORT	DP_TBTSNK1_ML_P<3..0>	NO_TEST=TRUE	28
I523	DP_TBTSNK1_ML	DP_85D	DISPLAYPORT	DP_TBTSNK1_ML_N<3..0>	NO_TEST=TRUE	28
I522		TBTDP_90D	DISPLAYPORT	DP_TBTSNK1_AUXCH_C_P		28 83
I524		TBTDP_90D	DISPLAYPORT	DP_TBTSNK1_AUXCH_C_N		28 83
I526	DP_TBTSNK1_AUX	TBTDP_90D	DISPLAYPORT	DP_TBTSNK1_AUXCH_P		28
I525	DP_TBTSNK1_AUX	TBTDP_90D	DISPLAYPORT	DP_TBTSNK1_AUXCH_N		28
I603	DP_INTPNL_ML	DP_85D	DISPLAYPORT	DP_TBTSRC_ML_P<3..0>	NO_TEST=TRUE	43
I604	DP_INTPNL_ML	DP_85D	DISPLAYPORT	DP_TBTSRC_ML_N<3..0>	NO_TEST=TRUE	43
I529	DP_INTPNL_ML_AUX	TBTDP_90D	DISPLAYPORT	DP_TBTSRC_AUXCH_P		43
I530	DP_INTPNL_ML_AUX	TBTDP_90D	DISPLAYPORT	DP_TBTSRC_AUXCH_N		43
I601	DP_INTPNL_ML_AUX	TBTDP_90D	DISPLAYPORT	DP_TBTSRC_AUX_C_P		43
I602	DP_INTPNL_ML_AUX	TBTDP_90D	DISPLAYPORT	DP_TBTSRC_AUX_C_N		43
I533	TBT_SPI_CLK	TBT_SPI_55S	TBT_SPI	TBT_SPI_CLK		28
I534	TBT_SPI_MOSI	TBT_SPI_55S	TBT_SPI	TBT_SPI_MOSI		28
I538	TBT_SPI_MISO	TBT_SPI_55S	TBT_SPI	TBT_SPI_MISO		28
I535	TBT_SPI_CS_L	TBT_SPI_55S	TBT_SPI	TBT_SPI_CS_L		28
I703		TBT_MISC_55S	TBT_MISC	DP_TBTSNK0_HPD		28 84
I704		TBT_MISC_55S	TBT_MISC	DP_TBTSNK1_HPD		28 84
I705		TBT_MISC_55S	TBT_MISC	DP_TBTSRC_HPD		28 43
I706		TBT_MISC_55S	TBT_MISC	DP_TBTPA_HPD		28 31
I707		TBT_MISC_55S	TBT_MISC	DP_TBTPB_HPD		28 32
I708		TBT_MISC_55S	TBT_MISC	TBT_A_HPD		31
I709		TBT_MISC_55S	TBT_MISC	TBT_B_HPD		32

*: Only used on hosts supporting T29 video-in

DisplayPort

	Electrical Contraint Set	Physical	Spacing	Net		
	SLAVE NETS - GPU/PANEL					
I701	DP_INTPNL_SL_AUX	TBTDP_90D	DISPLAYPORT	DP_INT_EG_SL_AUX_P		42 83
I702	DP_INTPNL_SL_AUX	TBTDP_90D	DISPLAYPORT	DP_INT_EG_SL_AUX_N		42 83
I695	DP_INTPNL_SL_AUX	TBTDP_90D	DISPLAYPORT	DP_INT_EG_SL_AUX_C_P		42
I692	DP_INTPNL_SL_AUX	TBTDP_90D	DISPLAYPORT	DP_INT_EG_SL_AUX_C_N		42
I699	DP_INTPNL_SL	DP_85D	DISPLAYPORT	DP_INT_EG_SL_P<3..0>	NO_TEST=TRUE	42 83
I700	DP_INTPNL_SL	DP_85D	DISPLAYPORT	DP_INT_EG_SL_N<3..0>	NO_TEST=TRUE	42 83
I693	DP_INTPNL_SL	DP_85D	DISPLAYPORT	DP_INT_EG_SL_C_P<3..0>	NO_TEST=TRUE	42
I694	DP_INTPNL_SL	DP_85D	DISPLAYPORT	DP_INT_EG_SL_C_N<3..0>	NO_TEST=TRUE	42
I720		TBTDP_90D	DISPLAYPORT	DP_INT_EG_SL_DDC_CLK		83
I721		TBTDP_90D	DISPLAYPORT	DP_INT_EG_SL_DDC_DAT		83
	MASTER NETS - GPU-MUX					
I642	DP_INTPNL_ML	DP_85D	DISPLAYPORT	DP_INT_EG_ML_P<3..0>	NO_TEST=TRUE	43 83
I643	DP_INTPNL_ML	DP_85D	DISPLAYPORT	DP_INT_EG_ML_N<3..0>	NO_TEST=TRUE	43 83
I644	DP_INTPNL_ML_AUX	TBTDP_90D	DISPLAYPORT	DP_INT_EG_ML_AUX_P		43 83
I645	DP_INTPNL_ML_AUX	TBTDP_90D	DISPLAYPORT	DP_INT_EG_ML_AUX_N		43 83
I646	DP_INTPNL_ML_AUX	TBTDP_90D	DISPLAYPORT	DP_INT_EG_ML_AUX_C_P		43
I647	DP_INTPNL_ML_AUX	TBTDP_90D	DISPLAYPORT	DP_INT_EG_ML_AUX_C_N		43
I722		TBTDP_90D	DISPLAYPORT	DP_INT_EG_ML_DDC_CLK		83
I723		TBTDP_90D	DISPLAYPORT	DP_INT_EG_ML_DDC_DAT		83
	MASTER NETS - MUX-PANEL					
I607	DP_INTPNL_ML	DP_85D	DISPLAYPORT	DP_INTPNL_ML_P<3..0>	NO_TEST=TRUE	43
I608	DP_INTPNL_ML	DP_85D	DISPLAYPORT	DP_INTPNL_ML_N<3..0>	NO_TEST=TRUE	43
I605	DP_INTPNL_ML	DP_85D	DISPLAYPORT	DP_INTPNL_ML_C_P<3..0>	NO_TEST=TRUE	42 43
I606	DP_INTPNL_ML	DP_85D	DISPLAYPORT	DP_INTPNL_ML_C_N<3..0>	NO_TEST=TRUE	42 43
I609	DP_INTPNL_ML_AUX	TBTDP_90D	DISPLAYPORT	DP_INTPNL_ML_AUX_P		43
I610	DP_INTPNL_ML_AUX	TBTDP_90D	DISPLAYPORT	DP_INTPNL_ML_AUX_N		43
I710	DP_INTPNL_ML_AUX	TBTDP_90D	DISPLAYPORT	DP_INTPNL_ML_AUX_C_P		42 43
I711	DP_INTPNL_ML_AUX	TBTDP_90D	DISPLAYPORT	DP_INTPNL_ML_AUX_C_N		42 43
	INTERNAL DP AUDIO					
I611			HDA	AUDIO_SCLK		42 96
I696			HDA	AUDIO_WS		42 96
I697			HDA	DP_INT_SPDIF_AUDIO		42 54
I698			HDA	AUDIO_MUTE_L		42 96

TBT/DP Net Properties

	Electrical Contraint Set	Physical	Spacing	Net		
	Port A					
I541	TBT_A_R2D1	TBTDP_90D	TBTDP	TBT_A_R2D_C_P<1>	NO_TEST=TRUE	28 31
I542	TBT_A_R2D1	TBTDP_90D	TBTDP	TBT_A_R2D_C_N<1>	NO_TEST=TRUE	28 31
I689	TBT_A_R2D0	TBTDP_90D	TBTDP	TBT_A_R2D_C_P<0>	NO_TEST=TRUE	28 31
I688	TBT_A_R2D0	TBTDP_90D	TBTDP	TBT_A_R2D_C_N<0>	NO_TEST=TRUE	28 31
I544		TBTDP_90D	TBTDP	TBT_A_R2D_P<1..0>	NO_TEST=TRUE	31
I543		TBTDP_90D	TBTDP	TBT_A_R2D_N<1..0>	NO_TEST=TRUE	31
I595	DP_TBTPA_ML1	DP_85D	DISPLAYPORT	DP_TBTPA_ML_C_P<1>	NO_TEST=TRUE	28 31
I596	DP_TBTPA_ML1	DP_85D	DISPLAYPORT	DP_TBTPA_ML_C_N<1>	NO_TEST=TRUE	28 31
I680	DP_TBTPA_ML3	DP_85D	DISPLAYPORT	DP_TBTPA_ML_C_P<3>	NO_TEST=TRUE	28 31
I681	DP_TBTPA_ML3	DP_85D	DISPLAYPORT	DP_TBTPA_ML_C_N<3>	NO_TEST=TRUE	28 31
I545		DP_85D	DISPLAYPORT	DP_TBTPA_ML_P<1>	NO_TEST=TRUE	31
I547		DP_85D	DISPLAYPORT	DP_TBTPA_ML_N<1>	NO_TEST=TRUE	31
I678		DP_85D	DISPLAYPORT	DP_TBTPA_ML_P<3>	NO_TEST=TRUE	31
I679		DP_85D	DISPLAYPORT	DP_TBTPA_ML_N<3>	NO_TEST=TRUE	31
I546		DP_85D	DISPLAYPORT	DP_A_LSX_ML_P<1>		31
I548		DP_85D	DISPLAYPORT	DP_A_LSX_ML_N<1>		31
I549		TBTDP_90D	TBTDP	TBT_A_D2R_C_P<1..0>	NO_TEST=TRUE	31
I550		TBTDP_90D	TBTDP	TBT_A_D2R_C_N<1..0>	NO_TEST=TRUE	31
I564	TBT_A_D2R1	TBTDP_90D	TBTDP	TBT_A_D2R_P<1>	NO_TEST=TRUE	28 31
I563	TBT_A_D2R1	TBTDP_90D	TBTDP	TBT_A_D2R_N<1>	NO_TEST=TRUE	28 31
I675	TBT_A_D2R0	TBTDP_90D	TBTDP	TBT_A_D2R_P<0>	NO_TEST=TRUE	28 31
I674	TBT_A_D2R0	TBTDP_90D	TBTDP	TBT_A_D2R_N<0>	NO_TEST=TRUE	28 31
I599	TBT_A_D2R1	TBTDP_90D	DISPLAYPORT	DP_TBTPA_AUXCH_C_P		28 31
I600	TBT_A_D2R1	TBTDP_90D	DISPLAYPORT	DP_TBTPA_AUXCH_C_N		28 31
I565		TBTDP_90D	DISPLAYPORT	DP_TBTPA_AUXCH_P		31
I566		TBTDP_90D	DISPLAYPORT	DP_TBTPA_AUXCH_N		31
I648	DP_A_AUXCH_DDC	TBTDP_90D	DISPLAYPORT	DP_A_AUXCH_DDC_P		
I649	DP_A_AUXCH_DDC	TBTDP_90D	DISPLAYPORT	DP_A_AUXCH_DDC_N		
I650	TBT_A_D2R1	TBTDP_90D	TBTDP	TBT_A_D2R1_AUXDDC_P		31
I651	TBT_A_D2R1	TBTDP_90D	TBTDP	TBT_A_D2R1_AUXDDC_N		31
I712	TBTPA_DDC_I2C	TBTDP_90D	TBT_I2C	DP_TBTPA_DDC_CLK		31 33
I713	TBTPA_DDC_I2C	TBTDP_90D	TBT_I2C	DP_TBTPA_DDC_DATA		31 33
	Port B					
I652	TBT_B_R2D1	TBTDP_90D	TBTDP	TBT_B_R2D_C_P<1>	NO_TEST=TRUE	28 32
I653	TBT_B_R2D1	TBTDP_90D	TBTDP	TBT_B_R2D_C_N<1>	NO_TEST=TRUE	28 32
I690	TBT_B_R2D0	TBTDP_90D	TBTDP	TBT_B_R2D_C_P<0>	NO_TEST=TRUE	28 32
I691	TBT_B_R2D0	TBTDP_90D	TBTDP	TBT_B_R2D_C_N<0>	NO_TEST=TRUE	28 32
I655		TBTDP_90D	TBTDP	TBT_B_R2D_P<1..0>	NO_TEST=TRUE	32
I654		TBTDP_90D	TBTDP	TBT_B_R2D_N<1..0>	NO_TEST=TRUE	32
I657	DP_TBTPB_ML1	DP_85D	DISPLAYPORT	DP_TBTPB_ML_C_P<1>	NO_TEST=TRUE	28 32
I658	DP_TBTPB_ML1	DP_85D	DISPLAYPORT	DP_TBTPB_ML_C_N<1>	NO_TEST=TRUE	28 32
I685	DP_TBTPB_ML3	DP_85D	DISPLAYPORT	DP_TBTPB_ML_C_P<3>	NO_TEST=TRUE	28 32
I684	DP_TBTPB_ML3	DP_85D	DISPLAYPORT	DP_TBTPB_ML_C_N<3>	NO_TEST=TRUE	28 32
I683		DP_85D	DISPLAYPORT	DP_TBTPB_ML_P<1>	NO_TEST=TRUE	32
I682		DP_85D	DISPLAYPORT	DP_TBTPB_ML_N<1>	NO_TEST=TRUE	32
I656		DP_85D	DISPLAYPORT	DP_TBTPB_ML_P<3>	NO_TEST=TRUE	32
I660		DP_85D	DISPLAYPORT	DP_TBTPB_ML_N<3>	NO_TEST=TRUE	32
I659		DP_85D	DISPLAYPORT	DP_B_LSX_ML_P<1>		32
I662		DP_85D	DISPLAYPORT	DP_B_LSX_ML_N<1>		32
I661		TBTDP_90D	TBTDP	TBT_B_D2R_C_P<1..0>	NO_TEST=TRUE	32
I663		TBTDP_90D	TBTDP	TBT_B_D2R_C_N<1..0>	NO_TEST=TRUE	32
I665	TBT_B_D2R1	TBTDP_90D	TBTDP	TBT_B_D2R_P<1>	NO_TEST=TRUE	28 32
I664	TBT_B_D2R1	TBTDP_90D	TBTDP	TBT_B_D2R_N<1>	NO_TEST=TRUE	28 32
I676	TBT_B_D2R0	TBTDP_90D	TBTDP	TBT_B_D2R_P<0>	NO_TEST=TRUE	28 32
I677	TBT_B_D2R0	TBTDP_90D	TBTDP	TBT_B_D2R_N<0>	NO_TEST=TRUE	28 32
I666	TBT_B_D2R1	TBTDP_90D	DISPLAYPORT	DP_TBTPB_AUXCH_C_P		28 32
I667	TBT_B_D2R1	TBTDP_90D	DISPLAYPORT	DP_TBTPB_AUXCH_C_N		28 32
I669		TBTDP_90D	DISPLAYPORT	DP_TBTPB_AUXCH_P		32
I670		TBTDP_90D	DISPLAYPORT	DP_TBTPB_AUXCH_N		32
I668	DP_B_AUXCH_DDC	TBTDP_90D	DISPLAYPORT	DP_B_AUXCH_DDC_P		
I672	DP_B_AUXCH_DDC	TBTDP_90D	DISPLAYPORT	DP_B_AUXCH_DDC_N		
I671	TBT_B_D2R1	TBTDP_90D	TBTDP	TBT_B_D2R1_AUXDDC_P		32
I673	TBT_B_D2R1	TBTDP_90D	TBTDP	TBT_B_D2R1_AUXDDC_N		32
I714	TBTPB_DDC_I2C	TBTDP_90D	TBT_I2C	DP_TBTPB_DDC_CLK		32 33
I715	TBTPB_DDC_I2C	TBTDP_90D	TBT_I2C	DP_TBTPB_DDC_DATA		32 33

SYNC_MASTER=J78_MLB SYNC_DATE=02/21/2014

PAGE TITLE: TBT/DP Constraints

Apple Inc.

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GDDR5

GDDR5-specific Physical Rules

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
GDDR_45S	*	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	100 MM	=STANDARD	=STANDARD
GDDR_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	100 MM	=STANDARD	=STANDARD
GDDR_80D	*	=80_OHM_DIFF	=80_OHM_DIFF	=80_OHM_DIFF	=80_OHM_DIFF	=80_OHM_DIFF	=80_OHM_DIFF
GDDR_BGA7MM_45S	*	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	7 MM	=STANDARD	=STANDARD
GDDR_BGA5MM_45S	*	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	5 MM	=STANDARD	=STANDARD
GDDR_BGA4MM_45S	*	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	4.9 MM	=STANDARD	=STANDARD
GDDR_BGA5MM_80D	*	=80_OHM_DIFF	=80_OHM_DIFF	=80_OHM_DIFF	4.9 MM	=80_OHM_DIFF	=80_OHM_DIFF
GDDR_BGA3MM_45S	*	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	3 MM	=STANDARD	=STANDARD
GDDR_BGA4MM_80D	*	=80_OHM_DIFF	=80_OHM_DIFF	=80_OHM_DIFF	4 MM	=80_OHM_DIFF	=80_OHM_DIFF

Physical Net Type to Rule Map

NET_PHYSICAL_TYPE	AREA_TYPE	PHYSICAL_RULE_SET
GDDR_MA_PHY	*	GDDR_45S
GDDR_ADBI_PHY	*	GDDR_45S
GDDR_CTRL_PHY	*	GDDR_45S
GDDR_CLK_PHY	*	GDDR_80D
GDDR_DQ_PHY	*	GDDR_45S
GDDR_EDC_PHY	*	GDDR_45S
GDDR_DBI_PHY	*	GDDR_45S
GDDR_WCK_PHY	*	GDDR_80D

NET_PHYSICAL_TYPE	AREA_TYPE	PHYSICAL_RULE_SET
GDDR_MA_PHY	BGA	GDDR_BGA7MM_45S
GDDR_ADBI_PHY	BGA	GDDR_BGA4MM_45S
GDDR_CTRL_PHY	BGA	GDDR_BGA7MM_45S
GDDR_CLK_PHY	BGA	GDDR_BGA5MM_80D
GDDR_DQ_PHY	BGA	GDDR_BGA7MM_45S
GDDR_EDC_PHY	BGA	GDDR_BGA4MM_45S
GDDR_DBI_PHY	BGA	GDDR_BGA4MM_45S
GDDR_WCK_PHY	BGA	GDDR_BGA5MM_80D

NET_PHYSICAL_TYPE	AREA_TYPE	PHYSICAL_RULE_SET
GDDR_MA_PHY	BGA_VRAM	GDDR_BGA3MM_45S
GDDR_ADBI_PHY	BGA_VRAM	GDDR_BGA3MM_45S
GDDR_CTRL_PHY	BGA_VRAM	GDDR_BGA3MM_45S
GDDR_CLK_PHY	BGA_VRAM	GDDR_BGA4MM_80D
GDDR_WCK_PHY	BGA_VRAM	GDDR_BGA4MM_80D

GDDR5-specific Spacing Definitions

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
GDDR_ISO	*	=5X_DIELECTRIC	?
GDDR_ISO	TOP,BOTTOM	=7X_DIELECTRIC	?
GDDR_MA2MA	*	=3X_DIELECTRIC	?
GDDR_MA2MA	TOP,BOTTOM	=5X_DIELECTRIC	?
GDDR_ADBI2ADBI	*	=5X_DIELECTRIC	?
GDDR_ADBI2ADBI	TOP,BOTTOM	=7X_DIELECTRIC	?
GDDR_CTRL2CTRL	*	=3X_DIELECTRIC	?
GDDR_CTRL2CTRL	TOP,BOTTOM	=3X_DIELECTRIC	?
GDDR_CLK2CLK	*	=5X_DIELECTRIC	?
GDDR_CLK2CLK	TOP,BOTTOM	=7X_DIELECTRIC	?

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
GDDR_DQ2DQ	*	=3x_DIELECTRIC	?
GDDR_DQ2DQ	TOP,BOTTOM	=5X_DIELECTRIC	?
GDDR_EDC_ISO	*	=7X_DIELECTRIC	?
GDDR_EDC_ISO	TOP,BOTTOM	=10X_DIELECTRIC	?
GDDR_EDC2EDC	*	=7X_DIELECTRIC	?
GDDR_EDC2EDC	TOP,BOTTOM	=10X_DIELECTRIC	?
GDDR_DBI2DBI	*	=5X_DIELECTRIC	?
GDDR_DBI2DBI	TOP,BOTTOM	=7X_DIELECTRIC	?
GDDR_WCK2WCK	*	=5X_DIELECTRIC	?
GDDR_WCK2WCK	TOP,BOTTOM	=7X_DIELECTRIC	?
GDDR_BGA_ISO	*	=1.3X_DIELECTRIC	?

Constraints (x in {A, B}, y in {0, 1})

Memory Address: MAxy[8:0]

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
GDDR_*_*_MA	*	*	GDDR_ISO
GDDR_*_*_MA	=SAME	*	GDDR_MA2MA

Address Dynamic Bus Inversion: ADBIxy

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
GDDR_*_*_ADBI	*	*	GDDR_ISO
GDDR_*_*_ADBI	=SAME	*	GDDR_ADBI2ADBI

Control: Reset, CKExy, CSxy, WExy, RASxy, CASxy

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
GDDR_CTRL	*	*	GDDR_ISO
GDDR_*_*_CTRL	*	*	GDDR_ISO
GDDR_*_*_CTRL	=SAME	*	GDDR_CTRL2CTRL

Clock: CKxy

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
GDDR_*_*_CLK	*	*	GDDR_ISO
GDDR_*_*_CLK	=SAME	*	GDDR_CLK2CLK

Data: DQxy[31:0]

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
GDDR_*_*_DQ	*	*	GDDR_ISO
GDDR_*_*_DQ	=SAME	*	GDDR_DQ2DQ

Error Detection: EDCxy[3:0]

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
GDDR_*_*_EDC	*	*	GDDR_EDC_ISO
GDDR_*_*_EDC	=SAME	*	GDDR_EDC2EDC

Data Dynamic Bus Inversion: DDBIxy[3:0]

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
GDDR_*_*_DBI	*	*	GDDR_ISO
GDDR_*_*_EDC	=SAME	*	GDDR_DBI2DBI

Forwarded Clock: WCKxy[1:0]

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
GDDR_*_*_WCK	*	*	GDDR_ISO
GDDR_*_*_WCK	=SAME	*	GDDR_WCK2WCK

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
GDDR_*	*	BGA	GDDR_BGA_ISO
GDDR_*	*	BGA_VRAM	GDDR_BGA_ISO

GPU

GPU-specific Physical Rules

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
CLK_GPU_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD

GPU-specific Spacing Definitions

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
CLK_GPU_ISO	*	=4:1_SPACING	?

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
CLK_GPU	*	*	CLK_GPU_ISO

GDDR5 Frame Buffer A

	Electrical Contraint Set	Physical	Spacing	Net		
Memory Address						
I642	GDDR_A0_MA	GDDR_MA_PHY	GDDR_A_0_MA	FB_A0_A<8..0>	NO_TEST=TRUE	77 79
I643	GDDR_A1_MA	GDDR_MA_PHY	GDDR_A_1_MA	FB_A1_A<8..0>	NO_TEST=TRUE	77 79
Address Dynamic Bus Inv						
I644	GDDR_A0_ADBI	GDDR_ADBI_PHY	GDDR_A_0_ADBI	FB_A0_ABI_L	NO_TEST=TRUE	77 79
I645	GDDR_A1_ADBI	GDDR_ADBI_PHY	GDDR_A_1_ADBI	FB_A1_ABI_L	NO_TEST=TRUE	77 79
Control						
I647	GDDR_A0_CTRL	GDDR_CTRL_PHY	GDDR_A_0_CTRL	FB_A0_CKE_L	NO_TEST=TRUE	77 79
I648	GDDR_A0_CTRL	GDDR_CTRL_PHY	GDDR_A_0_CTRL	FB_A0_CS_L	NO_TEST=TRUE	77 79
I649	GDDR_A0_CTRL	GDDR_CTRL_PHY	GDDR_A_0_CTRL	FB_A0_WE_L	NO_TEST=TRUE	77 79
I650	GDDR_A0_CTRL	GDDR_CTRL_PHY	GDDR_A_0_CTRL	FB_A0_CAS_L	NO_TEST=TRUE	77 79
I651	GDDR_A0_CTRL	GDDR_CTRL_PHY	GDDR_A_0_CTRL	FB_A0_RAS_L	NO_TEST=TRUE	77 79
I652	GDDR_A1_CTRL	GDDR_CTRL_PHY	GDDR_A_1_CTRL	FB_A1_CKE_L	NO_TEST=TRUE	77 79
I654	GDDR_A1_CTRL	GDDR_CTRL_PHY	GDDR_A_1_CTRL	FB_A1_CS_L	NO_TEST=TRUE	77 79
I653	GDDR_A1_CTRL	GDDR_CTRL_PHY	GDDR_A_1_CTRL	FB_A1_WE_L	NO_TEST=TRUE	77 79
I655	GDDR_A1_CTRL	GDDR_CTRL_PHY	GDDR_A_1_CTRL	FB_A1_CAS_L	NO_TEST=TRUE	77 79
I656	GDDR_A1_CTRL	GDDR_CTRL_PHY	GDDR_A_1_CTRL	FB_A1_RAS_L	NO_TEST=TRUE	77 79
Clock						
I657	GDDR_A0_CLK	GDDR_CLK_PHY	GDDR_A_0_CLK	FB_A0_CLK_P	NO_TEST=TRUE	77 79
I658	GDDR_A0_CLK	GDDR_CLK_PHY	GDDR_A_0_CLK	FB_A0_CLK_N	NO_TEST=TRUE	77 79
I660	GDDR_A1_CLK	GDDR_CLK_PHY	GDDR_A_1_CLK	FB_A1_CLK_P	NO_TEST=TRUE	77 79
I659	GDDR_A1_CLK	GDDR_CLK_PHY	GDDR_A_1_CLK	FB_A1_CLK_N	NO_TEST=TRUE	77 79
Data						
I661	GDDR_A0_DQ_BYTE0	GDDR_DQ_PHY	GDDR_A_0_DQ	FB_A0_DQ<7..0>	NO_TEST=TRUE	77 79
I663	GDDR_A0_DQ_BYTE1	GDDR_DQ_PHY	GDDR_A_0_DQ	FB_A0_DQ<15..8>	NO_TEST=TRUE	77 79
I662	GDDR_A0_DQ_BYTE2	GDDR_DQ_PHY	GDDR_A_0_DQ	FB_A0_DQ<23..16>	NO_TEST=TRUE	77 79
I664	GDDR_A0_DQ_BYTE3	GDDR_DQ_PHY	GDDR_A_0_DQ	FB_A0_DQ<31..24>	NO_TEST=TRUE	77 79
I666	GDDR_A1_DQ_BYTE0	GDDR_DQ_PHY	GDDR_A_1_DQ	FB_A1_DQ<7..0>	NO_TEST=TRUE	77 79
I665	GDDR_A1_DQ_BYTE1	GDDR_DQ_PHY	GDDR_A_1_DQ	FB_A1_DQ<15..8>	NO_TEST=TRUE	77 79
I667	GDDR_A1_DQ_BYTE2	GDDR_DQ_PHY	GDDR_A_1_DQ	FB_A1_DQ<23..16>	NO_TEST=TRUE	77 79
I668	GDDR_A1_DQ_BYTE3	GDDR_DQ_PHY	GDDR_A_1_DQ	FB_A1_DQ<31..24>	NO_TEST=TRUE	77 79
Error Detection						
I670	GDDR_A0_EDC0	GDDR_EDC_PHY	GDDR_A_0_EDC	FB_A0_EDC<0>	NO_TEST=TRUE	77 79
I669	GDDR_A0_EDC1	GDDR_EDC_PHY	GDDR_A_0_EDC	FB_A0_EDC<1>	NO_TEST=TRUE	77 79
I671	GDDR_A0_EDC2	GDDR_EDC_PHY	GDDR_A_0_EDC	FB_A0_EDC<2>	NO_TEST=TRUE	77 79
I672	GDDR_A0_EDC3	GDDR_EDC_PHY	GDDR_A_0_EDC	FB_A0_EDC<3>	NO_TEST=TRUE	77 79
I673	GDDR_A1_EDC0	GDDR_EDC_PHY	GDDR_A_1_EDC	FB_A1_EDC<0>	NO_TEST=TRUE	77 79
I675	GDDR_A1_EDC1	GDDR_EDC_PHY	GDDR_A_1_EDC	FB_A1_EDC<1>	NO_TEST=TRUE	77 79
I674	GDDR_A1_EDC2	GDDR_EDC_PHY	GDDR_A_1_EDC	FB_A1_EDC<2>	NO_TEST=TRUE	77 79
I676	GDDR_A1_EDC3	GDDR_EDC_PHY	GDDR_A_1_EDC	FB_A1_EDC<3>	NO_TEST=TRUE	77 79
Data Dynamic Bus Inv						
I677	GDDR_A0_DBI0	GDDR_DBI_PHY	GDDR_A_0_DBI	FB_A0_DBI_L<0>	NO_TEST=TRUE	77 79
I678	GDDR_A0_DBI1	GDDR_DBI_PHY	GDDR_A_0_DBI	FB_A0_DBI_L<1>	NO_TEST=TRUE	77 79
I680	GDDR_A0_DBI2	GDDR_DBI_PHY	GDDR_A_0_DBI	FB_A0_DBI_L<2>	NO_TEST=TRUE	77 79
I679	GDDR_A0_DBI3	GDDR_DBI_PHY	GDDR_A_0_DBI	FB_A0_DBI_L<3>	NO_TEST=TRUE	77 79
I682	GDDR_A1_DBI0	GDDR_DBI_PHY	GDDR_A_1_DBI	FB_A1_DBI_L<0>	NO_TEST=TRUE	77 79
I681	GDDR_A1_DBI1	GDDR_DBI_PHY	GDDR_A_1_DBI	FB_A1_DBI_L<1>	NO_TEST=TRUE	77 79
I683	GDDR_A1_DBI2	GDDR_DBI_PHY	GDDR_A_1_DBI	FB_A1_DBI_L<2>	NO_TEST=TRUE	77 79
I684	GDDR_A1_DBI3	GDDR_DBI_PHY	GDDR_A_1_DBI	FB_A1_DBI_L<3>	NO_TEST=TRUE	77 79
Forwarded Clock						
I685	GDDR_A0_WCK0	GDDR_WCK_PHY	GDDR_A_0_WCK	FB_A0_WCLK_P<0>	NO_TEST=TRUE	77 79
I687	GDDR_A0_WCK0	GDDR_WCK_PHY	GDDR_A_0_WCK	FB_A0_WCLK_N<0>	NO_TEST=TRUE	77 79
I686	GDDR_A0_WCK1	GDDR_WCK_PHY	GDDR_A_0_WCK	FB_A0_WCLK_P<1>	NO_TEST=TRUE	77 79
I688	GDDR_A0_WCK1	GDDR_WCK_PHY	GDDR_A_0_WCK	FB_A0_WCLK_N<1>	NO_TEST=TRUE	77 79
I689	GDDR_A1_WCK0	GDDR_WCK_PHY	GDDR_A_1_WCK	FB_A1_WCLK_P<0>	NO_TEST=TRUE	77 79
I690	GDDR_A1_WCK0	GDDR_WCK_PHY	GDDR_A_1_WCK	FB_A1_WCLK_N<0>	NO_TEST=TRUE	77 79
I692	GDDR_A1_WCK1	GDDR_WCK_PHY	GDDR_A_1_WCK	FB_A1_WCLK_P<1>	NO_TEST=TRUE	77 79
I691	GDDR_A1_WCK1	GDDR_WCK_PHY	GDDR_A_1_WCK	FB_A1_WCLK_N<1>	NO_TEST=TRUE	77 79

GDDR5 Frame Buffer B

	Electrical Contraint Set	Physical	Spacing	Net		
Memory Address						
I695	GDDR_B0_MA	GDDR_MA_PHY	GDDR_B_0_MA	FB_B0_A<8..0>	NO_TEST=TRUE	77 80
I694	GDDR_B1_MA	GDDR_MA_PHY	GDDR_B_1_MA	FB_B1_A<8..0>	NO_TEST=TRUE	77 80
Address Dynamic Bus Inv						
I696	GDDR_B0_ADBI	GDDR_ADBI_PHY	GDDR_B_0_ADBI	FB_B0_ABI_L	NO_TEST=TRUE	77 80
I697	GDDR_B1_ADBI	GDDR_ADBI_PHY	GDDR_B_1_ADBI	FB_B1_ABI_L	NO_TEST=TRUE	77 80
Control						
I698	GDDR_B0_CTRL	GDDR_CTRL_PHY	GDDR_B_0_CTRL	FB_B0_CKE_L	NO_TEST=TRUE	77 80
I701	GDDR_B0_CTRL	GDDR_CTRL_PHY	GDDR_B_0_CTRL	FB_B0_CS_L	NO_TEST=TRUE	77 80
I700	GDDR_B0_CTRL	GDDR_CTRL_PHY	GDDR_B_0_CTRL	FB_B0_WE_L	NO_TEST=TRUE	77 80
I702	GDDR_B0_CTRL	GDDR_CTRL_PHY	GDDR_B_0_CTRL	FB_B0_CAS_L	NO_TEST=TRUE	77 80
I703	GDDR_B0_CTRL	GDDR_CTRL_PHY	GDDR_B_0_CTRL	FB_B0_RAS_L	NO_TEST=TRUE	77 80
I704	GDDR_B1_CTRL	GDDR_CTRL_PHY	GDDR_B_1_CTRL	FB_B1_CKE_L	NO_TEST=TRUE	77 80
I705	GDDR_B1_CTRL	GDDR_CTRL_PHY	GDDR_B_1_CTRL	FB_B1_CS_L	NO_TEST=TRUE	77 80
I706	GDDR_B1_CTRL	GDDR_CTRL_PHY	GDDR_B_1_CTRL	FB_B1_WE_L	NO_TEST=TRUE	77 80
I708	GDDR_B1_CTRL	GDDR_CTRL_PHY	GDDR_B_1_CTRL	FB_B1_CAS_L	NO_TEST=TRUE	77 80
I707	GDDR_B1_CTRL	GDDR_CTRL_PHY	GDDR_B_1_CTRL	FB_B1_RAS_L	NO_TEST=TRUE	77 80
Clock						
I693	GDDR_B0_CLK	GDDR_CLK_PHY	GDDR_B_0_CLK	FB_B0_CLK_P	NO_TEST=TRUE	77 80
I710	GDDR_B0_CLK	GDDR_CLK_PHY	GDDR_B_0_CLK	FB_B0_CLK_N	NO_TEST=TRUE	77 80
I709	GDDR_B1_CLK	GDDR_CLK_PHY	GDDR_B_1_CLK	FB_B1_CLK_P	NO_TEST=TRUE	77 80
I711	GDDR_B1_CLK	GDDR_CLK_PHY	GDDR_B_1_CLK	FB_B1_CLK_N	NO_TEST=TRUE	77 80
Data						
I713	GDDR_B0_DQ_BYTE0	GDDR_DQ_PHY	GDDR_B_0_DQ	FB_B0_DQ<7..0>	NO_TEST=TRUE	77 80
I712	GDDR_B0_DQ_BYTE1	GDDR_DQ_PHY	GDDR_B_0_DQ	FB_B0_DQ<15..8>	NO_TEST=TRUE	77 80
I714	GDDR_B0_DQ_BYTE2	GDDR_DQ_PHY	GDDR_B_0_DQ	FB_B0_DQ<23..16>	NO_TEST=TRUE	77 80
I715	GDDR_B0_DQ_BYTE3	GDDR_DQ_PHY	GDDR_B_0_DQ	FB_B0_DQ<31..24>	NO_TEST=TRUE	77 80
I716	GDDR_B1_DQ_BYTE0	GDDR_DQ_PHY	GDDR_B_1_DQ	FB_B1_DQ<7..0>	NO_TEST=TRUE	77 80
I718	GDDR_B1_DQ_BYTE1	GDDR_DQ_PHY	GDDR_B_1_DQ	FB_B1_DQ<15..8>	NO_TEST=TRUE	77 80
I717	GDDR_B1_DQ_BYTE2	GDDR_DQ_PHY	GDDR_B_1_DQ	FB_B1_DQ<23..16>	NO_TEST=TRUE	77 80
I719	GDDR_B1_DQ_BYTE3	GDDR_DQ_PHY	GDDR_B_1_DQ	FB_B1_DQ<31..24>	NO_TEST=TRUE	77 80
Error Detection						
I720	GDDR_B0_EDC0	GDDR_EDC_PHY	GDDR_B_0_EDC	FB_B0_EDC<0>	NO_TEST=TRUE	77 80
I721	GDDR_B0_EDC1	GDDR_EDC_PHY	GDDR_B_0_EDC	FB_B0_EDC<1>	NO_TEST=TRUE	77 80
I723	GDDR_B0_EDC2	GDDR_EDC_PHY	GDDR_B_0_EDC	FB_B0_EDC<2>	NO_TEST=TRUE	77 80
I722	GDDR_B0_EDC3	GDDR_EDC_PHY	GDDR_B_0_EDC	FB_B0_EDC<3>	NO_TEST=TRUE	77 80
I725	GDDR_B1_EDC0	GDDR_EDC_PHY	GDDR_B_1_EDC	FB_B1_EDC<0>	NO_TEST=TRUE	77 80
I724	GDDR_B1_EDC1	GDDR_EDC_PHY	GDDR_B_1_EDC	FB_B1_EDC<1>	NO_TEST=TRUE	77 80
I726	GDDR_B1_EDC2	GDDR_EDC_PHY	GDDR_B_1_EDC	FB_B1_EDC<2>	NO_TEST=TRUE	77 80
I727	GDDR_B1_EDC3	GDDR_EDC_PHY	GDDR_B_1_EDC	FB_B1_EDC<3>	NO_TEST=TRUE	77 80
Data Dynamic Bus Inv						
I729	GDDR_B0_DBI0	GDDR_DBI_PHY	GDDR_B_0_DBI	FB_B0_DBI_L<0>	NO_TEST=TRUE	77 80
I728	GDDR_B0_DBI1	GDDR_DBI_PHY	GDDR_B_0_DBI	FB_B0_DBI_L<1>	NO_TEST=TRUE	77 80
I730	GDDR_B0_DBI2	GDDR_DBI_PHY	GDDR_B_0_DBI	FB_B0_DBI_L<2>	NO_TEST=TRUE	77 80
I731	GDDR_B0_DBI3	GDDR_DBI_PHY	GDDR_B_0_DBI	FB_B0_DBI_L<3>	NO_TEST=TRUE	77 80
I732	GDDR_B1_DBI0	GDDR_DBI_PHY	GDDR_B_1_DBI	FB_B1_DBI_L<0>	NO_TEST=TRUE	77 80
I733	GDDR_B1_DBI1	GDDR_DBI_PHY	GDDR_B_1_DBI	FB_B1_DBI_L<1>	NO_TEST=TRUE	77 80
I734	GDDR_B1_DBI2	GDDR_DBI_PHY	GDDR_B_1_DBI	FB_B1_DBI_L<2>	NO_TEST=TRUE	77 80
I735	GDDR_B1_DBI3	GDDR_DBI_PHY	GDDR_B_1_DBI	FB_B1_DBI_L<3>	NO_TEST=TRUE	77 80
Forwarded Clock						
I736	GDDR_B0_WCK0	GDDR_WCK_PHY	GDDR_B_0_WCK	FB_B0_WCLK_P<0>	NO_TEST=TRUE	77 80
I737	GDDR_B0_WCK0	GDDR_WCK_PHY	GDDR_B_0_WCK	FB_B0_WCLK_N<0>	NO_TEST=TRUE	77 80
I739	GDDR_B0_WCK1	GDDR_WCK_PHY	GDDR_B_0_WCK	FB_B0_WCLK_P<1>	NO_TEST=TRUE	77 80
I738	GDDR_B0_WCK1	GDDR_WCK_PHY	GDDR_B_0_WCK	FB_B0_WCLK_N<1>	NO_TEST=TRUE	77 80
I740	GDDR_B1_WCK0	GDDR_WCK_PHY	GDDR_B_1_WCK	FB_B1_WCLK_P<0>	NO_TEST=TRUE	77 80
I741	GDDR_B1_WCK0	GDDR_WCK_PHY	GDDR_B_1_WCK	FB_B1_WCLK_N<0>	NO_TEST=TRUE	77 80
I743	GDDR_B1_WCK1	GDDR_WCK_PHY	GDDR_B_1_WCK	FB_B1_WCLK_P<1>	NO_TEST=TRUE	77 80
I742	GDDR_B1_WCK1	GDDR_WCK_PHY	GDDR_B_1_WCK	FB_B1_WCLK_N<1>	NO_TEST=TRUE	77 80

GPU

Electrical Contraint Set	Physical	Spacing	Net	
Clocks				
	CLK_GPU_55S	CLK_GPU	TEST_GFX_JTAG_TCK	
	CLK_GPU_55S	CLK_GPU	GPU_OSC_27M_XTAL_IN	84
	CLK_GPU_55S	CLK_GPU	GPU_OSC_27M_XTAL_OUT	84
	CLK_GPU_55S	CLK_GPU	GFX_XTAL_IN_27M	84
	CLK_GPU_55S	CLK_GPU	GFX_XTAL_OUT_27M	84
SMB				
	SMB_PHY	SMB	GPU_SMB_CLK	85
	SMB_PHY	SMB	GPU_SMB_DAT	85
PCIe Compensation				
	PCIE_50S	COMP_PCIE	PCIE_CALR_RX	
	PCIE_50S	COMP_PCIE	PCIE_CALR_TX	
MISC				
	XDP_PHY	XDP	TEST_GFX_JTAG_TDO	
	XDP_PHY	XDP	TEST_GFX_JTAG_TDI	
	XDP_PHY	XDP	TEST_GFX_JTAG_TMS	
	XDP_PHY	XDP	TEST_GFX_JTAG_TRST_L	

Frame Buffer Reset

Electrical Contraint Set	Physical	Spacing	Net	
Reset				
GDDR_AB_RESET	GDDR_50S	GDDR_CTRL	FB_AB_RST_RR	77
GDDR_AB_RESET	GDDR_50S	GDDR_CTRL	FB_AB_RST_R	77
GDDR_AB_RESET	GDDR_50S	GDDR_CTRL	FB_AB_RESET_L	77 79 80
GDDR_CD_RESET	GDDR_50S	GDDR_CTRL	FB_CD_RST_RR	78
GDDR_CD_RESET	GDDR_50S	GDDR_CTRL	FB_CD_RST_R	78
GDDR_CD_RESET	GDDR_50S	GDDR_CTRL	FB_CD_RESET_L	78 81 82

SYNC_MASTER=J78_MLB SYNC_DATE=02/21/2014

PAGE TITLE

GDDR5/GPU Constraints

Apple Inc.

DRAWING NUMBER 051-1635 SIZE D

REVISION 5.0.0

BRANCH proto2

GDDR5 FRAME BUFFER C

	Electrical Contraint Set	Physical	Spacing			
	Memory Address					
I642	GDDR_C0_MA	GDDR_MA_PHY	GDDR_C_0_MA	FB_C0_A<8..0>	NO_TEST=TRUE	78 81
I643	GDDR_C1_MA	GDDR_MA_PHY	GDDR_C_1_MA	FB_C1_A<8..0>	NO_TEST=TRUE	78 81
	Address Dynamic Bus Inv					
I644	GDDR_C0_ADBI	GDDR_ADBI_PHY	GDDR_C_0_ADBI	FB_C0_ABI_L	NO_TEST=TRUE	78 81
I645	GDDR_C1_ADBI	GDDR_ADBI_PHY	GDDR_C_1_ADBI	FB_C1_ABI_L	NO_TEST=TRUE	78 81
	Control					
I647	GDDR_C0_CTRL	GDDR_CTRL_PHY	GDDR_C_0_CTRL	FB_C0_CKE_L	NO_TEST=TRUE	78 81
I648	GDDR_C0_CTRL	GDDR_CTRL_PHY	GDDR_C_0_CTRL	FB_C0_CS_L	NO_TEST=TRUE	78 81
I649	GDDR_C0_CTRL	GDDR_CTRL_PHY	GDDR_C_0_CTRL	FB_C0_WE_L	NO_TEST=TRUE	78 81
I650	GDDR_C0_CTRL	GDDR_CTRL_PHY	GDDR_C_0_CTRL	FB_C0_CAS_L	NO_TEST=TRUE	78 81
I651	GDDR_C0_CTRL	GDDR_CTRL_PHY	GDDR_C_0_CTRL	FB_C0_RAS_L	NO_TEST=TRUE	78 81
I652	GDDR_C1_CTRL	GDDR_CTRL_PHY	GDDR_C_1_CTRL	FB_C1_CKE_L	NO_TEST=TRUE	78 81
I654	GDDR_C1_CTRL	GDDR_CTRL_PHY	GDDR_C_1_CTRL	FB_C1_CS_L	NO_TEST=TRUE	78 81
I653	GDDR_C1_CTRL	GDDR_CTRL_PHY	GDDR_C_1_CTRL	FB_C1_WE_L	NO_TEST=TRUE	78 81
I655	GDDR_C1_CTRL	GDDR_CTRL_PHY	GDDR_C_1_CTRL	FB_C1_CAS_L	NO_TEST=TRUE	78 81
I656	GDDR_C1_CTRL	GDDR_CTRL_PHY	GDDR_C_1_CTRL	FB_C1_RAS_L	NO_TEST=TRUE	78 81
	Clock					
I657	GDDR_C0_CLK	GDDR_CLK_PHY	GDDR_C_0_CLK	FB_C0_CLK_P	NO_TEST=TRUE	78 81
I658	GDDR_C0_CLK	GDDR_CLK_PHY	GDDR_C_0_CLK	FB_C0_CLK_N	NO_TEST=TRUE	78 81
I660	GDDR_C1_CLK	GDDR_CLK_PHY	GDDR_C_1_CLK	FB_C1_CLK_P	NO_TEST=TRUE	78 81
I659	GDDR_C1_CLK	GDDR_CLK_PHY	GDDR_C_1_CLK	FB_C1_CLK_N	NO_TEST=TRUE	78 81
	Data					
I661	GDDR_C0_DQ_BYTE0	GDDR_DQ_PHY	GDDR_C_0_DQ	FB_C0_DQ<7..0>	NO_TEST=TRUE	78 81
I663	GDDR_C0_DQ_BYTE1	GDDR_DQ_PHY	GDDR_C_0_DQ	FB_C0_DQ<15..8>	NO_TEST=TRUE	78 81
I662	GDDR_C0_DQ_BYTE2	GDDR_DQ_PHY	GDDR_C_0_DQ	FB_C0_DQ<23..16>	NO_TEST=TRUE	78 81
I664	GDDR_C0_DQ_BYTE3	GDDR_DQ_PHY	GDDR_C_0_DQ	FB_C0_DQ<31..24>	NO_TEST=TRUE	78 81
I666	GDDR_C1_DQ_BYTE0	GDDR_DQ_PHY	GDDR_C_1_DQ	FB_C1_DQ<7..0>	NO_TEST=TRUE	78 81
I665	GDDR_C1_DQ_BYTE1	GDDR_DQ_PHY	GDDR_C_1_DQ	FB_C1_DQ<15..8>	NO_TEST=TRUE	78 81
I667	GDDR_C1_DQ_BYTE2	GDDR_DQ_PHY	GDDR_C_1_DQ	FB_C1_DQ<23..16>	NO_TEST=TRUE	78 81
I668	GDDR_C1_DQ_BYTE3	GDDR_DQ_PHY	GDDR_C_1_DQ	FB_C1_DQ<31..24>	NO_TEST=TRUE	78 81
	Error Detection					
I670	GDDR_C0_EDC0	GDDR_EDC_PHY	GDDR_C_0_EDC	FB_C0_EDC<0>	NO_TEST=TRUE	78 81
I669	GDDR_C0_EDC1	GDDR_EDC_PHY	GDDR_C_0_EDC	FB_C0_EDC<1>	NO_TEST=TRUE	78 81
I671	GDDR_C0_EDC2	GDDR_EDC_PHY	GDDR_C_0_EDC	FB_C0_EDC<2>	NO_TEST=TRUE	78 81
I672	GDDR_C0_EDC3	GDDR_EDC_PHY	GDDR_C_0_EDC	FB_C0_EDC<3>	NO_TEST=TRUE	78 81
I673	GDDR_C1_EDC0	GDDR_EDC_PHY	GDDR_C_1_EDC	FB_C1_EDC<0>	NO_TEST=TRUE	78 81
I675	GDDR_C1_EDC1	GDDR_EDC_PHY	GDDR_C_1_EDC	FB_C1_EDC<1>	NO_TEST=TRUE	78 81
I674	GDDR_C1_EDC2	GDDR_EDC_PHY	GDDR_C_1_EDC	FB_C1_EDC<2>	NO_TEST=TRUE	78 81
I676	GDDR_C1_EDC3	GDDR_EDC_PHY	GDDR_C_1_EDC	FB_C1_EDC<3>	NO_TEST=TRUE	78 81
	Data Dynamic Bus Inv					
I677	GDDR_C0_DBI0	GDDR_DBI_PHY	GDDR_C_0_DBI	FB_C0_DBI_L<0>	NO_TEST=TRUE	78 81
I678	GDDR_C0_DBI1	GDDR_DBI_PHY	GDDR_C_0_DBI	FB_C0_DBI_L<1>	NO_TEST=TRUE	78 81
I680	GDDR_C0_DBI2	GDDR_DBI_PHY	GDDR_C_0_DBI	FB_C0_DBI_L<2>	NO_TEST=TRUE	78 81
I679	GDDR_C0_DBI3	GDDR_DBI_PHY	GDDR_C_0_DBI	FB_C0_DBI_L<3>	NO_TEST=TRUE	78 81
I682	GDDR_C1_DBI0	GDDR_DBI_PHY	GDDR_C_1_DBI	FB_C1_DBI_L<0>	NO_TEST=TRUE	78 81
I681	GDDR_C1_DBI1	GDDR_DBI_PHY	GDDR_C_1_DBI	FB_C1_DBI_L<1>	NO_TEST=TRUE	78 81
I683	GDDR_C1_DBI2	GDDR_DBI_PHY	GDDR_C_1_DBI	FB_C1_DBI_L<2>	NO_TEST=TRUE	78 81
I684	GDDR_C1_DBI3	GDDR_DBI_PHY	GDDR_C_1_DBI	FB_C1_DBI_L<3>	NO_TEST=TRUE	78 81
	Forwarded Clock					
I685	GDDR_C0_WCK0	GDDR_WCK_PHY	GDDR_C_0_WCK	FB_C0_WCLK_P<0>	NO_TEST=TRUE	78 81
I687	GDDR_C0_WCK0	GDDR_WCK_PHY	GDDR_C_0_WCK	FB_C0_WCLK_N<0>	NO_TEST=TRUE	78 81
I686	GDDR_C0_WCK1	GDDR_WCK_PHY	GDDR_C_0_WCK	FB_C0_WCLK_P<1>	NO_TEST=TRUE	78 81
I688	GDDR_C0_WCK1	GDDR_WCK_PHY	GDDR_C_0_WCK	FB_C0_WCLK_N<1>	NO_TEST=TRUE	78 81
I689	GDDR_C1_WCK0	GDDR_WCK_PHY	GDDR_C_1_WCK	FB_C1_WCLK_P<0>	NO_TEST=TRUE	78 81
I690	GDDR_C1_WCK0	GDDR_WCK_PHY	GDDR_C_1_WCK	FB_C1_WCLK_N<0>	NO_TEST=TRUE	78 81
I692	GDDR_C1_WCK1	GDDR_WCK_PHY	GDDR_C_1_WCK	FB_C1_WCLK_P<1>	NO_TEST=TRUE	78 81
I691	GDDR_C1_WCK1	GDDR_WCK_PHY	GDDR_C_1_WCK	FB_C1_WCLK_N<1>	NO_TEST=TRUE	78 81

GDDR5 FRAME BUFFER D

	Electrical Contraint Set	Physical	Spacing			
	Memory Address					
I695	GDDR_D0_MA	GDDR_MA_PHY	GDDR_D_0_MA	FB_D0_A<8..0>	NO_TEST=TRUE	78 82
I694	GDDR_D1_MA	GDDR_MA_PHY	GDDR_D_1_MA	FB_D1_A<8..0>	NO_TEST=TRUE	78 82
	Address Dynamic Bus Inv					
I696	GDDR_D0_ADBI	GDDR_ADBI_PHY	GDDR_D_0_ADBI	FB_D0_ABI_L	NO_TEST=TRUE	78 82
I697	GDDR_D1_ADBI	GDDR_ADBI_PHY	GDDR_D_1_ADBI	FB_D1_ABI_L	NO_TEST=TRUE	78 82
	Control					
I698	GDDR_D0_CTRL	GDDR_CTRL_PHY	GDDR_D_0_CTRL	FB_D0_CKE_L	NO_TEST=TRUE	78 82
I701	GDDR_D0_CTRL	GDDR_CTRL_PHY	GDDR_D_0_CTRL	FB_D0_CS_L	NO_TEST=TRUE	78 82
I700	GDDR_D0_CTRL	GDDR_CTRL_PHY	GDDR_D_0_CTRL	FB_D0_WE_L	NO_TEST=TRUE	78 82
I702	GDDR_D0_CTRL	GDDR_CTRL_PHY	GDDR_D_0_CTRL	FB_D0_CAS_L	NO_TEST=TRUE	78 82
I703	GDDR_D0_CTRL	GDDR_CTRL_PHY	GDDR_D_0_CTRL	FB_D0_RAS_L	NO_TEST=TRUE	78 82
I704	GDDR_D1_CTRL	GDDR_CTRL_PHY	GDDR_D_1_CTRL	FB_D1_CKE_L	NO_TEST=TRUE	78 82
I705	GDDR_D1_CTRL	GDDR_CTRL_PHY	GDDR_D_1_CTRL	FB_D1_CS_L	NO_TEST=TRUE	78 82
I706	GDDR_D1_CTRL	GDDR_CTRL_PHY	GDDR_D_1_CTRL	FB_D1_WE_L	NO_TEST=TRUE	78 82
I708	GDDR_D1_CTRL	GDDR_CTRL_PHY	GDDR_D_1_CTRL	FB_D1_CAS_L	NO_TEST=TRUE	78 82
I707	GDDR_D1_CTRL	GDDR_CTRL_PHY	GDDR_D_1_CTRL	FB_D1_RAS_L	NO_TEST=TRUE	78 82
	Clock					
I693	GDDR_D0_CLK	GDDR_CLK_PHY	GDDR_D_0_CLK	FB_D0_CLK_P	NO_TEST=TRUE	78 82
I710	GDDR_D0_CLK	GDDR_CLK_PHY	GDDR_D_0_CLK	FB_D0_CLK_N	NO_TEST=TRUE	78 82
I709	GDDR_D1_CLK	GDDR_CLK_PHY	GDDR_D_1_CLK	FB_D1_CLK_P	NO_TEST=TRUE	78 82
I711	GDDR_D1_CLK	GDDR_CLK_PHY	GDDR_D_1_CLK	FB_D1_CLK_N	NO_TEST=TRUE	78 82
	Data					
I713	GDDR_D0_DQ_BYTE0	GDDR_DQ_PHY	GDDR_D_0_DQ	FB_D0_DQ<7..0>	NO_TEST=TRUE	78 82
I712	GDDR_D0_DQ_BYTE1	GDDR_DQ_PHY	GDDR_D_0_DQ	FB_D0_DQ<15..8>	NO_TEST=TRUE	78 82
I714	GDDR_D0_DQ_BYTE2	GDDR_DQ_PHY	GDDR_D_0_DQ	FB_D0_DQ<23..16>	NO_TEST=TRUE	78 82
I715	GDDR_D0_DQ_BYTE3	GDDR_DQ_PHY	GDDR_D_0_DQ	FB_D0_DQ<31..24>	NO_TEST=TRUE	78 82
I716	GDDR_D1_DQ_BYTE0	GDDR_DQ_PHY	GDDR_D_1_DQ	FB_D1_DQ<7..0>	NO_TEST=TRUE	78 82
I718	GDDR_D1_DQ_BYTE1	GDDR_DQ_PHY	GDDR_D_1_DQ	FB_D1_DQ<15..8>	NO_TEST=TRUE	78 82
I717	GDDR_D1_DQ_BYTE2	GDDR_DQ_PHY	GDDR_D_1_DQ	FB_D1_DQ<23..16>	NO_TEST=TRUE	78 82
I719	GDDR_D1_DQ_BYTE3	GDDR_DQ_PHY	GDDR_D_1_DQ	FB_D1_DQ<31..24>	NO_TEST=TRUE	78 82
	Error Detection					
I720	GDDR_D0_EDC0	GDDR_EDC_PHY	GDDR_D_0_EDC	FB_D0_EDC<0>	NO_TEST=TRUE	78 82
I721	GDDR_D0_EDC1	GDDR_EDC_PHY	GDDR_D_0_EDC	FB_D0_EDC<1>	NO_TEST=TRUE	78 82
I723	GDDR_D0_EDC2	GDDR_EDC_PHY	GDDR_D_0_EDC	FB_D0_EDC<2>	NO_TEST=TRUE	78 82
I722	GDDR_D0_EDC3	GDDR_EDC_PHY	GDDR_D_0_EDC	FB_D0_EDC<3>	NO_TEST=TRUE	78 82
I725	GDDR_D1_EDC0	GDDR_EDC_PHY	GDDR_D_1_EDC	FB_D1_EDC<0>	NO_TEST=TRUE	78 82
I724	GDDR_D1_EDC1	GDDR_EDC_PHY	GDDR_D_1_EDC	FB_D1_EDC<1>	NO_TEST=TRUE	78 82
I726	GDDR_D1_EDC2	GDDR_EDC_PHY	GDDR_D_1_EDC	FB_D1_EDC<2>	NO_TEST=TRUE	78 82
I727	GDDR_D1_EDC3	GDDR_EDC_PHY	GDDR_D_1_EDC	FB_D1_EDC<3>	NO_TEST=TRUE	78 82
	Data Dynamic Bus Inv					
I729	GDDR_D0_DBI0	GDDR_DBI_PHY	GDDR_D_0_DBI	FB_D0_DBI_L<0>	NO_TEST=TRUE	78 82
I728	GDDR_D0_DBI1	GDDR_DBI_PHY	GDDR_D_0_DBI	FB_D0_DBI_L<1>	NO_TEST=TRUE	78 82
I730	GDDR_D0_DBI2	GDDR_DBI_PHY	GDDR_D_0_DBI	FB_D0_DBI_L<2>	NO_TEST=TRUE	78 82
I731	GDDR_D0_DBI3	GDDR_DBI_PHY	GDDR_D_0_DBI	FB_D0_DBI_L<3>	NO_TEST=TRUE	78 82
I732	GDDR_D1_DBI0	GDDR_DBI_PHY	GDDR_D_1_DBI	FB_D1_DBI_L<0>	NO_TEST=TRUE	78 82
I733	GDDR_D1_DBI1	GDDR_DBI_PHY	GDDR_D_1_DBI	FB_D1_DBI_L<1>	NO_TEST=TRUE	78 82
I734	GDDR_D1_DBI2	GDDR_DBI_PHY	GDDR_D_1_DBI	FB_D1_DBI_L<2>	NO_TEST=TRUE	78 82
I735	GDDR_D1_DBI3	GDDR_DBI_PHY	GDDR_D_1_DBI	FB_D1_DBI_L<3>	NO_TEST=TRUE	78 82
	Forwarded Clock					
I736	GDDR_D0_WCK0	GDDR_WCK_PHY	GDDR_D_0_WCK	FB_D0_WCLK_P<0>	NO_TEST=TRUE	78 82
I737	GDDR_D0_WCK0	GDDR_WCK_PHY	GDDR_D_0_WCK	FB_D0_WCLK_N<0>	NO_TEST=TRUE	78 82
I739	GDDR_D0_WCK1	GDDR_WCK_PHY	GDDR_D_0_WCK	FB_D0_WCLK_P<1>	NO_TEST=TRUE	78 82
I738	GDDR_D0_WCK1	GDDR_WCK_PHY	GDDR_D_0_WCK	FB_D0_WCLK_N<1>	NO_TEST=TRUE	78 82
I740	GDDR_D1_WCK0	GDDR_WCK_PHY	GDDR_D_1_WCK	FB_D1_WCLK_P<0>	NO_TEST=TRUE	78 82
I741	GDDR_D1_WCK0	GDDR_WCK_PHY	GDDR_D_1_WCK	FB_D1_WCLK_N<0>	NO_TEST=TRUE	78 82
I743	GDDR_D1_WCK1	GDDR_WCK_PHY	GDDR_D_1_WCK	FB_D1_WCLK_P<1>	NO_TEST=TRUE	78 82
I742	GDDR_D1_WCK1	GDDR_WCK_PHY	GDDR_D_1_WCK	FB_D1_WCLK_N<1>	NO_TEST=TRUE	78 82

Backlight Controller

BLC-specific Physical Rules

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
BLC_P6MM	*	Y	0.600 MM	0.100 MM	3.0 MM	=STANDARD	=STANDARD
BLC_P3MM	*	Y	0.300 MM	0.100 MM	3.0 MM	=STANDARD	=STANDARD
BLC_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD
BLC_DIFFPAIR	*	Y	0.2 MM	=STANDARD	=STANDARD	0.1 MM	0.085 MM

Physical Net Type to Rule Map

NET_PHYSICAL_TYPE	AREA_TYPE	PHYSICAL_RULE_SET
POWER_BLC	*	BLC_P6MM
POWER_BLC_RET	*	BLC_P3MM
BLC_CTL_PHY	*	BLC_P3MM
BLC_MISC_PHY	*	BLC_55S
BLC_SNS_DIFF	*	BLC_DIFFPAIR

BLC-specific Spacing Definitions
BLC High Voltage Output

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
BLC_HV_ISO	*	0.45 MM	2000

Constraints
BLC High Voltage Output

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
BLC_HV	BLC_CTL	*	BLC_CTL_ISO
BLC_HV	BLC_HV	*	BLC_CTL_ISO
BLC_HV	*	*	BLC_HV_ISO

BLC Baddies

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
PHASE_ISO	*	=8:1_SPACING	3000
PHASE_SW2SW	*	=1:1_SPACING	1000
PHASE_SW2GND	*	=2:1_SPACING	1000

BLC Baddies

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
BLC_PHASE	*	*	PHASE_ISO
BLC_PHASE	BLC_PHASE	*	PHASE_SW2SW
BLC_PHASE	GND	*	PHASE_SW2GND

BLC Control

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
BLC_CTL_ISO	*	=3:1_SPACING	?

BLC Control

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
BLC_CTL	*	*	BLC_CTL_ISO

Is it chel'oh or sel'oh? Neither! It's SIT'arrr

	Physical	Spacing	Voltage	DIDT	NO_TEST	Net	Pages
Input Bus							
I802	POWER	POWER	12V			PP12V_BKLT_SNS	69
I749	POWER	POWER	12V			PP12V_BKLT_FUSED	69
I750	POWER	POWER	12V			PP12V_S0_BKLT_FILT	69
I772	POWER	POWER	12V			PP12V_S0_BKLT_PWR	69 70
I794	POWER	POWER	12V			PP12V_S0_BKLT_PWR_R	69
I795	POWER	POWER	5V			PP5V_S0_BKLT_VDDA	69
I813	POWER	POWER	5V			PP5V_S0_BKLT_VDDD	69
I796	POWER	POWER	3.3V			PP3V3_S0_BKLT_VDDIO	69
Local Ground							
I821	GND	GND	0V			BKLT_GNDA	69
I822	GND	GND	0V			BKLT_GNDD	69
I830	GND	GND	0V			BKLT_GND_BST_GD	69
Backlight							
I752	POWER_BLC	BLC_PHASE	80V	TRUE		BKLT_PHASE	69
I755	BLC_CTL_PHY	BLC_PHASE	80V	TRUE		BKLT_SNUBBER	69
I789	POWER_BLC_RET	BLC_HV				LED_RETURN_1_FB	69 70
I788	POWER_BLC_RET	BLC_HV				LED_RETURN_2_FB	69 70
I790	POWER_BLC_RET	BLC_HV				LED_RETURN_3_FB	69 70
I791	POWER_BLC_RET	BLC_HV				LED_RETURN_4_FB	69 70
I792	POWER_BLC_RET	BLC_HV				LED_RETURN_5_FB	69 70
I793	POWER_BLC_RET	BLC_HV				LED_RETURN_6_FB	69 70
I810	POWER_BLC_RET	BLC_HV				LED_RETURN_7_FB	69 70
I811	POWER_BLC_RET	BLC_HV				LED_RETURN_8_FB	69 70
I812	POWER_BLC_RET	BLC_HV				LED_RETURN_9_FB	69 70
I783	POWER_BLC_RET	BLC_HV				LED_RETURN_1	70
I782	POWER_BLC_RET	BLC_HV				LED_RETURN_2	70
I784	POWER_BLC_RET	BLC_HV				LED_RETURN_3	70
I785	POWER_BLC_RET	BLC_HV				LED_RETURN_4	70
I786	POWER_BLC_RET	BLC_HV				LED_RETURN_5	70
I787	POWER_BLC_RET	BLC_HV				LED_RETURN_6	70
I806	POWER_BLC_RET	BLC_HV				LED_RETURN_7	70
I807	POWER_BLC_RET	BLC_HV				LED_RETURN_8	70
I808	POWER_BLC_RET	BLC_HV				LED_RETURN_9	70
Output Bus							
I770	POWER_BLC	BLC_HV	67V			BKLT_BOOST	69 70
I773	POWER_BLC	BLC_HV	67V			BKLT_BOOST_1	70
I774	POWER_BLC	BLC_HV	67V			BKLT_BOOST_2	70

Backlight Current Sense

	Electrical Contraint Set	Physical	Spacing	Net	Pages
I832	BACKLIGHT_GD	BLC_SNS_DIFF	SENSE	BKLT_GD1	69 70
I831	BACKLIGHT_ISEN	BLC_SNS_DIFF	SENSE	BKLT_ISEN1_Q	69 70
I833	BACKLIGHT_GD	BLC_SNS_DIFF	SENSE	BKLT_GD2	69 70
I834	BACKLIGHT_ISEN	BLC_SNS_DIFF	SENSE	BKLT_ISEN2_Q	69 70
I835	BACKLIGHT_GD	BLC_SNS_DIFF	SENSE	BKLT_GD3	69 70
I836	BACKLIGHT_ISEN	BLC_SNS_DIFF	SENSE	BKLT_ISEN3_Q	69 70
I838	BACKLIGHT_GD	BLC_SNS_DIFF	SENSE	BKLT_GD4	69 70
I837	BACKLIGHT_ISEN	BLC_SNS_DIFF	SENSE	BKLT_ISEN4_Q	69 70
I839	BACKLIGHT_GD	BLC_SNS_DIFF	SENSE	BKLT_GD5	69 70
I840	BACKLIGHT_ISEN	BLC_SNS_DIFF	SENSE	BKLT_ISEN5_Q	69 70
I841	BACKLIGHT_GD	BLC_SNS_DIFF	SENSE	BKLT_GD6	69 70
I842	BACKLIGHT_ISEN	BLC_SNS_DIFF	SENSE	BKLT_ISEN6_Q	69 70
I843	BACKLIGHT_GD	BLC_SNS_DIFF	SENSE	BKLT_GD7	69 70
I844	BACKLIGHT_ISEN	BLC_SNS_DIFF	SENSE	BKLT_ISEN7_Q	69 70
I845	BACKLIGHT_GD	BLC_SNS_DIFF	SENSE	BKLT_GD8	69 70
I846	BACKLIGHT_ISEN	BLC_SNS_DIFF	SENSE	BKLT_ISEN8_Q	69 70
I847	BACKLIGHT_GD	BLC_SNS_DIFF	SENSE	BKLT_GD9	69 70
I848	BACKLIGHT_ISEN	BLC_SNS_DIFF	SENSE	BKLT_ISEN9_Q	69 70
I850		SNS_DIFF_PHY	SENSE	BKLT_SNSP	69
I849		SNS_DIFF_PHY	SENSE	BKLT_SNSN	69
I851		SNS_DIFF_PHY	SENSE	BKLT_SNSP_R	69
I852		SNS_DIFF_PHY	SENSE	BKLT_SNSN_R	69

Cello Miscellaneous

Electrical Contraint Set		Physical	Spacing	Net	Pages
SPI					
I564		SMB_PHY	SMB	BKLT_SCL	
I563		SMB_PHY	SMB	BKLT_SDA	
I814		SMB_PHY	SMB	I2C_DBG_BKLT_SDA_R	
I815		SMB_PHY	SMB	I2C_DBG_BKLT_SCL_R	
CONTROL					
I817		BLC_MISC_PHY	BLC_CTL	BLC_LSYNC	42 69
I816		BLC_MISC_PHY	BLC_CTL	BLC_LSYNC_R	69
I826		BLC_MISC_PHY	BLC_CTL	BLC_VSYNC	42 69
I829		BLC_MISC_PHY	BLC_CTL	BLC_VSYNC_R	69
I818		BLC_MISC_PHY	BLC_CTL	TCON_BLC_EN	42 69
I819		BLC_MISC_PHY	BLC_CTL	TCON_BLC_EN_R	
I827		BLC_MISC_PHY	BLC_CTL	TCON_BLC_EN_LED	3 42
I820		BLC_MISC_PHY	BLC_CTL	BLC_EN	42
I823		BLC_MISC_PHY	BLC_CTL	DP_INTPNL_ML_HPD	42 43 69
I828		BLC_MISC_PHY	BLC_CTL	DP_INTPNL_ML_HPD_R	69
I824		BLC_MISC_PHY	BLC_CTL	DP_INT_EG_SL_HPD	42 84
I825		BLC_MISC_PHY	BLC_CTL	DP_INT_EG_ML_HPD	43 84

GPU CORE PHASES

Electrical Contraint Set	Physical	Spacing	Voltage	DIDT	NO_TEST	Net	Pages
Input Bus	POWER	POWER	12V			PP12V_S0_GPUCORE_FLT	87 88 89
	POWER	POWER	5V			PP5V_S0_GPUCORE_VCC	
Local Ground	GND	GND	0V			AGND_GPU	
	POWER	POWER	1V			PPGPUCORE_S0_SENSE_1	88
	POWER	POWER	1V			PPGPUCORE_S0_SENSE_2	88
	POWER	POWER	1V			PPGPUCORE_S0_SENSE_3	88
	POWER	POWER	1V			PPGPUCORE_S0_SENSE_4	89
	POWER	POWER	1V			PPGPUCORE_S0_SENSE_5	89
	POWER	POWER	1V			PPGPUCORE_S0_SENSE_6	89
	VR_CTL_PHY	VR_CTL				REG_PWM_GPUCORE_1	87 88
	VR_CTL_PHY	VR_CTL				REG_PWM_GPUCORE_2	87 88
	VR_CTL_PHY	VR_CTL				REG_PWM_GPUCORE_3	87 88
	VR_CTL_PHY	VR_CTL				REG_PWM_GPUCORE_4	87 89
	VR_CTL_PHY	VR_CTL				REG_PWM_GPUCORE_5	87 89
	VR_CTL_PHY	VR_CTL				REG_PWM_GPUCORE_6	87 89
	VR_CTL_PHY	VR_CTL				VCR_TD_1	
	VR_CTL_PHY	VR_CTL				VCR_TD_2	
	VR_CTL_PHY	VR_CTL				VCR_TD_3	
	VR_CTL_PHY	VR_CTL				VCR_TD_4	
	VR_CTL_PHY	VR_CTL				VCR_TD_5	
	VR_CTL_PHY	VR_CTL				VCR_TD_6	
	VR_CTL_PHY	VR_CTL				VR_GPU_PWM1_R	
	VR_CTL_PHY	VR_CTL				VR_GPU_PWM2_R	
	VR_CTL_PHY	VR_CTL				VR_GPU_PWM3_R	
	VR_CTL_PHY	VR_CTL				VR_GPU_PWM4_R	
	VR_CTL_PHY	VR_CTL				VR_GPU_PWM5_R	
	VR_CTL_PHY	VR_CTL				VR_GPU_PWM6_R	
	VR_CTL_PHY	VR_SWITCH	12V	TRUE	TRUE	REG_SNUBBER_GPUCORE_1	88
	VR_CTL_PHY	VR_SWITCH	12V	TRUE	TRUE	REG_SNUBBER_GPUCORE_2	88
	VR_CTL_PHY	VR_SWITCH	12V	TRUE	TRUE	REG_SNUBBER_GPUCORE_3	88
	VR_CTL_PHY	VR_SWITCH	12V	TRUE	TRUE	REG_SNUBBER_GPUCORE_4	89
	VR_CTL_PHY	VR_SWITCH	12V	TRUE	TRUE	REG_SNUBBER_GPUCORE_5	89
	VR_CTL_PHY	VR_SWITCH	12V	TRUE	TRUE	REG_SNUBBER_GPUCORE_6	89
	VR_CTL_PHY	VR_SWITCH	12V	TRUE		REG_UGATE_GPUCORE_1	
	VR_CTL_PHY	VR_SWITCH	12V	TRUE		REG_UGATE_GPUCORE_2	
	VR_CTL_PHY	VR_SWITCH	12V	TRUE		REG_UGATE_GPUCORE_3	
	VR_CTL_PHY	VR_SWITCH	12V	TRUE		REG_UGATE_GPUCORE_4	
	VR_CTL_PHY	VR_SWITCH	12V	TRUE		REG_UGATE_GPUCORE_5	
	VR_CTL_PHY	VR_SWITCH	12V	TRUE		REG_UGATE_GPUCORE_6	
	VR_CTL_PHY	VR_SWITCH	12V	TRUE		REG_BOOT_GPUCORE_1	
	VR_CTL_PHY	VR_SWITCH	12V	TRUE	TRUE	REG_BOOT_GPUCORE_1_RC	
	VR_CTL_PHY	VR_SWITCH	12V	TRUE	TRUE	REG_BOOT_GPUCORE_2	
	VR_CTL_PHY	VR_SWITCH	12V	TRUE	TRUE	REG_BOOT_GPUCORE_2_RC	
	VR_CTL_PHY	VR_SWITCH	12V	TRUE	TRUE	REG_BOOT_GPUCORE_3	
	VR_CTL_PHY	VR_SWITCH	12V	TRUE	TRUE	REG_BOOT_GPUCORE_3_RC	
	VR_CTL_PHY	VR_SWITCH	12V	TRUE	TRUE	REG_BOOT_GPUCORE_4	
	VR_CTL_PHY	VR_SWITCH	12V	TRUE	TRUE	REG_BOOT_GPUCORE_4_RC	
	VR_CTL_PHY	VR_SWITCH	12V	TRUE	TRUE	REG_BOOT_GPUCORE_5	
	VR_CTL_PHY	VR_SWITCH	12V	TRUE	TRUE	REG_BOOT_GPUCORE_5_RC	
	VR_CTL_PHY	VR_SWITCH	12V	TRUE	TRUE	REG_BOOT_GPUCORE_6	
	VR_CTL_PHY	VR_SWITCH	12V	TRUE	TRUE	REG_BOOT_GPUCORE_6_RC	
	VR_CTL_PHY	VR_SWITCH	12V	TRUE		REG_LGATE_GPUCORE_1	
	VR_CTL_PHY	VR_SWITCH	12V	TRUE		REG_LGATE_GPUCORE_2	
	VR_CTL_PHY	VR_SWITCH	12V	TRUE		REG_LGATE_GPUCORE_3	
	VR_CTL_PHY	VR_SWITCH	12V	TRUE		REG_LGATE_GPUCORE_4	
	VR_CTL_PHY	VR_SWITCH	12V	TRUE		REG_LGATE_GPUCORE_5	
	VR_CTL_PHY	VR_SWITCH	12V	TRUE		REG_LGATE_GPUCORE_6	
	VR_CTL_PHY	VR_SWITCH	12V	TRUE		REG_PHASE_GPUCORE_1	
	VR_CTL_PHY	VR_SWITCH	12V	TRUE		REG_PHASE_GPUCORE_2	
	VR_CTL_PHY	VR_SWITCH	12V	TRUE		REG_PHASE_GPUCORE_3	
	VR_CTL_PHY	VR_SWITCH	12V	TRUE		REG_PHASE_GPUCORE_4	
	VR_CTL_PHY	VR_SWITCH	12V	TRUE		REG_PHASE_GPUCORE_5	
	VR_CTL_PHY	VR_SWITCH	12V	TRUE		REG_PHASE_GPUCORE_6	
	VR_CTL_PHY	VR_SWITCH	12V	TRUE		VR_PHASE_GPUCORE_1	88
	VR_CTL_PHY	VR_SWITCH	12V	TRUE		VR_PHASE_GPUCORE_2	88
	VR_CTL_PHY	VR_SWITCH	12V	TRUE		VR_PHASE_GPUCORE_3	88
	VR_CTL_PHY	VR_SWITCH	12V	TRUE		VR_PHASE_GPUCORE_4	89
	VR_CTL_PHY	VR_SWITCH	12V	TRUE		VR_PHASE_GPUCORE_5	89
	VR_CTL_PHY	VR_SWITCH	12V	TRUE		VR_PHASE_GPUCORE_6	89
ISNS_GPU_CORE	SNS_DIFF_PHY	SENSE				REG_GPUCORE_ISNS_1_N	87 88
ISNS_GPU_CORE	SNS_DIFF_PHY	SENSE				REG_GPUCORE_ISNS_1_P	87 88
ISNS_GPU_CORE	SNS_DIFF_PHY	SENSE				REG_GPUCORE_ISNS_2_N	87 88
ISNS_GPU_CORE	SNS_DIFF_PHY	SENSE				REG_GPUCORE_ISNS_2_P	87 88
ISNS_GPU_CORE	SNS_DIFF_PHY	SENSE				REG_GPUCORE_ISNS_3_N	87 88
ISNS_GPU_CORE	SNS_DIFF_PHY	SENSE				REG_GPUCORE_ISNS_3_P	87 88
ISNS_GPU_CORE	SNS_DIFF_PHY	SENSE				REG_GPUCORE_ISNS_4_N	87 89
ISNS_GPU_CORE	SNS_DIFF_PHY	SENSE				REG_GPUCORE_ISNS_4_P	87 89
ISNS_GPU_CORE	SNS_DIFF_PHY	SENSE				REG_GPUCORE_ISNS_5_N	87 89
ISNS_GPU_CORE	SNS_DIFF_PHY	SENSE				REG_GPUCORE_ISNS_5_P	87 89
ISNS_GPU_CORE	SNS_DIFF_PHY	SENSE				REG_GPUCORE_ISNS_6_N	87 89
ISNS_GPU_CORE	SNS_DIFF_PHY	SENSE				REG_GPUCORE_ISNS_6_P	87 89
ISNS_GPU_CORE		SENSE				VR_GPU_ISNS1_RR_2	
ISNS_GPU_CORE	SNS_DIFF_PHY	SENSE				VR_GPU_ISNS1_R_N	
ISNS_GPU_CORE	SNS_DIFF_PHY	SENSE				VR_GPU_ISNS1_R_P	
ISNS_GPU_CORE		SENSE				VR_GPU_ISNS2_RR_2	
ISNS_GPU_CORE	SNS_DIFF_PHY	SENSE				VR_GPU_ISNS2_R_N	
ISNS_GPU_CORE	SNS_DIFF_PHY	SENSE				VR_GPU_ISNS2_R_P	
ISNS_GPU_CORE		SENSE				VR_GPU_ISNS3_RR_2	
ISNS_GPU_CORE	SNS_DIFF_PHY	SENSE				VR_GPU_ISNS3_R_N	
ISNS_GPU_CORE	SNS_DIFF_PHY	SENSE				VR_GPU_ISNS3_R_P	
ISNS_GPU_CORE		SENSE				VR_GPU_ISNS4_RR_2	
ISNS_GPU_CORE	SNS_DIFF_PHY	SENSE				VR_GPU_ISNS4_R_N	
ISNS_GPU_CORE	SNS_DIFF_PHY	SENSE				VR_GPU_ISNS4_R_P	
ISNS_GPU_CORE		SENSE				VR_GPU_ISNS5_RR_2	
ISNS_GPU_CORE	SNS_DIFF_PHY	SENSE				VR_GPU_ISNS5_R_N	
ISNS_GPU_CORE	SNS_DIFF_PHY	SENSE				VR_GPU_ISNS5_R_P	
ISNS_GPU_CORE		SENSE				VR_GPU_ISNS6_RR_2	
ISNS_GPU_CORE	SNS_DIFF_PHY	SENSE				VR_GPU_ISNS6_R_N	
ISNS_GPU_CORE	SNS_DIFF_PHY	SENSE				VR_GPU_ISNS6_R_P	

GPU CORE CONTROLLER

Electrical Contraint Set	Physical	Spacing	Voltage	DIDT	NO_TEST	Net	Pages
IR3566BM							
	VR_CTL_PHY	VR_CTL				VRG_VDF_R1	
	VR_CTL_PHY	VR_CTL				VRG_VDF_R2	
	VR_CTL_PHY	VR_CTL				VR_GPU_COMP	
	VR_CTL_PHY	VR_CTL				VR_GPU_COMP_R	
	VR_CTL_PHY	VR_CTL				VR_GPU_COMP_RC	
	VR_CTL_PHY	VR_CTL				VR_GPU_DAC	
	VR_CTL_PHY	VR_CTL				VR_GPU_FAN	
	VR_CTL_PHY	VR_CTL				VR_GPU_FB	
	VR_CTL_PHY	VR_CTL				VR_GPU_FB_R	
	VR_CTL_PHY	VR_CTL				VR_GPU_FS	
	VR_CTL_PHY	VR_CTL				VR_GPU_IMON	50 51 87
	VR_CTL_PHY	VR_CTL				VR_GPU_IMON_R	87
	VR_CTL_PHY	VR_CTL				VR_GPU_IOUT_PD	
	VR_CTL_PHY	VR_CTL				VR_GPU_OFS	
	VR_CTL_PHY	VR_CTL				VR_GPU_APA	
		VR_CTL				VR_GPU_EN_VTT	
		VR_CTL				PM_EN_REG_GPUCORE_S0	72 87
		VR_CTL				PM_EN_REG_GPU_CORE_S0_R	
	VR_CTL_PHY	VR_CTL				VR_GPU_VRDHOT	
		VR_CTL				GPU_PSI_L	
		VR_CTL				GPU_PSI_L_R	
		VR_CTL				PM_PGOOD_REG_GPUCORE_S0	72 87
	VR_CTL_PHY	VR_CTL				VR_GPU_REF	
	VR_CTL_PHY	VR_CTL				VR_GPU_SS	
	VR_CTL_PHY	VR_CTL				VR_GPU_TCOMP	
	VR_CTL_PHY	VR_CTL				VR_GPU_TM	
	VR_CTL_PHY	VR_CTL				VR_GPU_VDIFF	
VR_GPU_VSEN	VR_GPU_VSEN	SENSE				VR_GPU_VSEN	87
VR_GPU_RGND	VR_GPU_VSEN	SENSE				VR_GPU_RGND	87
VR_GPU_RCSP	VR_SENSE	SENSE				VR_GPU_RCSP_R	87
VR_GPU_RCSM	VR_SENSE	SENSE				VR_GPU_RCSM_R	87
	VR_SENSE	SENSE				VR_GPU_RCSP_RT	
	VR_SENSE	SENSE				VR_GPU_RCSM_RT	
	SMB_PHY	SMB				SMB_VR_GPU_SDA_R	87
	SMB_PHY	SMB				SMB_VR_GPU_SCL_R	87
GPU SERIAL VID							
VR_GPU_SVC	VR_SVID_PHY	VR_SVID				VR_GPU_SVC	84 87
VR_GPU_SVD	VR_SVID_PHY					VR_GPU_SVD	84 87
VR_GPU_SVT	VR_SVID_PHY					VR_GPU_SVT	84 87
	VR_SVID_PHY	VR_SVID				VR_GPU_SVC_R	87
	VR_SVID_PHY					VR_GPU_SVD_R	87
	VR_SVID_PHY					VR_GPU_SVT_R	87
Output Bus	POWER	POWER	1V			PPGPUCORE_S0	94
VSNS_GPU_VDDCI	VR_SENSE	SENSE				SNS_GPU_VDDCI_P	75 91
VSNS_GPU_VDDCI	VR_SENSE	SENSE				SNS_GPU_VDDCI_N	75 91
ISNS_GPU_VDDCI	VR_SENSE	SENSE				GPU_ISNS_VDDCI_P	51 91
ISNS_GPU_VDDCI	VR_SENSE	SENSE				GPU_ISNS_VDDCI_N	51 91
VSNS_GPU_VDDCI_XW	VR_SENSE	SENSE				SNS_GPU_VDDCI_XW_P	91
VSNS_GPU_VDDCI_XW	VR_SENSE	SENSE				SNS_GPU_VDDCI_XW_N	91
VSNS_P0V95S0_XW	VR_SENSE	SENSE				SNS_P0V95S0_XW_P	92
VSNS_P0V95S0_XW	VR_SENSE	SENSE				SNS_P0V95S0_XW_N	92

GPU FBVDDQ

Electrical Contraint Set	Physical	Spacing	Voltage	DIDT	NO_TEST	Net	Pages
Input Bus	POWER	POWER	5V			REG_VCC_UA450	90
	POWER	POWER	5V			REG_PVCC_UA450	90
Local Ground	GND	GND	0V			AGND_FBVDDQ	90
FBVDDQ	POWER	VR_SWITCH	12V	TRUE		REG_PHASE_FBVDDQ	90
	POWER	VR_SWITCH	12V	TRUE		VR_PHASE_FBVDDQ	90
	VR_CTL_PHY	VR_SWITCH	12V	TRUE		REG_BOOT_FBVDDQ	90
	VR_CTL_PHY	VR_SWITCH	12V	TRUE	TRUE	REG_BOOT_FBVDDQ_RC	90
	VR_CTL_PHY	VR_SWITCH	12V	TRUE		REG_UGATE_FBVDDQ	90
	VR_CTL_PHY	VR_SWITCH	12V	TRUE		REG_LGATE_FBVDDQ	90
	VR_CTL_PHY	VR_SWITCH	12V	TRUE	TRUE	REG_SNUBBER_FBVDDQ	90
	VR_CTL_PHY	VR_CTL				REG_FBVDDQ_SET0	90
	VR_CTL_PHY	VR_CTL				REG_FBVDDQ_SET1	90
	VR_CTL_PHY	VR_CTL				REG_FBVDDQ_SET1_R	90
	VR_CTL_PHY	VR_CTL				REG_FBVDDQ_SREF	90
		VR_CTL				GPU_FBVDDQ_ALTV	84 90
		VR_CTL				FBVDDQ_ALTV0_R	90
		VR_CTL				FBVDDQ_ALTV1_R	90
		VR_CTL				PM_EN_REG_GPU_VDDQ_S0	72 90
		VR_CTL				PM_EN_REG_GPU_VDDQ_S0_R	
		VR_CTL				PM_PGOOD_REG_GPU_VDDQ_S0	72 90
	VR_CTL_PHY	VR_CTL				REG_FBVDDQ_OCSET	90
	VR_CTL_PHY	VR_CTL				REG_FBVDDQ_VO	90
	VR_CTL_PHY	VR_CTL				REG_FBVDDQ_FB	90
	VR_CTL_PHY	VR_CTL				REG_FBVDDQ_RTN	90
	VR_CTL_PHY	VR_CTL				REG_FBVDDQ_FB_R	90
	VR_CTL_PHY	VR_CTL				REG_FBVDDQ_RTN_R	90
Output Bus	POWER	POWER	1.5V			PPFBVDDQ_S0_GPU	94
	POWER	POWER	1.5V			VRDDQ_R	

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GPU VDDCI

Electrical Contraint Set	Physical	Spacing	Voltage	DIDT	NO_TEST		
Input Bus							
I145	POWER	POWER	5V			REG_VCC_UA550	91
I145	POWER	POWER	5V			REG_PVCC_UA550	91
Local Ground							
I145	GND	GND	0V			AGND_GPU_VDDCI	91
GPU VDDCI							
I145	POWER	VR_SWITCH	12V	TRUE		REG_PHASE_GPU_VDDCI	91
I151	POWER	VR_SWITCH	12V	TRUE		VR_PHASE_GPU_VDDCI	91
I145	VR_CTL_PHY	VR_SWITCH	12V	TRUE		REG_BOOT_GPU_VDDCI	91
I145	VR_CTL_PHY	VR_SWITCH	12V	TRUE	TRUE	REG_BOOT_GPU_VDDCI_RC	91
I145	VR_CTL_PHY	VR_SWITCH	12V	TRUE		REG_UGATE_GPU_VDDCI	91
I146	VR_CTL_PHY	VR_SWITCH	12V	TRUE		REG_LGATE_GPU_VDDCI	91
I145	VR_CTL_PHY	VR_SWITCH	12V	TRUE	TRUE	REG_SNUBBER_GPU_VDDCI	91
I151	VR_CTL_PHY	VR_CTL				REG_GPU_VDDCI_SET0	91
I151	VR_CTL_PHY	VR_CTL				REG_GPU_VDDCI_SET1	91
I150	VR_CTL_PHY	VR_CTL				REG_GPU_VDDCI_SET1_R	91
I150	VR_CTL_PHY	VR_CTL				REG_GPU_VDDCI_SREF	91
I146	VR_CTL_PHY	VR_CTL				REG_GPU_VDDCI_OCSET	91
I146	VR_CTL_PHY	VR_CTL				REG_GPU_VDDCI_VO	91
I146	VR_CTL_PHY	VR_CTL				REG_GPU_VDDCI_FB	91
I146	VR_CTL_PHY	VR_CTL				REG_GPU_VDDCI_RTN	91
I154		VR_CTL				PM_EN_GPU_VDDCI_S0	72 91
I152		VR_CTL				PM_EN_GPU_VDDCI_S0_R	91
I152		VR_CTL				PM_PGOOD_REG_GPU_VDDCI_S0	72 91
I153		VR_CTL				GPU_VDDCI_ALTV	84 91
I153		VR_CTL				GPU_VDDCI_ALTV0	91
I153		VR_CTL				GPU_VDDCI_ALTV1	91
I153	POWER	POWER	0.85V			VRVDDCI_R	91

GPU VDDC (0.95V)

Electrical Contraint Set	Physical	Spacing	Voltage	DIDT	NO_TEST		
Input Bus							
I149	POWER	POWER	5V			REG_VCC_UA660	92
I149	POWER	POWER	5V			REG_PVCC_UA660	92
Local Ground							
I149	GND	GND	0V			AGND_P0V95S0	92
GPU VDDC							
I149	POWER	VR_SWITCH	12V	TRUE		REG_PHASE_P0V95S0	92
I149	VR_CTL_PHY	VR_SWITCH	12V	TRUE		REG_BOOT_P0V95S0	92
I150	VR_CTL_PHY	VR_SWITCH	12V	TRUE	TRUE	REG_BOOT_P0V95S0_RC	92
I150	VR_CTL_PHY	VR_SWITCH	12V	TRUE		REG_UGATE_P0V95S0	92
I150	VR_CTL_PHY	VR_SWITCH	12V	TRUE		REG_LGATE_P0V95S0	92
I150	VR_CTL_PHY	VR_SWITCH	12V	TRUE	TRUE	REG_SNUBBER_P0V95S0	92
I150	VR_CTL_PHY	VR_CTL				REG_P0V95S0_OCSET	92
I150	VR_CTL_PHY	VR_CTL				REG_P0V95S0_VO	92
I150	VR_CTL_PHY	VR_CTL				REG_P0V95S0_FB	92
I150	VR_CTL_PHY	VR_CTL				REG_P0V95S0_RTN	92
I151	VR_CTL_PHY	VR_CTL				REG_P0V95S0_SREF	92
I154		VR_CTL				PM_EN_REG_P0V95_S0	72 92
I153		VR_CTL				PM_EN_REG_P0V95_S0_R	92
I153		VR_CTL				PM_PGOOD_REG_P0V95_S0	72 92

GPU VDD (1.8V)

Electrical Contraint Set	Physical	Spacing	Voltage	DIDT	NO_TEST		
GPU VDD							
I151	POWER	VR_SWITCH	12V	TRUE		REG_PHASE_PVDDR	93
I151	VR_CTL_PHY	VR_SWITCH	12V	TRUE	TRUE	REG_BOOT_PVDDR	93
I154	VR_CTL_PHY	VR_SWITCH	12V	TRUE	TRUE	REG_BOOT_PVDDR_R	93
I153	GND	GND	0V			AGND_PVDDR	93
I152	VR_CTL_PHY	VR_CTL				REG_VSNSR_PVDDR	93
I152	VR_CTL_PHY	VR_CTL				REG_VSNS_PVDDR	93
I152	VR_CTL_PHY	VR_CTL				REG_SSTR_PVDDR	93
I152	VR_CTL_PHY	VR_CTL				REG_COMP_PVDDR	93
I153		VR_CTL				PM_EN_REG_PVDDCT_S0	72 93
I153		VR_CTL				PM_EN_REG_PVDDCT_S0_R	93
I153		VR_CTL				PM_PGOOD_REG_PVDDCT_S0	72 93

GPU CORE (CONT.)

Electrical Contraint Set	Physical	Spacing	Voltage	DIDT	NO_TEST	
I154 ISNS_GPU_CORE	SNS_DIFF_PHY	SENSE				REG_GPUCORE_ISNS_1_XW_N
I154 ISNS_GPU_CORE	SNS_DIFF_PHY	SENSE				REG_GPUCORE_ISNS_1_XW_P
I154 ISNS_GPU_CORE	SNS_DIFF_PHY	SENSE				REG_GPUCORE_ISNS_2_XW_N
I154 ISNS_GPU_CORE	SNS_DIFF_PHY	SENSE				REG_GPUCORE_ISNS_2_XW_P
I154 ISNS_GPU_CORE	SNS_DIFF_PHY	SENSE				REG_GPUCORE_ISNS_3_XW_N
I155 ISNS_GPU_CORE	SNS_DIFF_PHY	SENSE				REG_GPUCORE_ISNS_3_XW_P
I155 ISNS_GPU_CORE	SNS_DIFF_PHY	SENSE				REG_GPUCORE_ISNS_4_XW_N
I154 ISNS_GPU_CORE	SNS_DIFF_PHY	SENSE				REG_GPUCORE_ISNS_4_XW_P
I155 ISNS_GPU_CORE	SNS_DIFF_PHY	SENSE				REG_GPUCORE_ISNS_5_XW_N
I155 ISNS_GPU_CORE	SNS_DIFF_PHY	SENSE				REG_GPUCORE_ISNS_5_XW_P
I155 ISNS_GPU_CORE	SNS_DIFF_PHY	SENSE				REG_GPUCORE_ISNS_6_XW_N
I155 ISNS_GPU_CORE	SNS_DIFF_PHY	SENSE				REG_GPUCORE_ISNS_6_XW_P

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4V5_*

	Physical	Spacing	Netname	
I2	POWER	POWER	4V5_REG_IN	54

ENET_*

	Physical	Spacing	Netname	
I104	XDP_PHY	XDP	ENET_LOW_PWR	20 37 39

GPU_*

	Physical	Spacing	Netname	
I105	SPI_50S	SPI	GPU_ROM_CS_L	84
I106	SPI_50S	SPI	GPU_ROM_CS_L_R	84 85
I107	SPI_50S	SPI	GPU_ROM_SI	84
I108	SPI_50S	SPI	GPU_ROM_SI_R	84 85
I109	SPI_50S	SPI	GPU_ROM_SO	84 85
I111	SPI_50S	SPI	GPU_ROM_SO_R	85
I110	SPI_50S	SPI	GPU_ROM_WP_L	85
I176	SPI_50S	SPI	GPU_ROM_SCLK	
I177	SPI_50S	SPI	GPU_ROM_SCLK_R	

JTAG_*

	Physical	Spacing	Netname	
I117	XDP_PHY	XDP	JTAG_TBT_TDI	20 28
I116	XDP_PHY	XDP	JTAG_TBT_TDO	20 28
I118	XDP_PHY	XDP	JTAG_TBT_TMS	20 28

ISNS_*

Physical	Spacing	Netname

SMC_*

	Physical	Spacing	Netname	
I134	CLK_XTAL	XTAL	SMC_XTAL_R	47

REG_*

Physical	Spacing	Netname

SMBUS_*

	Physical	Spacing	Netname	
I128	SMB_PHY	SMB	SMBUS_SMC_4_ASF_SCL	46 47
I129	SMB_PHY	SMB	SMBUS_SMC_4_ASF_SDA	46 47

SMC_*

	Physical	Spacing	Netname	
I130		SENSE	SMC_OOB1_RX_CN	36
I131		SENSE	SMC_OOB1_RX_FILT	36

TSNS_*

	Electrical	Physical	Spacing	Netname	
I137	SNS_TEMP	SNS_DIFF_PHY	SENSE	TSNS_1_1_N	52
I138	SNS_TEMP	SNS_DIFF_PHY	SENSE	TSNS_1_1_P	52
I139	SNS_TEMP	SNS_DIFF_PHY	SENSE	TSNS_1_2_N	52
I140	SNS_TEMP	SNS_DIFF_PHY	SENSE	TSNS_1_2_P	52
I141	SNS_TEMP	SNS_DIFF_PHY	SENSE	TSNS_1_3_N	52
I142	SNS_TEMP	SNS_DIFF_PHY	SENSE	TSNS_1_3_P	52
I143	SNS_TEMP	SNS_DIFF_PHY	SENSE	TSNS_2_1_N	52
I144	SNS_TEMP	SNS_DIFF_PHY	SENSE	TSNS_2_1_P	52
I145	SNS_TEMP	SNS_DIFF_PHY	SENSE	TSNS_2_2_N	52
I146	SNS_TEMP	SNS_DIFF_PHY	SENSE	TSNS_2_2_P	52
I147	SNS_TEMP	SNS_DIFF_PHY	SENSE	TSNS_2_3_N	52
I148	SNS_TEMP	SNS_DIFF_PHY	SENSE	TSNS_2_3_P	52
I149	SNS_TEMP	SNS_DIFF_PHY	SENSE	TSNS_2_4_N	52
I150	SNS_TEMP	SNS_DIFF_PHY	SENSE	TSNS_2_4_P	52
I151	SNS_TEMP	SNS_DIFF_PHY	SENSE	TSNS_2_5_N	52
I152	SNS_TEMP	SNS_DIFF_PHY	SENSE	TSNS_2_5_P	52
I153	SNS_TEMP	SNS_DIFF_PHY	SENSE	TSNS_2_6_N	52
I154	SNS_TEMP	SNS_DIFF_PHY	SENSE	TSNS_2_6_P	52
I155	SNS_TEMP	SNS_DIFF_PHY	SENSE	TSNS_2_7_N	52
I156	SNS_TEMP	SNS_DIFF_PHY	SENSE	TSNS_2_7_P	52

BLC_*

	Physical	Spacing	Voltage	Netname
I32	BLC_CTL_PHY	BLC_CTL	5V	BLC_MCU_AOUT_R
I36	SMB_PHY	SMB		BLC_MCU_B_SDA_CONN
I43	XDP_PHY	CLK_JTAG		BLC_MCU_RTCK
I45	XDP_PHY	XDP		BLC_MCU_TCK
I46	XDP_PHY	XDP		BLC_MCU_TDI
I47	XDP_PHY	XDP		BLC_MCU_TDO
I48	XDP_PHY	XDP		BLC_MCU_TMS
I49	XDP_PHY	XDP		BLC_MCU_TRST
I50	XDP_PHY	XDP		BLC_MCU_TXD0

TBT_*

	Physical	Spacing	Voltage	Netname	
I136	TBT_I2C_55S	TBT_I2C		TBT_A_CONFIG1_RC	31
I135	TBT_I2C_55S	TBT_I2C		TBT_B_CONFIG1_RC	32

AYNCHRONOUS

	Electrical	Physical	Spacing	Netname	
I159		CPU_ASYNC_PHY	CPU_ASYNC	CPU_PROCHOT_L	6 46 47 63
I160		CPU_ASYNC_PHY	CPU_ASYNC	CPU_PROCHOT_R_L	6
I162		CPU_ASYNC_PHY	CPU_ASYNC	SMC_PECI_L	46 47
I163		CPU_ASYNC_PHY	CPU_ASYNC	CPU_CATERR_L	6 47
I164		CPU_ASYNC_PHY	CPU_ASYNC	CPU_PWRGD	6 14 18
I165		CPU_ASYNC_PHY	CPU_ASYNC	PM_SYNC	6 12
I166		CPU_ASYNC_PHY	CPU_ASYNC	PM_THRMTRIP_L	14 47
I167		CPU_ASYNC_PHY	CPU_ASYNC	CPU_RESET_L	6 14 18
I169		CPU_ASYNC_PHY	CPU_ASYNC	SMC_CPU_CATERR_L	46 47
I170		CPU_ASYNC_PHY	CPU_ASYNC	PCH_PROCPWRGD	14
I171		PCH_ASYNC_PHY	PCH_ASYNC	GPU_GOOD	3 14 18

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
CPU_ASYNC_ISO	*	=3:1_SPACING	?
PCH_ASYNC_ISO	*	=3:1_SPACING	?

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
CPU_ASYNC	*	*	CPU_ASYNC_ISO
PCH_ASYNC	*	*	PCH_ASYNC_ISO

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
CPU_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=STANDARD	=STANDARD
PCH_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=STANDARD	=STANDARD

NET_PHYSICAL_TYPE	AREA_TYPE	PHYSICAL_RULE_SET
CPU_ASYNC_PHY	*	CPU_50S
PCH_ASYNC_PHY	*	PCH_50S

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